Self-Assessment in Adult Multiprofessional Critical Care

Ninth Edition

Society of Critical Care Medicine
The Intensive Care Proteeeienals

ANSWER SHEET

Instructions: Write the correct answer on the line provided.

Part 1 Renal

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Part 2 Electrolytes

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Part 3 Acid-Base Disorders

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Part 4 Endocrine

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Part 5 Cardiovascular Disorders

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Part 6 Pulmonary Disease

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Part 7 Infectious Disease

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Part 8 Gastrointestinal Disorders

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Part 9 Liver

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Part 10 Neurologic Disorders

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Part 11 Hematologic Disorder

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Part 12 Oncology

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Part 13 Surgery and Trauma

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Part 14 Transplantation

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Part 15 Environmental Injury and Disaster Management

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Part 16 Pharmacology and Toxicology

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Part 17 Obstetrics

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Part 18 Research, Administration, and Ethics

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Part 19 Critical Care Ultrasound

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Part 20 Epidemiology/Outcomes

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Contents

Questions

Part 1: Renal

Part 2: Electrolytes

Part 3: Acid-Base Disorders

Part 4: Endocrine

Part 5: Cardiovascular Disorders

Part 6: Pulmonary Disease

Part 7: Infectious Disease

Part 8: Gastrointestinal Disorders

Part 9: Liver

Part 10: Neurologic Disorders

Part 11: Hematologic Disorders

Part 12: Oncology

Part 13: Surgery and Trauma

Part 14: Transplantation

Part 15: Environmental Injury and Disaster Management

Part 16: Pharmacology and Toxicology

Part 17: Obstetrics

Part 18: Research, Administration, and Ethics

Part 19: Critical Care Ultrasound

Part 20: Epidemiology/Outcomes

Answers

Part 1: Renal

Part 2: Electrolytes

Part 3: Acid-Base Disorders

Part 4: Endocrine

Part 5: Cardiovascular Disorders

Part 6: Pulmonary Disease

Part 7: Infectious Disease

Part 8: Gastrointestinal Disorders

Part 9: Liver

Part 10: Neurologic Disorders

Part 11: Hematologic Disorders

Part 12: Oncology

Part 13: Surgery and Trauma

Part 14: Transplantation

Part 15: Environmental Injury and Disaster Management

Part 16: Pharmacology and Toxicology

Part 17: Obstetrics

Part 18: Research, Administration, and Ethics

Part 19: Critical Care Ultrasound

Part 20: Epidemiology/Outcomes

Part 1.

Renal

Instructions: For each question, select the most correct answer.

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1. A 55-year-old man who previously underwent aortic valve replacement is admitted with crushing chest pain and dyspnea. Cardiac catheterization reveals diffuse coronary artery disease, so he is scheduled for coronary artery bypass surgery. Past medical history is notable for active cigarette smoking, hypertension, and hyperlipidemia. Which of the following is a risk factor for the development of postoperative acute kidney insufficiency?

   A. Previous cardiac surgery

   B. Male gender

   C. Tobacco smoking

   D. Hyperlipidemia

2. A 63-year-old woman with a past medical history significant for cirrhosis, chronic back pain, and gastroesophageal reflux disease is admitted to the ICU with new-onset fever, abdominal pain, malaise, and hyperkalemia. Ten days ago, she was treated with trimethoprim/sulfamethoxazole for a urinary tract infection. Because of her abdominal pain, an abdominal CT with IV iodine radiocontrast was performed in the emergency department, which was negative for acute pathology. Her home medications include celecoxib and ranitidine. She is hemodynamically stable. Admission laboratory tests are significant for serum creatinine 2.5 mg/dL (baseline 0.8 mg/dL), BUN 32 mg/dL, serum sodium 137 mmol/L, and serum potassium 6.5 mmol/L.

Urinalysis reveals osmolality 283 mOsm/L, pH 6.0, glucose 2 mg/dL, predicted 24-hour protein 1 g/day, and microscopy WBC casts. Which of the following is the most likely etiology of her acute kidney injury?

A. Acute tubular necrosis

B. Acute interstitial nephritis

C. Hepatorenal syndrome

D. Acute contrast nephropathy

E. Obstructive nephropathy

3. A 49-year-old woman is admitted to the ICU with gross hematuria and hypotension. Within the past year she has had 4 episodes of gross hematuria, each lasting 1 to 3 days. The episodes do not coincide with menstruation. For each episode, she was treated with a course of antibiotics for a presumed urinary tract infection. She has no family history of kidney disease. On examination, she is afebrile and diaphoretic. Blood pressure is 74/46 mm Hg, heart rate 122 beats/min, and weight 68 kg (150 lb.). There is bilateral, mild, lower extremity pitting edema; the remainder of the physical examination is normal. After resuscitation with crystalloids, blood pressure is 105/62 mm Hg and heart rate 90 beats/min. Current laboratory results are: hemoglobin 11.8 g/dL, platelets 159,000/mm^3, and creatinine 3.6 mg/dL. Urine dipstick shows 3+ protein and 3+ blood. Urine sediment shows 51-100 RBCs/high-power field (reference range 0-2/high-power field). Ultrasound reveals normal-sized kidneys; there is no hydronephrosis. Which of the following is the most likely origin of the hematuria?

A. Glomeruli

B. Ureter

C. Bladder

D. Urethra

4. Which of the following is a characteristic of a drug that is likely to be effectively removed by continuous renal replacement therapy?

   A. Minimal protein binding

   B. Large volume of distribution

   C. Electronegative charge

   D. Large molecular weight

5. A 66-year-old, 102-kg (224-lb) man was evaluated in the emergency department for head trauma and a broken hand after a bicycling accident. He was transferred to the neuro-ICU for monitoring, where the nurse reports polyuria and increased thirst. Urine output for the most recent 8-hour shift was 2500 mL, and he did not receive diuretics. Past medical history includes hypertension. Laboratory results include sodium 156 mEq/L (from 151 mEq/L), potassium 3.2 mEq/L, BUN 18 mg/dL, creatinine 0.6 mg/dL, glucose 194 mg/dL, and urine specific gravity 1.010. Which of the following is the most appropriate treatment for this patient?

   A. Desmopressin

   B. Sterile water for IV bolus

   C. Furosemide

   D. Fludrocortisone

6. A 19-year-old man sustained a gunshot wound to the head approximately 4 days ago. Morning laboratory results are sodium 125 mEq/L (135 mEq/L yesterday), chloride 95 mEq/L (106 mEq/L yesterday), potassium 3 mEq/L, bicarbonate 20 mEq/L, BUN 56 mg/dL, serum creatinine 0.8 mg/dL, and glucose 185 mg/dL. Urinalysis shows sodium 55 mEq/L and osmolality 120 mOsm/kg. Serum osmolality is 280 mmol/kg.

Central venous pressure is 4 cm H_2O, and stroke volume variation is 16%. Heart rate is 120 beats/min, and systolic blood pressure is 95 mm Hg. Fluid balance is −1 L for the 24-hour period despite enteral nutrition at goal 50 mL/hr and no maintenance fluids. Which of the following diagnoses is most likely responsible for his hyponatremia?

A. Syndrome of inappropriate antidiuretic hormone secretion

B. Adrenal insufficiency

C. Diabetes insipidus

D. Cerebral salt wasting

7. A 60-year-old man in the ICU receiving twice-daily furosemide has developed oliguria. His serum creatinine level is 2.1 mg/dL. Serum and urine chemistries are pending. Which of the following calculations most accurately identifies prerenal azotemia as the etiology of oliguria in this patient?

A. Fractional excretion of sodium (FENa) of 1.1%

B. FENa of 2.0%

C. Fractional excretion of urea (FEU) of 30%

D. FEU of 55%

E. Urine sodium concentration of 22 mEq/L

8. A 38-year-old man with no past medical history was found unconscious after sustaining minor injuries in a traumatic fall the night before. Vital signs are heart rate 90 beats/min, blood pressure 203/101 mm Hg, oxygen saturation as measured by pulse oximetry 95% on FIO_2 1.0 via bag-valve-mask. He vomits and is intubated to secure his airway. Scans of head, chest, abdomen, and pelvis reveal no acute processes. Urinalysis reveals calcium oxalate crystals. Remarkable laboratory results include bicarbonate 5 mEq/L,

arterial pH 6.8, serum osmolality 366 mOsm/L, serum sodium 140 mEq/L, potassium 6.5 mEq/L, creatinine 3.18 mg/dL, glucose 209 mg/dL, and BUN 25 mg/dL. This patient's symptoms are most likely caused by ingestion of

A. methanol.

B. isopropyl alcohol.

C. ethylene glycol.

D. ethanol.

9. A 46-year-old man is admitted to the ICU with septic shock and respiratory failure. He is on mechanical ventilation with 40% FIO_2 and low tidal volume protocol (6 mL/kg). Despite adequate fluid resuscitation, he is receiving vasopressors. Off sedation, he wakes up and can follow simple commands. Urine output is 4 mL/kg/hr, BUN 60 mg/dL, creatinine 2 mg/dL, and potassium 4.3 mmol/L. On current ventilator settings, pH is 7.34, $PaCo_2$ 44 mm Hg, and PaO_2 88 mm Hg. Nephrology has been consulted for acute renal failure. The most appropriate next step is

   A. IV bicarbonate infusion to correct acidosis.

   B. catheter placement to start continuous dialysis.

   C. renal angiogram to evaluate kidney blood flow.

   D. continuation of current care while monitoring renal function.

Part 1 Answers:

Renal

1. Rationale

Answer: A

Acute kidney insufficiency is relatively common following cardiac surgery. It has been reported in 20% to 70% of patients during the postoperative period. Cardiopulmonary bypass, large volume infusions, blood transfusions, exogenous vasopressor use, ischemia-reperfusion injury, oxidative stress, and aortic cross-clamping all contribute to higher rates of acute kidney insufficiency among cardiac surgical patients than among the general surgical population. Risk factors for acute kidney insufficiency after cardiac surgery include female gender, congestive heart failure, left ventricular ejection fraction below 35%, preoperative intra-aortic balloon pump, chronic obstructive pulmonary disease, insulin-dependent diabetes mellitus, previous cardiac surgery, emergency surgery, elevated preoperative creatinine, and coronary bypass with valvular intervention.

References:

1. O'Neal JB, Shaw AD, Billings FT 4th. Acute kidney injury following cardiac surgery: current understanding and future directions. *Crit Care*. 2016 Jul;20(1):187.
2. Nadim MK, Forni LG, Bihorac A, et al. Cardiac and vascular surgery-associated acute kidney injury: the 20th International Consensus Conference of the ADQI (Acute Disease Quality Initiative) Group. *J Am Heart Assoc*. 2018 June 1;7(11): e008834.
3. Thakar CV, Arrigain S, Worley S, Yared JP, Paganini EP. A clinical score to predict acute renal failure after cardiac surgery. *J Am Soc Nephrol*. 2005 Jan;16(1):162-168.

2. Rationale

Answer: B

This patient most likely has acute interstitial nephritis (AIN), which is an immune-mediated cause of acute kidney injury characterized by inflammatory infiltration of the kidney interstitium. More than 75% of AIN is caused by drugs, with antibiotics (penicillins, cephalosporins, rifampin, and sulfonamides) the most common offending agents. Other drugs, including nonsteroidal anti-inflammatory drugs, allopurinol, proton pump inhibitors, furosemide, and acyclovir, have also been implicated. Although the classic triad of fever, rash, and eosinophilia is often emphasized in medical education, less than 10% to 15% of patients with AIN have all 3 of these features. In 82% of patients with AIN, urinalysis reveals leukocyturia and leukocyte casts. Nephrotic range proteinuria is rare (2.5%). Treatment is conservative and focuses on removal of the offending agent and supportive care. The role of steroids in drug-induced AIN is controversial. Although this patient is at risk for acute contrast nephropathy and hepatorenal syndrome, the presence of fever, malaise, WBC casts, and recent trimethoprim/sulfamethoxazole administration make AIN the most likely etiology of acute kidney injury. Acute tubular necrosis typically occurs in a patient with hemodynamic instability; epithelial cells and granular casts are often seen on urine microscopy. There is no suggestion of obstructive nephropathy.

References:

1. Praga M, Gonzalez E. Acute interstitial nephritis. *Kidney Int.* 2010 Jun;77(11):956-961.

2. Raghavan R, Eknoyan G. Acute interstitial nephritis: a reappraisal and update. *Clin Nephrol.* 2014 Sep;3(3):149-162.

3. Rationale

Answer: A

Hematuria can be caused by bleeding anywhere in the urinary tract, from the glomeruli to the urethra. Therefore, the differential diagnosis is broad and includes glomerular disease,

nephrolithiasis, urologic malignancy, renal papillary necrosis, prostate disorders, acute trauma, infection, hematologic disorders, and other forms of cystitis. Radiation to the pelvis may produce hemorrhagic cystitis; patients who are excessively anticoagulated may develop hematuria without identifiable structural disease. Abdominal aortic dissection may have associated hematuria due to renal ischemia. This patient has proteinuria and chronic kidney disease in addition to hematuria. Taken together, these findings suggest glomerular disease. Definitive diagnosis is determined only by kidney biopsy, which will help guide therapy.

References:

1. Bottini PV, Garlipp CR, Lauand JR, Lara Cioffi SG, Afaz SH, Lopes Prates R. Glomerular and non-glomerular haematuria: preservation of urine sediment. *Science*. 2005 Oct;36(10):647-649.

2. Hebert LA, Parikh S, Prosek J, Nadasdy T, Rovin BH. Differential diagnosis of glomerular disease: a systematic and inclusive approach. *Am J Nephrol*. 2013;38(3):253-266.

3. Yuste C, Gutierrez E, Sevillano AM, et al. Pathogenesis of glomerular haematuria. *World J Nephrol*. 2015 May 6;4(2):185-195.

4. Rationale

Answer: A

Drugs or toxins that are extensively large (monoclonal antibodies), extensively protein bound, with an extensive charge, or with a large volume of distribution are least likely to be effectively removed by continuous renal replacement therapy. Conversely, small, uncharged, unbound drugs that remain in the serum are most likely to be removed effectively.

Reference:

1. Pea F, Viale P, Pavan F, Furlanut M. Pharmacokinetic considerations for antimicrobial therapy in patients receiving renal replacement therapy. *Clin Pharmacokinet*. 2007;46(12):997-1038.

5. Rationale

Answer: A

This patient has diabetes insipidus; the clues are a head injury with excessive excretion of diluted urine with a low specific gravity. Diabetes insipidus is common in head trauma, especially if there is damage to the hypothalamus, which regulates antidiuretic hormone (ADH). The malfunction or inhibition of ADH can lead to excessive thirst to replete the fluids lost, which leads to hyponatremia. The first-line treatment for diabetes insipidus is desmopressin and fluid replacement. Fludrocortisone will only increase his sodium further. Sterile water for injection is not recommended for hypernatremia. Furosemide may cause excessive diuresis and will have variable effects on his sodium.

References:

1. Di Iorgi N, Napoli F, Allegri AEM, et al. Diabetes insipidus: diagnosis and management. *Horm Res Paediatr.* 2012;77(2):69-84.

2. Schreckinger M, Szerlip N, Mittal S. Diabetes insipidus following resection of pituitary tumors. *Clin Neurol Neurosurg.* 2013 Feb;115(2):121-126.

6. Rationale

Answer: D

Syndrome of inappropriate antidiuretic hormone secretion (SIADH) has similar biochemistry abnormalities to cerebral salt wasting (eg, hyponatremia, elevated urine osmolality, and sodium content) but can be distinguished by a normal-to-high effective circulating blood volume and low plasma osmolality. Normal-to-high fluid balance and central venous pressure (CVP) greater than 6 cm H_2O have also been suggested as markers of fluid/volume status in the setting of SIADH. This patient seems to be hypovolemic based on CVP, fluid balance, tachycardia/systolic blood pressure, and stroke volume variation, which rules out SIADH. Although signs of adrenal insufficiency include hyponatremia, there should also be

hyperkalemia, refractory hypotension, and hypoglycemia, which this patient does not have. Diabetes insipidus is associated with hypernatremia, serum hyperosmolarity, and low urine osmolality, which are not supported by laboratory results. Cerebral salt wasting is associated with hyponatremia, elevated heart rate, decreased volume status/negative fluid balance (eg, decreased CVP, increased stroke volume variation), increased urine osmolality, and urine sodium.

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1. Annane D, Pastores SM, Rochwerg B, et al. Guidelines for the diagnosis and management of critical illness-related corticosteroid insufficiency (CIRCI) in critically ill patients (part I): Society of Critical Care Medicine (SCCM) and European Society of Intensive Care Medicine (ESICM) 2017. *Crit Care Med*. 2017 Dec;45(12):2078-2088.
2. Palmer BF. Hyponatremia in a neurosurgical patient: syndrome of inappropriate antidiuretic hormone secretion versus cerebral salt wasting. *Nephrol Dial Transplant*. 2000 Feb;15(2):262-268.
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4. Tisdall M, Crocker M, Watkiss J, Smith M. Disturbances of sodium in critically ill adult neurologic patients: a clinical review. *J Neurosurg Anesthesiol*. 2006 Jan;18(1):57-63.
5. Upadhyay UM, Gormley WB. Etiology and management of hyponatremia in neurosurgical patients. *J Intensive Care Med*. 2012 May-Jun;27(3):139-144.

7. Rationale

Answer: C

In patients on long-term diuretics, the fractional excretion of sodium may be unreliable, whereas the fractional excretion of urea (FEU) generally remains accurate. An FEU of less than 35% indicates a prerenal cause, whereas FEU greater than 50%

indicates an intrinsic renal cause. Urine sodium level may be elevated secondary to diuretic use.

References:

1. Balogun R, Okusa M. Fractional excretion of sodium, urea, and other molecules in acute kidney injury (acute renal failure). UpToDate. Last updated May 26, 2017. Accessed February 23, 2022. https://utd.ungbuoubmt.com/d/topic.htm?path=fractional-excretion-of-sodium-urea-and-other-molecules-in-acute-kidney-injury-acuterenal-failure

2. Gotfried J, Wiesen J, Raina R, et al. Finding the cause of acute kidney injury: which index of fractional excretion is better? *Cleve Clin J Med*. 2012 Feb;79(2):121-126.

8. Rationale

Answer: C

Methanol and ethylene glycol toxicity presentations overlap; both can cause acute kidney injury and high osmolar gaps. The clinical giveaway that this patient ingested ethylene glycol is the presence in the urine of calcium oxalate crystals, which form only from the metabolites of ethylene glycol. Ethanol and isopropyl alcohol ingestions can cause severe central nervous system depression but are less likely to cause acute kidney injury this profound and metabolic disturbances so severe.

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1. Kraut JA. Diagnosis of toxic alcohols: limitations of present methods. *Clin Toxicol (Phila)*. 2015;53(7):589-595.

2. McMartin K, Jacobsen D, Hovda KE. Antidotes for poisoning by alcohols that form toxic metabolites. *Br J Clin Pharmacol*. 2016 Mar;81(3):505-515.

9. Rationale

Answer: D

This patient has septic shock and is being appropriately managed on mechanical ventilation with low tidal volume protocol. Low tidal volumes can lead to hypercarbia, which is

tolerated (permissive hypercarbia). For pH below 7.2, an IV bicarbonate infusion can be started to correct acidosis but his pH of 7.34 is adequate and needs no intervention. He has adequate urine output despite kidney injury, so a renal angiogram is not needed. Exposure to IV contrast should be avoided because it can worsen acute kidney injury. The AKIKI and IDEAL ICU trials randomized critically ill patients with acute kidney injury to early or delayed dialysis. In both trials, early intervention with dialysis did not result in improved outcomes. Approximately 25% to 35% of patients with acute renal failure who were in the delayed dialysis arm recovered their renal function and were able to avoid dialysis. These studies support close monitoring of critically ill patients with renal failure and initiating dialysis if the patient develops refractory volume overload leading to severe hypoxia, severe hyperkalemia, severe acidosis, and BUN greater than 100 mg/dL with neurologic symptoms. This patient does not meet any of the conditions required to start dialysis; he wakes up and follows commands off sedation, acidosis is mild, FIO_2 is 40%, and potassium is normal.

References:

1. Barbar SD, Clere-Jehl R, Bourredjem A, et al; IDEAL-ICU Trial Investigators and CRICS TRIGGERSEP Network. Timing of renal-replacement therapy in patients with acute kidney injury. *N Engl J Med.* 2018 Oct 11;379(15):1431-1442.

2. Gaudry S, Hajage D, Schortgen F, et al; AKIKI Study Group. Initiation strategies for renal-replacement therapy in the intensive care unit. *N Engl J Med.* 2016 Jul 14; 375(2):122-133.

Part 2.

Electrolytes

Instructions: For each question, select the most correct answer.

1. A 36-year-old man with nonischemic cardiomyopathy and heart failure with reduced ejection fraction requires progressive escalation in inotropes and eventually requires implantation of a left ventricular assist device. His postoperative course is complicated by hypervolemia with high pulmonary arterial pressures and moderate right ventricular dysfunction. A continuous infusion of high-dose furosemide is administered over several days until euvolemia is achieved. By postoperative day 2, he is extubated from mechanical ventilation to high-flow nasal cannula. He is awake but delirious and fails a formal swallow evaluation. Nutrition is provided with a post-pyloric nasogastric tube. Laboratory test results are shown below. Which of the following is the most appropriate treatment strategy for his hypernatremia?

Laboratory Parameter	Test Result
Sodium	159 mEq/L
Chloride	120 mEq/L
Potassium	4.2 mmol/L
BUN	42 mg/dL
Creatinine	1.2 mg/dL
Serum glucose	72 mg/dL
Urine specific gravity	1.020
Urine osmolality	610 mOsm/kg
Urine sodium	34 mmol/L

A. Continue diuretic and administer 5% dextrose in water infusion.

B. Discontinue diuretic and administer half-normal saline infusion.

C. Continue diuretic and administer enteral free water.

D. Discontinue diuretic and administer 5% dextrose in water infusion.

2. A 42-year-old woman with chronic pain, end-stage renal disease on outpatient hemodialysis, and hypertension presents to the emergency department after missing her most recent hemodialysis session. Serum potassium is 7.4 mmol/L, serum sodium 134 mEq/L, serum glucose 121 mg/dL, and serum creatinine 7.2 mg/dL. ECG has peaked T waves in multiple leads. If regular insulin, 5 units IV, is administered rather than regular insulin, 10 units IV, which of the following effects on serum potassium and risk of severe hypoglycemia will most likely result?

   A. Similar decrease in serum potassium and decreased risk of severe hypoglycemia

   B. Similar decrease in serum potassium and similar risk of severe hypoglycemia

   C. Smaller decrease in serum potassium and decreased risk of severe hypoglycemia

   D. Smaller decrease in serum potassium and similar risk of severe hypoglycemia

   E. Smaller decrease in serum potassium and increased risk of severe hypoglycemia

3. Which of the following is a consequence of low serum potassium?

   A. Increase in serum pH

   B. Increase in serum magnesium concentration

   C. Increased likelihood of ventricular arrhythmia

   D. Increased likelihood of diarrhea

E. Concurrent and proportional decrease in bound calcium

4. A patient presents with the following laboratory findings: serum creatinine 13.7 mg/dL on hemodialysis, calcium 7.5 mg/dL, phosphorus 8.2 mg/dL, and albumin 2.3 mg/dL. Which of the following is the most appropriate treatment for this patient's electrolyte disorder?

   A. Calcium carbonate

   B. Calcium acetate

   C. Sevelamer

   D. Corticosteroids

5. A 48-year-old man is admitted to the ICU for management of alcohol withdrawal, for which he was admitted to the ICU twice last year. He has no other significant medical history. On admission, he is poorly responsive. Vital signs are blood pressure 100/72 mm Hg, heart rate 110 beats/min, and respiratory rate 10 breaths/min. Laboratory findings are significant for potassium 3.1 mEq/L and magnesium 1.2 mg/dL. He is administered IV potassium chloride, 60 mEq, and IV magnesium sulfate, 6 g. The next day, magnesium level is 1.7 mg/dL. Which of the following is the most appropriate action in response to his magnesium level?

   A. Recheck magnesium level the next day.

   B. Start IV magnesium sulfate, 2 g daily for 3 days.

   C. Add 1 additional dose of IV magnesium sulfate, 2 g.

   D. Start magnesium chloride tablets, 200 mg twice daily.

6. A 78-year-old woman is admitted with hypotension and tachycardia following a 2-week history of diarrhea. She had previously been treated with

antibiotics for a urinary tract infection. Which of the following IV resuscitation fluids is most likely to reduce her likelihood of acute kidney injury?

A. Normal saline

B. Lactated Ringer solution

C. 2% potassium chloride

D. 5% dextrose in water

E. 5% albumin

7. A 78-year-old woman is admitted to the ICU from a nursing facility with altered mental status, vomiting, and hypotension. Serum sodium level is 161 mEq/L (normal range 135-145 mEq/L). Blood pressure is 78/46 mm Hg. Which of the following is the most appropriate initial fluid for this patient?

   A. Lactated Ringer solution

   B. 0.45% sodium chloride

   C. 5% dextrose in water

   D. 0.225% sodium chloride

8. A 78-year-old man with non-small cell lung cancer presents to the emergency department with confusion and weakness. Laboratory results reveal a serum calcium level of 14.6 mg/dL. Which of the following is the most appropriate initial treatment?

   A. 0.9% sodium chloride

   B. Zoledronic acid

   C. Calcitonin

   D. Furosemide

9. A 56-year-old male construction worker with a medical history of hypertension, chronic kidney disease, and daily alcohol use presents with multiple traumatic injuries after a structure collapsed on him.

Extrication took more than 6 hours. Primary injuries are to extremities, with significant crush injury to bilateral lower extremities. While awaiting operating room availability for wound debridement and fracture repair, he has a grand mal seizure. Relevant laboratory results are shown below. If he weighs 80 kg (176 lb), which of the following treatment regimens is optimal for correcting his hypocalcemia?

Laboratory Parameter	Test Result
Sodium	135 mEq/L
Potassium	6.3 mEq/L
Chloride	102 mEq/L
Bicarbonate	16 mEq/L
BUN	70 mg/dL
Creatinine	3.7 mg/dL
Calcium	4.6 mg/dL
Ionized calcium	0.5 mmol/L
Magnesium	0.7 mEq/L
Phosphorus	8.2 mEq/L
Creatine phosphokinase	125,000 U/L

A. 10% calcium chloride, 2 g through peripheral IV line, followed by calcium chloride infusion

B. 10% calcium gluconate, 2 g through peripheral IV line, followed by calcium gluconate infusion

C. 10% calcium chloride, 2 g through peripheral IV line, and IV magnesium sulfate, 6 g, followed by calcium chloride infusion

D. 10% calcium gluconate, 2 g through peripheral IV line, and IV magnesium sulfate, 6 g, followed by calcium gluconate infusion

Part 2 Answers:

Electrolytes

1. Rationale

Answer: D

This patient has euvolemic hypernatremia, the main cause of which is the infusion of a loop diuretic. The offending medication should be discontinued, and the free water deficit corrected.

Reference:

1. Rondon-Berrios H, Argyropoulos C, Ing TS, et al. Hypertonicity: clinical entities, manifestations, and treatment. *World J Nephrol*. 2017 Jan;6(1):1-13.

2. Rationale

Answer: A

IV administration of regular insulin can reduce serum potassium concentrations in the short term but carries a risk of hypoglycemia. Because effects on serum glucose are more pronounced than effects on serum potassium, the risk of hypoglycemia is increased with greater doses (3% vs. 6.8%, difference –3.8%, 95% CI, –7.4% to 0) but the effects on serum potassium are stable (–1 ± 0.8 mEq/L vs. –1 ± 0.7 mEq/L) between 5-unit and 10-unit doses.

Reference:

1. LaRue HA, Peksa GD, Shah SC. A comparison of insulin doses for the treatment of hyperkalemia in patients with renal insufficiency. *Pharmacotherapy*. 2017 Dec;37(12):1516-1522.

3. Rationale

Answer: A

Lower serum potassium levels result in increased reuptake of potassium in the distal nephron via the H-K exchanger,

resulting in loss of hydrogen ions and a resultant rise in serum pH. Low potassium levels are associated with lower magnesium levels. High potassium levels increase the likelihood of ventricular arrhythmias while low levels are more often associated with atrial arrhythmias. Low potassium can result in decreased gut motility, constipation, and nausea, but not diarrhea. Potassium has no significant effect on the proportion of bound calcium.

Reference:

1. Kardalas E, Paschou SA, Anagnostis P, Muscogiuri G, Siasos G, Vryonidou A. Hypokalemia: a clinical update. *Endocr Connect*. 2018 Apr;7(4):R135-R146.
2. Marino PL. *The ICU Book*. 3rd ed. Lippincott Williams & Wilkins; 2007:612-616.

4. Rationale

Answer: C

This patient's corrected calcium level is 8.9 mg/dL. The calcium-phosphorous product is 72.98 mg/dL. Since this product is greater than 55 mg/dL, a phosphate binder such as sevelamer is the most appropriate treatment. Calcium carbonate and calcium acetate are inappropriate; a calcium-based supplement is not needed because the corrected calcium level is within normal limits. Prednisone is not a treatment for hyperphosphatemia.

Reference:

1. Kraft MD, Btaiche IF, Sacks GS, Kudsk KA. Treatment of electrolyte disorders in adult patients in the intensive care unit. *Am J Health Syst Pharm*. 2005 Aug 15;62(16):1663-1682.

5. Rationale

Answer: B

This patient seems to have a significant history of heavy alcohol abuse, based on the current admission and previous ICU admissions for alcohol withdrawal. Alcohol abuse is associated

with hypomagnesemia due to increased renal excretion, impaired absorption, and poor nutritional status. Although the magnesium level obtained the next day is within normal range, it is probably falsely elevated, especially in a patient with alcohol abuse. It generally takes up to 48 hours for serum magnesium to redistribute intracellularly, which is the main reason for a falsely elevated level. Therefore, he requires additional magnesium replacement. In general, correction of low magnesium takes 2 to 5 days of magnesium replacement to replenish intracellular stores. Although oral magnesium may be appropriate in some patients to manage hypomagnesemia, IV magnesium sulfate can achieve a higher concentration of magnesium since it has a higher amount of magnesium compared with oral tablets. Further, magnesium chloride has a smaller amount of elemental magnesium compared with other magnesium salt formulations and thus would require many tablets to replace the necessary magnesium.

References:

1. Yu ASL. Hypomagnesemia: evaluation and treatment. UpToDate. Last updated May 25, 2021. Accessed February 23, 2022. https://www.uptodate.com/contents/hypomagnesemia-evaluation-and-treatment
2. Magnesium chloride. Lexicomp.

6. Rationale

Answer: B

Diarrhea commonly induces loss of electrolytes, particularly bicarbonate (with consequent metabolic acidosis) and potassium; chronic losses can lead to serious fluid depletion and prerenal impairment. This presentation could be secondary to *Clostridioides difficile* infection associated with antibiotic treatment of this patient's urinary tract infection. Fluid resuscitation is necessary, but significant chloride administration has been associated with acute kidney injury in several large-volume studies. A 5% dextrose in water solution has insufficient electrolytes to be useful, while 5% albumin and 2% potassium

chloride contain significant chloride. Lactated Ringer solution is converted to bicarbonate and allows some correction of the metabolic acidosis and base deficiency.

References:

1. Pfortmueller CA, Uehlinger D, von Haehling S, Schefold JC. Serum chloride levels in critical illness: the hidden story. *Intensive Care Med Exp.* 2018 Apr 13;6(1):10.

2. Marttinen M, Wilkman E, Petäjä L, Suojaranta-Ylinen R, Pettilä V, Vaara ST. Association of plasma chloride values with acute kidney injury in the critically ill: a prospective observational study. *Acta Anaesthesiol Scand*. 2016 Jul;60(6):790-799.

3. Self WH, Semler MW, Wanderer JP, et al; SALT-ED Investigators. Balanced crystalloids versus saline in noncritically ill adults. *N Engl J Med*. 2018 Mar 1;378(9):819-828.

7. Rationale

Answer: A

This patient is hypotensive. An isotonic crystalloid should be initially administered to restore hemodynamic stability, despite its sodium content. Hypotonic fluids (such as dextrose in water or 0.45% sodium chloride) could be used to replace the water deficit once she is hemodynamically stable. It has been shown that 0.225% sodium chloride causes hemolysis, and it is not routinely indicated for hypernatremia in critically ill adults.

References:

1. Dickerson RN, Maish 3rd GO, Weinberg JA, Croce MA, Minard G, Brown RO. Safety, and efficacy of intravenous hypotonic 0.225% sodium chloride infusion for the treatment of hypernatremia in critically ill patients. *Nutr Clin Pract*. 2013 Jun;28(3):400-408.

2. Kraft MD, Btaiche IF, Sacks GS, Kudsk KA. Treatment of electrolyte disorders in adult patients in the intensive care

unit. *Am J Health Syst Pharm*. 2005 Aug 15;62(16):1663-1682.

8. Rationale

Answer: A

Severe hypercalcemia (>14 mg/dL) should first be treated with aggressive hydration with 0.9% sodium chloride. After hydration, loop diuretics can be used to enhance urinary excretion of calcium, although their use in treating hypercalcemia is controversial. Calcitonin is likely to be initiated next because it has a faster onset than bisphosphonates (though tachyphylaxis can develop) but IV fluids are administered first. Bisphosphonates such as zoledronic acid are effective for hypercalcemia but are not used for initial emergent therapy because of their delayed onset.

References:

1. Canada TW, Lord LM. Fluids, electrolytes, and acid-base disorders. In: American Society for Parenteral and Enteral Nutrition. *The ASPEN Adult Nutrition Support Core Curriculum*. 3rd ed. American Society for Parenteral and Enteral Nutrition; 2017.

2. Minisola S, Pepe J, Piemonte S, Cipriani C. The diagnosis and management of hypercalcemia. *BMJ*. 2015 Jun 2;350:h2723.

9. Rationale

Answer: D

The optimal treatment regimen for this patient includes calcium gluconate and magnesium sulfate. Correction of hypocalcemia requires correction of hypomagnesemia. Hypomagnesemia decreases parathyroid hormone (PTH) secretion and increases PTH resistance. Solutions for replacement of hypocalcemia include 10% calcium gluconate, 1 g (90 mg elemental calcium) or 10% calcium chloride, 1 g (270 mg elemental calcium). Calcium chloride is usually administered via central line. Treatment for symptomatic hypocalcemia consists of 100 to 200 mg elemental calcium over

10 minutes followed by a continuous infusion of 0.5 to 1.5 mg/kg/hr.

References:

1. Aberegg SK. Ionized calcium in the ICU: Should it be measured and corrected? *Chest*. 2016 Mar;149(3):846-855.
2. Cherry RA, Bradburn E, Carney DE, Shaffer ML, Gabbay RA, Cooney RN. Do early ionized calcium levels really matter in trauma patients? *J Trauma*. 2006 Oct;61(4):774-779.
3. Kelly A, Levine MA. Hypocalcemia in the critically ill patient. *J Intensive Care Med*. 2013 May-Jun;28(3):166-177.
4. Steele T, Kolamunnage-Dona R, Downey C, Toh CH, Welters I. Assessment and clinical course of hypocalcemia in critical illness. *Crit Care*. 2013 Jun 4;17(3): R106.

Part 3.

Acid-Base Disorders

Instructions: For each question, select the most correct answer.

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1. A 45-year-old man was admitted to the ICU with uncontrolled hypertension. He was first diagnosed with hypertension 5 days ago and was started on nifedipine, 20 mg 3 times daily. He takes no other medications. On the day of admission, he had an intractable headache, nausea, and vomiting. Blood pressure is 260/120 mm Hg, without orthostatic changes. Laboratory studies show sodium 130 mEq/L, potassium 3.0 mEq/L, chloride 90 mEq/L, CO_2 32 mEq/L, BUN 14 mg/dL, and creatinine 0.8 mg/dL. Arterial blood gas analysis on room air shows pH 7.47, $PaCO_2$ 45 mm Hg, and PaO_2 74 mm Hg. Random urine chloride is 30 mEq/L. Which of the following is/are the best explanation(s) for this patient's acid-base abnormalities?

   A. Primary hypoventilation due to hypertensive encephalopathy

   B. Vomiting

   C. Primary hypoventilation and vomiting

   D. Mineralocorticoid excess

   E. Laboratory error

2. A 28-year-old man is brought to the hospital by ambulance after being found unconscious on the side of a road where he had been for an unknown period. Emergency medical services report that he mostly remained unconscious during transport to the hospital, and when they were able to arouse him, he

seemed intoxicated. On arrival, the patient reports blurred vision and abdominal pain. Laboratory results are sodium 145 mEq/L, chloride 105 mEq/L, BUN 9 mg/dL, glucose 92 mg/dL, potassium 3.6 mEq/L, bicarbonate 20 mEq/L, serum creatinine 0.8 mg/dL, serum osmolality 356 mOsm/kg, lactate 1 mmol/L, pH 7.2, PCO_2 38 mm Hg, PO_2 75 mm Hg, and arterial oxygen saturation 93%. Urine drug screen is negative. Ethanol level is 0, acetaminophen level is less than 1 μg/mL, and salicylate level is less than 1 mg/dL. Administration of which of the following agents is most appropriate?

A. Fomepizole

B. Flumazenil

C. Glucagon

D. Pralidoxime

3. A 25-year-old man is admitted to the ICU with an episode of uncontrolled type 1 diabetes mellitus. He has a 1-day history of nausea, vomiting, and diarrhea. He withheld his insulin because of poor oral intake. Blood pressure is 100/60 mm Hg, heart rate 104 beats/min, respiratory rate 28 breaths/min, and temperature 37.2°C (99°F). Physical examination reveals dry mucous membranes and mild diffuse abdominal tenderness. Laboratory tests show sodium 135 mEq/L, potassium 3.5 mEq/L, chloride 110 mEq/L, CO_2 8 mEq/L, BUN 22 mg/dL, creatinine 1.2 mg/dL, albumin 4.1 g/dL, and glucose 350 mg/dL. Arterial blood gas analysis on FIO_2 0.21 is: pH 7.25, $PaCO_2$ 20 mm Hg, and PaO_2 95 mm Hg. Serum ketones are present at a 1:8 dilution. Which of the following is/are the best explanation(s) for this patient's acid-base abnormality?

A. Diabetic ketoacidosis only

B. Diabetic ketoacidosis and hyperventilation

C. Diabetic ketoacidosis and diarrhea

D. Diabetic ketoacidosis and nausea/vomiting

E. Laboratory error

4. A 19-year-old female college student is brought to the emergency department by friends who state that she fainted several times today when attempting to stand. She denies nausea, emesis, or diarrhea. She says that she "watches what she eats." She takes no medications. Initial resting heart rate (HR) is 92 beats/min with blood pressure (BP) 98/52 mm Hg while in the supine position. When her legs dangle off the edge of the bed, she becomes "lightheaded." Heart rate is now 112 beats/min and blood pressure 75/38 mm Hg. Initial chemistry results are sodium 140 mEq/L, potassium 3 mEq/L, chloride 90 mEq/L, bicarbonate 34 mEq/L, glucose 115 mg/dL, BUN 25 mg/dL, and creatinine 0.7 mg/dL. Initial arterial blood gas analysis on room air shows: pH 7.48, $PaCO_2$ 47 mm Hg, PaO_2 89 mm Hg, and bicarbonate 34 mEq/L. She has which of the following acid-base abnormalities?

   A. Normal-gap metabolic acidosis

   B. Metabolic alkalosis only

   C. Anion-gap metabolic acidosis and respiratory alkalosis

   D. Metabolic alkalosis and compensatory respiratory acidosis

   E. Respiratory acidosis only

5. A 28-year-old man remains intubated in the ICU 3 days after an abdominal operation. In the operating room, he received more than 10 L of fluid and blood products and postoperatively has received aggressive

diuresis with furosemide. In the past 3 days, he has generated 7.5 L urine output. BUN has steadily increased to 40 mg/dL and serum creatinine has increased to 1.5 mg/dL. Urine chloride concentration was 9 mEq/L 24 hours after his most recent dose of furosemide. This morning, arterial blood gas analysis reveals pH 7.50, $PaCO_2$ 46 mm Hg, and bicarbonate 34 mEq/L. His acid-base status is

A. acute respiratory alkalosis.

B. chronic respiratory alkalosis.

C. chloride-sensitive metabolic alkalosis.

D. chloride-resistant metabolic alkalosis.

6. A 38-year-old man who has a history of poorly controlled diabetes is found hypothermic and is intubated. Initial vital signs in the emergency department are rectal temperature 31°C (87.8°F), blood pressure 153/94 mm Hg (mean arterial pressure 92 mm Hg), heart rate 161 beats/min, respiratory rate 10 breaths/min, arterial oxygen saturation 99% on FIO_2 1.0. He is chemically paralyzed. Initial arterial blood gas analysis shows: pH 6.682, PCO_2 42 mm Hg, PO_2 145 mm Hg, base deficit –30 mEq/L, sodium 132 mEq/L, potassium 6.4 mEq/L (no hemolysis), chloride 110 mEq/L, bicarbonate less than 5 mEq/L, ionized calcium 1.06 mg/dL, hemoglobin 11.9 g/dL, BUN 16 mg/dL, creatinine 1.3 mg/dL, blood glucose 552 mg/dL, anion gap 22 mEq/L, and lactate 2.8 mmol/L. In addition to anion gap metabolic acidosis, this patient has which of the following acid-base disorders?

A. Metabolic alkalosis and respiratory acidosis

B. Respiratory alkalosis and non-anion gap metabolic acidosis

C. Respiratory acidosis only

D. Respiratory acidosis and non-anion gap metabolic acidosis

7. Acetazolamide can be used for severe cases of which of the following acid-base disorders?

   A. Respiratory acidosis

   B. Metabolic acidosis

   C. Respiratory alkalosis

   D. Metabolic alkalosis

8. A patient has the following laboratory results: pH, 7.50; $PaCO_2$, 20 mm Hg; PaO_2, 80 mm Hg; bicarbonate, 15 mmol/L; sodium, 140 mmol/L; chloride, 103 mmol/L; potassium, 4.2 mmol/L; serum creatinine, 0.8 mg/dL; and glucose, 102 mg/dL. Which of the following is the patient's most likely metabolic state?

   A. Respiratory alkalosis with excess anion gap metabolic alkalosis

   B. Respiratory alkalosis with non-gap metabolic acidosis

   C. Respiratory alkalosis with anion gap metabolic acidosis

   D. Respiratory alkalosis only

9. A 74-year-old woman with advanced chronic obstructive pulmonary disease and atrial fibrillation that is treated with apixaban comes to the emergency department with acute agitation. To facilitate imaging, she is intubated for airway protection with propofol. Limited neurologic examination prior to intubation reveals agitation and no apparent difficulties in moving her limbs. Head CT reveals an acute subcortical intracerebral hemorrhage (ICH) that is 12 mL in volume and located in the right occipital lobe. One-

hour post-intubation, she is unable to arouse and has not received further sedatives. Repeat head CT reveals no changes to the ICH, and CT angiogram reveals no large vessel occlusion. Serum electrolytes are sodium 134 mEq/L, potassium 3.9 mEq/L, chloride 88 mEq/L, bicarbonate 38 mEq/L, BUN 24 mg/dL, and creatinine 1.1 mg/dL. Pulse oxygen sensor reads greater than 93% and end-tidal capnography is 34 mm Hg. What is the most likely cause of the neurologic decline?

A. Acute ischemic stroke

B. Acute respiratory alkalosis in a CO_2 retainer

C. Contrast-induced nephropathy

D. Hyponatremia leading to cerebral edema

E. Propofol-related seizure

Part 3 Answers:

Acid-Base Disorders

1. Rationale

Answer: D

The presence of alkalemia with increased bicarbonate indicates a primary metabolic alkalosis. The alkalemia leads to a direct suppression of respiration, and the $PaCO_2$ rises 0.5 to 0.7 mm Hg for each 1 mEq/L increase in plasma bicarbonate. Therefore, in this patient, respiratory compensation is appropriate for the degree of metabolic alkalosis. The pH, bicarbonate, and $PaCO_2$ are internally consistent, based on the Henderson-Hasselbalch equation, so a laboratory error does not explain the abnormalities. In addition to the history and physical examination, urine chloride concentration is useful to determine the cause of metabolic alkalosis. Although he has a history of vomiting, hypertension, hypokalemia, and urine chloride concentration greater than 20 mEq/L suggest mineralocorticoid excess as the cause of metabolic alkalosis. Vomiting, nasogastric suction, postdiuretic use, and posthypercapnic states are associated with chloride-responsive metabolic alkalosis, with urine chloride concentrations less than 15 mEq/L. In contrast, mineralocorticoid excess, current diuretic administration, milk-alkali syndrome, Bartter syndrome, and severe hypokalemia are associated with urine chloride concentrations over 20 mEq/L. Hypertension and hypokalemia in the absence of diuretic therapy should suggest mineralocorticoid excess, such as occurs in primary aldosteronism, exogenous corticosteroid use, and licorice use.

References:

1. Black RM. Metabolic acidosis and metabolic alkalosis. In: Rippe JM, Irwin RS, Fink MP, Cerra FB, eds. *Intensive Care Medicine*. 3rd ed. Little Brown; 1996:984.

2. Kamel KS, Ethier JH, Richardson RM, Bear RA, Halperin ML. Urine electrolytes and osmolality: when and how to use them. *Am J Nephrol*. 1990;10(2):89-102.

3. Latta K, Hisano S, Chan JC. Perturbations in potassium balance. *Clin Lab Med*. 1993 Mar;13(1):149-156.

4. Preuss HG. Fundamentals of clinical acid-base evaluation. *Clin Lab Med*. 1993 Mar;13(1):103-116.

2. Rationale

Answer: A

This patient has an anion gap (20 mEq/L) metabolic acidosis. Laboratory tests rule out ethanol, salicylates, uremia, lactic acidosis, and diabetic ketoacidosis as causes of the anion gap. Paraldehyde and isoniazid are unlikely causes. His osmolality is calculated as (2 × [sodium + potassium]) + (BUN/2.8) + (glucose/18) = 306 mOsm/kg. Therefore, his osmolality gap is 50 mOsm/kg. A normal osmolal gap is 10-20 mOsm/kg. Higher levels suggest accumulation of osmotically active substances, including toxic alcohols such as methanol or ethylene glycol, as the cause of the anion gap. It is highest shortly after ingestion and falls with time. Antidotes should be administered to patients with a documented history, a strong suspicion of ingesting a toxic alcohol, an osmolal gap exceeding 10 mOsm/kg, or a metabolic acidosis of unknown cause. Fomepizole strongly inhibits alcohol dehydrogenase, is an effective treatment for toxic alcohol ingestion with few adverse effects and does not require monitoring. Flumazenil is the antidote for benzodiazepines. Glucagon is administered to patients with a beta-blocker or calcium channel blocker overdose. He is not hypoglycemic and does not need it for such an indication. Pralidoxime is an antidote for cholinesterase inhibitor poisoning.

References:

1. Howland M. Antidotes in depth. In: Hoffman RS, Howland MA, Lewin NA, Nelson LS, Goldfrank LR, eds. *Goldfrank's Toxicologic Emergencies*. 10th ed. McGraw-Hill Education; 2015.

2. Howland M. Antidotes in depth. In: Hoffman RS, Howland M, Lewin NA, Nelson LS, Goldfrank LR. eds. *Goldfrank's Toxicologic Emergencies*. 10th ed. McGraw Hill Education; 2015. Accessed February 24, 2022. https://accesspharmacy.mhmedical.com/content.aspx?bookid=1163&sectionid=65088338

3. Kraut JA, Mullins, ME. Toxic alcohols. *New Eng J Med*. 2018 Jan 18;378(3):270-280.

3. Rationale

Answer: C

This patient's laboratory results show acidemia with decreased bicarbonate concentration, indicating metabolic acidosis. The degree of respiratory compensation is adequate (expected PCO_2 in a simple metabolic acidosis = [1.5 × bicarbonate] 8 ± 2), thereby excluding primary hyperventilation related to anxiety. The anion gap is increased and, considering the hyperglycemia and ketonemia, he has diabetic ketoacidosis. The anion gap is 17 mEq/L, which is 5 mEq/L higher than the upper limit of normal (12 mEq/L). In a pure anion gap metabolic acidosis, the serum bicarbonate decrease should approximate the anion gap increase. Thus, his expected bicarbonate level is 19 mEq/L (normal bicarbonate of 24 mEq/L minus the anion gap change of 5 mEq/L). The actual bicarbonate level of 8 mEq/L is much lower than anticipated, reflecting a bicarbonate change of 16 mEq/L. The difference between the bicarbonate changes and the anion gap change is called the delta gap; normal values are 0 ± 6. Because the serum bicarbonate is much lower than expected for a simple anion gap metabolic acidosis, his acid-base status is best described as a mixed anion gap metabolic acidosis (diabetic ketoacidosis) with a superimposed non-anion gap (hyperchloremic) metabolic acidosis. Diarrhea is the most likely explanation for the hyperchloremic metabolic acidosis. The serum bicarbonate is consistent with the pH and PCO_2 (based on the Henderson-Hasselbach equation); therefore, laboratory error does not account for the acid-base abnormality.

References:

1. Kellum JA. Diagnosis and treatment of acid-base disorders. In: Grenvik AG, Ayres SM, Holbrook PR, Shoemaker WC, eds. *Critical Care*. 4th ed. Saunders; 2000:834-853.
2. Paulsen WD, Gadallah MF. Diagnosis of mixed acid-base disorders in diabetic ketoacidosis. *Am J Med Sci*. 1993 Nov;306(5):295-300.
3. Ring T, Frische S, Nielsen S. Clinical review: renal tubular acidosis: a physicochemical approach. *Crit Care Med*. 2005;9(6):573-580.
4. Wrenn K. The delta (delta) gap: an approach to mixed acid-base disorders. *Ann Emerg Med*. 1990 Nov;19(11):1310-313.

4. Rationale

Answer: D

This patient has a hypochloremic metabolic alkalosis, probably secondary to self-induced vomiting. The diagnosis of metabolic alkalosis is defined by pH greater than 7.4 and serum bicarbonate level greater than 24 mEq/L. Partial respiratory compensation is defined by $PaCO_2$ greater than 40 mm Hg and establishes the presence of a respiratory acidosis. While metabolic acidosis may be categorized based on the presence or absence of an abnormal anion gap, no such categorization exists for metabolic alkalosis. Respiratory acidosis as an isolated disorder can be ruled out since pH is above 7.4 and pH would be expected to be below 7.4 if respiratory acidosis were the only disorder (pH reciprocally changes by 0.08 units for every 10-mm Hg change in $PaCO_2$).

Reference:

1. Ferrer L. Acid-base disorders. In: Marini J, Wheeler A, eds. *Critical Care Medicine: The Essentials*. 3rd ed. Lippincott Williams & Wilkins; 2006:213-229.

5. Rationale

Answer: C

This patient's metabolic alkalosis with urine chloride less than 10 mEq/L suggests chloride-sensitive metabolic alkalosis, not chloride-resistant metabolic alkalosis. The alkalosis is not respiratory.

Reference:

1. Berend K, de Vries AP, Gans RO. Physiological approach to assessment of acid-base disturbances. *N Engl J Med*. 2014 Oct 9;371(15):1434-1445.

6. Rationale

Answer: D

This patient has a primary metabolic acidosis, given the low pH (6.682) and low bicarbonate (< 5 mEq/L). He has an anion gap metabolic acidosis (AGMA) because the anion gap (AG) is 22 mEq/L (normal is 12 mEq/L). Given the mildly elevated lactic acid level, the cause of his AGMA is ketoacidosis. Since the primary disorder is AGMA, the appropriate compensation is respiratory alkalosis. The formula for expected PCO_2 is 1.5 × bicarbonate + 8. His expected PCO_2 for appropriately compensated AGMA is 8 mm Hg. His PCO_2 is 42 mm Hg, which means that, in addition to AGMA, he has severe respiratory acidosis, likely iatrogenic since he was paralyzed and, without any respiratory drive, his minute ventilation is fully ventilator dependent. Because he has AGMA, it is necessary to rule out a mixed acid-base disorder such as metabolic alkalosis or non-AG metabolic acidosis. First, delta AG is determined, then delta bicarbonate. Delta AG (excess AG) equals actual AG minus normal AG (22 mEq/L – 12 mEq/L = 10 mEq/L). Delta bicarbonate (bicarbonate deficit) equals normal bicarbonate minus measured bicarbonate (24 mEq/L – 0 mEq/L = 24 mEq/L). The ratio is 10/24 = 0.41, which indicates that he has a hidden non-AG metabolic acidosis in addition to a primary AGMA and a severe respiratory acidosis (delta gap > 2 indicates a mixed metabolic alkalosis). A simpler equation for determining delta gap is sodium minus chloride minus 36. A result less than –6 indicates a non-AG acidosis. His delta gap is 132 – 110 – 36 = –14.

References:

1. Maahs DM, West NA, Lawrence JM, Mayer-Davis EJ. Epidemiology of type I diabetes. *Endocrinol Metab Clin North Am*. 2010 Sep;39(3):481-497.
2. Tsapenko MV. Modified delta gap equation for quick evaluation of mixed metabolic acid-base disorders. *Oman Med J.* 2013 Jan;28(1):73-74.

7. Rationale

Answer: D

Acetazolamide is a carbonic anhydrase inhibitor. Via this mechanism, it increases urinary excretion of bicarbonate, which decreases serum bicarbonate concentration and subsequently lowers serum pH.

References:

1. Faisy C, Mokline A, Sanchez O, Tadié JM, Fagon JY. Effectiveness of acetazolamide for reversal of metabolic alkalosis in weaning COPD patients from mechanical ventilation. *Intensive Care Med*. 2010 May;36(5):859-863.
2. Gulsvik R, Skjørten I, Undhjem K, et al. Acetazolamide improves oxygenation in patients with respiratory failure and metabolic alkalosis. *Clin Respir J.* 2013 Oct;7(4):390-396.
3. Hamilton PK, Morgan NA, Connolly GM, Maxwell AP. Understanding acid-base disorders. *Ulster Med J.* 2017 Sep;86(3):161-166.
4. Soifer JT, Kim HT. Approach to metabolic alkalosis. *Emerg Med Clin North Am*. 2014 May;32(2):453-463.

8. Rationale

Answer: C

While the primary abnormality is a respiratory alkalosis, this patient also has an anion gap metabolic acidosis. Anion gap = (140 + 4.2) – (103 + 15) = 26.2 g/dL, indicating a metabolic

acidosis regardless of pH. The excess anion gap is 25 g/dL, indicating no additional metabolic abnormality.

Reference:

1. Sood P, Paul G, Puri S. Interpretation of arterial blood gas. *Indian J Crit Care Med*. 2010 Apr;14(2):57-64.

9. Rationale

Answer: B

In this patient with chronic obstructive pulmonary disease, the elevation in bicarbonate suggests chronic CO_2 retention. A rapid drop in CO_2 by mechanical ventilation can then lead to post-hypercapnic alkalosis and may cause acute rise in cerebral intracellular pH, which can lead to neurologic deterioration. She has no focal neurologic findings that would suggest acute ischemic stroke. Contrast-induced nephropathy would not present acutely and would manifest as elevated creatinine and azotemia that could then lead to altered mental status. Her sodium level of 134 mEq/L indicates that her neurologic decline is not likely due to hyponatremia. Propofol is an antiepileptic drug and unlikely to cause seizures.

Reference:

1. Rotheram Jr, EB, Safar P, Robin E. CNS disorder during mechanical ventilation in chronic pulmonary disease. *JAMA*. 1964 Sep 28; 189:993-996.

Part 4.

Endocrine

Instructions: For each question, select the most correct answer.

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1. A 32-year-old woman who is gravida 2 para 1 at 37 weeks' gestation presents to labor and delivery triage with progressive nausea, vomiting, and lethargy. Medical history is notable for insulin-dependent diabetes, which has been difficult to control during pregnancy. Fetal monitoring is not reassuring. Blood glucose is 190 mg/dL. Other significant laboratory results are pH 7.2, serum bicarbonate 13 mEq/L, anion gap 23 mEq/L, beta-hydroxybutyrate 4 mmol/L, and lactate 1.8 mmol/L. Vital signs are: temperature 37.8°C (100°F), heart rate 110 beats/min, respiratory rate 21 breaths/min, and blood pressure 125/89 mm Hg. Which of the following is the most likely diagnosis?

   A. Starvation ketoacidosis

   B. Hyperosmolar hyperglycemia state

   C. Diabetic ketoacidosis

   D. Sepsis

   E. Preeclampsia

2. A 46-year-old woman is admitted to the ICU for pneumonia and respiratory difficulty. She has no history of thyroid disease. Thyroid panel results are shown below. Which of the following is the most appropriate treatment?

Test	Value	Typical Range
Free thyroxine	0.6 ng/mL	0.7-1.9 ng/dL
Thyroid-stimulating hormone	0.4 mU/L	0.4-4.0 mU/L
Reverse triiodothyronine	23 ng/dL	10-24 ng/dL
Serum triiodothyronine	30 ng/dL	80-180 ng/dL

A. IV triiodothyronine

B. IV thyroxine

C. IV dexamethasone

D. Treatment of underlying illness

3. A 42-year-old man who has been on hemodialysis for 3 years is admitted to the ICU after a total parathyroidectomy. His postoperative course is uneventful except for persistently low levels of calcium, phosphorus, and magnesium. Ionized calcium is persistently less than 0.8 mmol/L (normal range 1.1-1.35 mmol/L). He undergoes aggressive replacement with IV calcium gluconate. Which of the following diagnoses is consistent with his clinical course?

   A. Milk-alkali syndrome

   B. Hungry bone syndrome

   C. Pseudohypoparathyroidism

   D. Secondary hyperparathyroidism

4. A 55-year-old man with no significant medical history undergoes an elective endoscopic transsphenoidal resection of a suprasellar tumor. He is neurologically intact and hemodynamically stable. Urine output has increased from 80-100 mL/hr to 400-600 mL/hr in the past 3 hours. Urine specific gravity has decreased from 1.02 to 1.005. Sodium level 6 hours ago was 142 mEq/L. Which of the following is the most appropriate treatment?

   A. Desmopressin

   B. Vasopressin

C. Tolvaptan

D. Conivaptan

5. A 67-year-old woman with a history of colon cancer, hypertension, and chronic obstructive pulmonary disease is admitted to the ICU with hypotension. She was in the ICU for 3 days last week following surgery for tumor resection. During the current admission, she is afebrile, with the following vital signs: blood pressure, 85/52 mm Hg; heart rate, 128 beats/min; and respiratory rate, 22 breaths/min. Laboratory results are: WBC count, 7,200/μL; hemoglobin, 10.2 g/dL; platelets, 285,000/μL; sodium, 134 mEq/L; chloride, 109 mEq/L; potassium, 4.1 mEq/L; BUN, 21 mg/dL; serum creatinine, 0.8 mg/dL; glucose, 158 mg/dL; and albumin, 3.6 mg/dL. Cultures are obtained, and 2 L normal saline and broad-spectrum antibiotics are administered. Norepinephrine is started and titrated up to 8 μg/min. On day 3, she continues to require norepinephrine, 4-6 μg/min, although no source of infection is identified, and she has clinically improved. Which of the following cortisol findings would suggest that she has relative adrenal insufficiency?

   A. Random plasma cortisol level of 12 μg/dL

   B. Change in plasma cortisol level of 5 μg/dL after cosyntropin test

   C. Morning plasma cortisol level of 15 μg/dL

   D. Plasma cortisol level of 15 μg/dL after cosyntropin test

6. A 28-year-old man who was recently diagnosed with pheochromocytoma presents to the emergency department with a severe headache, palpitations, flushing, and tremors. He says he has been nonadherent with his prescribed treatment due to financial reasons. Blood pressure is 208/110 mm Hg

and heart rate is 113 beats/min. Which of the following drugs is most appropriate for his initial treatment?

A. Nitroglycerin

B. Labetalol

C. Phenoxybenzamine

D. Propranolol

E. Valsartan

7. A 22-year-old man with obesity (body mass index 38 kg/m^2) is admitted to the ICU for observation after transnasal resection of a large craniopharyngioma. Neurosurgery requested that IV dexamethasone be administered for the next 72 hours. There is some concern for obstructive sleep apnea. He has no other significant medical history. Several hours postoperatively, he develops polyuria of 300-400 mL/hr, for which crystalloid fluid is administered to match the previous hour's urine output. On postoperative day 2, he is producing 500-600 mL/hr of urine. Heart rate is mildly elevated at 102 beats/min, but he remains normotensive. Blood glucose is 210 mg/dL, and sodium is 153 mEq/L (up from 138 mEq/L on admission). Which of the following is the most appropriate plan of care?

   A. Administer desmopressin, 2 μg, and encourage 2 to 3 L/day of oral water intake.

   B. Start vasopressin infusion at 0.04 U/min and stop IV fluids.

   C. Continue to match IV input to the previous hour's urine output.

   D. Restrict IV fluids to 150 mL/hr of crystalloid and administer desmopressin, 0.5 μg.

8. A 58-year-old woman with a history of chronic obstructive pulmonary disease, hypothyroidism, hypertension, and obesity is admitted from the emergency department with hypotension and lethargy and is found to have pneumonia. Vital signs are temperature 35.5°C (95.9°F), heart rate 52 beats/min, blood pressure 90/45 mm Hg, respiratory rate 8 breaths/min, and oxygen saturation 88% on 6 L/min face mask. Urinalysis findings show pH 5.0, specific gravity 1.006, urine osmolality 286 mOsm/kg, no ketones, leukocyte esterase negative, protein 9 mg/dL, and sodium 20 mEq/L. Serum laboratory findings show: sodium 132 mEq/L, potassium 3.2 mEq/L, BUN 16 mg/dL, creatinine 1.1 mg/dL, glucose 88 mg/dL, and thyroglobulin 18 μU/L. Which of the following is the most likely diagnosis?

   A. Cerebral salt wasting

   B. Diabetes insipidus

   C. Diabetic ketoacidosis

   D. Myxedema coma

   E. Syndrome of inappropriate antidiuretic hormone

9. Which of the following changes is least characteristic of protein metabolism in critical illness?

   A. Increased total body catabolism refractory to exogenous amino acids

   B. Decreased total body protein synthesis refractory to exogenous amino acids

   C. Reduced muscle amino acid uptake with increased amino acid release

   D. Increased hepatic amino acid uptake and synthesis of acute-phase reactant proteins

Part 4 Answers:

Endocrine

1. Rationale

Answer: C

Euglycemic diabetic ketoacidosis (DKA) is a life-threatening emergency and must be recognized early to allow for rapid treatment and favorable outcomes. Euglycemic DKA can occur at glucose levels below 200 mg/dL and must be considered in the differential diagnosis in all patients with diabetes who present with acidosis. Although euglycemic DKA occurs more frequently in patients with type 1 diabetes mellitus (DM), it can also occur in patients with type 2 DM (especially in patients taking sodium glucose transporter 2 inhibitors). Euglycemic DKA is associated with pregnancy, caloric starvation, heavy alcohol intake, cocaine abuse, pancreatitis, sepsis, and chronic liver disease. In pregnancy, the diagnostic threshold for bicarbonate is lower (typically ≤ 15 mEq/L) due to lower baseline bicarbonate in pregnancy. The priority in euglycemic DKA is rapid correction of the acidosis with fluids, insulin, and rarely sodium bicarbonate. In pregnancy, abnormalities in fetal tracings are typically due to maternal acidosis and usually improve with correction of maternal acidosis. Urgent maternal fetal medicine/obstetrics consultation is typically indicated. In this patient, euglycemic DKA is most likely due to her acidosis, elevated beta-hydroxybutyrate, and anion gap in the setting of known type 1 DM. Hyperosmolar hyperglycemia state does not present with acidosis. Although infection commonly precipitates DKA, she does not meet sepsis criteria. Preeclampsia does not present with elevated beta-hydroxybutyrate, and her history of insulin-dependent DM makes starvation ketoacidosis less likely.

References:

1. Barski L, Eshkoli T, Brandstaetter E, Jotkowitz A. Euglycemic diabetic ketoacidosis. *Eur J Intern Med*. 2019

May; 63:9-14.

2. De Veciana M. Diabetes ketoacidosis in pregnancy. *Semin Perinatol*. 2013 Aug;37(4):267-273.

2. Rationale

Answer: D

This patient's results are consistent with non-thyroid disease (euthyroid sick syndrome). Treatment is directed toward the underlying illness. No specific hormone replacement is indicated. IV dexamethasone is an appropriate treatment for thyroiditis; however, she has no history of thyroid disease, which, in the absence of symptoms, is highly unlikely, so IV dexamethasone is not indicated.

References:

1. Farwell A. Thyroid hormone therapy is not indicated in most patients with the sick euthyroid syndrome. *Endocr Pract*. 2008 Dec;14(9):1180-1187.

2. Hall JB, Schmidt GA, Kress JP. *Principles of Critical Care*. 4th ed. McGraw-Hill Education; 2015:987.

3. Rationale

Answer: B

In hungry bone syndrome there is a profound and sustained decrease in calcium, magnesium, and phosphorus due to the lack of parathyroid hormone (PTH) secretion after a parathyroidectomy, usually requiring aggressive replenishment. In pseudohypoparathyroidism, in which there is a resistance to PTH, PTH is elevated but calcium levels are low. In milk-alkali syndrome, calcium is elevated from inappropriately increased consumption of milk and calcium products. In secondary hyperparathyroidism, initially enlarged hyperactive parathyroid glands lead to low calcium level and high phosphorus level in the setting of elevated PTH level and later elevated calcium levels.

References:

1. Witteveen JE, van Thiel S, Romijn JA, Hamdy NA. Hungry bone syndrome: still a challenge in the post-operative management of primary hyperparathyroidism: a systematic review of the literature. *Eur J Endocrinol.* 2013 Feb 20;168(3):R45-R53.

2. Cunningham J, Locatelli F, Rodriguez M. Secondary hyperparathyroidism: pathogenesis, disease progression, and therapeutic options. *Clin J Am Soc Nephrol.* 2011 Apr;6(4):913-921.

4. Rationale

Answer: A

Patients who have had pituitary tumor resections are at risk for hypothalamic-pituitary axis complications due to surgical trauma to these structures. Patients should be closely monitored for central diabetes insipidus (DI), which results from insufficient production of vasopressin and can manifest as polyuria (> 40 mL/kg/24 hrs or > 250 mL/hr for > 2 hrs), low urine specific gravity, and increased serum sodium. Treatment of DI includes allowing patients to drink free water if they can keep up with their urinary losses. For those unable to keep up, 0.45% sodium chloride can be administered to replace losses. Additionally, desmopressin should be administered to prevent further free water loss. Desmopressin, a V2 receptor agonist, increases expression of the aquaporin 2 channel, which leads to reabsorption of water. This patient meets 2 of the 3 criteria for DI (urine output > 250 mL/hr and decreased urine specific gravity). Despite normal serum sodium level, the sample was drawn 6 hours ago and before the dramatic increase in urine output. Thus, it does not reflect his current serum sodium, which is likely increased. He should receive 0.45% sodium chloride to replace fluid losses, along with a dose of desmopressin to prevent further free water excretion. Vasopressin, a V1 and V2 receptor agonist, can also be used to treat refractory DI; however, its actions on V1 receptors result in arterial vasoconstriction and blood pressure augmentation. He has elevated blood pressures and has not yet failed any treatment, so he does not yet qualify for vasopressin.

Tolvaptan, a V2 receptor antagonist, and conivaptan, a V1 and V2 receptor antagonist, are used for the treatment of syndrome of inappropriate antidiuretic hormone secretion and would only worsen his free water excretion and hypernatremia. Providing no treatment is inappropriate because he is putting out excessive amounts of free water, as demonstrated by his urine output and urine specific gravity. If left untreated, his sodium may increase precipitously.

References:

1. Di Iorgi N, Napoli F, Allegri AE, et al. Diabetes insipidus: diagnosis and management. *Horm Res Paediatr.* 2012;77(2):69-84.

2. Schreckinger M, Szerlip N, Mittal S. Diabetes insipidus following resection of pituitary tumors. *Clin Neurol Neurosurg.* 2013 Feb;115(2):121-126.

5. Rationale

Answer: B

Although the 2008 guidelines suggested that the diagnosis of critical illness-related corticosteroid insufficiency (CIRCI) is best made by a delta total serum cortisol, the 2018 Society of Critical Care Medicine/European Society of Intensive Care Medicine guidelines provide no recommendations as to whether to use delta cortisol after cosyntropin administration or a random cortisol level; thus, both random cortisol level and change in cortisol in response to cosyntropin stimulation test can be used; however, according to the guidelines, relative adrenal insufficiency is diagnosed based on a change in baseline cortisol at 60 minutes of less than 9 μg/dL after cosyntropin administration or a random plasma cortisol level of less than 10 μg/dL. Measuring a morning cortisol level can be used for diagnosis, although measuring the level randomly is also acceptable; however, the diagnosis of CIRCI is based on a cortisol level of less than 10 μg/dL rather than less than 15 μg/dL. The assessment of adrenal insufficiency is based on the change in the pre- and post-cosyntropin test and not only on the post-cosyntropin level.

Reference:

1. Pastores SM, Annane D, Rochwerg G; Corticosteroid Guideline Task Force of SCCM and ESICM. Guidelines for the diagnosis and management of critical illness-related corticosteroid insufficiency (CIRCI) in critically ill patients (part II): Society of Critical Care Medicine (SCCM) and European Society of Intensive Care Medicine (ESICM) 2017. *Crit Care Med*. 2018 Jan;46(1):146-148.

6. Rationale

Answer: C

This patient presents with a pheochromocytoma crisis. Clinical presentation ranges from severe hypertension to circulatory failure and shock with the potential for involvement of multiple organ systems, including cardiovascular, pulmonary, neurologic, gastrointestinal, renal, and hepatic systems. Pheochromocytoma crisis can occur spontaneously or can be precipitated by manipulation of a tumor, trauma, certain medications (corticosteroids, beta-blockers, metoclopramide, and anesthetic agents), or stress from nonadrenal surgery. Initial management should be alpha blockade with a medication such as phenoxybenzamine, followed by beta blockade, if necessary, in combination with elective or urgent surgery. Emergency resection of pheochromocytoma is associated with high surgical morbidity and mortality and should be avoided.

Reference:

1. Scholten A, Cisco RM, Vriens MR, et al. Pheochromocytoma crisis is not a surgical emergency. *J Clin Endocrinol Metab*. 2013 Feb;98(2):581-591.

7. Rationale

Answer: D

Following pituitary resection, diabetes insipidus (DI) often presents in a triphasic pattern of altered posterior pituitary function, in an off/on/off sequence. This correlates to 1) endogenous vasopressin secretion ceasing due to local anterior pituitary trauma/resection, 2) recovery of posterior portal

circulation to release vesicle stores of hormone, and 3) exhaustion of vasopressin stores and hypothalamic failure to produce more. This is common particularly after craniopharyngioma. Patients may progress to any stage in this sequence to either recovery of function or permanent failure. Large-dose desmopressin and significant water intake risks hyponatremia, especially in the context of return of function. Vasopressin is used for shock at a dose of 0.04 U/min but would need to be titrated against urine output for adequate control of DI. The practice of matching urine output to IV intake can achieve euvolemia, especially in cases where vasopressin secretion is reduced, rather than arrested but, if continued, will eventually wash out the corticomedullary sodium gradient, compromising the kidney's concentrating ability. Administration of low-dose desmopressin (low dose to avoid excessive water retention and consequent hyponatremia), in conjunction with saline, is a reasonable initial strategy to maintain volume, avoid further losses, and avoid any pathology of hyponatremic change.

References:

1. Ghirardello S, Hopper N, Albanese A, Maghnie M. Diabetes insipidus in craniopharyngioma: postoperative management of water and electrolyte disorders. *J Pediatr Endocrinol Metab*. 2006 Apr;19 Suppl 1:413-421.
2. Kurbel S, Dodig K, and Radic R. The osmotic gradient in kidney medulla: a retold story. *Adv Physiol Educ.* 2002 Dec;26(1-4):278-281.

8. Rationale

Answer: D

Severe hypothyroidism, or myxedema coma, is an endocrine emergency that requires a high index of suspicion. Signs and symptoms include confusion, lethargy, bradycardia, hypotension, hyponatremia, hypoglycemia, and hypothermia. Patients can have severe manifestations, including impaired cardiac output, arrhythmias, and hypoxia, which can result in mortality. Precipitating factors include hypothermia, infections,

cerebrovascular accidents and, in critically ill patients, withdrawal from thyroid medications. Patients with cerebral salt wasting and syndrome of inappropriate antidiuretic hormone usually present with moderate to severe hyponatremia, hypovolemia, increased urinary osmolality (usually > 300 mOsm/kg), and increased urinary sodium (usually > 40 mEq/L). Patients with diabetes insipidus present with hypernatremia. Normal glucose and absence of ketones rule out diabetic ketoacidosis.

References:

1. Hoorn EJ, Zietse R. Diagnosis and treatment of hyponatremia: compilation of the guidelines. *J Am Soc Nephrol*. 2017 May;28(5):1340-1349.

2. Mathew V, Misgar RA, Ghosh S, et al. Myxedema coma: a new look into an old crisis. *J Thyroid Res*. 2011; 2011:493462.

9. Rationale

Answer: B

Nitrogen balance is estimated by subtracting nitrogen losses from all sources from nitrogen intake. A positive nitrogen balance of 2 to 4 g/day is desirable. Nitrogen balance and protein metabolism in critical illness are characterized by an increased total body catabolism that is refractory to exogenous amino acids. The decreased total body protein synthesis, however, is responsive to exogenous amino acids. There is reduced muscle amino acid uptake with increased amino acid release and an increased hepatic amino acid uptake and synthesis of acute-phase reactant proteins. This increased rate of protein synthesis by the liver is required for wound healing, granulocytic activity, and other immunologic defenses. The patient's preexistent nutritional and metabolic status is a major determinant of the response to critical injury or illness. Metabolic adaptation with a conversion to ketones as an energy substrate does not occur in critically ill patients. One result of this is an increase in muscle breakdown and the use of protein for fuel. The catabolic response peaks after 3 to 5 days, then

declines if there are no more insults to the organism. Protein catabolism is favored by increases in all catabolic hormones and peripheral resistance to insulin. Gluconeogenesis is not suppressed by exogenous glucose and protein catabolism proceeds. Protein synthesis is inadequate and negative nitrogen balance results. While protein catabolism cannot be stopped, hyper-catabolism can be prevented and a sparing effect on lean body mass effected by the administration of bicarbonate and fat as energy substrate. The administration of amino acids can support protein synthesis by ensuring adequate precursor availability.

References:

1. Abrams JH, Cerra F, Holcroft JW. Cardiopulmonary monitoring. In: Holcroft JW, Meakins JL, Wilmore DW, Brenna MF, Harken AH, eds. *Care of the Surgical Patient*. Scientific American; 1989:1-27.

2. Cerra FB, Upson D, Angelico R, et al. Branched chains support post operative protein synthesis. *Surgery*. 1982 Aug;92(2):192-199.

3. Hasselgren PO, Jagenburg R, Karlstrom L, Pedersen P, Seeman T. Changes of protein metabolism in liver and skeletal muscle following trauma complicated by sepsis. *J Trauma*. 1984 Mar;24(3):224-228.

4. Kinney JM, Furst P, Elwyn DH, et al. The intensive care patient. In: Kinney JM, Jeejeebhoy KN, Hill EL, Owen, eds. *Nutrition and Metabolism in Patient Care*. Saunders; 1988:656-671.

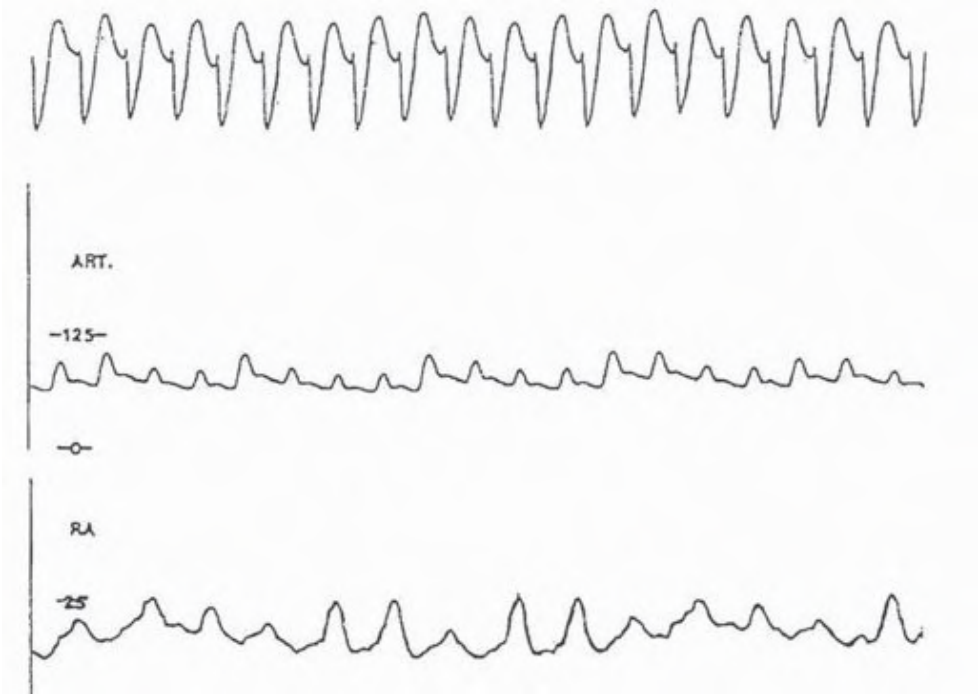
5. Long C, Kinney JM, Geiger JW. Nonsupressibility of gluconeogenesis by glucose in septic patients. *Metabolism*. 1976 Feb;25(2):193-201.

Part 5.

Cardiovascular Disorders

Instructions: For each question, select the most correct answer.

1. A 60-year-old woman is admitted with an acute anterior myocardial infarction. She has a history of paroxysmal supraventricular tachycardia and right bundle branch block. Her only medication is digoxin. She refuses thrombolytic therapy. Antianginal medications are begun. Four hours later, she develops congestive heart failure. A pulmonary artery thermodilution catheter is placed and appropriate therapy is administered, resulting in stable hemodynamics and remission of chest pain. Twelve hours later, she develops a wide-complex tachycardia. She denies chest pain or dyspnea. Blood pressure is 105/60 mm Hg, and heart rate is 160 beats/min and regular. The rhythm strip (MCL1 lead), arterial, and right atrial pressure tracings are shown below. Which of the following is the most appropriate initial treatment?



A. Verapamil, 5 mg IV

B. Lidocaine bolus followed by continuous infusion at 2 mg/min

C. Synchronized cardioversion at 100 J

D. Defibrillation at 200 J

2. An 80-year-old man with a history of type 2 diabetes, hypertension, peripheral artery disease, and cerebrovascular accident presents to the emergency department with a 1-hour history of severe pressure-like chest pain. He is administered aspirin. ECG shows ST elevation in leads V4–V6 with reciprocal ST depressions. He is transported emergently to the catheterization laboratory where a stent is placed in the left anterior descending artery, and he is transferred to the ICU. Which of the following medications is contraindicated for him?

   A. Clopidogrel

   B. Prasugrel

   C. Ticagrelor

   D. Cangrelor

3. A 40-year-old woman has had several episodes of exertional substernal chest pain with radiation to the jaw and shoulders during the past several days. She had a miscarriage 1 month ago. She does not smoke cigarettes or use illicit drugs. Family history is significant for Ehlers-Danlos syndrome. On physical examination, she is diaphoretic. Temperature is 36.6°C (97.8°F), heart rate 66 beats/min, blood pressure 141/85 mm Hg in both arms, respiratory rate 17 breaths/min, and oxygen saturation as measured by pulse oximetry 100% on ambient air. She has no carotid bruits or jugular venous distention. She has no cardiac murmur, rub, or gallop on auscultation. Her digits are hyperextensible. Troponin level is 0.21 ng/dL. Chest radiograph is normal. ECG shows normal sinus rhythm and submillimeter down-sloping ST segment depressions in leads V3, V4, and V5. Transthoracic echocardiography shows normal left ventricular and right ventricular systolic

function and a small focal area of hypokinesis in the left ventricular inferolateral wall. Which of the following is the most likely diagnosis?

A. Myopericarditis

B. Pulmonary embolism

C. Takotsubo cardiomyopathy

D. Aortic dissection

E. Spontaneous coronary artery dissection

4. A 67-year-old man was admitted to the ICU 1 hour ago after undergoing surgical aortic valve replacement and single-vessel coronary artery bypass grafting via midline sternotomy. He is being paced with epicardial ventricular wires at 80 beats/min and has a conventional pulmonary artery catheter in situ. Ventricular capture is suddenly lost and cannot be restored. Third-degree heart block with a malperfusing junctional escape rhythm at 40 beats/min is noted, with mean arterial pressure 45 mm Hg despite rapid titration of vasopressors and inotropes. Transcutaneous pacing is attempted without success. Soon thereafter asystole develops, and external chest compressions are begun. While continuing external compressions, which of the following is the most appropriate next step?

   A. Transvenous pacing

   B. Emergent re-sternotomy

   C. Atropine administration

   D. Focused echocardiography

5. A 58-year-old man with a past medical history of hypertension, type 2 diabetes mellitus, and coronary artery disease presents to the cardiovascular ICU intubated and mechanically ventilated after a 3-vessel coronary artery bypass graft. The surgery was notable for coagulopathy with substantial blood loss requiring massive transfusion efforts. Six hours after ICU arrival, he develops acute-onset hypotension and anuria despite

volume resuscitation. Hemodynamic parameter review findings are shown below. Which of the following is the most likely diagnosis?

Hemodynamic Parameter	Test Result
Blood pressure	62/39 mm Hg
Heart rate	113 beats/min
Central venous pressure	21 mm Hg
Pulmonary artery pressure	28/12 mm Hg
Pulmonary artery occlusion pressure	12 mm Hg
Thermodilution cardiac index	1.4 L/min/m^2

A. Hypovolemia

B. Vasoplegia

C. Cardiac tamponade

D. Right ventricular failure

E. Left ventricular failure

6. A 73-year-old man with a history of diastolic dysfunction has a myocardial infarction and undergoes emergent percutaneous angioplasty. Immediately afterward, his blood pressure falls to 93/70 mm Hg. Right heart catheterization reveals pulmonary capillary wedge pressure 17 mm Hg and low cardiac output. Which of the following is the most appropriate initial treatment?

   A. Dopamine

   B. Dobutamine

   C. Norepinephrine

   D. Fluid bolus

   E. IV furosemide

7. A 55-year-old man is being discharged from the ICU after admission for heart failure. Ejection fraction is 40%. Vital signs are temperature 37.2°C (98.9°F), heart rate 70 beats/min, respiratory rate 18 breaths/min, and blood pressure 120/76 mm Hg. His discharge medications are metoprolol, 50 mg twice daily; spironolactone, 25 mg twice daily; furosemide, 40 mg twice daily; aspirin, 160 mg once daily; and atorvastatin, 40 mg once daily. His 12-lead ECG is normal. Which of the following serum

parameters should be obtained before discharging him home?

A. B-type natriuretic peptide

B. Troponin I

C. Phosphorus

D. Creatinine kinase

8. A 70-year-old woman had a heart failure exacerbation and underwent right heart catheterization, which demonstrated reduced cardiac index of 1.8 L/min/m^2, elevated central venous pressure of 20 mm Hg, and pulmonary capillary wedge pressure of 22 mm Hg. She was started on a milrinone infusion, and diuresis was started with IV bumetanide, 2 mg/hr continuous infusion. The next day she remains fluid overloaded and is unable to generate a net negative fluid balance for the day. Hemodynamics are cardiac index 2.5 L/min/m^2, central venous pressure 18 mm Hg, pulmonary capillary wedge pressure 19 mm Hg, and blood pressure 100/80 mm Hg. Which of the following medications should be added to enhance her diuresis?

   A. Acetazolamide

   B. Spironolactone

   C. Metolazone

   D. Dobutamine

   E. Tolvaptan

9. A patient with a history of heart failure and ejection fraction of 32% is admitted to the ICU with increasing shortness of breath. The patient's current medications are captopril, aspirin, furosemide, and metoprolol. Potassium is 4.9 mEq/L, serum creatinine 0.8 mg/dL, blood glucose 154 mg/dL, and B-type natriuretic peptide 225 pg/mL. Chest radiograph shows perihilar infiltrates. Which of the following is recommended before discharge to reduce this patient's mortality and morbidity risks?

A. Replace captopril with an angiotensin receptor/neprilysin inhibitor.

B. Add an angiotensin receptor blocker to the angiotensin-converting enzyme regimen.

C. Discontinue furosemide.

D. Substitute lisinopril for captopril.

E. Discontinue metoprolol.

10. A 65-year-old man had an out-of-hospital cardiac arrest. Initial rhythm was asystole. Return of spontaneous circulation (ROSC) occurred after 10 minutes of resuscitation, and multiple epinephrine doses were administered. Defibrillation was not needed. Currently he is unresponsive in the ICU and not following any commands. ECG performed post-ROSC shows normal sinus rhythm with no acute ST segment changes. Telemetry strip shows some premature ventricular contractions. Head CT shows no acute abnormalities. Which of the following next steps is most likely to improve his outcome?

    A. Mild hypothermia protocol

    B. Emergent cardiac catheterization

    C. IV amiodarone

    D. Neurology consultation

11. A 46-year-old man with a past medical history significant for hypertension presents to the emergency department with retrosternal crushing chest pain with nausea and vomiting on the way to the hospital. Vital signs are heart rate 110 beats/min, blood pressure 180/100 mm Hg, temperature 37.7°C (100°F), respiratory rate 24 breaths/min in mild distress, and oxygen saturation 95% on room air. Physical examination reveals mild distress due to pain. His pain responds to IV morphine; blood pressure is now 160/90 mm Hg and heart rate 90 beats/min. ECG reveals nonspecific ST-T wave changes. CT is shown below. In addition to pain management, which of the

following is/are the most appropriate next step(s) in management?



A. Left heart catheterization alone

B. IV beta-blockers and left heart catheterization

C. IV beta-blockers and surgery consultation

D. IV beta-blockers alone

E. Pain management alone

12. A 65-year-old man is evaluated for new-onset seizures. His wife reports that he has been feeling ill for the past 3 weeks, with fever and poor appetite. She says that he has been short of breath and has started to sleep in a recliner. Vital signs are blood pressure 98/62 mm Hg, heart rate 112 beats/min, respiratory rate 18 breaths/min, and oxygen saturation as measured by pulse oximetry 90% on 4 L nasal cannula. Physical examination is notable for the finding shown below. Further history is most likely to reveal that this patient



A. has moderate mitral regurgitation and recently underwent endoscopic evaluation of his vocal cords.

B. had a bioprosthetic mitral valve replacement 1 year ago and recently had a dental cleaning.

C. had a heart transplant 5 years ago and recently underwent colonoscopy for routine screening.

D. had a coronary stent placed 8 months ago and recently underwent transurethral resection of his prostate.

13. A 55-year-old man is admitted to the ICU with a 4-day history of progressive dyspnea on exertion. He reports intermittent fevers and malaise for almost 3 weeks, starting several days after he had his first dental cleaning in several years. He says that he has a heart murmur from a childhood infection. Temperature is 38.8°C (101.8°F), heart rate 110 beats/min, and respiratory rate 30 breaths/min. He has a low, rumbling diastolic murmur and a loud holosystolic murmur. ECG reveals sinus tachycardia without heart block. Transthoracic echocardiography shows no vegetation. Two blood culture bottles are positive for gram-positive cocci in chains. Which of the following is the most appropriate next step in evaluation?

A. Antistreptolysin O titer

B. Dilated funduscopic examination

C. C-reactive protein level

D. Transesophageal echocardiography

E. Tagged WBC scan

14. A 35-year-old woman develops severe nausea and vomiting following a routine appendectomy. She has a history of motion sickness and often takes prochlorperazine when traveling long distances. During the procedure she was administered IV morphine, 6 mg. Now she is administered IV metoclopramide, 10 mg; IV ondansetron, 4 mg; and IV dexamethasone, 4 mg, for nausea and vomiting. Which of the following is the most appropriate next step for her continued postoperative nausea and vomiting?

    A. Administration of IV prochlorperazine, 5 mg

    B. ECG to look for QT interval prolongation

    C. Abdominal contrast CT to look for acute intestinal obstruction

    D. Administration of another 8 mg of IV ondansetron

15. A 45-year-old man is evaluated in the emergency department after a motor vehicle accident. History is significant for alcohol and drug abuse. He is hemodynamically unstable and is transferred to the ICU for advanced management. Heart rate is 150 beats/min, blood pressure 86/45 mm Hg, and oxygen saturation as measured by pulse oximetry 90%. Central access is obtained, and a pulmonary arterial catheter is placed. An arterial line is also placed, with telemetry tracing significant for exacerbated decrease in systolic pressure with inspiration. Central venous pressure is 21 mm Hg, and the tracing seems to be lacking Y descent. Cardiac output via Fick method is 2.6 L/min. Which of the following steps should be taken for which cause of shock?

    A. Initiate massive transfusion protocol for hypovolemic shock.

    B. Initiate antimicrobial therapy for acute pericarditis.

    C. Place a chest tube for a hemopneumothorax.

D. Perform pericardiocentesis for acute tamponade.

16. A 68-year-old man underwent endovascular repair of a ruptured infrarenal abdominal aortic aneurysm. He received 8 units packed RBCs, 6 units fresh frozen plasma, 2 apheresis units' platelets, and 1 L crystalloid solution before the repair was completed. He was admitted to the ICU intubated for respiratory distress. Initially, his hemodynamic status normalized after resuscitation. Twenty-four hours after ICU admission, blood pressure is 100/40 mm Hg, heart rate 120 beats/min, and urine output 20 mL/hr. His abdomen is distended but soft, with guarding on deep palpation to the left lower quadrant. Laboratory results show WBC count 18,000, hematocrit 24%, which is unchanged, and platelet count 170,000. Arterial blood gas analysis shows pH 7.23, PCO_2 35 mm Hg, PO_2 119 mm Hg, bicarbonate 17 mEq/L, and base deficit 10 mEq/L. Which of the following is/are the most appropriate next step(s) in management?

    A. Administer 1 L 0.9% normal saline and monitor urine output for the next 4 hours.

    B. Start broad-spectrum antibiotics because he meets criteria for sepsis.

    C. Obtain pathology consultation for possible transfusion-related acute lung injury.

    D. Perform flexible sigmoidoscopy to evaluate for colonic ischemia.

    E. Repeat CT angiogram of the chest, abdomen, and pelvis.

17. A 66-year-old man with medical history significant for severe coronary artery disease presents with acute onset of dyspnea. Echocardiography is significant for severe mitral valve regurgitation related to a ruptured posterior papillary muscle. Initiation of which of the following interventions is most likely to be an effective strategy for relieving pulmonary edema before surgical valve repair and revascularization?

A. Inhaled nitric oxide

B. Esmolol infusion

C. Vasopressin infusion

D. Intra-aortic balloon pump therapy

E. Therapeutic hypothermia

18. During hemodialysis with heparin, a well-dialyzed woman with chronic renal failure reports chest pain and dyspnea and develops hypotension (80/60 mm Hg) refractory to IV volume infusion. Lungs are clear and neck veins are distended. She has no cough or hemoptysis. Which of the following is the most important diagnostic test to perform?

    A. Ventilation/perfusion lung scan

    B. ECG followed by left heart catheterization

    C. ECG followed by right heart catheterization

    D. Thallium scan

    E. Chest radiograph

19. A 66-year-old man collapsed at work. He received immediate chest compressions by a coworker. When emergency medical services arrived 15 minutes later, he was in shockable rhythm. He received 1 shock and had return of spontaneous circulation. He was intubated and transferred to the ICU, where he was actively cooled to 33°C (91.4°F). The monitor showed tachycardic sinus rhythm. Transthoracic echocardiography showed moderate global dysfunction. EEG and ECG are normal. Electrolytes and liver function tests are within normal limits, but lactate level is elevated. After 16 hours of hypothermia, lactate remains elevated, and he continues to require vasopressors. Which of the following is the most appropriate next step to improve his condition?

    A. Discontinue the vasopressor.

    B. Warm him to 36°C (96.8°F).

    C. Decrease sedation.

D. Add neuromuscular blockade.

20. A 45-year-old man with a destination left ventricular assist device (LVAD) is admitted for pneumonia and found unresponsive in his room on hospital day 2. He has agonal respirations, and mask ventilation is initiated. End-tidal CO_2 by capnometry reads 15 mm Hg. Capillary refill is poor and mean arterial pressure by arterial line access is 30 mm Hg. Quick interrogation of the LVAD reveals a normal hum and no recent alarms. Which of the following is the most appropriate next step in management?

    A. Restart the LVAD.

    B. Initiate external chest compressions.

    C. Administer a crystalloid bolus for management of septic shock.

    D. Administer emergent systemic anticoagulation.

21. A 55-year-old man presents to the ICU following ventricular assist device implantation. Initially, hemodynamics were within goal; however, 6 hours postoperatively his urine output started to decline and he became more hypotensive. Hemodynamics are heart rate 110 beats/min, blood pressure 80/60 mm Hg, central venous pressure 20 mm Hg, pulmonary artery pressure 40/25 mm Hg, pulmonary capillary wedge pressure 20 mm Hg, and cardiac output 4 L/min. Laboratory findings are mostly unremarkable except for serum creatinine 1.8 mg/dL (increase from 1.1 mg/dL baseline), ALT 2000 U/L, and AST 1500 U/L. His current hemodynamic medications are: milrinone, 0.5 μg/kg/min; dobutamine, 5 μg/kg/min; norepinephrine, 0.1 μg/kg/min; and furosemide, 10 mg/hr. Which of the following medications is most appropriate at this point?

    A. IV nitroglycerin

    B. Inhaled epoprostenol

    C. IV nitroprusside

D. Oral sildenafil

22. A 73-year-old man with end-stage heart failure and dry body mass index (BMI) of 39 kg/m^2 is admitted to the ICU for medical management of cardiogenic shock. His 1-year survival is estimated to be 68% by the Seattle Heart Failure Model. He is taking dobutamine, 7 μg/kg/min, and is started on a bumetanide infusion. An intra-aortic balloon pump is inserted to further unburden his left ventricle. His pulmonary artery catheter shows central venous pressure 12 mm Hg, mixed venous oxygenation 58%, pulmonary vascular resistance (PVR) 2 Wood units, and pulmonary wedge pressure 15 mm Hg. Which of the following parameters precludes heart transplantation in this patient?

    A. Age 73 years

    B. Dry BMI 39 kg/m^2

    C. Estimated 1-year survival 68%

    D. PVR 2 Wood units

23. A 74-year-old man arrives in the ICU with chest pain and ST-segment elevation in leads V1 through V4 on ECG. Troponin elevation confirms the diagnosis of acute anterior myocardial infarction. He is treated with percutaneous coronary intervention (PCI) with a drug-eluting stent placed into his left anterior descending coronary artery. Further management includes aspirin, clopidogrel, metoprolol, heparin, and atorvastatin. On day 2, blood pressure decreases to 70/40 mm Hg. Pulmonary artery catheterization reveals cardiac index of 1.6 L/min/m^2 and pulmonary artery occlusion mean pressure of 28 mm Hg. No oxygen step-up is seen in right heart oximetry values. Echocardiogram reveals a large akinetic anterior wall with overall left ventricular ejection fraction of 20%. The echocardiogram shows no evidence of a ventricular septal defect or severe mitral regurgitation. Right ventricular function is normal. Which of the

following is true regarding cardiogenic shock following myocardial infarction?

A. Controlled clinical trials using nitric oxide inhibitors have shown improved mortality and shortened shock duration.

B. Systolic blood pressure, measured on vasopressor support, is a significant predictor of 30-day mortality.

C. Norepinephrine or dopamine therapy are comparable treatments for this disease.

D. APACHE II score does not predict outcome in this disease.

24. A 51-year-old woman with a history of breast cancer is brought to the emergency department by emergency medical services for chest pain and dyspnea. She reports sudden onset of dyspnea and pain approximately 30 minutes before arrival. Vital signs are respiratory rate 33 breaths/min, heart rate 127 beats/min, and blood pressure 87/46 mm Hg. Vasopressors are initiated to maintain mean arterial pressure above 65 mm Hg. She requires 10 liters of oxygen via nasal cannula to maintain oxygen saturation at 90%. She is awake but is becoming more confused and lethargic. Physical examination reveals distended neck veins and diminished breath sounds on the right. Bedside echocardiography shows an enlarged right ventricle. ECG shows an S wave in lead 1, Q wave in lead 3, and inverted T wave in lead 3. Which of the following is the next step in this patient's management?

A. Embolectomy

B. Catheter-directed thrombectomy

C. Systemic thrombolytics

D. Low-molecular-weight heparin

E. Warfarin

25. A 60-year-old man with a history of chronic obstructive pulmonary disease and congestive heart failure presents with shortness of breath, chest pain, productive cough,

and subjective fevers for 2 days. He says that he has quit smoking and has been adherent with his medications and inhalers. On ultrasound, A lines are present throughout the left lung and B lines diffusely on the right. These findings are most consistent with which of the following diagnoses?

A. Pneumothorax

B. Pulmonary edema

C. Pneumonia

D. Pulmonary embolism

26. A 29-year-old woman is admitted to the surgical ICU after giving birth to twins via cesarean delivery. History reveals that she underwent repair of tetralogy of Fallot when she was a child. Six hours later, she remains intubated, systolic blood pressure ranges from 110 to 120 mm Hg, heart rate is 100 beats/min, and she has minimal urine output. Which of the following is the immediate next step in her management?

A. Initiating furosemide to increase urine output

B. Adding a beta-blocker to slow her heart rate

C. Adding an inotrope to increase her heart rate

D. Infusing IV crystalloids to maintain high preload

27. An 82-year-old man is admitted to the ICU with new-onset shortness of breath and severe fatigue. Vital signs and laboratory values are within normal limits except for brain natriuretic peptide of 900 pg/mL. ECG shows repolarization abnormalities. A very focused echocardiogram is performed. Which of the following findings would suggest systolic failure on focal bedside echocardiography?

A. Pericardial fluid with diastolic compression of the right atria

B. No specific changes

C. E-wave-to-A-wave ratio greater than 2 and deceleration time less than 160 ms

D. Significant tricuspid regurgitation

E. Fractional shortening of 24% and E-point septal separation of 11 mm

28. A 51-year-old woman on postoperative day 1 after laparoscopic cholecystectomy presents to the emergency department for new-onset chest pain. Twelve-lead ECG is consistent with diffuse ST elevation. Troponin I is 4 ng/mL. She undergoes emergent left heart catheterization, and no significant plaque is found. Right heart catheterization is significant for pulmonary artery pressure 42/25 mm Hg, cardiac index 1.9 L/min/m^2, and central venous pressure 18 mm Hg. She is transferred to the ICU for further management. Vital signs are heart rate 140 beats/min, blood pressure 89/55 mm Hg, and oxygen saturation as measured by pulse oximetry 96%. Bedside surface echocardiography is significant for left ventricular ejection fraction 20%, left ventricular basal hyperkinesis, and dynamic left ventricular outflow tract obstruction. Which of the following is the most appropriate therapy for her?

A. Amiodarone

B. Nitric oxide

C. Esmolol

D. Normal saline infusion

29. A 67-year-old man with an automated implantable cardioverter defibrillator (AICD) develops cardiac arrest after routine colonoscopy. CPR is initiated. When the team arrive, they find the patient in ventricular fibrillation. According to guidelines, which of the following is the most appropriate next step?

A. Wait for the AICD to deliver the shock.

B. Administer a 1-mg IV bolus of epinephrine.

C. Administer a 300-mg IV bolus of amiodarone.

D. Deliver defibrillation with external pad.

30. A 68-year-old man with a past medical history significant for diabetes and hypertension is brought to the emergency department via ambulance for anxiety, chest pain, and dyspnea. His symptoms have lasted for approximately 1 hour and are not associated with any radiation. He describes his chest pain as a midsternal pressure. Vital signs are heart rate 109 beats/min, respiratory rate 28 breaths/min, temperature 36.4°C (97.5°F), blood pressure 143/79 mm Hg, with a systolic blood pressure of 119 mm Hg during inspiration, and oxygen saturations 93%. Examination reveals jugular venous distension in addition to a paradoxical pulse palpated radially. Chest radiograph reveals no acute findings. Laboratory data and ECG are unremarkable except for elevated D-dimer level. Chest CT angiography is negative. An emergent echocardiogram shows a 13-mm echo-free space around the heart. No valvular disease, calcifications, or emboli are seen. Which of the following is the first step in his treatment?

    A. Thrombectomy

    B. Pericardiocentesis

    C. Administration of an inotropic agent

    D. Heparin infusion

    E. Emergent heart catheterization

31. A 48-year-old woman is admitted to the ICU for management of an acute hemorrhagic stroke involving the left basal ganglia. Blood pressure is 180/98 mm Hg and heart rate 88 beats/min. Based on her acute presentation, which of the following medication regimens is most appropriate to manage her blood pressure?

    A. Bolus of IV hydralazine every 4 hours as needed

    B. Continuous infusion of IV nitroprusside

    C. Continuous infusion of IV nicardipine

    D. Continuous infusion of IV nitroglycerin

32. A 59-year-old man presents to the emergency department with blood pressure 230/135 mm Hg and heart rate 61 beats/min. His wife says that he has had significant confusion, which is worse this morning, and that he has had a severe headache for the past 2 days before this new onset of altered mental status. He takes lisinopril, 20 mg every 24 hours; amlodipine, 5 mg every 24 hours; hydrochlorothiazide, 25 mg every 24 hours; and metformin, 1000 mg twice daily. He has a soybean allergy. Initial laboratory results are sodium 140 mEq/L, chloride 103 mEq/L, potassium 4.6 mEq/L, CO_2 28 mEq/L, BUN 30 mg/dL, serum creatinine 0.98 mg/dL (baseline 0.8 mg/dL), and glucose 185 mg/dL. Lungs are clear to auscultation bilaterally, and he has +1 pitting edema. Chest radiograph is clear with slight cardiomegaly. Intracranial hemorrhage and acute ischemic stroke have been ruled out. Which of the following medications should be initiated?

    A. Esmolol

    B. Labetalol

    C. Clevidipine

    D. Nicardipine

33. Dopamine primarily causes stimulation of alpha-1 receptors at which of the following doses?

    A. 0-5 μg/kg/min

    B. 6-10 μg/kg/min

    C. 11-20 μg/kg/min

    D. 25-30 μg/kg/min

34. A 67-year-old man is in the ICU after undergoing liver transplantation. On postoperative day 3 he becomes short of breath and his heart rate increases to 190 beats/min. His blood pressure decreases to 78/50 mm Hg. ECG shows a regular monomorphic narrow-complex tachycardia. Which of the following is the most appropriate next step in management?

A. Begin synchronized cardioversion.

B. Administer adenosine.

C. Begin unsynchronized cardioversion.

D. Administer metoprolol.

E. Administer epinephrine.

35. A 22-year-old, 60-kg (132-lb) woman is admitted to the surgical ICU for postoperative management after laparoscopic appendectomy. She has hypoplastic left heart syndrome and underwent a Fontan procedure at age 8 years. During the 2-hour appendectomy, she had minimal urine output because of intraoperative negative fluid balance, which had been recommended by cardiology. She remains intubated when she arrives in the ICU with temperature 36.7°C (98°F), heart rate 120 beats/min, and systolic blood pressure 98 mm Hg, while on a continuous phenylephrine infusion to maintain mean arterial pressure above 65 mm Hg. She requires minimal ventilator support. Glasgow Coma Scale score is 11T after reversal of neuromuscular blockade. Lactate level at ICU admission is 2 mmol/L. Which of the following is the most appropriate immediate next step in management?

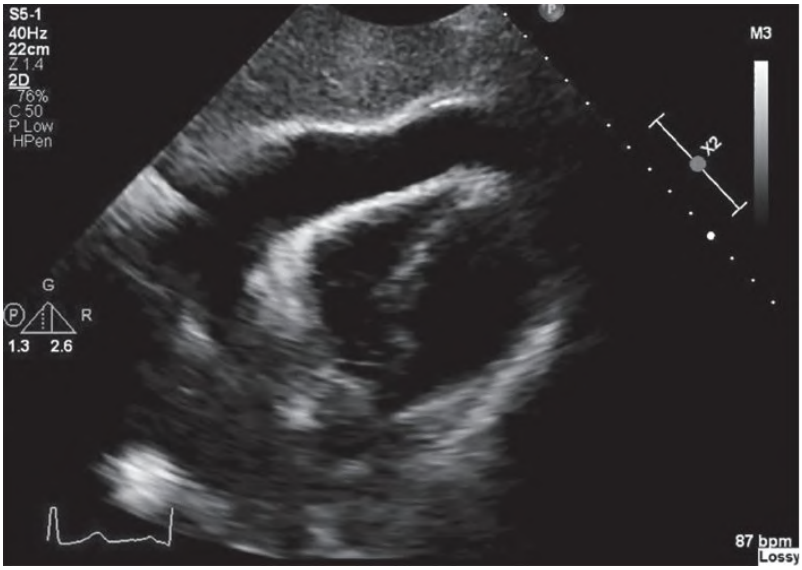
    A. Aggressive crystalloid administration

    B. Initiating IV furosemide infusion

    C. Discontinuing phenylephrine infusion

    D. Initiating norepinephrine in addition to the phenylephrine infusion

36. After receiving a first chemotherapy treatment, a 20-year-old woman with a recent diagnosis of lymphoma presents with shortness of breath and chest pain. The chest pain is worse with breathing. She is anxious, with heart rate 100 beats/min, blood pressure 90/60 mm Hg, and respiratory rate 20 breaths/min. Chest radiograph shows cardiomegaly with clear lungs. Echocardiogram is shown below. Which of the following is/are the most appropriate next step(s) in management?



A. Pericardial drain

B. IV fluids and vasopressors

C. Nonsteroidal anti-inflammatory drugs

D. Valve surgery

37. A 72-year-old woman presents with chest pain, shortness of breath, and a loud systolic murmur. Echocardiography shows left ventricular outflow track 2.4 cm, left ventricular outflow track peak velocity 100 cm/s, and aortic jet peak velocity 5.0 m/s. What is the aortic valve area?

A. 0.5 cm^2

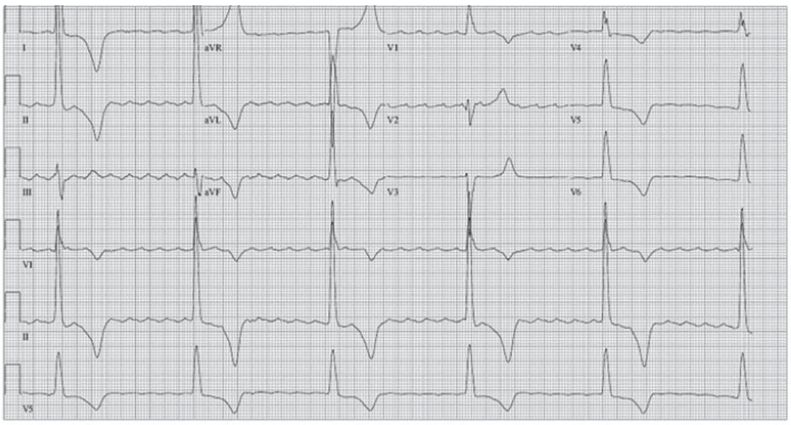
B. 0.9 cm^2

C. 1.6 cm^2

D. 3.6 cm^2

38. A 67-year-old man with a history of severe peripheral vascular disease and coronary artery disease presents to the emergency department with progressively worsening dyspnea over the past 24 hours. He underwent a transcatheter aortic valve replacement 4 days ago. He is tachypneic and diaphoretic. Vital signs are heart rate 32 beats/min, respiratory rate 30 breaths/min, blood pressure 92/50 mm Hg, and oxygen saturation 82% on room air,

improving to 92% on 5 L/min supplemental oxygen. Examination is remarkable for bradycardia without any significant murmurs, crackles in the lung bases bilaterally, and cyanotic digits in upper and lower extremities. Lactate level is 6.6 mmol/L. Chest radiograph reveals mild diffuse bilateral interstitial infiltrates. His medications include apixaban, clopidogrel, metoprolol, and atorvastatin. ECG is shown below. Which of the following is the most appropriate next step in his management?



A. Synchronized cardioversion

B. Pacemaker implant

C. Thrombolytic therapy

D. Cardiac catheterization

E. Continuation of current management

39. A 64-year-old man is admitted to the ICU with worsening shortness of breath over the past 4 days. Vital signs are temperature 38.05°C (100.5°F), heart rate 124 beats/min, respiratory rate 34 breaths/min, blood pressure 80/62 mm Hg supine, and oxygen saturation as measured by pulse oximetry 85% on room air. Urine output has declined from 600 mL/day to 100 mL/day despite no change in fluid intake. Peripheral extremities are cold and clammy. The family says that he has become more confused over the past week. Serum electrolyte levels are sodium 136 mEq/L, potassium 5.8 mEq/L, chloride 101 mEq/L, BUN 74 mg/dL, and creatinine 4.6 mg/dL. Central venous pressure is 28 mm Hg. After administering

oxygen, which of the following is the most appropriate next step to resolve his symptoms?

A. Bilateral chest tube placement

B. Pericardiocentesis with intrapericardial catheter left in place after echocardiographic confirmation

C. Tissue plasminogen activator, IV infusion of 0.9 mg/kg over 2 hours

D. Continuous venovenous hemofiltration with 2-mEq/L potassium bath at 20- to 30-mL/kg/hr target effluent flow rate

Part 5 Answers:

Cardiovascular Disorders

1. Rationale

Answer: B

It can be difficult to emergently differentiate wide-complex supraventricular tachycardia (SVT) from ventricular tachycardia (VT). ECG distinctions favoring a ventricular origin have been described (including QRS > 0.14 sec, left axis deviation, fusion beats), but therapeutic decisions are frequently based on rhythm strips. Atrioventricular (AV) dissociation is an important indication of the ventricular origin of an arrhythmia, although it may not always be appreciable on ECG. The right atrial pressure tracing demonstrates irregular cannon A waves due to AV dissociation during VT. Cannon waves result from atrial contraction against a closed tricuspid valve. The atria can contract against closed AV valves during VT, since atrial systole occurs without any relationship to ventricular systole. The cannon waves are associated with dips in the arterial pressure tracing because there is no atrial contribution to ventricular filling during these beats. During paroxysmal SVT, atrial and ventricular contractions remain associated, although the sequence of contraction is reversed so that cannon waves are present but are regular and of equal magnitude. When atrial contraction occurs with closed AV valves, the loudness of S1 is diminished; varying intensity of S1 is another indication of AV dissociation and VT. The most appropriate initial treatment for sustained VT without significant hemodynamic instability is a lidocaine bolus followed by a continuous infusion at 2 mg/min. Verapamil is commonly administered when VT is incorrectly diagnosed as SVT, resulting in hemodynamic deterioration. Synchronized cardioversion beginning at 50 J is the recommended initial treatment for sustained VT with a pulse when hemodynamic instability is present. Synchronized cardioversion with 75-100 J is the treatment for paroxysmal SVT that exacerbates cardiovascular dysfunction or

hypotension. Defibrillation is used in the treatment of ventricular fibrillation or pulseless VT.

References:

1. Miles WM, Prystowsky EN, Heger JJ, Zipes DP. Evaluation of the patient with wide QRS tachycardia. *Med Clin North Am*. 1984 Sep;68(5):1015-1038.
2. Sharkey SW. Beyond the wedge: clinical physiology and the Swan-Ganz catheter. *Am J Med*. 1987 Jul;83(1): III-122.
3. Steward RB, Bardy GH, Greene HL. Wide complex tachycardia: misdiagnosis and outcome after emergent therapy. *Ann Intern Med*. 1986 Jun;104(6):766-771.

2. Rationale

Answer: B

Prasugrel is contraindicated in patients with a history of stroke or transient ischemic attack or who are older than 75 years because of increased risk of bleeding.

Reference:

1. O'Gara PT, Kushner FG, Ascheim DD, et al; American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. 2013 ACCF/AHA guideline for the management of ST-elevation myocardial infarction: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2013 Jan 29;127(4): e362-e425.

3. Rationale

Answer: E

Several features of this patient's presentation are suspicious for spontaneous coronary artery dissection (SCAD), which accounts for up to 30% of acute coronary syndromes in women younger than 50 years. Clinical features consistent with SCAD include the absence of traditional cardiac risk factors, history of recent miscarriage, family history of Ehlers-Danlos syndrome, and hyperextensible digits. Certain hormonal influences are

thought to increase the risk of SCAD because many patients are diagnosed during late pregnancy or the early postpartum period. Increased progesterone levels during pregnancy could have led to increased fragility of the artery. SCAD should be a strong consideration in a young person who does not have traditional cardiac risk factors and who presents with an acute coronary syndrome.

References:

1. Henkin S, Negrotto SM, Tweet MS, et al. Spontaneous coronary artery dissection and its association with heritable connective tissue disorders. *Heart*. 2016 Jun 1;102(11):876-881.

2. Tinkle B, Castori M, Berglund B, et al. Hypermobile Ehlers-Danlos syndrome (a.k.a. Ehlers-Danlos syndrome type III and Ehlers-Danlos syndrome hypermobility type): clinical description and natural history. *Am J Med Genet C Semin Med Genet*. 2017 Mar;175(1):48-69.

3. Tweet MS, Gulati R, Hayes SN. What clinicians should know about spontaneous coronary artery dissection. *Mayo Clin Proc*. 2015 Aug;90(8):1125-1130.

4. Rationale

Answer: B

Third-degree heart block and severe bradycardia are common after aortic valve replacement due to anatomic disruption of the conduction system and numerous concurrent physiologic derangements associated with cardiopulmonary bypass and separation therefrom. Loss of effective epicardial ventricular pacing in the context of postoperative complete heart block is an emergency. If cardiac arrest occurs and efforts to restore capture and titrate pharmacologic adjuncts are unsuccessful, current guidelines for resuscitation after cardiac surgery call for emergent re-sternotomy within 5 minutes. Transvenous pacing can take time to establish, especially in the absence of a pulmonary artery catheter with a dedicated pacing port. Guidelines do not recommend delaying re-sternotomy to attempt transvenous pacing. Atropine is unlikely to be effective

in asystolic cardiac arrest. Focused echocardiography could aid in the detection of postoperative tamponade or other causes of cardiac arrest but are unlikely to have diagnostic value.

References:

1. Society of Thoracic Surgeons Task Force on Resuscitation After Cardiac Surgery. The Society of Thoracic Surgeons expert consensus for the resuscitation of patients who arrest after cardiac surgery. *Ann Thorac Surg*. 2017 Mar;103(3):1005-1020.
2. Brand J, McDonald A, Dunning J. Management of cardiac arrest following cardiac surgery. *BJA Educ*. 2018 Jan;18(1):16-22.
3. Link MS, Berkow LC, Kudenchuk PJ, et al. Part 7: adult advanced cardiovascular life support: 2015 American Heart Association guidelines update for cardiopulmonary resuscitation and emergency cardiovascular care. *Circulation*. 2015 Nov 3;132(18 suppl 2): S444-S464.

5. Rationale

Answer: D

Isolated right ventricular failure is characterized by hypotension, low cardiac index, elevated right atrial or central venous pressure, and normal or low pulmonary artery occlusion pressure. Specific pulmonary artery catheter values suggestive of right ventricular failure include central venous pressure greater than 20 mm Hg, central venous pressure greater than pulmonary artery occlusion pressure, and cardiac index less than 2.1 L/min/m^2. Hypovolemia is more likely to present with low central venous pressure, and this patient did not improve with ongoing volume resuscitation. Vasoplegia is more likely to present with a low central venous pressure and normal or high cardiac index. Cardiac tamponade is more likely to present with equalization of central venous and intracardiac pressures. Left ventricular failure is more likely to present with pulmonary artery occlusion pressure elevation.

References:

1. Bodson L, Bouferrache K, Vieillard-Baron A. Cardiac tamponade. *Curr Opin Crit Care*. 2011 Oct;17(5):416-424.

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3. Zochios V, Protopapas AD, Parhar K. Markers of right ventricular dysfunction in adult cardiac surgical patients. *J Cardiothorac Vasc Anesth*. 2017 Oct;31(5):1570-1574.

6. Rationale

Answer: D

It is important to preserve adequate preload in patients who have had an acute myocardial infarction, even in those with a history of heart failure. A volume challenge should be initiated. Although inotropes such as dobutamine may be required, they can, in the case of inadequate preload, worsen the hypotension. Vasopressors increase myocardial oxygen demand and are not first-line therapies. Furosemide may also worsen the hypotension, even in the face of relatively normal pulmonary capillary wedge pressure.

References:

1. Hall JB, Schmidt GA, Kress JP. *Principles of Critical Care*. 4th ed. New York, NY: McGraw Hill Education; 2015:256.

2. McMurray JJ, Adamopoulos S, Anker SD, et al; ESC Committee for Practice Guidelines. ESC guidelines for the diagnosis and treatment of acute and chronic heart failure 2012: the Task Force for the Diagnosis and Treatment of Acute and Chronic Heart Failure 2012 of the European Society of Cardiology. Developed in collaboration with the Heart Failure Association (HFA) of the ESC. *Eur J Heart Fail*. 2012 Aug;14(8):803-869.

7. Rationale

Answer: A

During a hospitalization for heart failure, a predischarge brain natriuretic peptide level (BNP) can be useful to establish post-

discharge prognosis. Predischarge BNP biomarker levels and their relative change during hospital treatment are strong predictors of the risk of death or hospital readmission for heart failure. Several studies have suggested that predischarge BNP biomarker levels had higher reclassification and discrimination value than clinical variables in predicting outcomes. Patients with higher predischarge levels and those with no decrease in BNP biomarker levels during hospitalization have worse outcomes. Although observational or retrospective studies have suggested that patients with BNP biomarker reduction had better outcomes than those without any changes or with a BNP biomarker rise, targeting a certain threshold, value, or relative change in these levels during hospitalization may not be practical or safe for every patient and have not been tested in a prospective large-scale trial. Clinical assessment and adherence to goal-directed medical therapy should be the emphasis, and the prognostic value of a predischarge level or relative change does not imply the necessity for serial and repeated BNP biomarker measurements during hospitalization.

References:

1. Felker GM, Hasselblad V, Hernandez AF, O'Connor CM. Biomarker-guided therapy in chronic heart failure: a meta-analysis of randomized controlled trials. *Am Heart J.* 2009 Sep;158(3):422-430.

2. Li P, Luo Y, Chen YM. B-type natriuretic peptide-guided chronic heart failure therapy: a meta-analysis of 11 randomized controlled trials. *Heart Lung Circ.* 2013 Oct;22(10):852-860.

3. Savarese G, Trimarco B, Dellegrottaglie S, et al. Natriuretic peptide-guided therapy in chronic heart failure: a meta-analysis of 2,686 patients in 12 randomized trials. *PLoS One.* 2013; 8(3): e58287.

4. Yancy CW, Jessup M, Bozkurt B, et al. 2017 ACC/AHA/HFSA focused update of the 2013 ACCF/AHA guideline for the management of heart failure: a report of the American College of Cardiology/American

Heart Association Task Force on Clinical Practice Guidelines and the Heart Failure Society of America. *Circulation*. 2017 Aug 8;136(6): e137-e161.

8. Rationale

Answer: C

According to the 2013 American College of Cardiology Foundation/American Heart Association guidelines for the management of heart failure, to enhance diuresis, either the loop diuretic dose should be increased, or a thiazide-type diuretic should be added (IIa, level of evidence B). Acetazolamide could be considered if this patient were developing contraction alkalosis, but no laboratory findings are presented. Spironolactone could be considered in patients who continue to develop hypokalemia while on a diuretic infusion and per guideline recommendations for long-term neurohormonal blockade for heart failure. Dobutamine is inappropriate because of her adequate cardiac index, and guidelines provide a lower recommendation for adding dopamine to enhance diuresis (IIb, level of evidence B). Tolvaptan could be considered for refractory diuresis and worsening hyponatremia, but no sodium level is provided.

References:

1. Writing Committee Members; Yancy CW, Jessup M, et al; American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. 2013 ACCF/AHA guideline for the management of heart failure: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2013 Oct 15;128(16):e240-e327.

2. Goyfman M, Zamudio P, Jang, K, et al. Combination aquaretic and diuretic therapy in acute heart failure. *Int J Nephrol Renovasc Dis*. 2017 Jun 6; 10:129-134.

3. Moranville MP, Choi S, Hogg J, Anderson AS, Rich JD. Comparison of metolazone versus chlorothiazide in acute

decompensated heart failure with diuretic resistance. *Cardiovasc Ther*. 2015 Apr;33(2):42-49.

9. Rationale

Answer: A

Patients with mild to moderate heart failure are characterized by either: 1) mildly elevated natriuretic peptide levels, B-type natriuretic peptide (BNP) levels greater than 150 pg/mL, or N-terminal pro B-type natriuretic peptide (NT-proBNP) levels greater than or equal to 600 pg/mL; or 2) BNP greater than or equal to 100 pg/mL or NT-proBNP greater than or equal to 400 pg/mL with a hospitalization in the preceding 12 months. For patients who were able to tolerate both a target dose of enalapril (10 mg twice daily) and a subsequent angiotensin receptor-neprilysin inhibitor (valsartan/sacubitril, 200 mg twice daily, with the angiotensin II receptor blocker component equivalent to valsartan, 160 mg), hospitalizations and mortality were significantly decreased with the valsartan/sacubitril compound compared with enalapril.

References:

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3. Yancy C, Jessup M, Bozkurt B, et al. 2017 ACC/AHA/HFSA focused update of the 2013 ACCF/AHA guideline for the management of heart failure: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines and the Heart Failure Society of America. *Circulation*. 2017 Aug 8;136(6): e137-e161.

10. Rationale

Answer: A

Survivors of out-of-hospital cardiac arrest have high mortality and poor neurologic outcomes. Mortality and neurologic outcomes in unresponsive patients after return of spontaneous circulation (ROSC) have been shown to improve with induced hypothermia, especially in those whose initial rhythm was ventricular fibrillation. This treatment is considered standard of care. Hypothermia is initiated as early as possible in unresponsive survivors. Hypothermia treatment is administered for 24 hours followed by gradual warming over 12 to 24 hours. Extending treatment to 48 hours has not been shown to improve outcomes. Mild hypothermia of 36°C (96.8°F) has been shown to be like moderate hypothermia of 33°C (91.4°F). Data are not robust for cardiac arrest due to rhythm other than ventricular fibrillation. The American Heart Association recommends hypothermia in such cases but supporting evidence is lacking. The HYPERION trial evaluated hypothermia after non-shockable rhythm. The hypothermia arm was associated with a significantly better neurologic outcome (10.2% vs. 5.7%), although mortality was the same in both arms. Cardiac catheterization is indicated if ECG post-ROSC shows ST-elevated myocardial infarction. Amiodarone to control premature ventricular contractions does not improve outcome. Neurology consultation can be sought, and EEG considered if the patient shows signs of seizure or myoclonus, but it will not improve outcome.

References:

1. Lascarrou JB, Merdji H, Le Gouge A, et al; CRICS-TRIGGERSEP Group. Targeted temperature management for cardiac arrest with nonshockable rhythm. *N Engl J Med*. 2019 Dec 12;381(24):2327-2337.

2. Nielsen N, Wetterslev J, Cronberg T, et al; TTM Trial Investigators. Targeted temperature management at 33°C versus 36°C after cardiac arrest. *N Engl J Med*. 2013 Dec 5;369(23):2197-2206.

11. Rationale Answer: C

This patient has DeBakey type I aortic dissection. For patients with ascending thoracic aortic dissection, guidelines suggest that the entire aneurysmal aorta and the proximal extent of the dissection should be resected. Thus, surgical consultation is indicated. Beta-blockers' ability to lower the rate of rise of ventricular force (dP/dt) and stress on the aorta make them the first choice. In addition, guidelines suggest lowering blood pressure to less than 140/90 mm Hg for a patient without diabetes and less than 130/80 mm Hg for a patient with diabetes and aortic dissection. Because this patient has aortic dissection, left heart catheterization is inappropriate. While pain management with heart rate and blood pressure control should be continued, definitive measures with surgical consultation should be considered at this point; therefore, neither IV beta-blockers nor pain management alone is appropriate.

References:

1. Hiratzka LF, Bakris GL, Beckman JA, et al; American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines; American Association for Thoracic Surgery; American College of Radiology; American Stroke Association; Society of Cardiovascular Anesthesiologists; Society for Cardiovascular Angiography and Interventions; Society of Interventional Radiology; Society of Thoracic Surgeons; Society for Vascular Medicine. 2010 ACCF/AHA/AATS/ACR/ASA/SCA/SCAI/SIR/STS/SVM guidelines for the diagnosis and management of patients with thoracic aortic disease: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines, American Association for Thoracic Surgery, American College of Radiology, American Stroke Association; Society of Cardiovascular Anesthesiologists, Society for Cardiovascular Angiography and Interventions, Society of Interventional Radiology, Society of Thoracic Surgeons, and Society for

Vascular Medicine. *Catheter Cardiovasc Interv*. 2010 Aug 1;76(2):E43-E86.

2. Braverman AC. Acute aortic dissection: clinician update. *Circulation*. 2010 Jul 13;122(2):184-188.

12. Rationale

Answer: B

This patient presents with infective endocarditis. The photograph shows erythematous macules characteristic of Janeway lesions. Fevers, anorexia, and neurologic symptoms are common features of infective endocarditis. Those at highest risk have prosthetic heart valves, prior history of infective endocarditis, and valve regurgitation following heart transplant. Antibiotic prophylaxis is recommended for patients before teeth cleaning, tooth extraction, respiratory tract procedures involving incision or biopsy, or surgeries involving infected skin or muscle. A patient with uncorrected mitral valve regurgitation does not need infective endocarditis prophylaxis. A patient with a heart transplant undergoing colonoscopy does not need prophylaxis either; there is no mention of valvular disorder after transplantation and colonoscopies are low-risk procedures for infective endocarditis. A patient with a coronary stent placed more than 6 months ago does not need prophylaxis, and urogenital procedures carry low risk for infective endocarditis.

References:

1. Cahill TJ, Prendergast BD. Infective endocarditis. *Lancet*. 2016 Feb 27;387(10021):882-893.

2. Pant S, Patel NJ, Deshmukh A, et al. Trends in infective endocarditis incidence, microbiology, and valve replacement in the United States from 2000 to 2011. *J Am Coll Cardiol*. 2015 May 19;65(19):2070-2076.

3. Wilson W, Taubert KA, Gewitz M, et al; American Heart Association Rheumatic Fever, Endocarditis, and Kawasaki Disease Committee; American Heart Association Council on Cardiovascular Disease in the Young; American Heart

Association Council on Clinical Cardiology; American Heart Association Council on Cardiovascular Surgery and Anesthesia; Quality of Care and Outcomes Research Interdisciplinary Working Group. Prevention of infective endocarditis: guidelines from the American Heart Association: a guideline from the American Heart Association Rheumatic Fever, Endocarditis, and Kawasaki Disease Committee, Council on Cardiovascular Disease in the Young, and the Council on Clinical Cardiology, Council on Cardiovascular Surgery and Anesthesia, and the Quality of Care and Outcomes Research Interdisciplinary Working Group. *Circulation*. 2007 Oct 9;116(15):1736-1754.

13. Rationale

Answer: D

Transesophageal echocardiography is the most appropriate diagnostic test for infective endocarditis; it is more sensitive and specific than transthoracic echocardiography. Infective endocarditis is diagnosed based on the modified Duke criteria. The diagnosis requires 2 major clinical criteria or 1 major criterion and 3 minor criteria or 5 minor criteria. Major criteria are: 1) 2 separate blood cultures positive for a microorganism that typically causes endocarditis, or a single positive blood culture for *Coxiella burnetii*, or immunoglobulin G titer greater than 1:800; and 2) evidence of endocardial involvement on echocardiography. Minor criteria are: 1) conditions predisposing toward endocarditis (eg, predisposing heart condition, injection-drug use); 2) fever; 3) vascular phenomena; 4) immunologic phenomena such as glomerulonephritis; and 5) microbiologic evidence not satisfying a major criterion. This patient's presentation is not typical for acute rheumatic fever, so antistreptolysin O titer for a recent group A streptococcus pharyngitis is not indicated. Acute rheumatic fever can manifest with arthritis, carditis, chorea, erythema marginatum, and subcutaneous nodules. C-reactive protein is a nonspecific marker of infection and inflammation; it is unlikely to be diagnostically informative in

this patient. A tagged WBC scan is sometimes used when looking for occult infections but is not indicated in this setting given the high suspicion for endocarditis. Dilated funduscopic examination would not be helpful because, even if it were normal, it would not rule out endocarditis.

References:

1. Baddour, Wilson WR, Bayer AS, et al; American Heart Association Committee on Rheumatic Fever, Endocarditis, and Kawasaki Disease of the Council on Cardiovascular Disease in the Young, Council on Clinical Cardiology, Council on Cardiovascular Surgery and Anesthesia, and Stroke Council. Infective endocarditis in adults: diagnosis, antimicrobial therapy, and management of complications: a scientific statement for healthcare professionals from the American Heart Association. *Circulation*. 2015 Oct 13;132(15):1435-1486.

2. Hoen B, Duval X. Clinical practice. Infective endocarditis. N Engl J Med. 2013 Apr 11;368(15):1425-1433. Erratum in: *N Engl J Med*. 2013 Jun 27;368(26):2536.

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14. Rationale — Answer: B

Antiemetics can notoriously prolong QTc interval on ECG, especially when administered together and intravenously. This can lead to fatal ventricular arrhythmias.

Reference:

1. Fazio G, Vernuccio F, Grutta G, Re GL. Drugs to be avoided in patients with long QT syndrome: focus on the anesthesiologic management. *World J Cardiol*. 2013 Apr 26;5(4):87-93.

15. Rationale — Answer: D

This patient presents with hypotension, tachycardia, and hypoxemia. He has pulsus paradoxus, which can be seen in tamponade. The central venous pressure tracing illustrates the lack of Y descent. With tamponade, the cardiac volume is fixed secondary to external obstruction, which limits venous return and filling, resulting in blunting of Y descent.

References:

1. Argulian E, Messerli F. Misconceptions and facts about pericardial effusion and tamponade. *Am J Med*. 2013 Oct;126(10):858-861.

2. Kuvin JT, Harati NA, Pandian NG, Bojar RM, Khabbaz KR. Postoperative cardiac tamponade in the modern surgical era. *Ann Thorac Surg*. 2002 Oct;74(4):1148-1153.

3. Perez-Casares A, Cesar S, Brunet-Garcia L, Sanchez-de-Toledo J. Echocardiographic evaluation of pericardial effusion and cardiac tamponade. *Front Pediatr*. 2017 April 24; 5:79.

16. Rationale — Answer: D

Repair of infrarenal aortic aneurysms requires coverage of the luminal orifice of the inferior mesenteric artery. While most patients have collateral flow that maintains perfusion to the

sigmoid colon, mesenteric ischemia of the colon is a well-known complication following both open and endovascular repair. The incidence is higher when the repair is done emergently for a ruptured aneurysm. Because of this, many experts recommend routine postoperative surveillance with flexible sigmoidoscopy. If patients have signs of worsening acidosis or sepsis following abdominal aortic aneurysm repair, this diagnosis must be considered. A crystalloid solution bolus may be appropriate, but it should not delay investigation and treatment of the underlying etiology. While this patient may meet criteria for sepsis, he is at low risk of infection this early in his hospital course. Furthermore, broad-spectrum antibiotics may mask symptoms of bowel ischemia. Nothing in the scenario specifically points to a diagnosis of transfusion-related acute lung injury. Repeating the CT will yield little, as detection of bowel ischemia is difficult, and it will further delay diagnosis.

References:

1. Becquemin JP, Majewski M, Fermani N, et al. Colon ischemia following abdominal aortic aneurysm repair in the era of endovascular abdominal aortic repair. *J Vasc Surg*. 2008 Feb;47(2):258-263; discussion 263.

2. Champagne BJ, Lee EC, Valerian B, Mulhotra N, Mehta M. Incidence of colonic ischemia after repair of ruptured abdominal aortic aneurysm. *J Am Coll Surg*. 2007 Apr;204(4):597-602.

17. Rationale

Answer: D

Acute ischemic mitral regurgitation is associated with rupture of the posterior papillary muscle due to its single blood supply. Immediate therapy includes afterload reduction and placement of an intra-aortic blood pump (IABP) for hemodynamic augmentation and coronary perfusion. Inotropic and vasoactive support are often necessary as well to maintain perfusion. Esmolol would decrease heart rate and inotropy, leading to elevated end-diastolic left ventricular pressure and higher

regurgitant volume across the mitral valve. Vasopressin is not a first-line vasoactive agent for cardiogenic shock; it would increase afterload and, in the absence of an IABP, would lead to a reduction in cardiac output. Although there may be a role for inhaled nitric oxide in refractory hypoxia, it would not lead to afterload reduction. It is important to note that an IABP is not a definitive management device but rather serves as a bridge through a period of critical illness.

References:

1. Patel AK, Hollenberg SM. Cardiovascular failure, and cardiogenic shock. *Semin Respir Crit Care Med*. 2011 Oct;32(5):598-606.

2. Sanborn TA, Feldman T. Management strategies for cardiogenic shock. *Curr Opin Cardiol*. 2004 Nov;19(6):608-612.

18. Rationale

Answer: C

Pericardial disease is common in patients with renal failure and can result in hypotension. Uremic pericarditis can occur before or during the first several weeks after initiating chronic dialysis. Patients well established on dialysis also can develop dialysis-associated pericarditis. In either case, 2 serious complications can lead to hypotension: cardiac tamponade or constrictive pericarditis. Most patients with uremic pericarditis respond well to the initiation or intensification of chronic dialysis therapy. Although BUN and creatinine concentrations are not significantly different from those of patients with chronic renal failure without pericarditis, the predictable response to dialysis strongly implicates retained uremic toxins as a causative factor. These patients are unlikely to present with chest pain or to develop serious complications. Chest pain, fever, and a scratchy or grating pericardial friction rub are usually present. Cardiomegaly with clear lung fields is seen on chest radiograph, sometimes accompanied by pleural effusion from concomitant pleuritis. Echocardiography reveals the presence and size of any pericardial effusion. Daily dialysis is instituted, without systemic

heparinization. Pericardiocentesis with local steroid instillation can be performed if no improvement occurs after 10 to 14 days of dialysis. If pericarditis fails to resolve or recurs after these measures, surgical pericardiectomy is indicated. Indomethacin may relieve chest pain but does not affect the overall course. Acute cardiac tamponade must be quickly recognized and treated. Hypotension with narrow pulse pressure, distended neck veins, dyspnea, tachypnea, and apprehension is typical. Pulsus paradoxus is often present but may be absent with severe left ventricular failure or extreme hypotension. Prompt right heart catheterization demonstrates the characteristic equilibration of atrial, ventricular, and pulmonary artery diastolic pressures. Pericardiocentesis should be performed acutely, with catheter drainage and the local instillation of corticosteroids. Some sources advocate moving invariably to open surgical procedures such as pericardiotomy or pericardiectomy. Others take a more conservative approach. Recurrent tamponade requires definitive surgery.

References:

1. Hateboer N, McGonigle RJ, Lewis CT. Pericardiectomy after two decades of constrictive pericarditis in a patient with chronic renal failure. *Nephrol Dial Transplant*. 1995 Oct;10(10):1935-1937.

2. Morimoto S, Katoh S, Inoue K, Terashima S, Tatsukawa H, Ida K. The efficacy of indomethacin in the treatment of uremic pericarditis. *Nippon Jinzo Gakkai Shi*. 1995 Feb;37(2):140-144.

3. Reddy PS, Curtis EI, O'Toole JD, Shaver JA. Cardiac tamponade: hemodynamic observations in man. *Circulation*. 1978 Aug;58(2):265-272.

4. Trotter MC, Chung KC, Ochsner JL, McFadden PM. Pericardiectomy for pericardial constriction. *Am Surg*. 1996 Apr;62(4):304-307.

19. Rationale

Answer: B

This patient has global cardiac dysfunction, so discontinuing the vasopressor will worsen his condition. Similarly, discontinuing hypothermia earlier than 24 hours will worsen his outcome. His persistent lactate elevation is not caused by sedation such as propofol; his liver function tests, and triglyceride level are normal. He is not seizing or shivering so adding neuromuscular blockade would not affect his condition. His persistent lactate elevation and requirement for a high vasopressor dose might be associated with hypothermia at 33°C (91.4°F). In a post hoc analysis of 940 patients in the target temperature management trial, undergoing hypothermia at 33°C (91.4°F) was associated with persistently higher lactate and higher vasopressor requirements. Therefore, warming him to 36°C (96.8°F) should be tried first to improve his vasopressor requirement and lactate clearance.

Reference:

1. Bro-Jeppesen J, Annborn M, Hassager C, et al; TTM Investigators. Hemodynamics and vasopressor support during targeted temperature management at 33°C versus 36°C after out-of-hospital cardiac arrest: a post hoc study of the target temperature management trial. *Crit Care Med*. 2015 Feb;43(2):318-327.

20. Rationale

Answer: B

Ventricular assist devices (VADs) are indicated for either bridge to transplantation or destination therapy. Acute management of hemodynamically unstable patients with VADs is challenging because of the difficulty of assessing perfusion status in patients with continuous-flow devices and broad differential diagnoses for acute disease states. American Heart Association guidelines encourage an algorithmic approach to unresponsive or mentally altered VAD patients. First, ventilation should be assisted, and perfusion assessed. Pulse oximetry may be challenging because of the lack of pulsatile flow. Instead, skin color, temperature, and capillary refill may inform patient perfusion status. If perfusion is adequate, nonmechanical (non-LVAD) causes for

decompensation should be assessed, including hypo- or hyperglycemia, stroke, hypoxemia, overdose, sepsis-mediated vasodilation, acute right ventricular dysfunction, tachyarrhythmias, bleeding due to over-anticoagulation or gastrointestinal malformations, and thromboembolic complications. The most common VAD-associated causes of decompensation are pump failure (ie, thrombosis), power failure, and driveline disconnection. Therefore, immediate interrogation of the device should include review of recent alarms and auscultation of normal VAD hum, since VAD restart or reconnection might be indicated. If VAD parameters are normal, end-tidal capnography and blood pressure may inform the next steps. If mean arterial pressure is less than 50 mm Hg or $ETCO_2$ is less than 20 mm Hg, external chest compressions are recommended. Other advanced cardiac life support goals of high-quality chest compressions and immediate defibrillation should also be considered, if indicated.

Reference:

1. Peberdy MA, Gluck JA, Ornato JP, et al; American Heart Association Emergency Cardiovascular Care Committee; Council on Cardiopulmonary, Critical Care, Perioperative, and Resuscitation; Council on Cardiovascular Diseases in the Young; Council on Cardiovascular Surgery and Anesthesia; Council on Cardiovascular and Stroke Nursing; and Council on Clinical Cardiology. Cardiopulmonary resuscitation in adults and children with mechanical circulatory support: A Scientific Statement from the American Heart Association. *Circulation*. 2017 Jun 13;135(24): e1115-e1134.

21. Rationale

Answer: B

Based on this patient's hemodynamics, he likely has right heart failure with pulmonary arterial hypertension. He has elevated pulmonary pressures, central venous pressure/right atrial pressure, and pulmonary capillary wedge pressure, in addition to reduced pulmonary artery pulsatility index and elevated

pulmonary vascular resistance. Inhaled epoprostenol has been demonstrated to be effective in reducing pulmonary vascular resistance and increasing right heart ejection. Nitroprusside and nitroglycerin are not recommended because he is on norepinephrine. Sildenafil is a potential option, but American College of Cardiology guidelines state that there are mixed anecdotal reports of efficacy. Additionally, the 2013 International Society for Heart and Lung Transplantation's mechanical circulatory support guidelines recommend using diuresis, inotropes, and pulmonary vasodilators, specifically either nitric oxide or epoprostenol for management of acute right ventricular dysfunction.

References:

1. Feldman D, Pamboukian SV, Teuteberg JJ, et al; International Society of Heart and Lung Transplantation. The 2013 International Society of Heart and Lung Transplantation guidelines for mechanical circulatory support. *J Heart Lung Transplant*. 2013 Feb;32(2):157-187.
2. Konstam MA, Kiernan MS, Bernstein D, et al; American Heart Association Council on Clinical Cardiology; Council on Cardiovascular Disease in the Young; Council on Cardiovascular Surgery and Anesthesia. Evaluation and management of right-sided heart failure: a scientific statement from the American Heart Association. *Circulation*. 2018 May 15;137(20):e578-e622.

22. Rationale

Answer: B

The International Society for Heart and Lung Transplantation's 2016 updated listing criteria for heart transplantation do not include an age cutoff. Patients older than 70 still derive benefit from heart transplantation and have lower incidences of rejection than younger patients, although they have higher mortality risk. The Seattle Heart Failure Score has been validated as a risk score for heart transplant listing. A 1-year survival of less than 80% is considered a reasonable point for listing. Pulmonary vascular resistance greater than 3 Wood units

that does not respond to pulmonary vasodilators, and left ventricular unloading is a contraindication to listing. The listing criteria set a limit for body mass index (BMI) of 35 kg/m^2. In published data, those with BMI greater than 35 kg/m^2 had longer wait times, limited suitable donors, and higher posttransplant mortality.

References:

1. Goldstein DJ, Bello R, Shin JJ, et al. Outcomes of cardiac transplantation in septuagenarians. *J Heart Lung Transplant*. 2012 Jul;31(7):679-685.
2. Mehra MR, Canter CE Hannan MM, et al; International Society for Heart Lung Transplantation (ISHLT) Infectious Diseases, Pediatric and Heart Failure and Transplantation Councils. The 2016 International Society for Heart Lung Transplantation listing criteria for heart transplantation: a 10-year update. *J Heart Lung Transplant*. 2016 Jan;35(1):1-23.
3. Weiss ES, Allen JG, Russell SD, Shah AS, Conte JV. Impact of recipient body mass index on organ allocation and mortality in orthotopic heart transplantation. *J Heart Lung Transplant*. 2009 Nov;28(11):1150-1157.

23. Rationale

Answer: B

This patient has developed cardiogenic shock secondary to myocardial infarction as evidenced by a low cardiac index and elevated left ventricular filling pressure (pulmonary artery occlusion pressure > 18 mm Hg). Echocardiogram excludes mechanical causes of cardiogenic shock and confirms a low ejection fraction as the cause of shock. Cardiogenic shock due to left ventricular dysfunction is the most common cause of in-hospital death in myocardial infarction. Analysis of potential predictors in cardiogenic shock has identified a low systolic blood pressure, measured on vasopressor support, as a powerful predictor of 30-day mortality. Other mortality predictors are low creatinine clearance and the number of vasopressors. In

other studies, APACHE II score has also been identified as a strong predictor of mortality in cardiogenic shock. Cardiogenic shock is generally considered a hemodynamic disease characterized by poor left ventricular function, low cardiac output, and high ventricular filling pressures, usually caused by acute ischemic heart disease. However, cardiogenic shock is also associated with an inflammatory response characterized by cytokine release (especially interleukin 6), other mediator expression (including receptor for advanced glycation end products [RAGE] expression on monocytes), and nitric oxide production. This inflammatory response may contribute to the pathogenesis of cardiogenic shock by causing inappropriate systemic vasodilatation, progressive systemic and coronary hypoperfusion, myocardial depression, and multiple organ system dysfunction. This hypothesis was partially tested in a randomized, multicenter, clinical trial using the nonselective nitric oxide inhibitor tilarginine; however, the trial result was negative, failing to demonstrate improved survival or shortened shock duration. A large clinical trial compared dopamine to norepinephrine in all forms of shock. The overall study showed no difference in death rates between patients treated with dopamine versus norepinephrine as a first-line vasopressor agent. However, dopamine was associated with more arrhythmic adverse events. Further, in the prospectively defined subgroup of patients with cardiogenic shock, norepinephrine was associated with a lower mortality. Based on this clinical trial, norepinephrine is considered a better initial vasopressor agent in cardiogenic shock.

References:

1. Geppert A, Steiner A, Zorn G, et al. Multiple organ failure in patients with cardiogenic shock is associated with high plasma levels of interleukin-6. *Crit Care Med*. 2002 Sep;30(9):1987-1994.

2. Hochman J, Sleeper L, Webb J, et al. Early revascularization in acute myocardial infarction complicated by cardiogenic shock. Should we emergently revascularize

occluded coronaries for cardiogenic shock. *N Engl J Med*. 1999 Aug 26;341(9):625-634.

3. Katz JN, Stebbins AL, Alexander JH, et al; TRIUMPH Investigators. Predictors of 30-day mortality in patients with refractory cardiogenic shock following acute myocardial infarction despite a patent infarct artery. *Am Heart J*. 2009 Oct;158(4):680-687.

4. Prondzinsky R, Lemm H, Swyter M, et al. Intra-aortic balloon counterpulsation in patients with acute myocardial infarction complicated by cardiogenic shock: the prospective, randomized IABP SHOCK trial for attenuation of multiorgan dysfunction syndrome. *Crit Care Med*. 2010 Jan;38(1):152-160.

5. TRIUMPH Investigators; Alexander J, Reynolds H, et al. Effects of tilarginine in patients with acute myocardial infarction and cardiogenic shock: the TRIUMPH randomized controlled trial. *JAMA*. 2007 Apr 18;297(15):1657-1666.

24. Rationale

Answer: C

This patient is in obstructive shock as manifested by hemodynamic instability requiring vasopressor support. She also needs a high concentration of oxygen. Acute pulmonary embolism (PE) is suspected, but transport to radiology is not recommended for patients who are hemodynamically unstable. Bedside echocardiography is strongly suggestive of PE as evidenced by right ventricular enlargement. ECG is suggestive of PE also, with an S1Q3T3 pattern. PE treatment in patients with malignancy is with low-molecular-weight heparin. However, data are for patients with malignancy who are hemodynamically stable. Vasopressor requirements and high levels of oxygen make this patient unstable, as does her deteriorating physical examination. Therefore, systemic thrombolytics are the treatment of choice; while systemic thrombolytic therapy may increase bleeding risk, morbidity is improved when it is used in patients with cardiogenic shock.

Catheter-directed thrombectomy is not indicated because she is too unstable to be transported for CT to determine the thrombus location. Embolectomy is indicated only in those for whom thrombolytics were not therapeutic. Warfarin is not indicated for acute management of PE.

References:

1. Kearon C, Akl EA, Comerota AJ, et al. Antithrombotic therapy for VTE disease: antithrombotic therapy and prevention of thrombosis, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. *Chest*. 2012 Feb;141(2 Suppl): e419S-e496S.
2. Konstantinides SV, Meyer G, Becattini C, et al; ESC Scientific Document Group. 2019 ESC guidelines for the diagnosis and management of acute pulmonary embolism developed in collaboration with the European Respiratory Society (ERS). *Eur Heart J*. 2020 Jan 21;41(4):543-603.

25. Rationale

Answer: C

On ultrasound, A lines are an indication of normal lung, with horizontal lines reverberating through the lung. B lines are hyperechoic, vertically oriented linear artifacts that originate at the pleural line and transverse the entire ultrasound screen, erasing A lines. The presence of B lines indicates fluid in the interstitium. Bilateral B lines would be expected with pulmonary edema and unilateral B lines would be expected with pneumonia.

Reference:

1. Lichtenstein DA, Meziere GA. Relevance of lung ultrasound in the diagnosis of acute respiratory failure: the BLUE protocol. *Chest*. 2008 Jul;134(1);117-125.

26. Rationale

Answer: B

For this patient, who has undergone repair of tetralogy of Fallot resulting in a biventricular system with systemic left ventricle,

maintaining normal central venous pressure between 8 and 12 cm H_2O is recommended. Management of patients with this condition involves avoiding tachycardia and bradycardia. Therefore, the first step is to add a short-acting beta-blocker to maintain her heart rate between 60 and 80 beats/min. For this patient, a normal heart rate would increase cardiac output (CO). Tachycardia, on the other hand, would decrease CO. Similarly, bradycardia should be avoided in these patients. It is recommended that they maintain a normal preload, with central venous pressure between 8 and 12 cm H_2O. High preload may cause them to have right heart failure.

Reference:

1. Shaddy RE, Webb G. Applying heart failure guidelines to adult congenital heart disease patients. *Expert Rev Cardiovasc Ther.* 2008 Feb;6(2):165-174.

27. Rationale

Answer: E

A rapid way to diagnose systolic dysfunction, which can be present in various types of shock such as cardiogenic shock or in severe systolic heart failure and is considered a surrogate of ejection fraction, is the fractional shortening of the left ventricle and the E-point septal separation (EPSS) of the mitral valve and septum. Fractional shortening of the left ventricle is measured on parasternal long axis view. On M-mode view, the cursor is placed across the distal tip of the mitral leaflets, and the diameter is measured at that level at the end of systole and the end of diastole. The formula is (EDD – ESD/EDD) × 100 of the ventricular walls, where EDD is end-diastolic dimension and ESD is end-systolic dimension. A normal shortening is above 35% to 45%. Anything lower implies poor systolic function of the left ventricle. The EPSS is measured with M-mode cursor over the tip of the anterior mitral leaflet. As the mitral valve moves during diastole there are 2 repeating waves; the first is the E wave, which reflects the initial and maximal opening of the valve to allow passive filling of the left ventricle, followed by the A wave, which reflects the atrial contraction.

EPSS is the minimal distance between the E wave and the septum and should be less than 7 mm. If it is more than 1 cm it reliably reflects poor ejection fraction.

Reference:

1. Seif D, Perera P, Mailhot T, Riley D, Mandavia D. Bedside ultrasound in resuscitation and the rapid ultrasound in shock protocol. *Crit Care Res Pract*. 2012; 2012:503254.

28. Rationale

Answer: C

This patient has Takotsubo cardiomyopathy with left ventricular outflow tract (LVOT) obstruction, based on echocardiographic findings. She has cardiogenic shock based on data obtained from vital signs and the pulmonary artery catheter. The cardiomyopathy is nonischemic as evidenced by the cardiac catheterization data. The most appropriate therapy is beta blockade to relieve the LVOT obstruction by slowing the heart rate and allowing the left ventricle to fill in.

References:

1. Bybee KA, Prasad A. Stress-related cardiomyopathy syndromes. *Circulation*. 2008 Jul 22;118(4):397-409.

2. Fernández-Pérez GC, Aguilar-Arjona JA, de la Fuente GT, et al. Takotsubo cardiomyopathy: assessment with cardiac MRI. *AJR Am J Roentgenol*. 2010 Aug;195(2): W139-W145.

3. Kawaji T, Shiomi H, Morimoto T, Clinical impact of left ventricular outflow tract obstruction in Takotsubo cardiomyopathy. *Circ J*. 2015;79(4):839-846.

4. Virani SS, Khan AN, Mendoza CE, Ferreira AC, de Marchena E. Takotsubo cardiomyopathy, or broken-heart syndrome. *Tex Heart Inst J*. 2007;34(1):76-79.

29. Rationale

Answer: D

Two case series reported pacemaker or implantable cardioverter defibrillator (ICD) malfunction after external defibrillation when the pads were placed near the device generator (level of evidence 4). One small study on atrial cardioversion found that positioning the pads on the chest at least 8 cm from the device generator did not produce significant damage to pacing sensing and capturing (level of evidence 4). One case report suggested that pacemaker spikes generated by devices programmed to unipolar pacing may confuse automated external defibrillator software and emergency personnel and may prevent the detection of ventricular fibrillation (level of evidence 4). In patients with an ICD or permanent pacemaker, the placement of paddles or pads should not delay defibrillation. When treating an adult with an ICD or permanent pacemaker, the defibrillator paddle or pad should be placed on the chest wall ideally at least 8 cm from the generator position. The anterior-posterior and anterior-lateral paddle or pad placements on the chest are acceptable in patients with an ICD or permanent pacemaker. Epinephrine and amiodarone are indicated after defibrillation.

References:

1. Link MS, Berkow LC, Kudenchuk PJ, et al. Part 7: Adult advanced cardiovascular life support: 2015 American Heart Association guidelines update for cardiopulmonary resuscitation and emergency cardiovascular care. *Circulation*. 2015 Nov 3;132(18 Suppl 2): S444-S464.

2. Manegold JC, Israel CW, Ehrlich JR, et al. External cardioversion of atrial fibrillation in patients with implanted pacemaker or cardioverter-defibrillator systems: a randomized comparison of monophasic and biphasic shock energy application. *Eur Heart J*. 2007 Jul;28(14):1731-1738.

30. Rationale

Answer: B

This patient has acute cardiac tamponade as indicated by the echo-free space around the heart. Despite the classic symptoms of pulmonary embolism (PE), he remains hemodynamically

stable, and his chest CT is negative for PE, which makes thrombectomy inappropriate currently. Hypovolemic shock can produce similar symptoms of dyspnea and paradoxical pulse; his normal blood pressure and jugular venous distension do not suggest hypovolemia. Currently, he is awake and alert with adequate oxygen saturations of 93%, so mechanical ventilation is not necessary. Echocardiography does not show any structural defects to his heart except for cardiac tamponade. Checking the coronary arteries at some point may be beneficial; it is not necessary to perform a heart catheterization emergently when pericardiocentesis is the gold standard treatment for cardiac tamponade, the diagnosis of which is evidenced by the negative CT, echocardiography results, and classic clinical finding of paradoxical pulse.

References:

1. Adler Y, Charron P, Imazio M, et al; ESC Scientific Document Group. 2015 ESC guidelines for the diagnosis and management of pericardial diseases: the Task Force for the Diagnosis and Management of Pericardial Diseases of the European Society of Cardiology (ESC) endorsed by: the European Association for Cardio-Thoracic Surgery (EACTS). *Eur Heart J*. 2015 Nov 7;36(42):2921-2964.
2. Kearon C, Akl EA, Comerota AJ, et al. Antithrombotic therapy for VTE disease: antithrombotic therapy and prevention of thrombosis, 9th ed: American College of Chest Physicians evidence-based clinical practice guidelines. *Chest*. 2012 Feb;141(2 Suppl): e419S-e496S.
3. Little WC, Freeman GL. Pericardial disease. *Circulation*. 2006 Mar 28;113(12):1622-1632.

31. Rationale

Answer: C

The choice of medication to quickly lower blood pressure in a hypertensive emergency depends on the site of end-organ damage; in this patient presenting with a neurologic emergency, initiating a nicardipine infusion will result in quick onset but gradual reduction in blood pressure. Nicardipine is a

dihydropyridine calcium channel blocker with selectivity for cerebral and coronary arteries. Unlike nitroprusside and nitroglycerin, nicardipine is safe to use in neurologic hypertensive emergencies since it does not increase intracranial pressure. Hydralazine can also potentially increase intracranial pressure since it has greater vasodilatory effect on the cerebral arteries than the cerebral veins. Hydralazine can also cause a sudden, precipitous drop in blood pressure. Given its prolonged and often unpredictable effect, hydralazine is inappropriate for this patient.

References:

1. Hemphill 3rd JC, Greenberg SM, Anderson CS, et al; American Heart Association Stroke Council; Council on Cardiovascular and Stroke Nursing; Council on Clinical Cardiology. Guidelines for the management of spontaneous intracerebral hemorrhage: a guideline for healthcare professionals from the American Heart Association/American Stroke Association. *Stroke*. 2015 Jul;46(7):2032-2060.
2. Ohwaki K, Yano E, Nagashima H, Hirata M, Nakagomi T, Tamura A. Blood pressure management in acute intracerebral hemorrhage: relationship between elevated blood pressure and hematoma enlargement. *Stroke*. 2004 Jun;35(6):1364-1367.
3. Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA guideline for the prevention, detection, evaluation, and management of high blood pressure in adults: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2018 May 15;71(19): e127-e248.

32. Rationale

Answer: D

The ideal agent to manage a hypertensive emergency would be rapid acting and titratable and have no side effects. Since no

agent with these properties exists, pharmacotherapy should be dictated by the clinical picture. Based on the 2017 hypertension guidelines, this patient should be admitted to the ICU because he has end-organ damage (encephalopathy). Esmolol and labetalol should be avoided because his heart rate is already in the 60s beats/min and esmolol could further lower his heart rate via its beta-1 selectivity and should be reserved for those with concomitant elevated heart rate and blood pressure. Clevidipine is contraindicated because of his soybean allergy; it is a lipid formulation. Nicardipine has been recommended in the setting of hypertensive encephalopathy because of its minimal effect on intracranial pressure and reduction in cerebral ischemia.

References:

1. Marik PE, Varon J. Hypertensive crises: challenges and management. *Chest*. 2007 Jun;131(6):1949-1962.
2. Papadopoulos D, Mourouzis I, Thomopoulos C, Makris T, Papademetriou V. Hypertension crisis. *Blood Press*. 2010 Dec;19(6):328-336.
3. Rhoney D, Peacock WF. Intravenous therapy for hypertensive emergencies, part 1. *Am J Health Syst Pharm*. 2009 Aug 1;66(15):1343-1352.
4. Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA guideline for the prevention, detection, evaluation, and management of high blood pressure in adults: executive summary: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Hypertension*. 2018 Jun;71(6):1269-1324.

33. Rationale

Answer: C

Dopamine exhibits dose-related pharmacology, although effects at receptors may occur across doses; primary effects on target receptors are associated with specific dose ranges.

Vasoconstriction, mediated through alpha-1 stimulation, occurs primarily at doses of 10 to 20 μg/kg/min. Doses above 20 μg/kg/min do not provide additional vasoconstriction but can result in untoward chronotropic or arrhythmic effects. Doses of 5 to 10 μg/kg/min primarily cause stimulation of beta-1 receptors. Doses of 5 μg/kg/min or less primarily stimulate dopamine receptors. Doses above 20 μg/kg/min provide no additional beneficial effect of vasoconstriction in a patient with shock but do cause substantial tachycardia and arrhythmias via beta-1 stimulation.

Reference:

1. Westfall TC, Macarthur H, Westfall DP. Adrenergic agonists and antagonists. In: Brunton L, Knollmann B, Hilal-Dandan R, eds. Goodman & Gilman's *The Pharmacological Basis of Therapeutics*. 13th ed. McGraw-Hill Education; 2018.

34. Rationale

Answer: A

This patient has an unstable supraventricular tachycardia (SVT) since he is hypotensive and dyspneic. SVTs appear as regular monomorphic narrow-complex tachyarrhythmias on ECG, typically with heart rate faster than 150 beats/min. Appropriate treatment is synchronized cardioversion. Stable SVTs may be treated with vagal maneuvers, adenosine, beta blockers (such as metoprolol), or calcium channel blockers. Because he is unstable, these treatments are not recommended as first-line interventions. SVTs are not managed with unsynchronized cardioversion; it is appropriate for pulseless ventricular tachycardia and ventricular fibrillation. Epinephrine is not indicated for unstable SVT; it is used in the treatment of asystole and pulseless electrical activity and in ventricular fibrillation/pulseless ventricular tachycardia.

Reference:

1. Sinz E, Navarro K, Soderberg ES, eds. *Advanced Cardiovascular Life Support Provider Manual*. American Heart Association; 2011:61, 127-130.

35. Rationale

Answer: C

This patient has congenital heart disease with structural defects. Her condition caused a univentricular heart. Patients with a single ventricle have a much higher probability of heart failure than patients with other types of heart defect of the same age. While this patient's history and clinical picture strongly suggest hypovolemia, infusing aggressive crystalloids immediately is contraindicated. A gentler approach with multiple doses of crystalloids is recommended for her. She has low urine output most likely because she was hypovolemic based on her clinical signs and because she did not receive fluid intraoperatively. Therefore, initiating furosemide to increase urine output is contraindicated. She has only 1 ventricle to supply blood flow to both arterial and venous systems. Phenylephrine, which increases afterload, is contraindicated, and should be discontinued. In patients with only 1 ventricle, afterload reduction is most important. Adding norepinephrine to phenylephrine is contraindicated. While she may need IV crystalloids, the most appropriate immediate intervention is to discontinue phenylephrine to reduce afterload.

Reference:

1. Shaddy RE, Webb G. Applying heart failure guidelines to adult congenital heart disease patients. *Expert Rev Cardiovasc Ther.* 2008 Feb;6(2):165-174.

36. Rationale

Answer: A

The echocardiogram reveals a massive malignant pericardial effusion that will develop cardiac tamponade if not drained. The effusion should be drained as soon as possible. This is not pericarditis to be treated with nonsteroidal anti-inflammatory drugs. No valvular problem is seen.

Reference:

1. Peter McCanny, Frances Colreavy, Echocardiographic approach to cardiac tamponade in critically ill patients. *Crit*

Care. 2017 Jun;39:271-277.

37. Rationale

Answer: B

Aortic valve area (AVA) can be determined by measuring left ventricular outflow tract (LVOT) diameter, then converting to cross-sectional area (CSALVOT); LVOT peak velocity (VTILVOT where VTI is velocity-time integral); and aortic valve (AV) jet peak velocity (VTIAV), using the continuity equation AVA = (CSALVOT × VTILVOT)/VTIAV.

Reference:

1. Baumgartner H, Hung J, Bermejo J, et al. Recommendations on the echocardiographic assessment of aortic valve stenosis: a focused update from the European Association of Cardiovascular Imaging and the American Society of Echocardiography. *J Am Soc Echocardiogr* 2017 Apr;30(4):372-392.

38. Rationale

Answer: B

Although this patient's background atrial rhythm is atrial flutter, the primary rhythm is complete heart block with slow escape. Left bundle branch block and high-degree atrioventricular block are known complications after transcatheter aortic valve replacement. Although these rhythms are more common within the first 24 hours post-procedure, such conduction abnormalities can occur within 7 days post-procedure. The European Society of Cardiology recommends a period of careful observation and EEG monitoring up to 7 days before considering pacemaker implant; however, in cases of complete atrioventricular block with slow escape rhythm, the guidelines recommend earlier intervention.

References:

1. Auffret V, Puri R, Urena M, et al. Conduction disturbances after transcatheter aortic valve replacement:

status and future perspectives. *Circulation*. 2017 Sep 12;136(11):1049-1069.

2. Brignole M, Auricchio A, Baron-Esquivias G, et al. 2013 ESC guidelines on cardiac pacing and cardiac resynchronization therapy: the task force on cardiac pacing and resynchronization therapy of the European Society of Cardiology (ESC). *Eur Heart J*. 2013 Aug;34(29):2281-2329

39. Rationale

Answer: B

This patient presents with cardiac tamponade, which is a clinical syndrome caused by the accumulation of fluid in the pericardial space resulting in reduced ventricular filling and subsequent hemodynamic compromise. It is a medical emergency, complications of which include pulmonary edema, shock, and death. Symptoms vary with the acuteness and underlying cause of the tamponade. Patients with acute tamponade may present with dyspnea, tachycardia, and tachypnea. Cold and clammy extremities from hypoperfusion are observed in some patients. Prompt diagnosis is key to reducing mortality risk. Although cardiac tamponade is a clinical diagnosis, further assessment of the underlying cause of the tamponade can be obtained through laboratory studies, imaging studies, electrocardiography, and echocardiography. Echocardiography can be used to visualize ventricular and atrial compression abnormalities as blood cycles through the heart, while laboratory studies can demonstrate signs of myocardial infarction, cardiac trauma, and infectious disease. Removal of pericardial fluid is the definitive therapy for tamponade and can be accomplished with the following 3 methods: emergency subxiphoid percutaneous drainage, pericardiocentesis (with or without echocardiographic guidance), or percutaneous balloon pericardiotomy.

References:

1. Mujovic N, Marinkovic M, Markovic N, et al. Management and outcome of periprocedural cardiac perforation and tamponade with radiofrequency catheter

ablation of cardiac arrhythmias: a single medium-volume center experience. *Adv Ther.* 2016 Oct;33(10):1782-1796.

2. AD, Imazio M, Adler Y, et al. Triage strategy for urgent management of cardiac tamponade: a position statement of the European Society of Cardiology Working Group on Myocardial and Pericardial Diseases. *Eur Heart J.* 2014 Sep 7;35(34):2279-2284.

3. Shiota T. Role of echocardiography for catheter-based management of valvular heart disease. *J Cardiol.* 2017 Jan;69(1):66-73.

Part 6.

Pulmonary Disease

Instructions: For each question, select the most correct answer.

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1. A 66-year-old man is admitted to the ICU after undergoing pericardiectomy for constrictive pericarditis. He has no preexisting lung problems and does not smoke. Shortly after surgery, he develops persistent hypercapnic respiratory failure, and a tracheostomy is placed. During his hospital stay, he is treated with inhalers, pulsed steroids, and antibiotics. At 4 weeks into his stay, arterial blood gas analysis results remain approximately: pH 7.44, PCO_2 55 mm Hg, and PO_2 72 mm Hg on FIO_2 0.50 via tracheal collar. He can wear the tracheal collar for only 5 hours daily because of anxiety and air hunger. He has no fever or new secretions. Which of the following is the most useful test for determining the etiology of his newly acquired chronic pulmonary failure?

   A. Sniff test

   B. Serum albumin level

   C. Noncontrast chest CT

   D. Serum rheumatoid factor

   E. Spirometry

2. A 58-year-old woman is admitted to the ICU for respiratory failure. She does not smoke and has no significant past medical history. She is up to date on immunizations and denies any incarceration history or travel outside the United States. Her symptoms started with fevers and chills. She also has bilateral knee pain and intermittent headaches. She recently started

working on a cattle farm. She has a 3-week history of progressive dyspnea and increased work of breathing. She continues to have a dry nonproductive cough but is now afebrile. Which of the following treatments is considered first line for her condition?

A. Levofloxacin

B. Prednisone

C. Rituximab

D. Pirfenidone

3. A 56-year-old man is admitted to the ICU postoperatively and requires continued intubation and mechanical ventilation for 2 days. A neuromuscular blocking agent is administered briefly but otherwise he is sedated with propofol, fentanyl, and as-needed midazolam. His Preoperative Prediction of Obstructive Sleep Apnea (SPOSA) score is 20. A score less than 24 suggests a low risk for obstructive sleep apnea (OSA). The amount of which of the following drugs or drug classes administered within 24 hours before extubation most closely correlates with an increased likelihood of developing OSA within 24 hours after extubation?

   A. Opioids

   B. Propofol

   C. Neuromuscular blockers

   D. Benzodiazepines

4. A 23-year-old man with sickle cell disease has been receiving treatment in the hospital for a vaso-occlusive pain episode. He develops a fever and dyspnea acutely. Oxygen saturation is initially 88% on room air and increases to 95% on supplemental oxygen at 4 L/min via nasal cannula; however, oxygenation subsequently worsens, and he becomes confused. He is transferred

to the ICU, where he is intubated. Chest radiograph reveals bilateral lung opacities. Relevant laboratory findings are shown below. Which of the following is the most appropriate intervention?

Study	Baseline level	ICU test result
Hemoglobin (g/dL)	8.0	6.2
Platelet count (cells/µL)	300,000	150,000
Creatinine (mg/dL)	0.9	1.0
ALT level (U/L)	34	37

A. Hydroxyurea

B. Inhaled nitric oxide

C. Dexamethasone

D. Exchange transfusion

5. Which of the following pulmonary function measures is most appropriate for assessing a patient with myasthenia gravis?

   A. Residual volume

   B. Total lung capacity

   C. Forced expiratory volume

   D. Maximal respiratory pressure

   E. Diffusing capacity of the lungs for carbon monoxide

6. A 75-year-old man with known adenocarcinoma of the lung is admitted to the ICU with respiratory distress. Previous chest radiograph showed a large left-sided pleural effusion. Now, thoracentesis with ultrasound guidance is performed, and 1750 mL of total fluid is drained. Initial pleural pressure is 5 cm H_2O. After 1500 mL of pleural fluid is removed, pleural pressure is –5 cm H_2O and eventually –12 cm H_2O. The patient reports chest discomfort and the procedure is terminated. His dyspnea improves only

slightly following drainage of the fluid. Which of the following is the most likely cause of his symptoms?

A. Pneumothorax

B. Lung entrapment

C. Trapped lung

D. Re-expansion of normal lung

7. A 59-year-old man with no known medical history presents to the emergency department for acute onset of shortness of breath. He has a 3-day history of chest tightness and has had an intermittent cough for 2 months that is productive of clear sputum. CT shows emphysematous changes without any evidence of pulmonary embolism. He is started on noninvasive positive pressure ventilation and admitted to the hospital. Which of the following vaccines should he receive before hospital discharge?

   A. Influenza and pneumococcal polysaccharide

   B. Pneumococcal conjugate and Haemophilus influenzae type B (HIB)

   C. Meningococcal

   D. Diphtheria, tetanus, and pertussis

   E. HIB only

8. A 65-year-old man is discharged from the ICU after recovering from a lower respiratory tract infection. His past medical history includes coronary artery disease, oxygen-dependent chronic obstructive pulmonary disease (COPD) (2 L/min), and diabetes. Vital signs are blood pressure 120/80 mm Hg, heart rate 85 beats/min, and respiratory rate 18 breaths/min on home oxygen requirements. He has a low-grade fever of 37.7°C (99.86°F). He received a pneumococcal polysaccharide vaccine 7 years ago and asks whether he needs any other vaccines before going home.

Which of the following is the most appropriate response?

A. No pneumococcal vaccines should be administered until he is afebrile.

B. Both pneumococcal conjugate and pneumococcal polysaccharide vaccines should be administered today.

C. Only the pneumococcal conjugate vaccine should be administered.

D. The pneumococcal conjugate vaccine should be administered today and the pneumococcal polysaccharide vaccine in 1 year.

9. A 64-year-old man with a history of myocardial infarction, hypertension, and hyperlipidemia is admitted for sepsis secondary to community-acquired pneumonia. He is fluid resuscitated with 5 L crystalloid and, according to dynamic indices, his stroke volume variation is 5%. He is currently on norepinephrine, 30 μg/min, and vasopressin, 0.03 U/min. Mean arterial pressure is 65 mm Hg, and he is showing signs of end-organ damage due to elevated creatinine and liver function tests. Central venous saturation is 45%. Which of the following is the most reasonable next step in his hemodynamic management?

   A. Increase vasopressin.

   B. Increase norepinephrine.

   C. Initiate angiotensin II.

   D. Initiate dobutamine.

10. A 64-year-old man with a medical history significant for diabetes and tobacco abuse is admitted to the ICU for acute hypoxemic respiratory failure secondary to influenza. Arterial blood gas analysis obtained 4 hours

after initiation of mechanical ventilation shows pH 7.59, PCO_2 31 mm Hg, PaO_2 93 mm Hg, and bicarbonate level 20 mEq/L. Ventilator settings are: tidal volume 480 mL, respiratory rate 22 breaths/min, positive end-expiratory pressure (PEEP) 5 cm H_2O, and FIO_2 0.80. Which of the following ventilator adjustments is needed?

A. Increase tidal volume to 550 mL.

B. Decrease respiratory rate to 18 breaths/min.

C. Increase FIO_2 to 1.0.

D. Decrease PEEP to 0.

11. A 62-year-old man is admitted to the ICU with acute pancreatitis complicated by acute respiratory distress syndrome. He requires intubation for acute hypoxemic respiratory failure. He is ventilated with positive end-expiratory pressure (PEEP) 5 cm H_2O and FIO_2 0.4. He is stable on these ventilator settings for 4 days. On ventilator day 5, he becomes more hypoxemic and requires increased PEEP and FIO_2. Based on the CDC's National Healthcare Safety Network, which of the following parameters qualifies his condition as a ventilator-associated condition?

A. Requiring increased PEEP to 7 cm H_2O and FIO_2 to 0.5 for 1 day

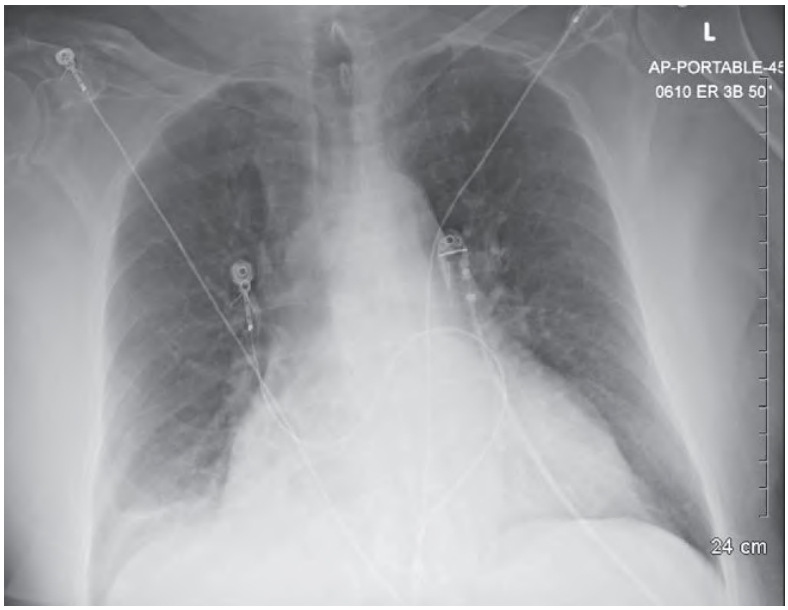
B. Requiring increased PEEP to 6 cm H_2O and FIO_2 to 0.6 for 2 days

C. Requiring increased PEEP to 7 cm H_2O and FIO_2 to 0.45 for 3 days

D. Requiring increased PEEP to 6 cm H_2O and FIO_2 to 0.55 for 2 days

E. Requiring increased PEEP to 7 cm H_2O and FIO_2 to 0.55 for 1 day

12. A 67-year-old woman with moderate chronic obstructive lung disease and ischemic cardiomyopathy presents to the emergency department with a 24-hour history of progressively worsening dyspnea and fever. Her most recent left ventricular ejection fraction was 35%. She is started on bilevel noninvasive positive pressure ventilation (NIPPV) for acute-on-chronic hypercapnic respiratory failure. Per hospital guidelines for management of suspected sepsis, 30 mL/kg ideal body weight isotonic IV fluid is administered and, per hospital antibiogram for community-acquired pneumonia, antibiotics are administered. Additional pertinent findings include WBC count of 18,700 cells/μL and procalcitonin level of 3.7 ng/mL. Chest radiograph is shown below. Two hours after ICU admission, clinical evidence of respiratory distress and worsened encephalopathy remains. Which of the following is the most appropriate next step in this patient's management?



A. Sedation

B. Arterial blood gas analysis

C. Optimization of NIPPV settings

D. Endotracheal intubation

E. Broadening antimicrobial coverage

13. A 63-year-old woman is admitted to the ICU for fever of 39.3°C (102.9°F), nonproductive cough, and tachypnea lasting 6 days. She had been treated at home with antibiotics for possible bronchitis but did not improve. Initial laboratory testing shows WBC count 14.7 × 10^9/L, erythrocyte sedimentation rate 42 mm/hr, and C-reactive protein level 9 mg/L. Chest radiograph shows diffuse ground-glass opacities with small bilateral pleural effusions. On ICU arrival, physical examination reveals bibasilar inspiratory crackles and hypoxic respiratory failure with oxygen saturation 75% on 15-L nonrebreather mask, requiring intubation. After intubation, bronchoscopy with bronchoalveolar lavage (BAL) shows no malignancy or bacteria, but eosinophils on BAL are 32%. Peripheral blood eosinophils are 1%. Which of the following is the most appropriate treatment for this patient?

A. IV cyclophosphamide

B. Oseltamivir via feeding tube

C. IV methylprednisolone

D. IV immunoglobulin

14. A 66-year-old man with non-small cell lung cancer was initiated on checkpoint blockade immunotherapy with pembrolizumab in combination with pemetrexed and carboplatin. After 3 weeks of treatment, he suddenly develops worsening dyspnea. He is brought to the emergency department where he has acute hypoxemic respiratory failure. Noninvasive positive pressure ventilation is initiated for severe hypoxemia. Further workup shows no pulmonary embolism but bilateral infiltrates and previously known lung cancer

findings with no significant change. Bronchoalveolar lavage (BAL) shows mild lymphocytosis but no significant neutrophilia. In 24 hours, BAL *Pneumocystis carinii* stain and Gram stain are negative. During the next few days, he does not improve and continues to have bilateral infiltrates. In addition to stopping immunotherapy, which of the following is the most appropriate next step in management?

A. IV sulfamethoxazole/trimethoprim

B. Medium-dose corticosteroids with prednisone

C. Infliximab with or without cyclophosphamide for immunosuppression

D. No additional treatment

15. A patient is somnolent because of a narcotic overdose, causing an acute 50% decrease in alveolar ventilation. If the patient previously had normal arterial blood gas values while breathing room air, which of the following sets of values most likely corresponds to the patient's current arterial blood gas values while breathing room air?

A. pH 7.40, $PaCO_2$ 40 mm Hg, PaO_2 100 mm Hg

B. pH 7.40, $PaCO_2$ 80 mm Hg, PaO_2 100 mm Hg

C. pH 7.08, $PaCO_2$ 80 mm Hg, PaO_2 100 mm Hg

D. pH 7.08, $PaCO_2$ 80 mm Hg, PaO_2 50 mm Hg

16. A 72-year-old man is evaluated in the emergency department for progressive shortness of breath. He has a history of coronary artery disease, heart failure, and atrial fibrillation. His medications are aspirin, warfarin, lisinopril, carvedilol, atorvastatin, and furosemide. On physical examination, he is in respiratory distress. Temperature is 37°C (98.6°F), blood pressure 156/98 mm Hg, heart rate 73 beats/min and irregular, and respiratory rate 32

breaths/min. Oxygen saturation as measured by pulse oximetry is 92% on 5 L oxygen via nasal cannula. Jugular venous pressure is elevated. Cardiac examination reveals an irregular rhythm. Pulmonary examination reveals bibasilar crackles. Chest radiograph shows vascular congestion. A dose of IV furosemide is administered. Which of the following is the most appropriate next step?

A. IV furosemide infusion

B. Intubation and mechanical ventilation

C. Noninvasive positive pressure ventilation

D. Ultrafiltration

17. A 65-year-old woman with a history of deep vein thrombosis has increasing dyspnea, weight gain, and lower extremity edema. She is using accessory muscles to breathe. Oxygen saturation is 85% on room air. She is started on noninvasive ventilation and admitted to the ICU. Bedside ultrasound reveals enlarged right ventricle (RV), with bowing of the interventricular septum. She is diuresed aggressively and anticoagulated and she improves. Toward the end of her admission, echocardiography reveals moderate to severe reduction in RV function and elevated RV systolic pressure. Left-sided ejection fraction is normal. Workup for pulmonary hypertension is started. Which of the following is the most sensitive method for diagnosing chronic thromboembolic pulmonary hypertension?

A. Lower extremity Doppler ultrasound

B. Ventilation/perfusion lung scan

C. Chest CT with IV contrast

D. Speckle-tracking echocardiography

E. Right heart catheterization

18. A 39-year-old woman who has been bed bound presents with tachypnea and shortness of breath. She is intubated and point-of-care ultrasound shows a massively dilated right ventricle with decreased right ventricular function. The left ventricle is hyperdynamic but has normal function. There is bowing of the ventricular septum into the left ventricle. Her blood pressure continues to drop. The emergency department physician is concerned that the patient will have a cardiac arrest. Which of the following is the most appropriate management approach?

    A. Administer alteplase, 100 mg over 2 hours, and start an unfractionated heparin (UFH) infusion at 18 units/kg after the alteplase infusion has completed.

    B. Administer tenecteplase, 40 mg IV push over 5 seconds, and start a UFH infusion at 18 units/kg at the same time.

    C. Start UFH at 18 units/kg and obtain chest CT angiogram.

    D. Obtain chest CT angiogram before starting antithrombotic therapy.

19. A 65-year-old woman with a past medical history significant for chronic obstructive pulmonary disease and tobacco use presents to the emergency department with hemoptysis. She reports coughing up nearly half a cup of blood in the past 30 minutes. Chest radiograph is normal, and laboratory findings are unremarkable, with prothrombin time 12.1 seconds and INR 0.9. Vital signs are blood pressure 128/79 mm Hg, heart rate 82 beats/min, oxygen saturation 91% on room air, and respiratory rate 17 breaths/min. She is admitted to the ICU for close airway monitoring and management of hemoptysis.

Which of the following is the most appropriate next step in assessment?

A. Bronchoscopy

B. Surgical intervention

C. CT

D. Angiography

20. A 65-year-old man is admitted to the ICU with a 4-day history of productive cough, subjective fevers, pleuritic right-sided chest pain, and progressive dyspnea. Vital signs are temperature 39.1°C (102.5°F), heart rate 125 beats/min, respiratory rate 22 breaths/min, blood pressure 95/40 mm Hg, and oxygen saturation 94% on 4 L/min oxygen by nasal cannula. Crackles and egophony are auscultated at the right lung base. WBC count is 26,000/μL with 20% band forms. Chest CT with IV contrast demonstrates a right lower lobe consolidation and a moderate-sized, loculated pleural effusion with pleural thickening. IV piperacillin-tazobactam and a bolus of IV crystalloid fluid are administered. A chest tube is placed, and 750 mL of purulent fluid is drained. Which of the following is expected after the administration of intrapleural DNase and tissue plasminogen activator twice daily for 3 days?

A. Markedly elevated risk of hemorrhage

B. Increased need for surgical referral

C. Reduction in mortality

D. Improved drainage of infected pleural fluid

21. A 63-year-old woman presents to the emergency department with worsening dyspnea, fevers, and chills. Her family reports a recent hospitalization at an outside hospital for an exacerbation of underlying chronic obstructive pulmonary disease. Respiratory

rate is 40 breaths/min and oxygen saturation on room air is 70%, requiring intubation. Post-intubation chest radiograph is notable for scattered bilateral consolidations and a large right pleural effusion. She undergoes diagnostic thoracentesis and is started on a regimen of vancomycin and piperacillin/tazobactam. Pleural fluid results are notable for pH 7.10, glucose 40 mg/dL, and 50,000 neutrophils/μL. Which of the following is the most appropriate next step in management?

A. Placement of a 14-French chest tube

B. Consultation for video-assisted thoracoscopic surgery

C. Discontinuing piperacillin/tazobactam and starting meropenem

D. Repeat diagnostic thoracentesis to trend the pleural fluid pH and cell count

22. A 74-year-old man with heart failure and metastatic melanoma to the liver and bone presents to the ICU with worsening shortness of breath. Initial examination is significant for tachypnea, increased work of breathing, and hypoxia. Pulmonary examination reveals dullness to percussion and reduced breath sounds over the right hemithorax. Ultrasound examination confirms a large right pleural effusion. Thoracentesis is performed. Gross visual examination of the aspirated pleural fluid is shown below. Which of the following is the most likely diagnosis?



A. Empyema

B. Pleural effusion secondary to heart failure

C. Thoracic metastasis of malignant melanoma

D. Iatrogenic hemothorax

23. A 58-year-old man with a history significant for alcoholic cirrhosis with portal hypertension is admitted to the ICU with progressive dyspnea,

fatigue, and worsening ascites. Oxygen saturation is 91% on room air. Arterial blood gas analysis reveals PaO_2 70 mm Hg. His dyspnea is worsened by sitting and improves when he returns to a recumbent position. In addition to this symptom, which of the following would confirm the diagnosis of hepatopulmonary syndrome?

A. Orthodeoxia with a decrease in PaO_2 of 5% or more from supine to sitting position

B. Intrapulmonary vasodilatation on echocardiography

C. Spider nevi on hands and feet

D. Forced expiratory volume in 1 second less than 50% predicted

E. Portopulmonary hypertension on right heart catheterization

24. A 27-year-old man with no reported past medical history presents to the emergency department with intermittent shortness of breath during the past 12 hours. He denies history of trauma. Examination reveals a tall and healthy-appearing man in no acute distress. He is afebrile with blood pressure 134/66 mm Hg, heart rate 68 beats/min, respiratory rate 14 breaths/min, and oxygen saturation as measured by pulse oximetry 99% on room air. Clear bilateral breath sounds are noted on auscultation. Chest radiograph reveals a 1-cm left-sided apical pneumothorax with no other abnormalities. The most appropriate next step in management is

A. placement of a 14-French pigtail chest tube with CT guidance.

B. thoracic surgery consultation for video-assisted thoracoscopic surgery with pleurodesis.

C. direct aspiration of pleural air via 8.5-French catheter with ultrasound guidance.

D. admission to the ICU for close monitoring and serial chest radiographs.

E. observation in the emergency department with repeat chest radiograph in 6 hours.

25. A 45-year-old man presents to the emergency department after crashing his motorcycle. He has shortness of breath and decreased lung sounds on the left. A left-sided chest tube is placed, and 250 mL of blood is drained immediately. CT reveals that his left 1-9 ribs are fractured. He is admitted to the trauma ICU for closer monitoring. During the next 6 hours, more than 2 L of blood is drained from the chest tube. He received 2 units packed RBCs in the ICU. Which of the following is the most appropriate next step in management?

A. Transfuse 2 units packed RBCs and continue to monitor chest tube output.

B. Transfuse 2 units packed RBCs and repeat chest CT.

C. Perform urgent thoracotomy.

D. Check coagulation studies and correct any coagulopathy.

26. A 32-year-old man is admitted to the ICU after a motor vehicle accident in which he sustained a traumatic brain injury. He was intubated on admission because his Glasgow Coma Scale score was 6 and he was unable to protect his airway. On day 4 of admission, he develops a fever of 39.1°C (102.3°F), and the nurse notes thick, purulent secretions from endotracheal suctioning. WBC count is 18.2 cells/mm^3. Chest radiograph shows left lower lobe consolidation. Blood and urine cultures are negative; respiratory culture is pending. Nasal swab screen is positive for methicillin-resistant *Staphylococcus aureus*.

Two months ago, he was treated with amoxicillin/clavulanate for sinusitis and had an anaphylactic reaction. Which of the following is the most appropriate empiric treatment regimen for him?

A. Levofloxacin alone

B. Ceftriaxone plus azithromycin

C. Piperacillin/tazobactam plus vancomycin

D. Aztreonam plus tobramycin plus vancomycin

E. Piperacillin/tazobactam plus tobramycin plus vancomycin

27. A 39-year-old woman presents with altered mental status, hypercapnic respiratory failure, and elevated blood glucose after feeling ill for 2 days. She is initiated on broad-spectrum antibiotics, intubated for pneumonia and acute respiratory distress syndrome, and mechanically ventilated on pressure-regulated volume-control mode with tidal volume 6 mL/kg, respiratory rate 22 breaths/min, positive end-expiratory pressure 20 cm H_2O, and FIO_2 1.0. The next day, arterial blood gas analysis reveals PaO_2 80 mm Hg on the same ventilator settings. Twelve hours later she desaturates, and an emergent arterial blood gas analysis reveals PaO_2 65 mm Hg. Which of the following interventions should be initiated next to improve her lung mechanics and oxygenation?

A. High-frequency oscillatory ventilation

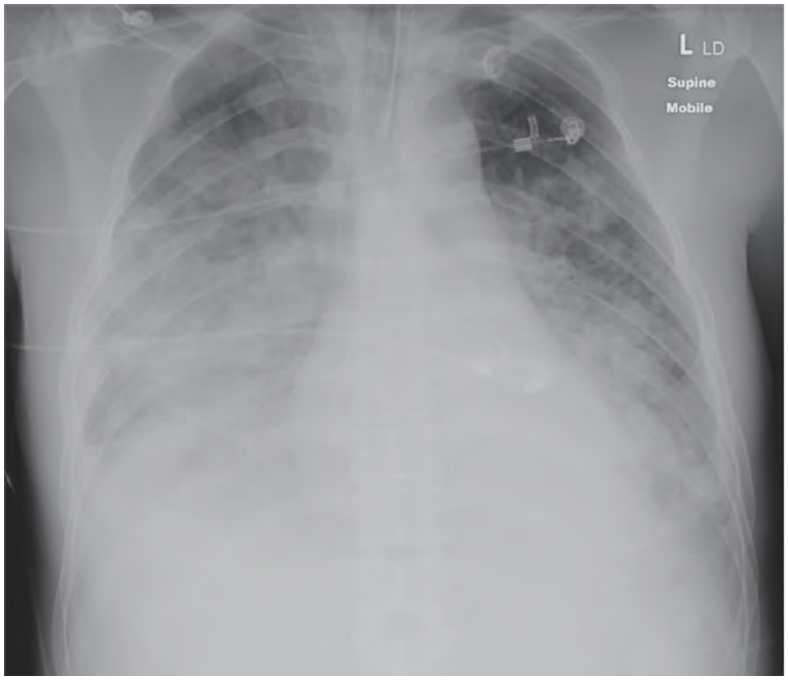
B. Inhaled epoprostenol

C. Inhaled nitric oxide

D. IV cisatracurium

28. A 55-year-old man with a history of hypertension and hyperlipidemia was recently diagnosed with influenza A. He presents to the emergency department with

worsening shortness of breath, fever, productive cough, and new onset altered mental status. On primary assessment, oxygen saturation is 61% on room air. Respiration is labored, with abdominal accessory muscle use. Bilevel positive airway pressure is started but his respiratory status continues to deteriorate, and he is intubated and sedated. His oxygenation worsens, with PaO_2 48 mm Hg on 4 mL/kg ideal body weight, FIO_2 1.0, positive end-expiratory pressure (PEEP) 18 cm H_2O, with peak pressure of 36 cm H_2O. Plateau pressure is 33 cm H_2O. Chest radiograph is shown below. Which of the following interventions is most likely to have the greatest mortality benefit?



A. Recruitment maneuver

B. High-frequency oscillatory ventilation

C. Prone positioning

D. Increased PEEP

29. A 67-year-old woman requires mechanical ventilation for acute hypoxemic respiratory failure. Chest

radiograph shows diffuse bilateral infiltrates without significant pleural effusions. Bedside point-of-care cardiac ultrasound shows normal left ventricular size and function. Ventilator settings are volume assist control mode, tidal volume 360 mL (6 mL/kg predicted body weight), rate 16 breaths/min, positive end-expiratory pressure (PEEP) 15 cm H_2O, and FIO_2 0.80. Ventilator synchrony appears adequate. Arterial blood gas analysis reveals PaO_2 60 mm Hg. In addition to initiating prone ventilation for at least 12 hours, which of the following is the most appropriate management recommendation?

A. High-frequency oscillatory ventilation

B. Increased PEEP

C. Extracorporeal membrane oxygenation

D. IV beta-agonist

E. Keratinocyte growth factor

30. A 69-year-old man with a history of chronic obstructive pulmonary disease (COPD) presents to the emergency department with increasing shortness of breath. Initial arterial blood gas analysis shows: pH, 7.29; PCO_2, 90 mm Hg; PO_2, 85 mm Hg; and bicarbonate, 22 mEq/L. Initial vital signs are: heart rate, 102 beats/min; blood pressure, 148/88 mm Hg; respiratory rate, 24 breaths/min; temperature, 37.2°C (98.6°F); and oxygen saturation as measured by pulse oximetry, 90% on room air. He is started on bilevel positive airway pressure (BiPAP) and transferred to the step-down unit. His vital signs are now: heart rate, 51 beats/min; blood pressure, 64/40 mm Hg; respiratory rate, 12 breaths/min; and oxygen saturation as measured by pulse oximetry, 88%. Which of the following is the most appropriate next step?

A. Intubate.

B. Increase inspiratory pressure.

C. Administer normal saline, 1 L.

D. Administer epinephrine.

31. A 44-year-old woman presents to the emergency department with dyspnea, cough, hemoptysis, and fever. Past medical history includes systemic lupus erythematosus. She is intubated and admitted to the ICU with hypoxemic respiratory failure. Chest CT shows diffuse, bilateral ground-glass opacities. Bronchoscopy shows no evidence of active infection. Sequential bronchoalveolar lavage of the right middle lobe reveals progressively more hemorrhagic returns. Cytologic staining of the lavage shows hemosiderin-laden macrophages. Which of the following is the most appropriate initial intervention for this patient?

    A. IV tranexamic acid

    B. IV methylprednisolone

    C. IV piperacillin/tazobactam

    D. Platelet transfusion

    E. IV recombinant factor VIIa

32. A 72-year-old woman is postoperative after 4-vessel coronary artery bypass grafting. Right and left ventricular function are normal, as they were preoperatively. Intraoperative course was uncomplicated, and she arrives in the ICU hemodynamically stable on no infusions. Shortly thereafter, blood pressure is 82/46 mm Hg, dual-chamber pacing with dual-chamber sensing and dual inhibition is paced at 90 beats/min, central venous pressure is 15 mm Hg, pulmonary artery pressure is 53/27 mm Hg, and cardiac index is 2.0. Chest tube output is minimal. Cardiology is notified to perform emergent echocardiography. Which of the following

interventions is most appropriate until further information is available?

A. Phenylephrine infusion

B. Milrinone infusion

C. Intra-aortic balloon pump

D. Inhaled nitric oxide

E. Pericardiocentesis

33. A 70-year-old man with chronic obstructive pulmonary disease and renal cell cancer presents to the emergency department with acute-onset shortness of breath and exercise intolerance. On initial examination, he is tachypneic and hypoxic and is using his accessory muscles of respiration. Pulmonary examination is significant for wheezing. He is started on inhaled bronchodilator therapy, steroids, and antibiotics. Initial arterial blood gas results on room air are pH 7.43, PO_2 70 mm Hg, PCO_2 33 mm Hg, bicarbonate 22 mEq/L, and lactate 1.5 mg/dL. Twenty-four hours later, he shows no improvement. Which of the following is the most appropriate next test for this patient?

    A. Transthoracic echocardiography

    B. Renal ultrasound

    C. CT pulmonary angiogram

    D. Pulmonary function test

34. A 78-year-old man arrives intubated to the ICU after a 6-hour spine procedure. He is hemodynamically stable; however, the nurse has noticed that his oxygen saturation as measured by pulse oximetry has dropped from 95% to 88%. His chest radiograph is pending. On auscultation, no breath sounds are heard on the left hemithorax, and peak airway pressures are slightly

elevated. He is slightly agitated. Which of the following is the most appropriate next step?

A. Obtain emergent chest CT.

B. Obtain arterial blood gas analysis.

C. Suction the endotracheal tube (ETT).

D. Check the ETT for optimal placement.

E. Perform thoracic ultrasound.

35. A 67-year-old woman underwent an exploratory laparotomy because of a bowel perforation; she was profoundly hypotensive after arrival in the ICU, requiring massive volume resuscitation and two vasopressors. Late that night her initially stable blood gases changed dramatically, and she became significantly hypoxic despite higher FIO_2. Chest radiograph demonstrates bilateral infiltrates. Bedside ultrasound shows a collapsible inferior vena cava with inspiration. Which of the following is the single most important treatment to improve her long-term respiratory outcome?

    A. Aggressive diuresis

    B. Inhaled nitric oxide

    C. Low-tidal-volume ventilation

    D. Pulse-dose steroids

    E. Early neuromuscular blockade

36. A 40-year-old man presents with acute onset of shortness of breath. History is significant for a recent trans-Atlantic flight. He denies any cough, fevers, chills, or chest pain. He has not had any recent episodes of bleeding. On examination, he is in mild distress. Vital signs are temperature 36.9°C (98.5°F), heart rate 110 beats/min, respiratory rate 28 breaths/min, blood pressure 130/80 mm Hg, and

oxygen saturation as measured by pulse oximetry 88% on room air. Examination is significant for left calf swelling. Lungs are clear to auscultation bilaterally. Which of the following is the most appropriate next step?

A. Parenteral therapeutic anticoagulation

B. CT pulmonary angiogram

C. Systemic thrombolytic therapy

D. D-dimer test

37. A 28-year-old man sustained a gunshot wound to the right chest 3 days ago. He is intubated in the ICU and has developed a large leak from 2 chest tubes properly positioned in the right hemithorax. Plain chest radiograph shows a refractory right pneumothorax and progressively worsening diffuse infiltrates in the left lung field. Much of the tidal volume delivered is not returned. Which of the following is the most appropriate next step in management?

A. Placement of another right thoracostomy tube

B. Initiation of independent lung ventilation

C. High positive end-expiratory pressure/low-tidal-volume ventilation

D. Inhaled nitric oxide therapy

38. A 39-year-old man with a history of alcoholic cirrhosis has worsening ascites and new dyspnea on exertion. Blood pressure is 100/65 mm Hg, heart rate 78 beats/min, respiratory rate 23 breaths/min, and oxygen saturation 88%. When standing, his oxygen saturation decreases to 83%. Room air arterial blood gas analysis reveals pH 7.46, $PaCO_2$ 47 mm Hg, PaO_2 52 mm Hg, and alveolar-arterial oxygen gradient 18 mm Hg. Which of the following is the most effective treatment for his hypoxia?

A. Supplemental oxygen

B. Norfloxacin

C. Octreotide

D. Indomethacin

E. Inhaled epoprostenol

39. A 58-year-old woman has been intubated in the ICU for 3 days after open cardiac surgery. She is recovering well from surgery but on hospital day 4 develops a fever of 38.8°C (101.8°F) and increased purulent secretions. Tracheal culture is positive for methicillin-sensitive *Staphylococcus aureus* at a concentration greater than 10^5 colony-forming units/mL. Urine and blood cultures are negative for any organisms. Chest radiograph shows no pulmonary infiltrates. Which of the following is the most appropriate treatment strategy?

    A. No treatment needed

    B. Vancomycin

    C. Inhaled tobramycin

    D. Inhaled colistin

    E. Nafcillin

40. An otherwise healthy 26-year-old man is admitted to the trauma ICU after sustaining injuries in a fall. He has no past medical history. He denies illicit drug use. He has smoked 1 pack of cigarettes per day for 4 years and drinks beer approximately once per month. The hospital has increasingly emphasized quality improvement measures meant to increase patient pneumococcal immunization rates before discharge. According to the Centers for Disease Control and Prevention, the most appropriate pneumococcal vaccination schedule for this patient is 1 dose of

A. pneumococcal polysaccharide vaccine (PPSV23) at age 65.

B. pneumococcal conjugate vaccine (PCV13) before hospital discharge and 1 dose of PPSV23 at ages 27, 32, and 65.

C. PPSV23 before hospital discharge and another dose at age 65.

D. PCV13 before hospital discharge and 1 dose of PPSV23 at ages 27 and 65.

41. An 82-year-old woman presents to the emergency department from home with altered mental status, productive cough, and shortness of breath. Vital signs are: heart rate 115 beats/min, respiratory rate 40 breaths/min, blood pressure 90/50 mm Hg, and oxygen saturation 78% on room air. She is awake and alert but confused. On examination, she is tachypneic and has diffuse rhonchi. She is started on a nonrebreather mask, and oxygen saturation improves to 94%. Portable chest radiograph shows multifocal areas of consolidation. WBC count is 19,000/μL with 10% bands. Arterial blood gas analysis shows: pH, 7.35; $PaCO_2$, 35 mm Hg; PaO_2, 65 mm Hg; bicarbonate, 18 mEq/L; and oxygen saturation, 90% on 100% nonrebreather mask. She is transferred to the ICU. Which of the following is the most appropriate next step in managing her acute hypoxic respiratory failure?

A. Continue nonrebreather mask.

B. Start humidified high-flow oxygen.

C. Start noninvasive positive pressure ventilation.

D. Intubate and start mechanical ventilation.

Part 6 Answers:

Pulmonary Disease

1. Rationale

Answer: A

This patient has postoperative failure secondary to damage to the diaphragmatic nerve, which is sometimes injured during surgery involving the pericardium. His arterial blood gas analysis shows typical features of hypercapnic failure, although PaO_2 is slightly borderline but does not point toward purely hypoxic pathologies. The most useful test to establish or exclude a diaphragmatic injury is a sniff test: paradoxical motion of a hemidiaphragm when a patient sniffs vigorously, demonstrating phrenic nerve paralysis or paresis of the hemidiaphragm. Patients with prolonged ICU stays are frequently malnourished and exhibit generalized myopathy. Serum albumin can be used to detect severe malnourishment, but this pathology should be secondary to the diaphragmatic injury in this clinical scenario. Serum rheumatoid factor level can be tested if an autoimmune disease is suspected. Pericarditis is one of the manifestations of autoimmune disease leading to lung demise. However, rheumatoid factor is not the most useful screening test. Chest CT scan should be performed in the case of suspected pneumonia, atelectasis, or bronchitis, but this patient exhibits no other features of infection.

References:

1. Qureshi A. Diaphragm paralysis. *Semin Respir Crit Care Med*. 2009 Jun;30(3):315-320.
2. Verhey PT, Gosselin MV, Primack SL, Kraemer AC. Differentiating diaphragmatic paralysis and eventration. *Acad Radiol*. 2007 Apr;14(4):420-42.

2. Rationale

Answer: B

This patient has farmer's lung, a type of hypersensitivity pneumonitis (HP). Environmental and exposure control are the cornerstones of treatment. If elimination of antigen exposure does not result in full disease regression, corticosteroids are recommended. Levofloxacin is an antibiotic for treating bacterial pneumonia, which is unlikely in this patient with a nonproductive cough and no fever. Rituximab is a B-cell-depleting monoclonal antibody that has been used in a few cases of HP that are refractory to steroids and at least 1 other immunosuppressant. Pirfenidone is an antifibrotic drug being studied for fibrotic HP but is not a first-line agent or standard of care.

References:

1. Hsieh C. Hypersensitivity pneumonitis questions & answers. Medscape. Updated October 8, 2020. Accessed February 27, 2022. https://emedicine.medscape.com/article/299174-questions-and-answers
2. King Jr TE. Hypersensitivity pneumonitis (extrinsic allergic alveolitis): treatment, prognosis, and prevention. UpToDate. Last updated October 22, 2021. Accessed February 27, 2022. https://www.uptodate.com/contents/hypersensitivity-pneumonitis-extrinsic-allergic-alveolitis-treatment-prognosis-and-prevention

3. Rationale

Answer: A

Several studies have suggested acute development of obstructive sleep apnea (OSA) soon after extubation. OSA potentially increases the risk of further respiratory complications if not recognized or prepared for. One study looked at risk factors for developing OSA after extubation in patients thought to be at low risk for OSA. The investigators prospectively performed polysomnography during the first night after extubation in patients who had been mechanically ventilated for 24 hours or more. They found that 71% of the patients had an Apnea-

Hypopnea Index (AHI) greater than 5, with 38% having moderate to severe OSA. The number of opioids given within 24 hours before extubation was significantly correlated with post-extubation AHI. Further, each 10 mg of morphine-equivalent increased the risk of having more severe OSA following extubation by 15% (odds ratio 1.15; 95% CI, 1.01-1.32). Neither propofol, neuromuscular blockers, nor benzodiazepines showed this correlation, although benzodiazepine doses were minimal in the study.

References:

1. Chung F, Liao P, Elsaid H, Shapiro CM, Kang W. Factors associated with postoperative exacerbation of sleep-disordered breathing. *Anesthesiology*. 2014 Feb;120(2):299-311.

2. Timm FP, Zaremba S, Grabitz SD, et al. Effects of opioids given to facilitate mechanical ventilation on sleep apnea after extubation in the intensive care unit. *Sleep*. 2018 Jan 1;41(1).

4. Rationale

Answer: D

Despite a lack of randomized clinical trial data, guidelines recommend exchange transfusion over simple transfusion for patients with severe acute chest syndrome (ACS). Hydroxyurea can reduce the incidence of ACS but is not useful for acute episodes. Steroid therapy may decrease hospital length of stay but has been associated with rebound vaso-occlusive episodes after treatment is completed. Although subgroup analysis suggests that inhaled nitric oxide may decrease treatment failure in patients with severe ACS, it was not superior to placebo in the overall study population of patients with ACS.

References:

1. Sarode R, Ballas SK, Garcia A, et al. Red blood cell exchange: 2015 American Society for Apheresis consensus conference on the management of patients with sickle cell disease. *J Clin Apher.* 2017 Oct;32(5):342-367.

2. National Heart, Lung, and Blood Institute. Evidence-based management of sickle cell disease: expert panel report, 2014. September 2014. Accessed February 27, 2022. https://www.nhlbi.nih.gov/health-topics/evidence-based-management-sickle-cell-disease.

5. Rationale

Answer: D

Maximal respiratory pressure is the most appropriate measure for quantifying the degree of neuromuscular weakness in patients with diseases such as myasthenia gravis. Neither residual volume, total lung capacity, forced expiratory volume, nor diffusing capacity of the lungs for carbon monoxide are useful for assessing neuromuscular disease.

References:

1. Dempsey TM, Scanlon PD. Pulmonary function tests for the generalist: a brief review. *Mayo Clin Proc*. 2018 Jun;93(6):763-771.

2. Hyatt RE, Scanlon PD, Nakamura M. *Interpretation of Pulmonary Function Tests*. 4th ed. Lippincott Williams & Wilkins; 2014.

3. American Thoracic Society/European Respiratory Society. ATS/ERS statement on respiratory muscle testing. *Am J Respir Crit Care Med*. 2002 Aug 15;166(4):518-624.

6. Rationale

Answer: B

This patient likely has lung entrapment, in which the lung does not fully expand because of an active inflammatory condition (eg, malignancy) that prevents the visceral pleura from expanding. As fluid is removed by thoracentesis, the pleural pressure gradually decreases and a steep decrease in pressure occurs when minimal fluid remains in the pleural space, as occurs in this patient.

References:

1. Doelken P, Huggins JT, Pastis NJ, Sahn S. Pleural manometry: technique and clinical implications. *Chest*. 2004 Dec;126(6):1764-1769.

2. Josephson T, Nordenskjold CA, Larsson J, Rosenberg LU, Kaijser M. Amount drained at ultrasound-guided thoracentesis and risk of pneumothorax. *Acta Radiol*. 2009 Jan;50(1):42-47.

7. Rationale

Answer: A

This patient's CT shows evidence of emphysema. His symptoms are slowly improving with noninvasive positive pressure ventilation. His clinical diagnosis is chronic obstructive pulmonary disease (COPD) exacerbation. An official diagnosis of COPD can be made as an outpatient with pulmonary function testing. Appropriate vaccinations must be administered to prevent episodic exacerbations. He should be vaccinated before hospital discharge. For someone younger than 65 years, the diagnosis of COPD makes the pneumococcal polysaccharide vaccine an appropriate choice. Additionally, the influenza vaccine should be administered annually to patients with chronic lung disease.

References:

1. Kobayashi M, Bennett NM, Gierke R, et al. Intervals between PCV13 and PPSV23 vaccines: recommendations of the Advisory Committee on Immunization Practices (ACIP). *MMWR Morb Mortal Wkly Rep*. 2015 Sep 4;64(34):944-947.

2. Walters JA, Tang JN, Poole P, Wood-Baker R. Pneumococcal vaccines for preventing pneumonia in chronic obstructive pulmonary disease. *Cochrane Database Syst Rev*. 2017 Jan 24;1:CD001390.

3. Wongsurakiat P, Maranetra KN, Wasi C, Kositanont U, Dijsomritrutai W, Charoenratanakus S. Acute respiratory illness in patients with COPD and the effectiveness of

influenza vaccination: a randomized controlled study. *Chest*. 2004 Jun;125(6):2011-2020.

8. Rationale

Answer: D

Pneumococcal vaccines do not contain live organisms and can be administered to a febrile patient if benefits are deemed to exceed risk. All patients aged 65 and older ideally should receive the pneumococcal conjugate vaccine first, followed by the pneumococcal polysaccharide vaccine 1 year later and at least 5 years after a first pneumococcal polysaccharide dose.

Reference:

1. Centers for Disease Control and Prevention. Pneumococcal vaccine recommendations. Page last reviewed January 24, 2022. Accessed February 27, 2022. https://www.cdc.gov/vaccines/vpd/pneumo/hcp/recommendations.html

9. Rationale

Answer: D

The Surviving Sepsis Campaign guidelines suggest administering dobutamine for patients who show evidence of persistent hypoperfusion despite adequate fluid loading and vasopressor agents (weak recommendation, low quality of evidence).

Reference:

1. Rhodes A, Evans LE, Alhazzani W, et al. Surviving Sepsis Campaign: international guidelines for management of sepsis and septic shock: 2016. *Crit Care Med*. 2017 Mar;45(3):486-552.

10. Rationale

Answer: B

This patient has respiratory alkalosis related to the ventilator settings. Increasing tidal volume would not help because it might increase CO_2 elimination, which is already low.

Changing FIO_2 and positive end-expiratory pressure will alter the oxygenation, and his oxygen saturations are adequate on the current settings. Decreasing the respiratory rate will decrease the rate of CO_2 elimination and will likely allow the pH and PCO_2 to normalize.

Reference:

1. Tobin MJ. Mechanical ventilation. *N Engl J Med.* 1994 Apr 14;330(15):1056-1061.

11. Rationale

Answer: B

The CDC's National Healthcare Safety Network implemented ventilator-associated event (VAE) surveillance in 2013. The term VAE was proposed to provide a more uniform and consistent manner of reporting cases of ventilator-associated complications. The VAE system is a 3-tier surveillance definition that uses objective, readily available data to identify complications, including ventilator-associated pneumonia (VAP), that occur in mechanically ventilated adult patients: First tier definition: Ventilator-associated condition (VAC) applies to patients with a period of sustained respiratory deterioration (changes in positive end-expiratory pressure [PEEP] ≥ 3 cm H_2O or FIO_2 ≥ 0.2 [20 points] for 2 days) following a sustained period of stability or improvement on the ventilator (≥ 2 days). Second tier definition: Infection-related ventilator-associated complication (IVAC) requires that patients with VAC also have abnormal temperature (< 36°C [96.8°F]) or > 38°C [100.4°F]) or WBC count ≤ 4000 cells/mm^3 or ≥ 12,000 cells/mm^3) and start on 1 or more new antibiotics that continue for 4 or more days. Third tier definitions: Possible and probable VAP require that patients with IVAC also have laboratory and/or microbiologic evidence of respiratory infection. Possible VAP is defined as Gram stain evidence of purulent pulmonary secretions or a pathogenic pulmonary culture in a patient with IVAC. Probable VAP is defined as Gram stain evidence of purulence plus quantitative or

semiquantitative growth of a pathogenic organism beyond specified thresholds. Probable VAP can also be triggered by positive tests for respiratory viruses, Legionella species, pleural fluid cultures, and suggestive histopathology with or without an abnormal Gram stain result.

References:

1. Boyer AF, Schoenberg N, Babcock H, McMullen KM, Micek ST, Kollef MH. A prospective evaluation of ventilator-associated conditions and infection-related ventilator-associated conditions. *Chest*. 2015 Jan;147(1):68-81.
2. Klompas M. Complications of mechanical ventilation: the CDC's new surveillance paradigm. *N Engl J Med*. 2013 Apr 18;368(16):1472-1475.
3. Magill SS, Klompas M, Balk R, et al. Developing a new, national approach to surveillance for ventilator-associated events. *Crit Care Med*. 2013 Nov;41(11):2467-2475.

12. Rationale

Answer: D

Patients most likely to benefit from noninvasive positive pressure ventilation (NIPPV) include those with acute hypercapnic exacerbations of chronic obstructive lung disease and acute cardiogenic pulmonary edema. In these patients, early use of NIPPV has resulted in shorter hospital stays and improved outcomes. Relative exclusion therapy includes patients with hemodynamic instability, central neurologic dysfunction, or inability to protect the upper airway. This patient is an excellent candidate for NIPPV based on presentation including hypercapnic respiratory failure, radiographic evidence of cardiogenic pulmonary edema, and no contraindications to therapy. A favorable response to NIPPV, however, is usually apparent within the initial 2 hours of treatment. The absence of improvement in dyspnea, respiratory rate, and gas exchange during this period strongly suggests a need for endotracheal intubation. Failure to

recognize appropriate response with use of NIPPV and postponing endotracheal intubation may be deleterious.

References:

1. Brochard L, Akoumianaki E, Cordioli RL. Noninvasive ventilation. In: Hall JB, Schmidt GA, Kress JP. *Principles of Critical Care*. 4th ed. McGraw-Hill Education; 2015:2489-2527.

2. Wright BJ, Slesinger TL. Noninvasive positive pressure ventilation. In: Farcy D, Shiu W, Marshall J, Osborn T. *Critical Care Emergency Medicine*. 2nd ed. McGraw-Hill Education; 2016:59-72.

13. Rationale

Answer: C

Based on acute presentation of hypoxemic respiratory failure, diffuse infiltrates, and bronchoalveolar lavage showing more than 25% eosinophils without other cause such as drugs or infection, this patient meets criteria for acute eosinophilic pneumonia (AEP), for which corticosteroid treatment is first line. Cyclophosphamide is an appropriate treatment for granulomatosis with polyangiitis; however, the finding of significantly elevated eosinophils excludes this diagnosis. IV immunoglobulin has been used off label for treating some interstitial lung diseases, but no definitive studies have shown that it should be used as an adjunct therapy. High-dose corticosteroids are the first-line treatment for AEP. IV methylprednisolone, 125 mg every 6 hours, should be used. Oseltamivir is an antiviral drug for influenza and would not be helpful.

References:

1. Hallowell RW, Amariei D, Danoff SK. Intravenous immunoglobulin as potential adjunct therapy for interstitial lung disease. *Ann Am Thorac Soc*. 2016 Oct;13(10):1682-1688.

2. Sohn JW. Acute eosinophilic pneumonia. *Tuberc Respir Dis (Seoul)*. 2013 Feb;74(2):51-55.

14. Rationale

Answer: B

This patient has immunotherapy-related pneumonitis. The overall incidence of pneumonitis is approximately 5%. The incidence of pneumonitis is similar in patients treated for melanoma and non-small cell lung cancer. The most common presenting symptoms are dyspnea and cough (53% and 35%, respectively), while one-third of patients are asymptomatic. No characteristic radiographic or pathologic features are associated with pneumonitis due to checkpoint inhibitor immunotherapy. The 5 grades of severity based on the Common Terminology Criteria for Adverse Events are shown below. He has grade 3 symptoms, which require steroids. Only medium-dose steroids have been studied, not high-dose steroids. Immunosuppression may be indicated for grade 4 symptoms.

Grade	Severity
Grade 1	Asymptomatic, clinical or diagnostic observations only, intervention not indicated
Grade 2	Symptomatic, medical intervention indicated, limiting instrumental ADL
Grade 3	Severe symptoms, limiting self-care ADL, oxygen indicated.
Grade 4	Life-threatening respiratory compromise, urgent intervention indicated (eg, tracheostomy or intubation)
Grade 5	Death

ADL = activities of daily living

References:

1. Naidoo J, Wang X, Woo KM, et al. Pneumonitis in patients treated with anti-programmed death-1/programmed death ligand 1 therapy. *J Clin Oncol*. 2017 Mar;35(7):709-717.

2. Nishino M, Ramaiya NH, Awad MM, et al. PD-1 inhibitor-related pneumonitis in advanced cancer patients: radiographic patterns and clinical course. *Clin Cancer Res.* 2016 Dec 15;22(24):6051-6060.

3. National Cancer Institute. Division of Cancer Treatment and Diagnosis. Cancer Therapy Evaluation Program. Common terminology criteria for adverse events (CTCAE) v6.0. Last updated September 21, 2020. Accessed February 28, 2022. https://ctep.cancer.gov/protocoldevelopment/electronic_applications/ctc.htm

15. Rationale

Answer: D

The narcotic has caused respiratory depression, leading to acute respiratory acidosis as evidenced by low pH, high $PaCO_2$, and inadequate arterial oxygenation.

Reference:

1. Boyer E. Management of opioid analgesic overdose. *N Engl J Med*. 2012 Jul 12;367(2):146-155.

16. Rationale

Answer: C

Noninvasive positive pressure ventilation decreases the need for intubation in patients with hypoxic respiratory failure due to cardiogenic pulmonary edema. Since this patient was administered a dose of IV furosemide, it would be premature to start him on a diuretic infusion before assessing his response to the furosemide. Intubation should be avoided as a first-line treatment for a patient who has no contraindication to noninvasive ventilation. It would be inappropriate to start ultrafiltration for volume removal before assessing his response to diuretics.

Reference:

1. Weng CL, Zhao YT, Liu QH, et al. Meta-analysis: noninvasive ventilation in acute cardiogenic pulmonary edema. *Ann Intern Med*. 2010 May 4;152(9):590-600. Erratum in: *Ann Intern Med*. 2010 Aug 17;153(4):280.

17. Rationale

Answer: B

This patient is admitted with right heart failure from an acute pulmonary embolism. After diuresis, evidence remains of chronic right-sided dysfunction and possible pulmonary hypertension on echocardiography (elevated right ventricular systolic pressure). The most likely etiology of her pulmonary hypertension is chronic thromboembolic pulmonary hypertension (CTEPH) from recurrent venous thromboemboli.

In contrast to diagnosing an acute pulmonary embolism, ventilation/perfusion (V/Q) lung scan is the most sensitive test for diagnosing chronic embolic disease. Several guidelines recommend V/Q lung scan for diagnosing CTEPH. One single-center study showed that V/Q scan had 97% sensitivity for diagnosing CTEPH, compared to 51% for CT. Right heart catheterization will help confirm the pulmonary hypertension diagnosis and provide useful information for treatment (eg, surgical endarterectomy) but is not used for diagnosing CTEPH by itself.

References:

1. Bajc M, Neilly JB, Miniati M, Schuemichen C, Meignan M, Jonson B. EANM guidelines for ventilation/perfusion scintigraphy: Part 2. Algorithms and clinical considerations for diagnosis of pulmonary emboli with V/P(SPECT) and MDCT. *Eur J Nucl Med Mol Imaging*. 2009 Sep;36(9):1528-1538.
2. Memon HA, Lin CH, Guha A. Chronic thromboembolic pulmonary hypertension: pearls and pitfalls of diagnosis. *Methodist Debakey Cardiovasc J*. 2016 Oct-Dec;12(4):199-204.
3. Tunariu N, Gibbs SJ, Win Z, et al. Ventilation-perfusion scintigraphy is more sensitive than multidetector CTPA in detecting chronic thromboembolic pulmonary disease as a treatable cause of pulmonary hypertension. *J Nucl Med*. 2007 May;48(5):680-684.

18. Rationale

Answer: B

This patient's presentation is most consistent with massive pulmonary embolism. Bedside ultrasound highly suggests acute right ventricular dysfunction. Given her deteriorating hemodynamics and concern for impending cardiac arrest, tenecteplase, 40 mg IV push over 5 seconds, is most appropriate. An alternative is to administer alteplase, 50 mg, over 2 minutes. If she showed no signs of impending cardiac arrest, then administering alteplase, 100 mg over 2 hours,

would be most appropriate. Because of her instability, starting unfractionated heparin, 18 units/kg, and obtaining chest CT angiogram or obtaining chest CT angiogram before starting antithrombotic therapy are inappropriate.

References:

1. Fengler BT, Brady WJ. Fibrinolytic therapy in pulmonary embolism: an evidence-based treatment algorithm. *Am J Emerg Med*. 2009;27(1):84-95.
2. Kearon C, Akl EA, Ornelas J, et al. Antithrombotic therapy for VTE disease: CHEST guideline and expert panel report. *Chest*. 2016 Feb;149(2):315-352.
3. Soar J, Donnino MW, Maconochie I, et al; ILCOR Collaborators. 2018 international consensus on cardiopulmonary resuscitation and emergency cardiovascular care science with treatment recommendations summary. *Resuscitation*. 2018 Dec; 133:194-206.

19. Rationale

Answer: C

This patient has hemoptysis but is stable from both hemodynamic and respiratory standpoints. She will need bronchoscopy at some point but not emergently because her airway is currently stable. CT should be performed to check for masses or nodules that may not be visible on chest radiograph, especially given her history of tobacco use. After CT, bronchoscopy can be performed in a controlled environment.

Reference:

1. Jean-Baptiste E. Clinical assessment and management of massive hemoptysis. *Crit Care Med*. 2000 May;28(5):1642-1647.

20. Rationale

Answer: D

In the MIST2 trial, patients with empyema were randomized to intrapleural placebo, tissue plasminogen activator (tPA) alone, DNase alone, or a combination of tPA and DNase twice

daily for 3 days. The tPA-DNase group had a significant reduction in pleural opacity and improved drainage of the infected pleural fluid compared to the other groups. The frequency of surgical referral was also lower in the tPA-DNase group. There was no difference in mortality among treatment groups. Adverse events, including pleural hemorrhage, were similar among the treatment groups.

Reference:

1. Rahman NM, Maskell NA, West A, et al. Intrapleural use of tissue plasminogen activator and DNase in pleural infection. *N Engl J Med*. 2011 Aug 11;365(6):518-526.

21. Rationale

Answer: A

This patient presents with healthcare-associated pneumonia. Her pleural fluid characteristics suggest complicated parapneumonic effusion and possibly even an empyema. In addition to initiating broad-spectrum antibiotics appropriately targeting hospital-based organisms, prompt drainage of the pleural space is recommended as first-line treatment. Video-assisted thoracoscopic surgery, while indicated in some cases, is not first-line therapy unless antibiotic treatment and chest tube drainage have failed. There is no indication for adding meropenem. The ideal chest tube size in this clinical scenario is not definitively clear. While older literature more strongly supported larger-bore tubes, several newer studies have indicated that smaller drains have success rates of approximately 80%, which are like success rates reported by studies using larger tubes.

References:

1. Colice GL, Curtis A, Deslauriers J, et al. Medical and surgical treatment of parapneumonic effusions: an evidence-based guideline. *Chest*. 2000 Oct;118(4):1158-1171.

2. Rahman NM, Maskell NA, Davies CW, et al. The relationship between chest tube size and clinical outcome in pleural infection. *Chest*. 2010 Mar;137(3):536-543.

3. Shen KR, Bribriesco A, Crabtree T, et al. The American Association for Thoracic Surgery consensus guidelines for the management of empyema. *J Thorac Cardiovasc Surg.* 2017 Jun;153(6): e129-e146.

22. Rationale

Answer: C

Gross macroscopic examination of aspirated pleural fluid can often suggest the underlying etiology. Thoracentesis on this patient reveals opaque, black-colored pleural fluid. Black pleural effusions are rare; they can be caused by direct invasion of the pleural space by metastatic melanoma. Other causes of black pleural effusion include fungal infections (*Aspergillus niger* and *Rhizopus oryzae*) and charcoal-containing empyema. Given this patient's history, the most likely diagnosis is thoracic metastasis of malignant melanoma. Pleural cytology is confirmatory and reveals abundant melanocytes. Empyema, or pus in the pleural cavity, can also be diagnosed by visual examination of aspirated pleural fluid. Gross examination, however, reveals a cloudy yellow fluid with a foul odor. Effusions from heart failure are usually straw colored, not black. A hemothorax would appear red.

References:

1. Chhabra A, Mukherjee V, Chowdhary M, Danckers M, Fridman D. Black pleural effusion: a unique presentation of metastatic melanoma. *Case Rep Oncol.* 2015 May 7;8(2):222-225.

2. Saraya T, Light RW, Takizawa H, Goto H. Black pleural effusion. *Am J Med.* 2013 Jul;126(7): 641.e1-641.e6.

23. Rationale

Answer: B

This patient has hepatopulmonary syndrome (HPS), which is a pulmonary complication of cirrhosis with concomitant portal hypertension. Patients with HPS develop progressive dyspnea secondary to alterations in the pulmonary vascular physiology leading to intrapulmonary vasodilation and impaired oxygen

exchange. Although the pathophysiology has yet to be elucidated, nitric oxide and pulmonary angiogenesis likely play a role. Diagnostic criteria for HPS include: 1) impaired oxygenation (PaO_2 < 80 mm Hg or increased age-corrected alveolar-arterial oxygen gradient), 2) intrapulmonary vasodilation (confirmed by contrast echocardiography or lung perfusion scanning), and 3) liver disease and/or portal hypertension. Although pulmonary vasodilator therapy is often used in the management of patients with HPS, liver transplantation is the definitive treatment. Although orthodeoxia and spider nevi are often present in HPS, they are not pathognomonic for HPS and are not sensitive indicators of the disease. Spirometric results are typically within normal limits in patients with HPS in the absence of comorbid lung disease. Portopulmonary hypertension can coexist with HPS but is not a diagnostic criterion.

References:

1. Koch DG, Fallon MB. Hepatopulmonary syndrome. *Clin Liver Dis.* 2014 May;18(2):407-420.
2. Rodriguez-Roisin R, Krowka MJ, Herve Ph, Fallon MB; ERS Task Force Pulmonary-Hepatic Vascular Disorders (PHD) Scientific Committee. Pulmonary-hepatic vascular disorders (PHD). *Eur Respir J.* 2004 Nov;24(5):861-880.

24. Rationale

Answer: E

This patient has a primary spontaneous pneumothorax, based on his unremarkable medical history and lack of trauma. He is clinically stable and minimally symptomatic, so observation for a set period in the emergency department followed by repeat chest radiograph to confirm no further expansion of the pneumothorax is the most appropriate next step in management. If repeat chest radiograph is stable or improved, discharge home with outpatient follow-up would be reasonable. An increasing body of evidence shows that observation alone is a valid and effective initial treatment strategy for asymptomatic patients with primary spontaneous

pneumothorax. Up to 80% of spontaneous pneumothoraces are estimated to be smaller than 15% of total lung volume, and recurrence in those managed with observation alone is less than in those treated by chest drainage. Supplemental oxygen should be administered because it has been shown to result in an increased rate of pneumothorax resolution. If this patient were symptomatic, intervention with both aspiration and the placement of a small chest tube would be indicated. Video-assisted thoracoscopic surgery with pleurodesis is usually reserved for a persistent pneumothorax that does not resolve with either aspiration or chest tube placement.

References:

1. Kelly AM, Kerr D, Clooney M. Outcomes of emergency department patients treated for primary spontaneous pneumothorax. *Chest*. 2008 Nov;134(5):1033-1036.

2. MacDuff A, Arnold A, Harvey J; BTS Pleural Disease Guideline Group. Management of spontaneous pneumothorax: British Thoracic Society Pleural Disease Guideline 2010. *Thorax*. 2010 Aug;65 Suppl 2: ii18-ii31.

25. Rationale

Answer: C

Thoracotomy is the most appropriate procedure for surgical exploration of the chest when massive hemothorax or persistent bleeding is present. Traditional criteria indicating the need for urgent thoracotomy are more than 1500 mL blood immediately evacuated by tube thoracostomy; persistent bleeding from the chest, defined as 150 mL/hr to 200 mL/hr for 2 to 4 hours; and persistent blood transfusion required to maintain hemodynamic stability. According to the Eastern Association for the Surgery of Trauma 2011 practice management guidelines, patient physiology should be used as the primary indications for surgical intervention rather than absolute numbers of initial or persistent output (level 2). Regardless of mechanism, 1500 mL of blood loss via a chest tube in any 24-hour period should prompt consideration for surgical exploration (level 2).

References:

1. Boersma WG, Stigt JA, Smit HJ. Treatment of hemothorax. *Respir Med*. 2010 Nov;104(11):1583-1587.
2. Mowery NT, Gunter OL, Collier BR, et al. Practice management guidelines for management of hemothorax and occult pneumothorax. *J Trauma*. 2011 Feb;70(2):510-518.

26. Rationale

Answer: D

This patient has signs and symptoms of ventilator-associated pneumonia, which is diagnosed based on the presence of infiltrates and 2 out of 3 of the following clinical features: fever above 38.1°C (100.6°F), purulent secretions, and leukocytosis or leukopenia in the absence of pneumonia on admission and time of onset more than 48 hours after intubation. The typical treatment for hospital- or ventilator-acquired pneumonia is an antipseudomonal beta-lactam, aminoglycoside, and glycopeptide; however, beta-lactams are contraindicated because of his history of severe penicillin allergy.

References:

1. Kalil AC, Metersky ML, Klompas M, et al. Management of adults with hospital-acquired and ventilator-associated pneumonia: 2016 clinical practice guidelines by the Infectious Diseases Society of America and the American Thoracic Society. *Clin Infect Dis.* 2016 Sep 1;63(5):e61-e111.
2. Rioux J, Edwards J, Bresee L, et al. Nasal-swab results for methicillin-resistant Staphylococcus aureus and associated infections. *Can J Hosp Pharm*. 2017 Mar-Apr;70(2):107-112.
3. Sager R, Kutz A, Mueller B, Schuetz P. Procalcitonin-guided diagnosis, and antibiotic stewardship revisited. *BMC Med*. 2017 Jan 24;15(1):15.

27. Rationale

Answer: D

High-frequency oscillatory ventilation is unlikely to improve this patient's chance of survival; in the OSCILLATE trial it was found to increase mortality in patients with acute respiratory distress syndrome (ARDS). The OSCAR trial showed no difference in mortality. Inhaled epoprostenol is unlikely to improve her chance of survival; it has shown improvement in oxygenation with no change in mortality. Atorvastatin is unlikely to improve her chance of survival because statin therapy has shown no mortality benefit and potential harm from adverse drug effects in patients with ARDS. The ACURASYS trial found that early (within 72 hours) initiation of cisatracurium continuous infusion reduced mortality in patients with ARDS.

References:

1. Ferguson ND, Cook DJ, Guyatt GH, et al; OSCILLATE Trial Investigators; Canadian Critical Care Trials Group. High-frequency oscillation in early acute respiratory distress syndrome. *N Engl J Med*. 2013 Feb 28;368(9):795-805.
2. Pacheco J, Arnold H, Skrupky L, Watts P, Micek ST, Kollef MH. Predictors of outcome in 216 subjects with ARDS treated with inhaled epoprostenol. *Respir Care*. 2014 Aug;59(8):1178-1185.
3. Papazian L, Forel JM, Gacouin A, et al; ACURASYS Study Investigators. Neuromuscular blockers in early acute respiratory distress syndrome. *N Engl J Med*. 2010 Sep 16;363(12):1107-1116.
4. Nagendran M, McAuley DF, Kruger PS, et al. Statin therapy for acute respiratory distress syndrome: an individual patient data meta-analysis of randomized clinical trials. *Intensive Care Med*. 2017 May;43(5):663-671.

28. Rationale

Answer: C

Prone positioning has been evaluated in 8 randomized controlled trials (RCTs) comprising a total of 2129 patients. There was no significant difference in mortality between the

prone and supine groups (RR 0.84; 95% CI, 0.68-1.04; moderate confidence). However, in prespecified subgroup analyses (based on proning duration, acute respiratory distress syndrome [ARDS] severity, and concomitant low tidal volume ventilation), prone positioning reduced mortality in patients with moderate or severe ARDS and prone duration greater than 12 hours per day. A patient-level meta-analysis of 4 earlier RCTs demonstrated lower mortality in patients with severe ARDS at baseline, with subsequent confirmation of this finding in the Proning Severe ARDS Patients (PROSEVA) RCT (baseline P/F ratio 100 ± 30 in the prone group). Prone positioning was significantly associated with higher rates of endotracheal tube obstruction and pressure sores. There was no significant difference in barotrauma between groups. PROSEVA included 466 patients with severe ARDS (P/F ratio < 150 mm Hg). The protocol consisted of eligible patients stabilized over 12 to 24 hours and proned for at least 16 hours. Regular hospital beds were used. Results showed reduced 28-day mortality (16% vs. 33%).

References:

1. Fan E, Del Sorbo L, Goligher EC, et al; American Thoracic Society; European Society of Intensive Care Medicine; Society of Critical Care Medicine. An official American Thoracic Society/European Society of Intensive Care Medicine/Society of Critical Care Medicine clinical practice guideline: mechanical ventilation in adult patients with acute respiratory distress syndrome. *Am J Respir Crit Care Med*. 2017 May 1;195(9):1253-1263.
2. Guerin C, Réignier J, Richard JC, et al; PROSEVA Study Group. Prone positioning in severe acute respiratory distress syndrome. *N Engl J Med*. 2013 Jun 6;368(23):2159-2168.

29. Rationale

Answer: B

The 2017 American Thoracic Society, European Society of Intensive Care Medicine, and Society of Critical Care Medicine clinical practice guidelines for the management of

patients with acute respiratory distress syndrome strongly recommend lower tidal volumes (4-8 mL/kg predicted body weight), lower inspiratory pressures (plateau pressure < 30 cm H_2O), and prone positioning for more than 12 hours per day. Conditional recommendations included higher positive end-expiratory pressure and recruitment maneuvers for those with moderate to severe acute respiratory distress syndrome, which this patient has, with a P/F ratio less than 100. The guidelines strongly recommend against the use of high-frequency oscillatory ventilation. They make no definitive recommendation for or against the use of extracorporeal membrane oxygenation because of insufficient data. Investigators tested various pharmacologic therapies to find additional efficacious therapies. Although a single-center trial showed some efficacy of IV beta-agonist, a multicenter randomized controlled trial was prematurely stopped because of increased 28-day mortality in the beta-agonist group. Inhaled beta-agonists have also been shown to be ineffective. Keratinocyte growth factor is important in the repair of the alveolar epithelium, but a randomized controlled trial did not show any benefit and led to fewer ventilator-free days and increased mortality.

References:

1. Fan E, Del Sorbo L, Goligher EC, et al; American Thoracic Society; European Society of Intensive Care Medicine; Society of Critical Care Medicine. An official American Thoracic Society/European Society of Intensive Care Medicine/Society of Critical Care Medicine clinical practice guideline: mechanical ventilation in adult patients with acute respiratory distress syndrome. *Am J Respir Crit Care Med*. 2017 May 1;195(9):1253-1263.

2. Gao Smith F, Perkins GD, Gates S, et al; BALTI-2 Study Investigators. Effect of intravenous β-2 agonist treatment on clinical outcomes in acute respiratory distress syndrome (BALTI-2): a multicentre, randomised controlled trial. *Lancet* 2012 Jan 21;379(9812):229-235.

3. Perkins GD, McAuley DF, Thickett DR, Gao F. The beta-agonist lung injury trial (BALTI): a randomized placebo-controlled clinical trial. *Am J Respir Crit Care Med* 2006 Feb 1;173(3):281-287.

4. National Heart, Lung, and Blood Institute Acute Respiratory Distress Syndrome (ARDS) Clinical Trials Network, Matthay MA, Brower RG, et al. Randomized, placebo-controlled clinical trial of an aerosolized β2-agonist for treatment of acute lung injury. *Am J Respir Crit Care Med* 2011 Sep 1;184(5):561-568.

5. Ware LB, Matthay MA. Keratinocyte and hepatocyte growth factors in the lung: roles in lung development, inflammation, and repair. *Am J Physiol Lung Cell Mol Physiol*. 2002 May;282(5): L924-L940.

30. Rationale

Answer: A

This patient with chronic obstructive pulmonary disease is hemodynamically unstable and has air trapping. His intrathoracic pressure is transiently increasing since he was started on positive pressure ventilation. His preload has been affected and has caused hypotension.

Reference:

1. Critical Matters. Sound Physicians. Noninvasive ventilation and high-flow oxygen nasal cannula. Apple Podcast. https://podcasts.apple.com/us/podcast/critical-matters/id1335759655?i=1000442612520

31. Rationale

Answer: B

This patient has diffuse alveolar hemorrhage (DAH). Initial treatment for DAH is systemic glucocorticoids. While tranexamic acid has been reported in a few cases to temporize bleeding in patients with DAH, it is not used widely. Piperacillin/tazobactam is a broad-spectrum antibiotic with pseudomonal coverage often used for healthcare-acquired

pneumonia. Bronchoscopy did not show infection, but findings were consistent with DAH. There is no role for routine platelet transfusion in DAH. Finally, recombinant factor VIIa is a form of blood factor VII that is necessary for blood coagulation. Case reports have highlighted its topical use to control refractory bleeding in DAH, although it is not considered primary therapy.

References:

1. Danoff SK, Hallowell R. The diffuse alveolar hemorrhage syndromes. UpToDate. Last updated January 24, 2022.
2. Kazzaz NM, Coit P, Lewis EE, McCune WJ, Sawalha AH, Knight JS. Systemic lupus erythematosus complicated by diffuse alveolar haemorrhage: risk factors, therapy and survival. *Lupus Sci Med*. 2015 Sep 23;2(1) e000117.
3. Ta R, Celli R, West AB. Diffuse alveolar hemorrhage in systemic lupus erythematosus: histopathologic features and clinical correlations. *Case Rep Pathol*. 2017; 2017:1936282.

32. Rationale

Answer: D

This patient has signs of right ventricular dysfunction or failure as evidenced by high pulmonary artery pressures, high central venous pressure, and low cardiac index (CI). Milrinone, a phosphodiesterase-3 inhibitor, is a potent inotrope and causes pulmonary vasodilation but also tends to cause further afterload reduction and thus should not be the first choice because of her hypotension. Sometimes vasopressin in conjunction with milrinone can counteract the hypotension. Phenylephrine would cause an increase in afterload that would be deleterious. An intra-aortic balloon pump is best used for left-sided heart failure and would not benefit this patient with a normal CI. Pericardiocentesis is not indicated without further echocardiographic information. Inhaled nitric oxide (iNO) is most appropriate because it works only in the pulmonary circulation and can profoundly decrease pulmonary pressures, thus increasing CI. iNO is a naturally occurring vasodilator. Nitric oxide activates soluble guanylyl cyclase to produce cyclic

guanosine monophosphate (cGMP), which then activates cGMP-dependent protein kinase, resulting in the activation of several regulatory mechanisms that decrease intracellular calcium and decrease vascular smooth muscle tone (ie, vasodilation) in precapillary resistance arterioles. It selectively dilates the pulmonary vasculature when administered by inhalation. Its onset of action is rapid, and its effects are short-lived due to its short half-life of 15 to 30 seconds.

References:

1. Archer SL, Huang JM, Hampl V, Nelson DP, Schultz PJ, Weir EK. Nitric oxide and cGMP cause vasorelaxation by activation of a charybdotoxin-sensitive K channel by cGMP-dependent protein kinase. *Proc Natl Acad Sci USA*. 1994 Aug 2;91(16):7583-7587.
2. Gaston B, Drazen JM, Loscalzo J, Stamler JS. The biology of nitrogen oxides in the airways. *Am J Respir Crit Care Med*. 1994 Feb;149(2 Pt 1):538-551.

33. Rationale

Answer: C

This patient with chronic obstructive pulmonary disease (COPD) and renal cell cancer presents with unexplained shortness of breath and wheezing. Up to 25% of patients with unexplained COPD exacerbations are found to have a pulmonary embolism (PE) on further investigation. Clinical risk factors for PE include alkalosis, malignant disease, and previous thromboembolic disease. He has 2 of these 3 risk factors, so PE should be high on the differential diagnosis. There is no evidence of heart failure, so transthoracic echocardiography (TTE) is not the next step. TTE might be useful for risk stratification if PE is diagnosed since the presence of right ventricular strain is a predictor of mortality in these patients. Renal ultrasound can be performed to follow the renal cell cancer but would not help to diagnose the cause of his acute respiratory failure. Pulmonary function tests are useful in patients with COPD but should not be performed in the

setting of an acute exacerbation. CT pulmonary angiogram is the most appropriate next step for this patient.

References:

1. Stergiopoulos K, Bahrainy S, Strachan P, Kort S. Right ventricular strain rate predicts clinical outcomes in patients with acute pulmonary embolism. *Acute Card Care*. 2011 Sep;13(3):181-188.
2. Tillie-Leblond I, Marquette CH, Perez T, et al. Pulmonary embolism in patients with unexplained exacerbation of chronic obstructive pulmonary disease: prevalence and risk factors. *Ann Intern Med*. 2006 Mar 21;144(6):390-396.

34. Rationale

Answer: D

A common problem in the ICU is the position of the endotracheal tube (ETT), especially after a long surgical procedure where the patient was prone. This patient has a right mainstem intubation that could be diagnosed by paying attention to the depth of the ETT at the lip and by the absence of breath sounds on the lung field. Arterial blood gas analysis will confirm the hypoxia observed due to the shunt created. The clinician should not have to wait for a radiograph if the patient is not stable. Suctioning the ETT could help only if there were secretions but will not help the left-sided atelectatic lung. CT is not an option for a patient who is agitated. Thoracic ultrasonography would not show the position of the ETT in the trachea, nor would it differentiate between pneumothorax and lung collapse.

References:

1. Ng A, Swanevelder J. Hypoxaemia during one-lung anaesthesia. *Contin Educ Anaesth Crit Care Pain*. 2010 Aug;10(4)117-122.
2. Shafiepour D. Troubleshooting one-lung ventilation. In: Slinger P, ed. *Principles and Practice of Anesthesia for Thoracic Surgery*. 2nd ed. Springer; 2019:429-436.

3. Weizman L, Tabrikian J, Cohen A. Detection of one lung intubation by monitoring lungs sounds. *Conf Proc IEEE Eng Med Biol Soc*. 2004; 2004:917-920.

35. Rationale

Answer: C

This patient has been resuscitated rapidly, remains hemodynamically unstable, is septic, and has a possible positive P/F ratio (hypoxic with high oxygen intake). She is probably in the initial stages of acute respiratory distress syndrome (ARDS). A low-tidal-volume strategy has been pinpointed as the most important therapy to prevent its full development. It could have a significant impact on her respiratory long-term outcome. Forced diuresis has been shown to be beneficial but does not improve mortality. If diuretics cannot be used, a fluid restriction strategy can be useful. Even when a pulmonary embolism can be considered in the differential diagnosis, the prior history of fluids and sepsis supports a diagnosis of ARDS. High-dose and short-course treatment with steroids does not improve the outcomes of patients with ARDS. Paralytic agents can sometimes be used when physiologically and clinically indicated rather than being applied uniformly to all patients.

References:

1. Bourdeaux CP, Thomas MJ, Gould TH, et al. Increasing compliance with low tidal volume ventilation in the ICU with two nudge-based interventions: evaluation through intervention time-series analyses. *BMJ Open*. 2016 May 26;6(5): e010129.

2. Hough CL. Steroids for acute respiratory distress syndrome? *Clin Chest Med*. 2014 Dec;35(4):781-795.

3. National Heart, Lung, and Blood Institute PETAL Clinical Trials Network, Moss M, Huang DT, et al. Early neuromuscular blockade in the acute respiratory distress syndrome. *N Engl J Med*. 2019 May 23;380(21):1997-2008.

36. Rationale

Answer: A

This patient has a high pretest probability of acute pulmonary embolism (PE). His risk factors (recent air travel), symptoms (acute onset shortness of breath) and signs (tachycardia, tachypnea, unilateral leg edema) suggest treating with parenteral therapeutic anticoagulation while awaiting diagnostic test results. Anticoagulation has been shown to reduce mortality in patients with PE and should be initiated as early as possible to prevent clot propagation. In patients with high clinical suspicion of PE and without high risk of bleeding, consensus guidelines suggest beginning treatment with parenteral anticoagulants while awaiting diagnostic test results. In the absence of contraindications, CT pulmonary angiography should be performed to confirm the diagnosis of PE, but systemic anticoagulation should start before the CT. Systemic thrombolytic therapy is reserved for patients who are hypotensive (systolic blood pressure < 90 mm Hg). In hemodynamically stable patients such as this one, the risks of bleeding outweigh the benefits of clot lysis. D-dimer testing is reserved for patients with a low probability of PE.

References:

1. Kearon C, Akl A, Comerota AJ, et al. Antithrombotic therapy for VTE disease: antithrombotic therapy and prevention of thrombosis, 9th ed: American College of Chest Physicians evidence-based clinical practice guidelines. *Chest*. 2012 Feb;141(2 Suppl): e419S-e496S.

2. Smith SB, Geske JB, Maguire JM, Zane NA, Carter RE, Morgenthaler TI. Early anticoagulation is associated with reduced mortality for acute pulmonary embolism. *Chest*. 2010 Jun;137(6):1382-1390.

3. PREPIC Study Group. Eight-year follow-up of patients with permanent vena cava filters in the prevention of pulmonary embolism: the PREPIC (Prevention du Risque d'Embolie Pulmonaire par Interruption Cave) randomized study. *Circulation*. 2005 Jul 19;112(3):416-422.

37. Rationale

Answer: B

Traumatic bronchopleural fistula (BPF) represents an especially difficult ventilatory management problem. Once the alveolar wall has been disrupted, large tidal volumes delivered by positive pressure ventilation allow air to pass into the bronchoalveolar sheath and root of the lung, from which air may enter the pleural space and result in pneumothorax. While potentially lifesaving, tube thoracostomy to drain the pneumothorax allows a continuous leak of air from the tracheal tree to the external world. The resulting loss of airway pressure leads to progressive alveolar collapse. The traditional treatment for acute lung injury and acute respiratory distress syndrome, increased tidal volumes and positive end-expiratory pressure, may serve only to worsen the magnitude of the fistula. BPF management should therefore focus on decreasing airway pressure and minimizing pleural suction to decrease the air leak and promote healing. Weaning from positive pressure ventilation and avoiding alveolar hyperinflation are most advantageous in decreasing the air leak. Independent lung ventilation has been reported by numerous authors as one method by which to restore alveolar volume and oxygenation and promote healing of a BPF.

References:

1. Anantham D, Jagadesan R, Tiew PE. Clinical review: independent lung ventilation in critical care. *Crit Care*. 2005;9(6):594-600.
2. Cheatham ML, Promes JT. Independent lung ventilation in the management of traumatic bronchopleural fistula. *Am Surg*. 2006 Jun;72(6):530-533.
3. Hillman KM, Barber JD. Asynchronous independent lung ventilation (AILV). *Crit Care Med*. 1980 Jul;8(7):390-395.
4. Thomas AR, Bryce TL. Ventilation in the patient with unilateral lung disease. *Crit Care Clin*. 1998 Oct;14(4):743-773.

38. Rationale

Answer: A

Hepatopulmonary syndrome (HPS) is a pathophysiologic process in patients with cirrhosis that carries a poor prognosis without liver transplantation. Median survival is 24 months in patients with cirrhosis and HPS. It is characterized by platypnea and orthodeoxia and worsening oxygenation when standing secondary to severe pulmonary vasodilation that occurs in liver disease. It is diagnosed by increased room air alveolar arterial oxygen gradient and decreased PO_2. While many treatments have improved oxygenation in animal studies, the only proven effective treatment for HPS in humans, aside from supplemental oxygen, is liver transplantation.

References:

1. Soulaidopoulos S, Cholongitas E, Giannakoulas G, Vlachou M, Goulis I. Review article: update on current and emergent data on hepatopulmonary syndrome. *World J Gastroenterol*. 2018 Mar 28;24(12):1285-1298.
2. Shah R, Mousa O, John S. Hepatopulmonary syndrome: a rare diagnosis or easily missed entity? *QJM*. 2014 Jul;107(7):565-566.

39. Rationale

Answer: E

This patient is at risk for ventilator-associated tracheitis (VAT) and ventilator-associated pneumonia (VAP) based on her age, intubation, and recent surgery. Fevers, purulent secretions, and positive tracheal culture are clinical signs of infection. These findings are consistent with the diagnosis of VAT (rather than VAP, which would include infiltrate on chest radiograph). Treating VAT significantly reduces progression to VAP because the trachea is a route to progression of bronchitis and pneumonia. The preferred treatment for methicillin-sensitive *Staphylococcus aureus* (MSSA) tracheitis is nafcillin.

References:

1. Craven DE, Hudcova J, Craven KA, Scopa C, Lei Y. Antibiotic treatment of ventilator-associated

tracheobronchitis: to treat or not to treat? *Curr Opin Crit Care*. 2014 Oct;20(5):532-541.

2. Dallas J, Skrupky L, Abebe N, Boyle WA 3rd, Kollef MH. Ventilator-associated tracheobronchitis in a mixed surgical and medical ICU population. *Chest*. 2011 Mar;139(3):513-518.

3. Martin-Loeches I, Povoa P, Rodríguez A, et al; TAVeM study. Incidence and prognosis of ventilator-associated tracheobronchitis (TAVeM): a multicentre, prospective, observational study. *Lancet Respir Med*. 2015 Nov;3(11):859-868.

40. Rationale

Answer: C

The Centers for Disease Control and Prevention recommends administration of pneumococcal polysaccharide (PPSV23) and pneumococcal conjugate (PCV13) vaccines in patients younger than 65 years who have certain high-risk conditions. Patients aged 19 to 64 years who smoke or have alcoholism, diabetes, chronic heart (excluding hypertension), lung, or liver disease should receive 1 dose of PPSV23 at the time of diagnosis of the high-risk condition, and a 1-time revaccination dose of PPSV23 can be administered at age 65 if it has been at least 5 years since the previous dose. Patients 19 years and older with cerebrospinal fluid leaks or cochlear implants should receive 1 dose of PCV13 at the time of diagnosis and PPSV23 at least 8 weeks, but ideally 1 year, later and a second PPSV23 dose at age 65 if it has been at least 5 years since the previous dose. Immunocompromised adults or those with anatomic or functional asplenia should receive 1 dose of PCV13 at the time of diagnosis and a dose of PPSV23 at least 8 weeks, but ideally 1 year, afterward and another PPSV23 dose 5 years after the first dose and, if applicable, another dose of PPSV23 at age 65 years or 5 years after the second dose, whichever comes later.

Reference:

1. Centers for Disease Control and Prevention. Adult immunization schedule. Recommendations for ages 19

years or older, United States, 2022. Page last reviewed February 17, 2022. Accessed March 1, 2022. https://www.cdc.gov/vaccines/schedules/hcp/imz/adult.htm

41. Rationale

Answer: B

The most appropriate next step for this patient is to start humidified high-flow oxygen, which has been shown to reduce mortality risk and the need for mechanical ventilation in acute hypoxia respiratory failure. It also has the physiologic properties of reducing work of breathing (increasing tidal volume and reducing respiratory rate), allowing for bronchodilation and improved mobilization of secretions, and reducing dead space with washout and reduction of PCO_2 within the upper airway.

Continuing the nonrebreather mask is inappropriate because it will not help reduce work of breathing and carries a higher risk of needing mechanical ventilation. Noninvasive ventilation might help reduce work of breathing but will make it difficult for her to clear secretions from the airway. Intubation is a reasonable option but is invasive and can be avoided with early implementation of humidified high-flow oxygen.

References:

1. Ferreyro BL, Angriman F, Munshi L, et al, Association of noninvasive oxygenation strategies with all-cause mortality in adults with acute hypoxemic respiratory failure: a systematic review and meta-analysis. *JAMA*. 2020 Jul 7;324(1):57-67.

2. Frat JP, Thille AW, Mercat A, et al; FLORALI Study Group; REVA Network. High-flow oxygen through nasal cannula in acute hypoxemic respiratory failure. *N Engl J Med*. 2015 Jun 4;372(23):2185-2196.

Part 7.

Infectious Disease

Instructions: For each question, select the most correct answer.

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1. A 64-year-old man was admitted to the surgical ICU 3 weeks ago with acute severe pancreatitis. His admission has been complicated by multiorgan dysfunction requiring mechanical ventilation and continuous renal replacement therapy. He has been receiving total parenteral nutrition for 7 days because of an ileus. He acutely decompensates and now requires multiple vasopressors for hemodynamic support. Blood cultures are drawn. Empiric vancomycin, meropenem, and micafungin are initiated. Preliminary blood culture results demonstrate *Candida krusei* fungemia. Which of the following is the most appropriate next step in this patient's management?

   A. Consult ophthalmology for a dilated ophthalmologic examination.

   B. De-escalate micafungin to fluconazole.

   C. Continue treatment for 4 weeks from documented clearance of *Candida*.

   D. Repeat blood cultures.

2. A blood sample is sent for rapid diagnostic testing and shows preliminary *Escherichia coli* findings. The need for carbapenem therapy would be indicated by detection of

   A. the mecA gene.

   B. the KPC gene.

   C. the ampC gene.

D. VanA resistance.

3. A 75-year-old man presents to the emergency department with back pain and fever. Past medical history is significant for diabetes. Vital signs are temperature 40°C (103.4°F), heart rate 122 beats/min, blood pressure 132/84 mm Hg, and respiratory rate 23 breaths/min. WBC count is 16,000 cells/μL. Abdominal CT identifies a saccular 5-cm infrarenal aortic aneurysm with surrounding stranding. Which of the following is the most appropriate next step in management?

   A. Consult vascular surgery for open repair.

   B. Consult interventional radiology for endovascular repair.

   C. Begin a 3-week course of IV vancomycin

   D. Begin a 6-week course of IV ceftriaxone.

   E. Begin a 6-week course of IV micafungin.

4. A 29-year-old woman presents to the emergency department (ED) with a 4-day history of headache and photo-phobia. She has been to the ED with similar issues twice during the past 4 days. She has no past medical history. On admission she is found to be HIV positive, with a CD4 count of 43 cells/μL. Head CT is negative for acute findings or lesions. Multiple attempts at bedside lumbar puncture are unsuccessful. Throughout the day her mental status declines and breathing becomes labored. Shortly after broad-spectrum antibiotics are initiated with cefepime, ampicillin, vancomycin, and acyclovir, an interventional radiology consult is requested for lumbar puncture. She is transferred to the medical ICU for closer monitoring, where, on arrival, blood culture stain is positive for encapsulated yeast suggestive of *Cryptococcus*. Which of the following therapy regimens should be initiated?

A. IV amphotericin B deoxycholate plus oral flucytosine

B. IV amphotericin B deoxycholate only

C. IV caspofungin only

D. IV amphotericin B deoxycholate plus IV fluconazole

5. An 84-year-old man is transferred from a skilled nursing facility to the ICU for management of severe sepsis. He was recently discharged from a hospital after being treated with IV antibiotics for a urinary tract infection. Empiric antibiotics, including vancomycin and piperacillin-tazobactam, are administered; IV fluid resuscitation is initiated; and diagnostic studies, including blood cultures, urinalysis, and chest radiograph, are obtained. Chest radiograph shows right lower lobe consolidation. He deteriorates clinically and requires central venous catheter placement for vasopressor administration. Vancomycin and piperacillin/tazobactam are discontinued, and he is transitioned to meropenem mono-therapy. Blood cultures are positive for meropenem-resistant *Klebsiella pneumoniae*. Which of the following is most likely true regarding his condition?

A. Amikacin should be added to his current antibiotic regimen.

B. He received a carbapenem during his most recent hospitalization.

C. He has an extended-spectrum beta-lactamase-producing strain.

D. Susceptibility testing is not required because several antibiotics are highly active against most strains of the pathogen.

E. The plasmid conferring resistance can be spread from one bacterial species to another from a

completely different genus.

6. A 65-year-old man presents to the emergency department for a 5-day history of fatigue. He says that he usually injects heroin at least twice a day and noticed 4 days ago that his left arm had become swollen. The swelling and redness became worse and prompted him to seek medical care. Temperature is 38°C (100.4°F), heart rate 120 beats/min, blood pressure 80/40 mm Hg, and respiratory rate 24 breaths/min. Physical examination is significant for tenderness and crepitus at the left upper arm. Laboratory results are significant for WBC count 26/μL and creatinine 3 mg/dL. After receiving 3 L crystalloids, his blood pressure improves to 110/60 mm Hg. Which of the following is/are most likely to improve his overall outcome?

   A. Vancomycin, piperacillin/tazobactam, and clindamycin

   B. Plasmapheresis

   C. IV immunoglobulin

   D. Hyperbaric oxygen treatment

   E. Wound debridement

7. A 55-year-old female mailroom worker is brought to the emergency department 4 days after being exposed to a suspicious white powdery substance in a package. She reports generalized fatigue, myalgias, and subjective fevers during the past day. She then develops severe dyspnea, nausea, vomiting, chest pain, and hemoptysis. Temperature is 38.3°C (101.1°F), heart rate 115 beats/min, respiratory rate 24 breaths/min, blood pressure 105/60 mm Hg, and oxygen saturation as measured by pulse oximetry 92% on 4-L nasal cannula. Heart sounds are regular but tachycardic, crackles are heard throughout both lungs, and no lower extremity edema is noted. WBC count

is 18,000/μL, sodium 140 mEq/L, potassium 4.1 mEq/L, chloride 105 mEq/L, bicarbonate 25 mEq/L, BUN 15 mg/dL, and creatinine 1.2 mg/dL. Chest radiograph shows bilateral infiltrates, left-sided pleural effusion, and widened mediastinum. Which of the following is the most appropriate treatment regimen for her condition?

A. Vancomycin, cefepime, and metronidazole

B. Ciprofloxacin, linezolid, and meropenem

C. Isoniazid, rifampin, pyrazinamide, and ethambutol

D. Quinidine, doxycycline, and clindamycin

8. A 60-year-old man with a history of diabetes mellitus, ulcerative colitis, and primary sclerosing cholangitis undergoes a total abdominal colectomy with ileorectal anastomosis. On postoperative day 8, he is transferred to the ICU for altered mental status, hypoxia, hypotension not responsive to fluid boluses, and abdominal distention with guarding. Abdominal CT reveals pneumoperitoneum; he is scheduled for emergent exploratory laparotomy. Which of the following is the most appropriate initial antibiotic regimen?

   A. Piperacillin/tazobactam and vancomycin

   B. Levofloxacin, metronidazole, and vancomycin

   C. Cefepime and metronidazole

   D. Meropenem

9. Three cases of hospital-acquired *Clostridioides difficile* infection (CDI) are diagnosed in an ICU during a 1-week period. While the dates of admission and some care team members overlapped for all 3 patients, none of the patients occupied the same room, and each had different bedside nurses. Which of the following

interventions is most likely to be effective in decreasing the risk of CDI in this ICU in the future?

A. Avoid the use of bedside ultrasonography on patients with CDI.

B. Perform C difficile testing on every patient admitted to the ICU.

C. Add ultraviolet light to bleach decontamination of rooms occupied by patients with CDI.

D. Ensure that patients with CDI remain on contact isolation for at least 48 hours after resolution of diarrhea.

10. A 57-year-old man with a history of ventriculoperitoneal shunt (VPS) placement 2 months ago presents to the emergency department with fevers, chills, vomiting, and worsening headache. Oral temperature is 38.6°C (101.4°F) and heart rate is 119 beats/min. Laboratory tests reveal elevated WBC count of 18 × 10^3/μL and serum glucose of 118 mg/dL. Urinalysis is negative, and he denies any cough or increased sputum production. Lumbar puncture results are consistent with ventriculitis. Culture is pending. Which of the following is the most appropriate empiric antimicrobial regimen?

A. Vancomycin plus ceftriaxone

B. Vancomycin plus cefepime

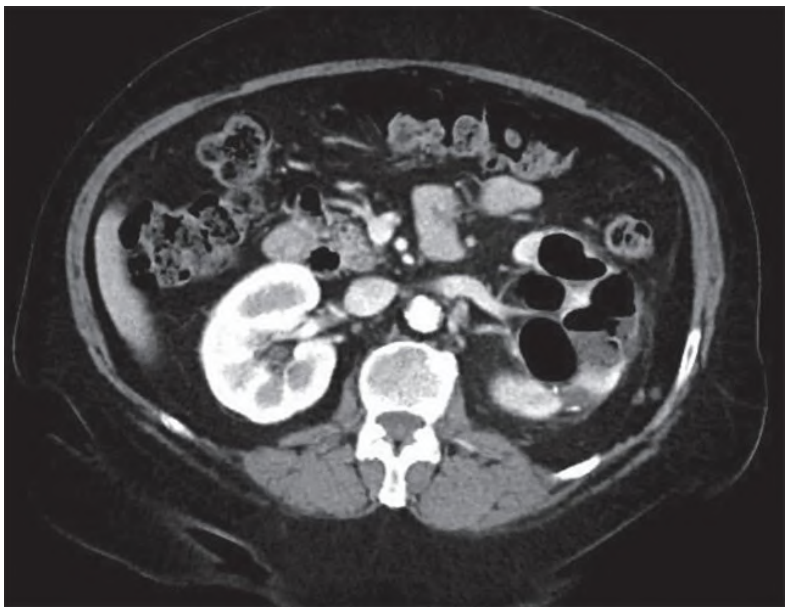
C. Vancomycin plus ceftriaxone plus rifampin

D. Vancomycin plus piperacillin/tazobactam

E. Ampicillin plus cefepime

11. A 74-year-old woman is admitted to the ICU for severe sepsis and is treated with broad-spectrum antibiotic therapy for 3 days without clinical improvement. She has a history of diabetes mellitus and requires an insulin infusion to maintain glycemic

control. Urinalysis reveals positive leukocyte esterase and urine culture grows multiple organisms suggestive of contamination. Noncontrast abdominal CT scan is shown below. Which of the following is the most likely diagnosis?



A. Perinephric abscess

B. Ureteral obstruction

C. Tuberculous pyelonephritis

D. Emphysematous pyelonephritis

E. Xanthogranulomatous pyelonephritis

12. A 36-year-old man presents to the emergency department with fever, jaw pain below his tongue, and difficulty breathing. On examination, he is sitting upright with his neck extended upward. His mouth is open due to his protruding tongue, and he is having difficulty speaking. Vital signs are temperature 38.9°C (102°F), heart rate 111 beats/min, respiratory rate 24 breaths/min, and blood pressure 135/82 mm Hg. Emergent CT is performed, and a submandibular space infection is diagnosed. Which of the following criteria is/are required to make this diagnosis?

    A. Infection beginning in the roof of the mouth

B. Unilateral infection

C. Involvement of submental and sublingual spaces

D. Slow-spreading infection

E. Lymphatic involvement and abscess

13. A 21-year-old man with a grade 2 astrocytoma is transferred to the ICU with suspected sepsis. Two days ago, his tumor was resected and a ventriculoperitoneal shunt was placed. When he arrives in the ICU, he is awake but confused and in respiratory distress. Vital signs are temperature 38.9°C (102°F), blood pressure 100/56 mm Hg, heart rate 110 beats/min, and respiratory rate 36 breaths/min. Laboratory findings obtained a few hours earlier were significant for WBCs 11,000/μL, platelets 95,000/μL, potassium 3.2 mEq/L, and glucose 225 mg/dL. Morning chest radiograph is unremarkable. Blood, sputum, urine, and cerebrospinal fluid cultures are obtained. The most appropriate empiric antimicrobial regimen should include

    A. cefepime and vancomycin.

    B. ceftriaxone and tobramycin.

    C. meropenem and amikacin.

    D. piperacillin/tazobactam and vancomycin.

14. A 48-year-old woman was admitted to the medical ward 2 days ago and started on oseltamivir, 75 mg twice daily. She is transferred to the ICU with respiratory failure secondary to H1N1 pneumonia. She is intubated, and samples are obtained for blood, urine, and respiratory cultures. Vital signs are temperature 38.8°C (102°F), blood pressure 110/60 mm Hg, and heart rate 120 beats/min. Laboratory results are significant for WBC count 11,000/μL and serum creatinine 1.6 mg/dL. She is started on broad-spectrum antibiotics for possible secondary bacterial

infection. The most appropriate next step in therapy for H1N1 pneumonia is to

A. discontinue oseltamivir and start peramivir and hydrocortisone.

B. continue oseltamivir and start methylprednisolone.

C. discontinue oseltamivir and start peramivir.

D. continue oseltamivir.

15. A 67-year-old man was admitted on the night shift with chronic obstructive pulmonary disease exacerbation and possible pneumonia. He was administered IV cefepime, 2 g, and IV vancomycin, 1 g, on admission. Morning laboratory results show WBC count 7.6 cells/μL, lactate 0.5 mmol/L, and procalcitonin 0.08 ng/mL. Which of the following correctly identifies the significance of obtaining procalcitonin levels for this patient?

A. Decisions about continued antibiotic use can be based solely on procalcitonin levels.

B. Procalcitonin can be used to reduce the risk of *Clostridioides difficile*.

C. Procalcitonin levels can help support discontinuation of empiric antibiotics.

D. There is little association between antibiotic de-escalation and mortality.

Part 7 Answers:

Infectious Disease

1. Rationale

Answer: A

Up to 16% of patients with candidemia have some manifestation of ocular involvement; therefore, it is recommended that all non-neutropenic patients have a dilated ophthalmologic examination within a week of candidemia diagnosis. *Candida krusei* has a natural resistance to fluconazole. Treatment duration in patients without metastatic disease is 2 weeks after documented clearance of *Candida*. Blood cultures should be repeated every 24 to 48 hours to establish a time point at which candidemia has cleared.

Reference:

1. Pappas PG, Kauffman CA, Andes DR, et al. Clinical practice guideline for the management of candidiasis: 2016 update by the Infectious Diseases Society of America. *Clin Infect Dis*. 2016 Feb 15;62(4):e1-e50.

2. Rationale

Answer: C

The mecA gene is responsible for methicillin resistance via alterations in penicillin-binding protein. KPC production implies carbapenem resistance. VanA resistance is seen with gram-positive organisms indicating resistance to vancomycin. The ampC gene encodes for clinically important cephalosporinases and has the potential to readily mutate to develop resistance to other beta-lactams.

Reference:

1. Jacoby GA. AmpC beta-lactamases. *Clin Microbiol Rev* 2009 Jan;22(1):161-182.

3. Rationale

Answer: A

This patient's presentation suggests a mycotic aneurysm. Endovascular aneurysm repair for a mycotic aneurysm can be considered in high-risk patients but is not the standard of care in the United States. Open surgical repair remains the standard treatment, along with antibiotics for at least 6 weeks. Antibiotics should be broad-spectrum initially and de-escalated according to culture results. The prevalent offending bacteria are *Staphylococcus aureus* and *Salmonella*, so initial antibiotic treatment for this patient could include both vancomycin and ceftriaxone. Treatment with antibiotics alone results in a high mortality rate.

References:

1. Deipolyi A, Czaplicki C, Oklu R. Inflammatory and infectious aortic diseases. *Cardiovasc Diagn Ther*. 2018 Apr;8(Suppl 1): S61-S70.
2. Hsu RB, Chang CI, Wu IH, Lin FY. Selective medical treatment of infected aneurysms of the aorta in high-risk patients. *J Vasc Surg*. 2009 Jan;49(1):66-70.
3. Lau C, Gaudino M, de Biasi AR, Munjal M, Girardi LN. Outcomes of open repair of mycotic descending thoracic and thoracoabdominal aortic aneurysms. *Ann Thorac Surg*. 2015 Nov;100(5):1712-1717.
4. Sorelius K, Mani K, Bjorck M, et al; Euroopean MAA collaborators. Endovascular treatment of mycotic aortic aneurysms: a European multicenter study. *Circulation*. 2014 Dec 9;130(24):2136-2142.

4. Rationale

Answer: A

This patient's most likely diagnosis is cryptococcal meningitis. Amphotericin in combination with flucytosine is most appropriate in this setting. Amphotericin B in combination with flucytosine has been shown to be superior in clinical outcome than either amphotericin alone, amphotericin in combination with fluconazole, or a 3-drug regimen including amphotericin, flucytosine, and fluconazole. Caspofungin lacks

appreciable cryptococcal coverage and thus is inappropriate for empiric coverage in this patient.

References:

1. Abruzzo GK, Flattery AM, Gill CJ, et al. Evaluation of the echinocandin antifungal MK-0991 (L-743,872): efficacies in mouse models of disseminated aspergillosis, candidiasis, and cryptococcosis. *Antimicrob Agents Chemother.* 1997 Nov;41(11):2333-2338.
2. Perfect JR, Dismukes WE, Dromer F, et al. Clinical practice guidelines for the management of cryptococcal disease: 2010 update by the Infectious Diseases Society of America. *Clin Infect Dis*. 2010 Feb 1;50(3):291-322.

5. Rationale

Answer: E

Three main classes of carbapenemase have been identified. In the United States the most common is the *Klebsiella pneumoniae* carbapenemase (KPC) group. Despite its name, the plasmid-borne genetic element coding for this protein can be transmitted to a number of *Enterobacteriaceae*. About 50% of pathogens producing this enzyme are resistant to aminoglycosides. Because pathogens demonstrate variable susceptibility to the few agents that are potentially useful for treatment, in vitro testing is recommended to help guide management. These pathogens can emerge in patients treated with beta-lactam antibiotics in addition to carbapenems. Carbapenemase is different from extended-spectrum beta-lactamases (ESBLs), although some pathogens coproduce them. Carbapenems are generally considered the treatment of choice for ESBL-producing pathogens, while colistin, tigecycline, and ceftazidime-avibactam are options for KPC producers.

References:

1. Bassetti M, Giacobbe DR, Giamarellou H, et al; Critically Ill Patients Study Group of the European Society of Clinical Microbiology and Infectious Disease (ESCMID); Hellenic Society of Chemotherapy (HSC); Societa Italiana di Terapia

Antinfettiva (SITA). Management of KPC-producing Klebsiella pneumoniae infections. *Clin Microbiol Infect*. 2018 Feb;24(2):133-144.

2. Rodriguez-Bano J, Gutierrez-Gutierrez B, Machuca I, Pascual A. Treatment of infections caused by extended-spectrum-beta-lactamase-, AmpC-, and carbapenemase-producing Enterobacteriaceae. *Clin Microbiol Rev.* 2018 Feb 14;31(2): e00079-e000117.

6. Rationale

Answer: E

This patient has necrotizing soft tissue infection (NSTI) based on his physical examination and laboratory results. Although broad-spectrum antibiotics are important, they would not provide adequate source control and thus would be insufficient for improving his outcome. Plasmapheresis is an adjuvant intervention that has been shown to be beneficial in patients with severe sepsis or septic shock, but it has not been studied in NSTI. While IV immunoglobulin might reduce shock in NSTI, it is an adjunct intervention, and its effectiveness has not been proven in high-quality studies. While hyperbaric oxygen might offer some benefit in patients with polymicrobial NSTI, it is an adjunct intervention and should be administered after surgical debridement. Surgical debridement is the definitive measure to remove the source of infection. Early and adequate surgical debridement in the emergency department has been associated with improved outcome.

References:

1. Cocanour CS, Chang P, Huston JM, et al. Management and novel adjuncts of necrotizing soft tissue infections. *Surg Infect (Larchmt)*. 2017 Apr;18(3):250-272.

2. Hong YC, Chou MH, Liu EH, et al. The effect of prolonged ED stays on outcome in patients with necrotizing fasciitis. *Am J Emerg Med*. 2009 May;27(4):385-390.

7. Rationale

Answer: B

Inhalational anthrax is caused by exposure to aerosolized bacterial spores that later germinate in the lung. The intubation period is typically several days but can range from 1 day to many weeks. Initial symptoms often mimic an influenza-like illness. As the disease progresses, fever, shock, and respiratory distress ensue. Hemorrhagic mediastinitis results in the characteristic widened mediastinum noted on chest radiograph. In patients with inhalational anthrax, a 3-drug regimen should be used until meningitis can be excluded by cerebrospinal fluid testing. The preferred antibiotic regimen is ciprofloxacin, the bactericidal agent meropenem, and the protein synthesis inhibitor linezolid.

References:

1. Adalja AA, Toner E, Inglesby TV. Clinical management of potential bioterrorism-related conditions. *N Engl J Med*. 2015 Mar 5;372(10):954-962.

2. Kyriacou DN, Stein AC, Yarnold PR, et al. Clinical predictors of bioterrorism-related inhalational anthrax. *Lancet*. 2004 Jul 31-Aug 6;364(9432):449-452.

8. Rationale

Answer: A

Intra-abdominal infections are generally caused by a disruption of the gastrointestinal mucosa that results in bowel flora inoculating the abdominal cavity. Considerations in selecting an antibiotic regimen include the source of infection (ie, gastrointestinal tract location), likelihood of antibiotic-resistant pathogens, clinical risk factors for treatment failure, and severity of infection. In this postoperative patient with multiple gastrointestinal disorders, the source of infection could be any point in the gastrointestinal tract. Coverage must account for colonic (*Escherichia coli*, *Klebsiella*, *Proteus*, *Enterobacter*, streptococci, enterococci, and anaerobic bacteria) and upper gastrointestinal pathogens (aerobic and anaerobic gram-positive bacteria or *Candida*). Exposure to the healthcare system increases his risk of antibiotic-resistant pathogens, including methicillin-resistant *Staphylococcus aureus* (MRSA), vancomycin-

resistant *Enterococcus* species (VRE), carbapenem-resistant *Pseudomonas aeruginosa*, extended-spectrum β-lactamase-producing *E coli* and *Klebsiella* species, and multidrug-resistant *Acinetobacter* species. An appropriate antibiotic regimen should adequately address resistant organisms. Piperacillin/tazobactam provides broad-spectrum coverage, including *Pseudomonas*. Vancomycin provides coverage for MRSA, which is recommended for patients with healthcare-associated intra-abdominal infection. Although his MRSA status is unknown, it is not an unreasonable empiric therapy. Although a combination of levofloxacin, metronidazole, and vancomycin provides broad-spectrum coverage, there is increasing *E coli* resistance to fluoroquinolones, so this is not the best option. Neither a combination of cefepime and metronidazole nor meropenem would provide MRSA coverage. Empiric VRE coverage is not recommended; it is usually reserved for patients at high risk for VRE (liver transplant recipients with intra-abdominal infection of hepatobiliary source and patients known to be colonized).

References:

1. Baron EJ, Miller JM, Weinstein MP, et al. A guide to utilization of the microbiology laboratory for diagnosis of infectious diseases: 2013 recommendations by the Infectious Diseases Society of America (IDSA) and the American Society for Microbiology (ASM). *Clin Infect Dis*. 2013 Aug;57(4): e22-e121.
2. Solomkin JS, Mazuski JE, Bradley JS, et al. Diagnosis and management of complicated intra-abdominal infection in adults and children: guidelines by the Surgical Infection Society and the Infectious Diseases Society of America. *Clin Infect Dis*. 2010 Jan 15;50(2):133-164.
3. Wong PF, Gilliam AD, Kumar S, Shenfine J, O'Dair GN, Leaper DJ. Antibiotic regimens for secondary peritonitis of gastrointestinal origin in adults. *Cochrane Database Syst Rev*. 2005 Apr 18;(2) CD004539.

9. Rationale

Answer: D

Guidelines for prevention of *Clostridioides difficile* infection (CDI) in acute care settings recommend maintaining contact precautions for at least 48 hours after resolution of diarrhea in patients with CDI. When possible, disposable, or dedicated equipment should be used for such patients. When this approach is not feasible, equipment should be sterilized according to manufacturers' specifications. Screening to identify asymptomatic *C difficile* carriers is not recommended and has not been shown to decrease the incidence of CDI. Likewise, the addition of ultraviolet light to standard bleach decontamination has not been proven to decrease the incidence of CDI in acute care settings.

References:

1. Banach DB, Bearman G, Barnden M, et al. Duration of contact precautions for acute-care settings. *Infect Control Hosp Epidemiol*. 2018 Feb;39(2):127-144.
2. Dubberke ER, Carling P, Carrico R, et al. Strategies to prevent Clostridium difficile infections in acute care hospitals: 2014 update. *Infect Control Hosp Epidemiol*. 2014 Jun;35(6):628-645.
3. McDonald LC, Gerding DN, Johnson S, et al. Clinical practice guidelines for Clostridium difficile Infection in adults and children: 2017 update by the Infectious Diseases Society of America (IDSA) and Society for Healthcare Epidemiology of America (SHEA). *Clin Infect Dis*. 2018 Mar;66(7): e1-e48.

10. Rationale

Answer: B

Most shunt infections occur in the first few weeks to several months after ventriculoperitoneal shunt (VPS) placement. The most likely microorganisms associated with healthcare-acquired cerebrospinal fluid (CSF) shunt and drain infections are coagulase-negative *Staphylococci* (especially *Staphylococcus epidermidis*), *Staphylococcus aureus*, *Propionibacterium acnes*, and

gram-negative bacilli (including *Escherichia coli*, *Enterobacter* species, *Citrobacter* species, *Serratia* species, and *Pseudomonas aeruginosa*). The 2017 Infectious Diseases Society of America guidelines for healthcare-associated ventriculitis or meningitis recommend vancomycin and an antipseudomonal beta-lactam (such as cefepime, ceftazidime, or meropenem) for empiric therapy. Ceftriaxone and ampicillin lack pseudomonal coverage. Piperacillin/tazobactam is an antipseudomonal beta-lactam but has limited penetration in the CSF. Rifampin is recommended as part of combination therapy for any patient with intracranial or spinal hardware such as a CSF shunt or drain, but it is generally held until a pathogen is identified. The addition of rifampin is warranted if 1) *Staphylococci* isolate is susceptible and prosthetic material is in place or 2) *Streptococcus pneumoniae* minimum inhibitory concentration for ceftriaxone is 2 μg/mL. The combination of ampicillin and a fourth-generation cephalosporin is inappropriate as an empiric regimen.

References:

1. Paff M, Alexandru-Abrams D, Muhonen M, Loudon W. Ventriculoperitoneal shunt complications: a review. *Inter-disc Neurosurg*. 2018 Sep; 13:66-70.
2. Tunkel AR, Hasbun R, Bhimraj A, et. al. 2017 Infectious Diseases Society of America's clinical practice guidelines for healthcare-associated ventriculitis and meningitis. *Clin Infect Dis.* 2017 Mar 15;64(6): e34-e65.

11. Rationale

Answer: D

The abdominal CT demonstrates several air fluid levels that are pathognomonic for emphysematous pyelonephritis, which most often develops in patients with poorly controlled diabetes mellitus. Antibiotic therapy alone is often ineffective. The CT does not demonstrate hydronephrosis, making ureteral obstruction less likely. The lesions on CT are air-filled, inconsistent with a perinephric abscess. Tuberculous pyelonephritis also would not have this radiographic appearance, and the positive urine culture (although

nondiagnostic at this point) makes this diagnosis less likely. Xanthogranulomatous pyelonephritis is characterized by an enlarged kidney with hydronephrosis in the setting of chronic infection, and staghorn calculi are usually present.

Reference:

1. Yu M, Robinson K, Siegel C, Menias C. Complicated genitourinary tract infections and mimics. *Curr Probl Diagn Radiol*. 2017 Jan-Feb;46(1)74-83.

12. Rationale

Answer: C

Ludwig angina is a potentially life-threatening infection that affects the airway. It typically begins with a dental infection that rapidly spreads. It is a bilateral infection that affects both the submental and sublingual spaces. Since it is typically associated with dental infections, it begins in the floor of the mouth. With this type of submandibular space infection, there is usually no lymphatic involvement or abscess formation.

References:

1. Barton ED, Bair AE. Ludwig's angina. *J Emerg Med*. 2008 Feb;34(2):163-169.

2. Rizzo P, Da Mosto MC. Submandibular space infection: a potentially lethal infection. *Int J Infect Dis*. 2009 May;13(3):327-333.

3. Pak S, Cha D, Meyer C, Dee C, Fershko A. Ludwig's angina. *Cureus*. 2017 Aug 21;9(8): e1588.

13. Rationale

Answer: A

This patient's clinical picture suggests sepsis, which is likely meningitis secondary to ventriculoperitoneal shunt infection. Infections at other sites are also possible pending cerebrospinal fluid (CSF) analysis. As with all healthcare-associated infections, initial empiric therapy includes broad-spectrum antibiotics; however, when meningitis is part of the differential diagnosis, the antibiotics chosen should have good CSF penetration.

Guidelines for the management of healthcare-associated ventriculitis and meningitis recommend that empiric therapy should include vancomycin plus an antipseudomonal beta-lactam (eg, cefepime, ceftazidime, or meropenem). Although piperacillin/tazobactam has antipseudomonal activity, tazobactam's CSF penetration is poor and is therefore not recommended for meningitis.

References:

1. Rhodes A, Evans LE, Alhazzani W, et al. Surviving Sepsis Campaign: international guidelines for management of sepsis and septic shock: 2016. *Crit Care Med*. 2017 Mar;45(3):486-552.

2. Tunkel AR, Hasbun R, Bhimraj A, et al. 2017 Infectious Diseases Society of America's clinical practice guidelines for healthcare-associated ventriculitis and meningitis. *Clin Infect Dis.* 2017 Mar 15;64(6): e34-e65.

14. Rationale

Answer: D

The therapy recommended by the CDC for H1N1 infection in hospitalized patients is oral or enterically administered oseltamivir. IV peramivir is not recommended as first-line treatment because of insufficient data on its efficacy in hospitalized patients. This patient's worsening condition is not likely related to oseltamivir resistance but most likely reflects disease progression. Although her serum creatinine is elevated at 1.6 mg/dL, suggesting possible acute kidney injury, oseltamivir is not contraindicated even though it requires dose adjustments based on her creatinine clearance. Peramivir is not indicated because there is nothing in the scenario to suggest that she cannot tolerate or absorb oral oseltamivir (eg, gastric obstruction, gastrointestinal bleeding, malabsorption). The CDC recommendations are based on literature suggesting that oseltamivir administered orally or by oro- or nasogastric tube is well absorbed in critically ill ICU patients with influenza. The most recent guidelines on the diagnosis and management of critical illness-related corticosteroid insufficiency (CIRCI) do

not recommend corticosteroids in patients with influenza. This recommendation is based on studies that reported a higher incidence of mortality in patients with influenza who received corticosteroids.

References:

1. Pastores SM, Annane D, Rochwerg B; Corticosteroid Guideline Task Force of SCCM and ESICM. Guidelines for the diagnosis and management of critical illness-related corticosteroid insufficiency (CIRCI) in critically ill patients (part II): Society of Critical Care Medicine (SCCM) and European Society of Intensive Care Medicine (ESICM) 2017. *Crit Care Med*. 2018 Jan;46(1):146-148.
2. Rodrigo C, Leonardi-Bee J, Nguyen-Van-Tam J, Lim WS. Corticosteroids as adjunctive therapy in the treatment of influenza. *Cochrane Database Syst Rev.* 2016 Mar 7;3:CD010406.
3. Centers for Disease Control and Prevention. Influenza antiviral medications: summary for clinicians. Undated. Accessed March 1, 2022. https://www.cdc.gov/flu/pdf/professionals/antivirals/antiviral-summary-clinician.pdf

15. Rationale

Answer: C

The 2016 Surviving Sepsis Campaign guidelines suggest testing procalcitonin levels to help shorten the duration of antibiotic use and to support the discontinuation of empiric antibiotics. While there are several algorithms for suggested use with procalcitonin testing, no single algorithm is suggested over another. Research has shown some conflicting evidence between early antimicrobial de-escalation and survival in sepsis and septic shock. Procalcitonin testing provides additional information to aid clinical decision-making, but levels should not be used solely for antibiotic decision-making.

References:

1. Dickstein PM, Raz-Pasteur A. Antibiotic de-escalation for bloodstream infections and pneumonia: systematic review and meta-analysis. *Clin Microbiol Infect*. 2016 Dec;22(12):960-967.

2. Liu D, Su L, Han G, Yan P, Xie L. Prognostic value of procalcitonin in adult patients with sepsis: a systematic review and meta-analysis. *PLoS One*. 2015 Jun 15;10(6) e0129450.

3. Rhodes A, Evans LE, Alhazzani W, et al. Surviving Sepsis Campaign: international guidelines for the management of sepsis and septic shock: 2016. *Crit Care Med*. 2017 Mar;45(3):486-552.

Part 8.

Gastrointestinal Disorders

Instructions: For each question, select the most correct answer.

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1. A 44-year-old man with no known medical history was admitted 48 hours ago for acute pancreatitis. In the past 24 hours, he has required intubation and vasopressor therapy initiation. Base deficit on admission was 4.5 mEq/L, which had improved with initial fluid resuscitation. Despite receiving 8 L lactated Ringer solution in the past 24 hours, the most recent arterial blood gas analysis shows pH 7.19, $PaCO_2$ 65 mm Hg, PaO_2 55 mm Hg, lactate 9 mmol/L, and base deficit 12 mEq/L. The noninvasive cardiac monitor shows stroke volume variation 9% and cardiac index 2.2 L/min/m^2. He has been anuric for the past 8 hours, and ultrasound shows no urine in the bladder. He is receiving norepinephrine, 1.5 μg/kg/min, and vasopressin, 0.03 U/min. Bladder pressure is 24 mm Hg. Peak airway pressure has increased to 40 cm H_2O. Which of the following is the most appropriate management approach?

   A. Increase sedation, paralyze the patient and in reverse Trendelenburg position, reassess bladder pressure hourly, and obtain emergent surgical evaluation.

   B. Administer albumin, 50 g followed by IV furosemide, 40 mg and attempt paracentesis.

   C. Increase cardiovascular support with an inotrope, administer a 1-L fluid challenge, and increase maintenance IV fluids.

D. Obtain CT scan to evaluate for infected necrotizing pancreatitis and retroperitoneal abscess.

2. A 70-year-old man presents to the emergency department with melena, coffee-ground vomitus, and generalized weakness. He has coronary artery disease and hypertension and recently underwent percutaneous coronary angioplasty and atrial fibrillation. He takes aspirin, warfarin, lisinopril, and metoprolol. Blood pressure is 80/60 mm Hg, heart rate 120 beats/min and irregular, respiratory rate 20 breaths/min, and temperature 36.5°C (97.7°F). He is pale and lethargic. Nasogastric tube is placed with return of coffee-ground material. Laboratory studies show hemoglobin 7.5 g/dL, hematocrit 25%, platelets 200×10^9/L, INR 2.5, BUN 30 mg/dL, and creatinine 1.2 mg/dL. He is administered 2 liters normal saline, 1-unit packed RBCs, and a pantoprazole infusion before being transferred to the ICU. Which of the following is the most appropriate next step?

   A. Early esophagogastroduodenoscopy

   B. Octreotide infusion

   C. Transfusion of RBCs to maintain hemoglobin above 11 g/dL

   D. Ceftriaxone

   E. Colonoscopy

3. A 78-year-old man with a history of congestive heart failure, hypertension, diabetes, and hypothyroidism is admitted to the ICU with right upper quadrant abdominal pain and some nausea and vomiting. On physical examination, he has isolated right upper quadrant tenderness to palpation. He has normal breath sounds bilaterally with mild pitting edema of the bilateral lower extremities. Creatinine level is

elevated at 1.5 mg/dL (baseline 0.9 mg/dL). WBC count is elevated at 14,000/μL. Abdominal ultrasound findings are consistent with acute cholecystitis secondary to cholelithiasis. He is admitted to the ICU for initial hypotension, which is fluid responsive. Current vital signs are temperature 38.4°C (101.1°F), heart rate 109 beats/min, and blood pressure 109/64 mm Hg. Acute Physiologic Assessment and Chronic Health Evaluation II score is 8. In addition to antibiotics, which of the following treatment strategies is most appropriate for this patient?

A. Close observation

B. Laparoscopic cholecystectomy

C. Percutaneous cholecystostomy tube

D. Lithotripsy

4. A 45-year-old man with no past medical history presents after drinking a large amount of alcohol. He has isolated upper abdominal tenderness to palpation with serum lipase level 3 times the upper limit of normal. Subsequent CT with IV shows acute necrotizing pancreatitis. Now, 72 hours after ICU admission, he is intubated and has been started on norepinephrine for refractory shock. Vital signs are temperature 38.9°C (102°F), heart rate 131 beats/min, and blood pressure 94/60 mm Hg. He is on a high and escalating norepinephrine dose, which has been increased from yesterday. He has also been started on carbapenem for presumed infected necrotizing pancreatitis. CT shows air bubbles within a peripancreatic fluid collection. Which of the following is the most appropriate next step in management?

   A. Vancomycin to broaden antibiotic coverage

   B. Open necrosectomy

   C. Video-assisted retroperitoneal debridement

D. Interventional radiology catheter drainage

E. Laparoscopic necrosectomy

5. A medical resident has read about specialty tube feedings designed for chronic obstructive pulmonary disease patients and asks about using this tube feeding for ventilated patients to shorten time on the ventilator. Although routine use is not recommended, what is the rationale behind this request?

   A. It has lower free water content than traditional formulas, which decreases pulmonary edema.

   B. It has antioxidant supplements, which decrease free radicals.

   C. It is enriched with eicosatetraenoic acid, which decreases airway inflammation.

   D. It has a high fat-to-carbohydrate ratio to reduce respiratory quotient.

6. A 60-year-old woman is evaluated in the emergency department for an episode of hematemesis that occurred 1 hour ago. She has a recent ankle injury, for which she has been taking ibuprofen. She is feeling weak and lightheaded. She does not smoke or drink alcohol. She has a history of peptic ulcer disease, for which she uses over-the-counter antacids as needed, and hypertension, for which she takes hydrochlorothiazide. On physical examination, she appears pale and diaphoretic. Temperature is 36.5°C (97.7°F), blood pressure 95/64 mm Hg supine and 73/40 mm Hg sitting, and heart rate 106 beats/min supine and 128 beats/min sitting. Respiratory rate is 15 breaths/min. Abdominal examination reveals no tenderness or organomegaly. The remainder of the examination is unremarkable. Laboratory studies and typing and cross-matching are pending. Which of the following is the most appropriate IV access for this patient?

A. Intraosseous catheter

B. Large-caliber peripheral IV catheter

C. Single-lumen peripherally inserted central venous catheter

D. Triple-lumen internal jugular venous catheter

7. A 24-year-old woman who is 7 weeks pregnant presents to the emergency department with a 3-day history of difficulty swallowing and chest pain. Her pregnancy has been complicated by hyperemesis gravidarum, for which she has been admitted to the hospital for IV fluid administration. Examination reveals crepitus in the neck region and the anterior chest wall. Vital signs are stable; she can vocalize without difficulty and has no difficulty breathing. Noncontrast chest CT does not show any airway injury but does show pneumomediastinum. She is admitted to the ICU. Which of the following is the most appropriate management approach?

   A. Broad-spectrum antibiotics including fluconazole

   B. Swallow study and broad-spectrum antibiotics including fluconazole

   C. Close observation

   D. Endoscopy

8. A 79-year-old man is admitted to the hospital for recurrent *Clostridioides difficile* infection. He is started on oral vancomycin, but his condition worsens over the first 6 hours. He is transferred to the ICU with septic shock. Vital signs are blood pressure 79/38 mm Hg after 2 L crystalloids administered for fluid resuscitation, heart rate 128 beats/min, oral temperature 38.7°C (101.6°F), and respiratory rate 22 breaths/min. WBC count is 18,100/μL. He reports increasing abdominal pain, and CT shows colonic

dilation at 8 cm. The diagnosis of toxic megacolon includes which of the following criteria?

A. Radiographic evidence, fever, tachycardia, leukocytosis, and hypotension

B. Fever, tachycardia, and anemia

C. Radiographic evidence of colonic distension only

D. Radiographic evidence, fever, tachycardia, and leukocytosis

E. Fever, hypotension, leukocytosis, and abnormal electrolytes

9. A 69-year-old woman with a history of coronary artery disease, type 2 diabetes, and peripheral vascular disease is admitted to the ICU with a 2-day history of palpitations and acute onset of abdominal pain and bloody stools. On examination, she is ill appearing. Heart rate is 110 beats/min and irregular, and blood pressure is 100/60 mm Hg. Abdomen is diffusely tender but soft and without rebound tenderness. Dark, bloody stool is in the rectal vault. INR is 1.0. Other laboratory findings are hemoglobin 12 g/dL, leukocyte count 13,000/mm^3, platelet count 225,000/mm^3, BUN 18 mg/dL, and creatinine 1.1 mg/dL. ECG shows atrial fibrillation. Which of the following tests is most appropriate in her initial evaluation?

   A. Upper endoscopy

   B. Colonoscopy

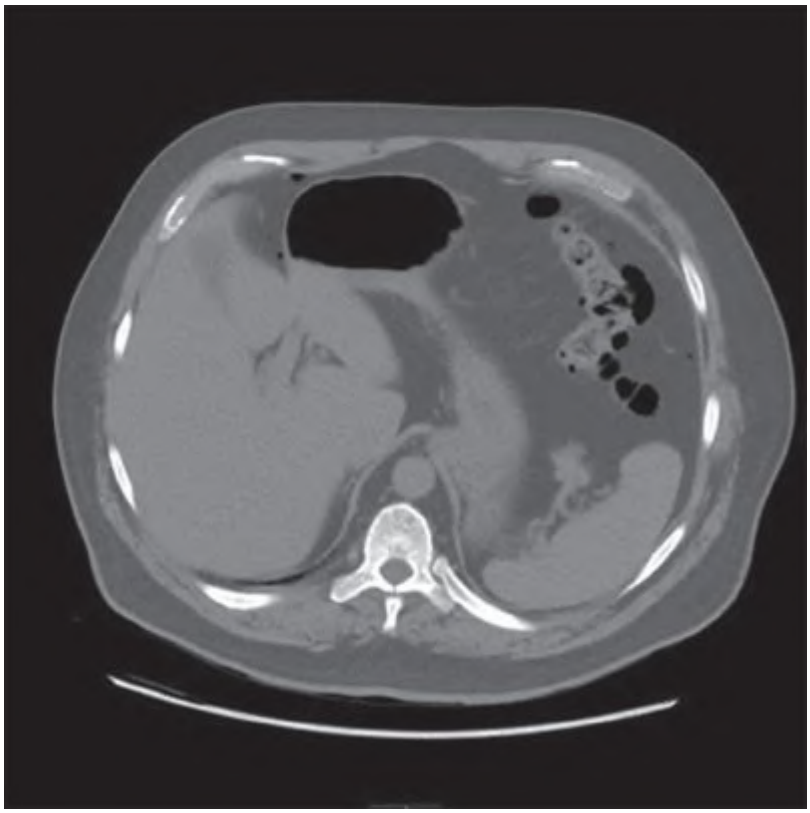
   C. Mesenteric magnetic resonance angiography

   D. CT mesenteric angiography

   E. Mesenteric duplex ultrasound

10. A 75-year-old man with a history of end-stage renal disease requiring intermittent hemodialysis is admitted

to the ICU with severe abdominal pain and hypotension. Vital signs are blood pressure 80/50 mm Hg, heart rate 130 beats/min, and respiratory rate 22 breaths/min. On physical examination, his abdomen is diffusely tender to palpation. He is started on broad-spectrum antibiotics. Blood pressure does not improve despite 2 L normal saline. A norepinephrine infusion is started to maintain mean arterial pressure 65 mm Hg. CT is shown below. Which of the following is the most appropriate next step in management?



A. Continue current treatment with antibiotics and close monitoring.

B. Consult interventional radiology for percutaneous drain placement.

C. Consult general surgery for exploratory laparotomy.

D. Check serum lactate and consult general surgery if elevated.

11. A 35-year-old man with pan-ulcerative colitis who was previously maintained in remission on oral and rectal mesalamine is hospitalized for a severe ulcerative colitis (UC) flare. After 4 days of treatment with IV hydration; IV methylprednisolone, 60 mg daily; and supportive blood transfusions, his symptoms have not improved, and he is admitted to the ICU. Flexible sigmoidoscopy shows severe UC. The next day, his abdominal pain becomes worse. Temperature is 38.9°C (102°F), heart rate 115 beats/min, and blood pressure 90/60 mm Hg. On physical examination, his abdomen is distended and diffusely tender to palpation with mild guarding. Laboratory results are hemoglobin 9.6 g/dL, leukocyte count 14,800/mm^3, creatinine 0.8 mg/dL, and albumin 3.0 g/dL. Abdominal radiograph reveals a dilated transverse colon to about 9 cm but no free air. Which of the following management approaches is most appropriate for this patient?

    A. Initiate IV ciprofloxacin and metronidazole.

    B. Perform urgent colectomy with end ileostomy.

    C. Increase the IV methylprednisolone dose.

    D. Monitor closely and intensify hydration.

    E. Initiate IV infliximab.

12. A 34-year-old man with a past medical history significant for advanced cirrhosis, hepatic encephalopathy, and spontaneous bacterial peritonitis presents with altered mental status, malaise, and pallor and is admitted to the ICU. Serum hemoglobin is 5.5 g/dL. Emergency esophagogastroduodenoscopy reveals 2 bleeding gastric ulcers, which are clipped. Subsequently, IV infusion of pantoprazole is initiated. Which of the following interventions can improve the risk of mortality in the acute setting?

    A. IV ceftriaxone

B. IV octreotide

C. IV albumin 5%

D. Oral nadolol

13. An intubated 55-year-old man is admitted to the ICU to undergo treatment for acute respiratory distress syndrome. On hospital day 15, he develops increased abdominal distention over 4 hours. His most recent bowel movement was 5 days ago. Soapsuds enemas were administered without symptom resolution. Abdominal plain radiograph shows diffuse colonic dilation of the colon consistent with ileus, without an identifiable transition point. Physical examination reveals a rotund abdomen without a fluid wave or shifting dullness. Digital rectal examination shows no stool in the rectal vault. Which of the following is the most appropriate immediate next step in management?

    A. Repeat soapsuds enema

    B. Therapeutic enteroclysis

    C. Colonoscopic decompression

    D. IV neostigmine

    E. Surgical consultation for colectomy

14. A 50-year-old man with end-stage alcoholic liver disease is admitted to the ICU with severe respiratory distress. Heart rate is 110 beats/min, respiratory rate 35 breaths/min with use of accessory muscles of respiration, and blood pressure 120/60 mm Hg. He has a markedly distended abdomen with shifting dullness to percussion. He has bilateral pitting edema up to his thighs. Initial laboratory findings are significant for platelet count 50,000/μL and INR 2.2. Chest radiograph shows low lung volumes and small bilateral pleural effusions. Which of the following is the most appropriate next step?

A. Therapeutic thoracentesis only

B. Transfusion of fresh frozen plasma followed by therapeutic thoracentesis

C. Therapeutic paracentesis only

D. Transfusion of fresh frozen plasma followed by therapeutic paracentesis

E. IV furosemide

15. A 47-year-old man presents to the emergency department with alcoholic pancreatitis. Abdominal and pelvic CT shows multiple peripancreatic fluid collections and 40% pancreatic necrosis. He is admitted to the surgical ICU. After initial resuscitation and stabilization, he cannot tolerate an oral diet. Which of the following is the most appropriate nutritional strategy?

A. Parenteral nutrition started at 50% goal rate and advanced to goal over the next 2 days

B. Polymeric enteral nutrition (either gastric or jejunal) started at a trophic rate and advanced to goal as tolerated

C. Immune-enhancing enteral nutrition (either gastric or jejunal) started at a trophic rate and advanced to goal as tolerated

D. Maintaining nothing-by-mouth status for the first 72 hours

E. Mixture of enteral nutrition and partial parenteral nutrition to obtain total caloric goal

Part 8 Answers:

Gastrointestinal Disorders

1. Rationale

Answer: A

Intra-abdominal hypertension (IAH) is intra-abdominal pressure (IAP) greater than 12 mm Hg. Abdominal compartment syndrome (ACS) is sustained IAP greater than 20 mm Hg that is associated with new organ dysfunction or failure. ACS can be due to primary or secondary causes. In this patient with severe pancreatitis who has received large fluid volumes, ACS is due to bowel edema, possible ascites, and decreased abdominal wall compliance. He has worsening organ failure. The initial approach is to maximize medical management with sedation and paralytics since the pressure is greater than 20 mm Hg. If these measures do not reduce the IAP below 20 mm Hg, he will need an emergent decompressive laparotomy, so emergent surgical evaluation is also warranted. Additional volume will only exacerbate his condition. Transport to CT will not add any benefit at this time and, in the setting of acute renal failure exacerbated by ACS, a furosemide challenge is of little benefit.

References:

1. Kirkpatrick AW, Roberts DJ, de Waele J, et al; Pediatric Guidelines Sub-Committee for the World Society of the Abdominal Compartment Syndrome. Intra-abdominal hypertension and the abdominal compartment syndrome: updated consensus definitions and clinical practice guidelines from the World Society of the Abdominal Compartment Syndrome. *Intensive Care Med*. 2013 Jul;39(7):1190-1206.

2. Popescu GA, Bara T, Rad P. Abdominal compartment syndrome as a multidisciplinary challenge. A literature reviews. *J Crit Care Med (Targu Mures)*. 2018 Oct 1;4(4):114-119.

2. Rationale

Answer: A

The most appropriate next step is continuation of volume resuscitation and urgent referral to gastroenterology for early esophagogastroduodenoscopy. Guidelines for management of nonvariceal hemorrhage strongly recommend prompt intravascular volume replacement with crystalloid fluids and early (< 24 hours) upper gastrointestinal endoscopy. Very early (< 12 hours) upper gastrointestinal endoscopy may be considered in patients with high-risk clinical features, such as hemodynamic instability, eg, hypotension and tachycardia that persist despite ongoing attempts of volume resuscitation, in-hospital bloody emesis/nasogastric aspirate, or contraindication for interruption of anticoagulation. Proton pump inhibitor should not delay the early endoscopy. The guideline for transfusion strategy for upper gastrointestinal bleeding is a restrictive cell transfusion strategy aiming for a target hemoglobin between 7 g/dL and 9 g/dL. A higher target should be considered in patients with significant comorbidity such as ischemic cardiovascular disease. There is no need to increase hemoglobin to 11 g/dL in this patient before endoscopy. Octreotide and antibiotic would be appropriate to use empirically in a patient with known or suspected liver disease, but this patient meets neither criterion. Octreotide is effective in variceal hemorrhage by decreasing portal venous inflow and intra-variceal pressure. He does not need an urgent colonoscopy; this is not the priority. Furthermore, he has melena and hematemesis, indicating significant upper gastrointestinal hemorrhage.

References:

1. IM, Dumonceau JM, Kuipers EJ, et al. Diagnosis and management of nonvariceal upper gastrointestinal hemorrhage: European Society of Gastrointestinal Endoscopy (ESGE) guideline. *Endoscopy.* 2015 Oct; 47(10): a1-a46.

2. Hwang JH, Fisher DA, Ben-Menachem T, et al; Standards of Practice Committee of the American Society for

Gastrointestinal Endoscopy. The role of endoscopy in the management of acute non-variceal upper GI bleeding. *Gastrointest Endosc.* 2012 Jun;75(6):1132-1138.

3. Villanueva C, Colomo A, Bosch A, et al. Transfusion strategies for acute upper gastrointestinal bleeding. *N Engl J Med.* 2013 Jan 3;368(1):11-21.

3. Rationale

Answer: B

Laparoscopic cholecystectomy is the treatment of choice for acute calculous cholecystitis. However, management of high-risk patients with serious comorbidities is unclear, ranging from antibiotics only to percutaneous cholecystostomy to operative intervention. Past international guidelines suggested invasive radiologic procedures as appropriate treatment for high-risk patients with moderate or severe cholecystitis; however, more recently the randomized controlled CHOCOLATE trial showed decreased major complications (eg, infectious, cardiopulmonary, endoscopic) with laparoscopic cholecystectomy compared to percutaneous drainage.

References:

1. Loozen CS, van Santvoort HC, van Duijvendijk, et al. Laparoscopic cholecystectomy versus percutaneous catheter drainage for acute cholecystitis in high-risk patients (CHOCOLATE): multicenter randomized clinical trial. *BMJ.* 2018 Oct 8;363: k3965.

2. Miura F, Takada T, Strasberg SM, et al; Tokyo Guidelines Revision Committee. TG13 flowchart for the management of acute cholangitis and cholecystitis. *J Hepatobiliary Pancreat Sci.* 2013 Jan;20(1)47-54.

4. Rationale

Answer: D

This patient has necrotizing pancreatitis, which occurs in approximately 20% of patients with pancreatitis. Secondary infection may occur, as it has in this patient. A hallmark of this

diagnosis is air within the retroperitoneal fluid collection. Historically patients underwent open necrosectomy with mortality rate 11% to 39%; however, a step-up approach first using less invasive techniques such as interventional radiology-directed catheter drainage followed only if it fails by a minimally invasive surgical intervention such as a laparoscopic approach or video-assisted retroperitoneal debridement was found to significantly decrease major complications (eg, perforation, enterocutaneous fistula, bleeding and/or organ failure) leaving open necrosectomy as a last resort. In a randomized controlled trial of 43 patients assigned to a step-up approach, 35% of the patients survived without need for surgical necrosectomy.

Reference:

1. Van Santvoort H, Besselink MG, Bakker OJ, et al; Dutch Pancreatitis Study Group. A step-up approach or open necrosectomy for necrotizing pancreatitis. *N Engl J Med*. 2010 Apr 22;362(16):1491-1502.

5. Rationale

Answer: D

Tube feedings specifically designed for chronic obstructive pulmonary disease patients have a high-calorie, modified carbohydrate and fat enteral formula that may help diet-induced CO_2 production. Tube feedings enriched with antioxidants and eicosatetraenoic acid are tailored for surgery and septic patients with conflicting benefits. The Society of Critical Care Medicine and American Society of Parenteral and Enteral Nutrition do not recommend routine use of any specialized formula.

Reference:

1. McClave SA, Taylor BE, Martindale RG, et al; Society of Critical Care Medicine; American Society for Parenteral and Enteral Nutrition. Guidelines for the provision and assessment of nutrition support therapy in the adult critically ill patient: Society of Critical Care Medicine

(SCCM) and American Society for Parenteral and Enteral Nutrition (A.S.P.E.N.). *JPEN J Parenter Enteral Nutr.* 2016 Feb;40(2):159-211.

6. Rationale

Answer: B

This patient has an acute upper gastrointestinal bleed and requires infusion of a large volume of crystalloids and blood quickly. A large-caliber IV catheter is the most appropriate access for this indication. Central access (triple-lumen or peripherally inserted central catheter) is not recommended for large-volume infusion. Intraosseous access is used in emergent situations or trauma when no other access can be secured.

Reference:

1. Lewis GC, Crapo SA, Williams JG. Critical skills, and procedures in emergency medicine: vascular access skills and procedures. *Emerg Med Clin North Am.* 2013 Feb;31(1):59-86.

7. Rationale

Answer: B

This patient has a high risk of Boerhaave syndrome (also known as spontaneous esophageal rupture or effort rupture of the esophagus) in the setting of hyperemesis gravidarum. Boerhaave syndrome is caused by a sudden increase in intra-esophageal pressure, leading to a complete transmural tear through the esophagus. Although vomiting is thought to be the most common cause, other causes include weightlifting, defecation, epileptic seizures, abdominal trauma, compressed air injury, and childbirth, all of which can increase pressure in the esophagus and cause a barogenic esophageal rupture. CT findings can be suggestive of the diagnosis, but the diagnostic test of choice is the contrast esophagogram (swallow study) using a water-soluble contrast agent such as diatrizoate meglumine and diatrizoate sodium solution. This will show extravasation of contrast material at the site of the perforation. Endoscopy should be performed with caution because of the

risk of further esophageal perforation. The mainstay of treatment is volume replacement, broad-spectrum antibiotic coverage, and surgical evaluation.

References:

1. Lindenmann J, Matzi V, Neuboeck N, et al. Management of esophageal perforation in 120 consecutive patients: clinical impact of a structured treatment algorithm. *J Gastrointest Surg*. 2013 Jun;17(6):1036-1043.
2. He X, He Z, Li H. Boerhaave syndrome: challenges in diagnosis and treatment of the early presentation and its complication. *J Trauma Acute Care Surg*. 2018 Jun;84(6):1030-1032.
3. Turner AR, Turner SD. Boerhaave syndrome. *StatPearls Publishing.* 2022 Jan. 2021 Dec 15.

8. Rationale

Answer: A

Diagnostic criteria for toxic megacolon depend on radiographic evidence along with at least 3 of the following findings: fever higher than 38°C (100.4°F), heart rate higher than 120 beats/min, leukocytosis with more than 10,500/μL, and anemia plus at least 1 of the following findings: dehydration, altered sensorium, electrolyte disturbances, or hypotension. This patient has radiographic evidence, fever, leukocytosis, tachycardia, plus hypotension in the setting of recurrent *Clostridioides difficile* infection. Therefore, he can be diagnosed with toxic megacolon.

References:

1. Autenrieth DM, Baumgart DC. Toxic megacolon. *Inflamm Bowel Dis*. 2012 Mar;18(3):584-591.
2. Sheth SG, LaMont JT. Toxic megacolon. *Lancet*. 1998 Feb 14;351(9101):509-513.

9. Rationale

Answer: D

The most likely diagnosis in a patient with atrial fibrillation who presents with acute severe abdominal pain and a tender soft abdomen with no rebound tenderness is acute mesenteric ischemia (AMI). Most commonly secondary to acute embolism to the superior mesenteric artery, AMI is a medical emergency. Patients with AMI classically have abdominal pain out of proportion to examination findings. AMI carries a mortality risk ranging from 30% to 90%. Arterial emboli, predominantly of cardiac origin, represent approximately 40% to 50% of AMI cases. The most appropriate first-line diagnostic evaluation for a patient with suspected mesenteric ischemia is CT mesenteric angiography. Neither upper endoscopy nor colonoscopy are preferred first-line evaluations for patients with features of mesenteric ischemia because these procedures do not provide opportunities for therapeutic intervention and would cause unnecessary delays. Magnetic resonance angiography is a lengthy examination with limited utility in the acute setting. Mesenteric duplex ultrasound is highly specific but has lower sensitivity than angiography due to overlying bowel gas.

References:

1. Clair DG, Beach JM. Mesenteric ischemia. *N Engl J Med*. 2016 Mar 10;374(10):959-968.
2. Oliva IB, Davarpanah AH, Rybicki FJ, et al. ACR appropriateness criteria imaging of mesenteric ischemia. *Abdom Imaging*. 2013 Aug;38(4):714-719. Erratum in: *Abdom Imaging*. 2014 Aug;39(4):937-939.
3. Wang JM, Chang SC. Images in clinical medicine. Acute mesenteric infarction associated with atrial fibrillation. *N Engl J Med*. 2011 Apr 7;364(14):1349.

10. Rationale

Answer: C

This patient has perforated diverticulitis. CT shows free air adjacent to the thickened sigmoid colon with fat stranding. Initial plans for antibiotic treatment would be reasonable if his abdomen were not diffusely tender and his vital signs were normal. But he is diffusely tender and hypotensive and is

developing septic shock. He needs urgent general surgery consultation for source control. CT shows no abscess or other fluid collection amenable to drainage by interventional radiology.

References:

1. Regenbogen SE, Hardiman KM, Hendren S, Morris AM. Surgery for diverticulitis in the 21st century: a systematic review. *JAMA Surg*. 2014 Mar;149(3):292-303.
2. Sartelli, M, Catena, F, Ansaloni, L, et al. WSES guidelines for the management of acute left sided colonic diverticulitis in the emergency setting. *World J Emerg Surg*. 2016 Jul 29; 11:37.

11. Rationale

Answer: B

Toxic megacolon complicating ulcerative colitis (UC) is defined as nonobstructive total or segmental colonic dilation (> 5.5 cm) associated with systemic toxicity. Appropriate management includes urgent surgical consultation, with urgent subtotal or total colectomy and end ileostomy typically the procedures of choice. Other indications for operative intervention in severe UC include severe hemorrhage, perforation, peritonitis, and failure of rescue therapy with infliximab or cyclosporine. In the setting of toxic megacolon, neither increasing the methylprednisolone dose nor adding infliximab confer advantage, and further delay in surgical therapy may result in increased risk for morbidity and mortality. The addition of antibiotics alone would be insufficient to prevent morbidity and mortality from toxic megacolon.

References:

1. Danese S, Fiocchi C. Ulcerative colitis. *N Engl J Med*. 2011 Nov 3;365(18):1713-1725.
2. Harbord M, Eliakim R, Bettenworth D, et al; European Crohn's and Colitis Organisation [ECCO]. Third European evidence-based consensus on diagnosis and management of

ulcerative colitis. Part 2: current management. *J Crohns Colitis*. 2017 Jul 1;11(7):769-784.

3. Strong SA. Management of acute colitis and toxic megacolon. *Clin Colon Rectal Surg*. 2010 Dec;23(4):274-284.

12. Rationale

Answer: A

Multiple antibiotics, including fluoroquinolone and third-generation cephalosporin antibiotics, have been shown to decrease the risk of mortality in the acute setting of an upper gastrointestinal bleed of esophageal variceal and stomach origins. IV octreotide is unlikely to decrease the risk of mortality because it has a role in management of esophageal variceal bleeding, but no mortality benefit has been found in upper gastrointestinal bleeding of a non-esophageal variceal source. IV albumin 5% has a role in acute spontaneous bacterial peritonitis and hepatorenal syndrome but no specific role in upper gastrointestinal bleeding. Oral nadolol has long-term mortality benefits in esophageal variceal bleeding but no such benefits in gastrointestinal bleeding from a non-esophageal variceal source.

References:

1. Chavez-Tapia NC, Barrientos-Gutierrez T, Tellez-Avila F, et al. Meta-analysis: antibiotic prophylaxis for cirrhotic patients with upper gastrointestinal bleeding: an updated Cochrane review. *Aliment Pharmacol Ther*. 2011 Sep;34(5):509-518.

2. Fernandez J, Tandon P, Mensa J, Garcia-Tsao G. Antibiotic prophylaxis in cirrhosis: good and bad. *Hepatology*. 2016 Jun;63(6):2019-2031.

13. Rationale

Answer: D

Acute colonic pseudo-obstruction, also known as Ogilvie syndrome, arises spontaneously in critically ill patients. Signs

and symptoms of bowel obstruction are typically present but there is no actual mechanical obstruction. Symptoms include distention, diarrhea, nausea, vomiting, abdominal pain, and tenderness to palpation on examination. Although its etiology is incompletely understood, it may derive from the dysregulation of the autonomic nervous system combined with pharmacologic or metabolic factors. The risk of spontaneous colonic perforation is 3%, with an ensuing mortality rate of up to 50%. The cecum should be carefully evaluated for size, and concern should be raised if the acute distention results in cecal diameter greater than 12 cm. Mechanical obstruction should be ruled out. Most patients respond within 3 days to conservative treatment, including nasogastric tube decompression, promotility agents, and enemas. The patient is monitored closely with serial abdominal examinations. If the condition persists, IV neostigmine, 2 mg, may be administered as first-line therapy. It can have significant side effects such as bradycardia, bronchial secretions, and asthma. If still unresolved, colonoscopic decompression is attempted. Surgical treatment is indicated for refractory pseudo-obstruction or spontaneous perforation.

References:

1. Saunders MD. Acute colonic pseudo-obstruction. *Best Pract Res Clin Gastroenterol*. 2007;21(4):671-687.
2. Vogel JD, Feingold DL, Stewart DB, et al. Clinical practice guidelines for colon volvulus and acute colonic pseudo-obstruction. *Dis Colon Rectum*. 2016 Jul;59(7):589-600.

14. Rationale

Answer: C

The most appropriate next step for this patient is therapeutic paracentesis. He has respiratory distress, most likely secondary to elevated abdominal pressures. He is tachypneic and is using accessory muscles of respiration. If unaddressed, this can quickly decompensate into respiratory arrest. Treating the underlying cause, in this case massive ascites, would improve his respiratory system compliance and reduce his work of

breathing. A trial of diuresis is a consideration in patients who are stable and not on the verge of respiratory arrest. Resolution of massive ascites with diuresis takes longer to mobilize ascitic fluid than paracentesis; in this setting this delay might worsen his respiratory status. The presence of mild coagulopathy should not delay paracentesis. Even in the setting of coagulopathy, the frequency of significant bleeding is extremely low. Additionally, the administration of blood products will worsen fluid overload and delay the time to paracentesis. He has small bilateral effusions, and therapeutic thoracentesis is unlikely to improve his breathing.

Reference:

1. McVay PA, Toy PT. Lack of increased bleeding after paracentesis and thoracentesis in patients with mild coagulation abnormalities. *Transfusion*. 1991 Feb;31(2):164-171.

15. Rationale

Answer: B

This patient has severe acute pancreatitis based on the CT severity index. CT grade E score is 4 points for multiple or extensive fluid collections, necrosis score is 4 points for one-third but less than one-half pancreatic necrosis, for a total score of 8 points, indicating severe pancreatitis. If a patient with severe acute pancreatitis cannot tolerate an oral diet, the preferred nutritional strategy is to place a naso-/oroenteric tube and start enteral nutrition at a trophic rate, increasing to goal within 24 to 48 hours of admission. Enteral nutrition is superior to parenteral nutrition in such patients. Data are insufficient to recommend the use of immune-enhancing formulas in such patients.

References:

1. Balthazar EJ, Robinson DL, Megibow AJ, Ranson JH. Acute pancreatitis: value of CT in establishing prognosis. *Radiology*. 1990 Feb;174(2):331-336.

2. McClave SA, Chang WK, Dhaliwal R, Heyland DK. Nutrition support in acute pancreatitis: a systematic review of the literature. *JPEN J Parenter Enteral Nutr.* 2006 Mar-Apr;30(2):143-156.

3. McClave SA, Taylor BE, Martindale RG, et al; Society of Critical Care Medicine; American Society for Parenteral and Enteral Nutrition. Guidelines for the provision and assessment of nutrition support therapy in the adult critically ill patient: Society of Critical Care Medicine (SCCM) and American Society for Parenteral and Enteral Nutrition (A.S.P.E.N.). *JPEN J Parenter Enteral Nutr.* 2016 Feb;40(2):159-211.

4. Pearce CB, Sadek SA, Walters AM, et al. A double-blind, randomised, controlled trial to study the effects of an enteral feed supplemented with glutamine, arginine, and omega-3 fatty acid in predicted acute severe pancreatitis. *JOP.* 2006 Jul 10;7(4):361-371.

5. Windsor AC, Kanwar S, Li AG, et al. Compared with parenteral nutrition, enteral feeding attenuates the acute phase response and improves disease severity in acute pancreatitis. *Gut.* 1998 Mar;42(3):431-435.

Part 9.

Liver

Instructions: For each question, select the most correct answer.

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1. A 28-year-old man is admitted to the surgical ICU with septic shock from a perforated viscus. Past medical history is significant for untreated Crohn disease and a seizure disorder controlled with valproic acid. He undergoes emergent exploratory laparotomy with resection of transverse colon and is left in discontinuity. Postoperatively, he is hypoglycemic despite multiple doses of dextrose 50% with increasing lactate level. ALT is 3984 U/L and AST greater than 10,000 U/L. Ammonia is 540 umol/L. He is somnolent but arousable. Which of the following is the most appropriate pharmacologic treatment of his hyperammonemia?

   A. Lactulose per rectum

   B. Oral rifaximin

   C. IV levocarnitine

   D. No pharmacologic treatment

2. A 58-year-old man with a history of chronic hepatitis C-related cirrhosis is admitted to the ICU for hematemesis. Vital signs are blood pressure 78/42 mm Hg, pulse rate 125 beats/min, respiratory rate 18 breaths/min, and temperature 37.3°C (99°F). Hemoglobin is 5.5 g/dL and lactic acid 4.3 mg/dL. He is administered 1 L 0.9% normal saline; 2 units packed RBCs; and octreotide, 50 μg/hr, and is started on a proton pump inhibitor infusion. Esophagogastroduodenoscopy (EGD) is performed with band ligation of multiple bleeding varices in the

distal third of the esophagus, after which bleeding is controlled. In addition to EGD, which of the following interventions has/have been shown to improve survival in a patient such as this one?

A. Urgent transjugular intrahepatic portosystemic shunt

B. Albumin, 1 g/kg, and norepinephrine

C. Transfusion of RBCs to target hemoglobin greater than 10 g/dL

D. Repeat EGD in 48 hours

E. Ceftriaxone

3. A 37-year-old woman undergoing liver transplant has hemoglobin 6.2, INR 1.7, and platelets 150,000/mL. Rapid thromboelastography values are R time 7 minutes (normal range 3.8-9.8 minutes), α-angle 34 degrees (normal range 47.8-77.7 degrees), MA 40 mm (normal range 49.7-72.7 mm), and LY30 2.5% (normal range 2.3-5.77%). Which of the following resuscitation strategies is optimal for her transplant?

   A. Cryoprecipitate and fresh frozen plasma (FFP)

   B. Cryoprecipitate and platelets

   C. FFP and platelets

   D. Cryoprecipitate and tranexamic acid bolus

4. A 56-year-old man with a history of chronic hepatitis B is admitted to the ICU with septic shock. A medical intern sustains a needle stick injury while placing a central venous catheter. The intern says that he has received 1 dose of the hepatitis B vaccine but that he may have forgotten to receive any additional doses. Laboratory studies reveal that the intern's hepatitis B surface antibody is negative. Which of the following is the most appropriate postexposure management for the intern?

A. Administer hepatitis B immunoglobulin and hepatitis B vaccine.

B. Administer hepatitis B vaccine only.

C. Express fluid by squeezing the wound and then cleansing with antiseptic solution.

D. Administer prophylactic ledipasvir and sofosbuvir.

5. A 47-year-old woman with a past medical history significant for ethanol-associated cirrhosis presents febrile with diffuse abdominal pain. Laboratory findings are significant for WBC count 26 × 10^9 cells/L, platelet count 68 × 10^3/μL, sodium 123 mEq/L, BUN 65 mg/dL, serum creatinine 1.3 g/dL, total bilirubin 2.4 mg/dL, and ammonia 78 μg/dL. Paracentesis with fluid analysis reveals total polymorphonuclear cell count 430 cells/mm^3. She has no known allergies. Which of the following therapies is most appropriate for management of this patient's spontaneous bacterial peritonitis?

   A. IV ceftriaxone, 1 g

   B. IV ceftriaxone, 1 g; oral midodrine, 7.5 mg 3 times daily; and subcutaneous octreotide, 50 μg 3 times daily

   C. Octreotide, 50 μg bolus followed by 50 μg/hr infusion plus 5 doses of IV ceftriaxone, 1 g

   D. Norepinephrine infusion plus albumin, 1 g/kg on days 1 and 2, followed by 20-40 g daily

   E. IV ceftriaxone, 1 g, plus albumin, 1.5 g/kg on day 1 and 1 g/kg on day 3 after recognition

Part 9 Answers:

Liver

1. Rationale

Answer: C

Because this patient is in discontinuity, neither oral nor per-rectum lactulose titrated to 2 to 3 bowel movements per day is appropriate. Rifaximin must be given orally and is typically reserved for patients with insufficient response or intolerance to lactulose. Valproic acid can induce hyperammonemic encephalopathy, which can occur due to carnitine deficiency.

References:

1. Lheureux P, Penaloza A, Zahir S, Gris M. Science review: carnitine in the treatment of valproic acid-induced toxicity: what is the evidence? *Crit Care.* 2005 Oct 5;9(5):431-440.
2. Valproate. In: *LiverTox: Clinical and Research Information on Drug-Induced Liver Injury [Internet].* National Institute of Diabetes and Digestive and Kidney Diseases; 2012. Valproate. Last update July 31, 2020.

2. Rationale

Answer: E

Therapy goals in patients with bleeding esophageal varices are to restore and maintain hemodynamic instability, maintain adequate oxygenation, control bleeding with early (< 12 hours) esophagogastroduodenoscopy by sclerotherapy or band ligation, and prevent complications. Prophylactic antibiotics are preferably administered before endoscopy (although effectiveness has also been shown when administered after endoscopy). A broad-spectrum antibiotic such as IV ceftriaxone, 1 g daily for 7 days, is recommended. For patients who are discharged before 7 days of IV antibiotic therapy, transitioning to an oral antibiotic such as ciprofloxacin, 500 mg every 12 hours, to complete a total of 7 days of antibiotic

therapy is recommended. Prophylactic antibiotics reduce the risk of mortality, infection (eg, spontaneous bacterial peritonitis, urinary tract infection), and rebleeding. In a meta-analysis of 12 trials including more than 1200 patients with cirrhosis and gastrointestinal bleeding, antibiotic prophylaxis decreased mortality and rebleeding compared with either placebo or no intervention. Blood transfusion is warranted if hemoglobin is less than 7 g/dL for most patients, with a goal of maintaining hemoglobin level at greater than or equal to 7 g/dL and less than 9 g/dL. However, for patients at increased risk of adverse events in the setting of anemia (ie, those with unstable coronary artery disease or ongoing active bleeding), the goal is to maintain hemoglobin level at greater than or equal to 9 g/dL. Intravascular volume status must be monitored to avoid volume overload because of the risk of rebound portal hypertension and rebleeding. Albumin and norepinephrine are administered as part of resuscitation to maintain hemodynamic stability and may be used to treat hepatorenal syndrome. In patients who have rebleeding after initial control of bleeding with endoscopic therapy, a second session of endoscopic treatment is advocated rather than proceeding to transjugular intrahepatic portosystemic shunt or surgery.

References:

1. Chavez-Tapia NC, Barrientos-Gutierrez T, Tellez-Avila F, et al. Meta-analysis: antibiotic prophylaxis for cirrhotic patients with upper gastrointestinal bleeding: an updated Cochrane review. *Aliment Pharmacol Ther.* 2011 Sep;34(5):509-518.

2. Hou MC, Lin HC, Liu TT, et al. Antibiotic prophylaxis after endoscopic therapy prevents rebleeding in acute variceal hemorrhage: a randomized trial. *Hepatology*. 2004 Mar;39(3):746-753.

3. Villanueva C, Colomo A, Bosch A, et al. Transfusion strategies for acute upper gastrointestinal bleeding. *N Engl J Med.* 2013 Jan 3;368(1):11-21.

3. Rationale

Answer: B

In patients with chronic liver disease, traditional coagulation tests can indicate liver dysfunction but do not quantify bleeding risk. Thromboelastography (TEG) has been shown to reduce blood product administration in perioperative liver transplant patients. This patient's TEG results seem to indicate coagulopathy of fibrinogen and platelet function whereas the traditional coagulation parameters would indicate the need for fresh frozen plasma and no need for platelets.

Reference:

1. Hawkins RB, Raymond SL, Hartjes T, et al. Review: the perioperative use of thromboelastography for liver transplant patients. *Transplant Proc.* 2018 Dec;50(10):3552-3558.

4. Rationale

Answer: A

The intern's hepatitis B antibody is negative, suggesting that he does not have hepatitis B immunity. Because the source patient has known chronic hepatitis B, the intern should receive hepatitis B immunoglobulin, which provides passive protection against hepatitis B for approximately 3 to 6 months. In addition, the intern should receive 1 dose of the hepatitis B vaccine as soon as possible after the exposure and should subsequently complete the rest of the 3-dose vaccination schedule (0, 1, and 6 months). Wounds that have been in contact with body fluid should be washed with soap and water. Antiseptics and wound expression have not been shown to reduce hepatitis B virus transmission. Ledipasvir and sofosbuvir are antivirals used for hepatitis C but are not used for hepatitis B postexposure prophylaxis.

Reference:

1. Schillie S, Murphy TV, Sawyer M, et al; Centers for Disease Control and Prevention (CDC). CDC guidance for evaluating health-care personnel for hepatitis B virus protection and for administering protection for and

administering postexposure management. *MMWR Recomm Rep*. 2013 Dec 20;62(RR10):1-19.

5. Rationale

Answer: E

Ceftriaxone in combination with albumin on days 1 and 3 is the most appropriate therapy for this patient with spontaneous bacterial peritonitis. Guidelines recommend a third-generation cephalosporin or equivalent for gram-negative coverage. In addition, there is a high risk of hepatorenal syndrome and subsequent decline. Research has identified a protective effect of early upfront albumin in patients with serum creatinine level greater than 1 mg/dL, BUN greater than 30 mg/dL, or total bilirubin greater than 4 mg/dL. IV ceftriaxone, 1 g, is appropriate antimicrobial therapy but incomplete for the therapeutic requirements of this high-risk patient because it does not include albumin. Norepinephrine or midodrine in combination with antimicrobials or albumin is recommended for patients with hepatorenal syndrome; however, this therapy has not shown any benefit in the treatment of spontaneous bacterial peritonitis. The combination of ceftriaxone and octreotide is appropriate therapy for the treatment of a variceal bleed in this population but, based on her status, it is inappropriate.

References:

1. Garcia-Tsao G, Abraldes JG, Berzigotti A, Bosch J. Portal hypertensive bleeding in cirrhosis: risk stratification, diagnosis, and management: 2016 practice guidance by the American Association for the Study of Liver Diseases. *Hepatology*. 2017 Jan;65(1):310-335.

2. Runyon BA; Practice Guidelines Committee, American Association for the Study of Liver Diseases (AASLD). Management of adult patients with ascites due to cirrhosis. *Hepatology*. 2004 Mar;39(3):841-856.

3. Sigal SH, Stanca CM, Fernandez J, Arroyo V, Navasa M. Restricted use of albumin for spontaneous bacterial peritonitis. *Gut*. 2007 Apr;56(4):597-599.

4. Sort P, Navasa M, Arroyo V, et al. Effect of intravenous albumin on renal impairment and mortality in patients with cirrhosis and spontaneous bacterial peritonitis. *N Engl J Med*. 1999 Aug 5;341(6):403-409.

Part 10.

Neurologic Disorders

Instructions: For each question, select the most correct answer.

1. A 36-year-old man was admitted to the ICU after a motorcycle accident in which he sustained a severe traumatic brain injury with bilateral frontal contusions and traumatic subarachnoid hemorrhage. On initial examination, Glasgow Coma Scale score is 7. An intracranial pressure (ICP) monitor is placed. His head of bed is maintained at 30 degrees, sodium maintained at 145 to 150 mmol/L, and CO_2 has been consistently between 35- and 40-mm Hg. During the first 3 days of admission, ICP has remained stable at 19 mm Hg. His most recent ICP pressure waveform is shown below. Which of the following is the most appropriate next step in management?



A. Administer 200-mL bolus of 6% sodium chloride to increase sodium to new goal of 150-155 mmol/L.

B. Continue to monitor ICP and maintain all current therapies.

C. Consult neurosurgery for emergent decompressive craniectomy.

D. Increase respiratory rate to decrease his CO_2 to 30 to 35 mm Hg.

E. Increase sedation to increase his comfort.

2. A 34-year-old man with injuries sustained during a high-speed motor vehicle crash has hemorrhagic shock and is undergoing fluid resuscitation. Head CT shows bilateral frontal contusion with cerebral edema and decreased ventricular size. Which of the following is the most appropriate agent for treating his intracranial hypertension?

   A. Mannitol

   B. Dexamethasone

   C. Pentobarbital

   D. 3% saline

3. A 66-year-old woman underwent a minor orthopedic procedure under nerve block and propofol sedation. She has schizophrenia, for which she takes chlorpromazine. After the procedure, she has a fever of 39.4°C (102.9°F), heart rate of 121 beats/min, muscle rigidity, and confusion. Laboratory analysis shows: WBC count 9000/μL, hemoglobin 13.2 g/dL, and creatinine phosphokinase 3600 U/L. Which of the following is the most likely diagnosis?

   A. Serotonin syndrome

   B. Malignant hyperthermia

   C. Neuroleptic malignant syndrome

   D. Central anticholinergic syndrome

4. A 90-kg (198-lb), 45-year-old man is admitted to the neurotrauma ICU after falling off his roof while hanging Christmas lights. His injuries include a C3 fracture, multiple bilateral rib fractures, and an open right femur fracture. He has significant deficits in all 4 extremities. His only past medical history is hyperlipidemia. Blood pressure is 110/50 mm Hg, with mean arterial pressure (MAP) 70 mm Hg, and heart rate 55 beats/min. Which of the following treatments is most appropriate?

   A. Norepinephrine titrated to MAP 85 mm Hg

   B. Dopamine titrated to MAP 80 mm Hg

   C. Phenylephrine titrated to MAP 85 mm Hg

   D. No treatment

5. Which of the following outcomes are most likely for patients who are deeply sedated during their ICU stays?

   A. Increased hospital and 6-month mortality, longer ICU length of stay (LOS), longer course of mechanical ventilation, and increased risk of post-intensive care syndrome (PICS)

   B. Increased hospital mortality, decreased 6-month mortality, longer ICU LOS, longer course of mechanical ventilation, and decreased risk of PICS

   C. Decreased hospital and 6-month mortality, shorter ICU LOS, shorter course of mechanical ventilation, and decreased risk of PICS

   D. Decreased hospital mortality, increased 6-month mortality, longer ICU LOS, longer course of mechanical ventilation, and increased risk of PICS

6. A 65-year-old man is evaluated in the emergency department for altered mentation and difficulty speaking. He was last seen normal before he went to sleep at 10:00 p.m. last night. This morning when he awoke at 6:00 a.m., he had difficulty speaking, right-sided weakness, and confusion. Noncontrast head CT shows no bleeding or acute pathology. Blood pressure is 170/85 mm Hg and heart rate is 110 beats/min and irregular. Telemetry strip shows atrial fibrillation. He takes no anticoagulant medications, and his INR is normal. Diffusion-weighted MRI shows a lesion consistent with a middle cerebral artery ischemic stroke, but this lesion is not seen on fluid-attenuated inversion recovery MRI. The most appropriate next step is

   A. full anticoagulation with IV heparin including initial bolus dose.

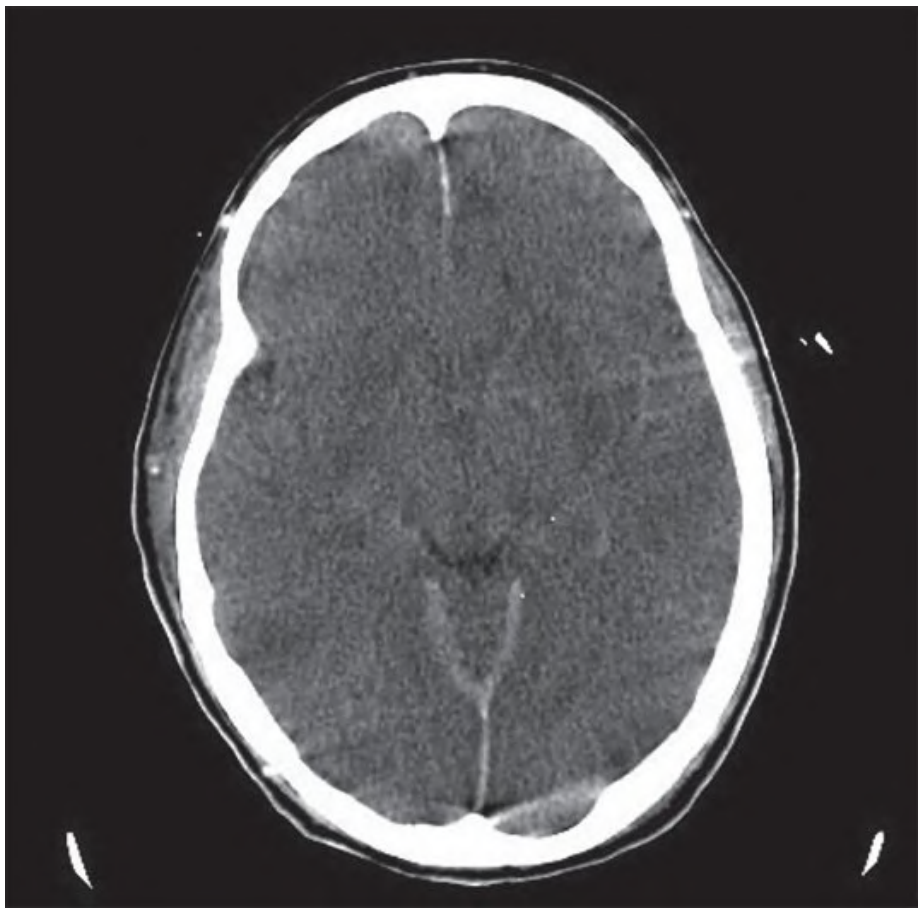
   B. full anticoagulation with oral apixaban, 5 mg twice daily.

   C. IV thrombolytic therapy with recombinant human tissue plasminogen activator.

   D. IV nicardipine infusion for reducing blood pressure to 140/80 mm Hg.

7. A 70-year-old man with a history of coronary artery disease and hypertension was having his morning coffee at home when he suddenly slumped in his chair at the table. His wife called emergency medical services, who arrived 5 minutes later and performed CPR. Initial rhythm was ventricular fibrillation. Return of spontaneous circulation was achieved in 15 minutes. Therapeutic hypothermia was initiated, and he was cooled to 36°C (96.8°F) for the next 24 hours. He was sedated with propofol. Blood pressure is

120/70 mm Hg on norepinephrine infusion and his sedating medications are stopped. About 80 hours after rewarming, he is still unresponsive to painful stimuli. Pupillary and corneal reflexes are absent, as are oculocephalic and oculovestibular reflexes and cough and gag reflexes. Head CT without contrast is shown below. Brain death examination is planned. Which of the following would preclude proceeding with the examination?



A. Core body temperature 36°C (96.8°F)

B. All sedating drugs discontinued for more than 5 half-lives

C. Serum sodium 175 mEq/L

D. Hypotension requiring vasopressors to maintain systolic blood pressure above 100 mm Hg

8. A 44-year-old woman with a previous medical history significant for depression treated with citalopram is brought to the ICU with difficulty breathing. The decision is made to secure the airway. She is successfully intubated using midazolam, fentanyl, propofol, and succinylcholine. Approximately 30 minutes later, the resident reports that the physical examination has changed as follows: temperature 38.8°C (102°F), heart rate 130 beats/min, increased muscle rigidity, hyper-deep tendon reflexes in the lower extremity,

profound diaphoresis, and end-tidal CO_2 47 mm Hg. Which of the following drugs most likely contributed to these changes?

A. Propofol

B. Fentanyl

C. Midazolam

D. Succinylcholine

9. A 39-year-old man is admitted to the ICU after a motor vehicle collision. He is intubated and ventilated on minimal sedation. He is abnormally flexing with upper limbs bilaterally and extending his lower limbs. He has a dysconjugate gaze, but his pupils are reactive to light. Blood pressure is 118/46 mm Hg (mean arterial pressure 70 mm Hg), and heart rate 120 beats/min. Head CT shows diffuse swelling and small petechial hemorrhages in bilateral basal ganglia, with 0.3-mm midline shift from left to right. Neurosurgery has placed an intraparenchymal pressure monitor, which shows an intracranial pressure (ICP) of 32 mm Hg. Which of the following treatments is the most appropriate next step?

   A. Induced barbiturate coma for 72 hours

   B. Sedation with propofol to maintain ICP below 20 mm Hg

   C. Emergent bilateral decompressive craniectomy

   D. Administration of IV dexamethasone

   E. Administration of IV bolus of 3% saline

10. A 68-year-old woman with a past medical history of poorly controlled diabetes, liver failure, and osteoporosis experienced dizziness followed by clenched teeth, irregular breathing, loss of consciousness, convulsions, and incontinence. In the emergency department, she is administered lorazepam. Home medications include metformin, alendronate, and selegiline. Her aid witnessed the seizure and said that it lasted 8 minutes. EEG reveals seizure activity. ECG shows first-degree atrioventricular block. Current heart rate is 55 beats/min. AST is 359 IU/L, ALT 162 IU/L, and sodium 129 mEq/L. Which of the following medications is most appropriate to treat her status epilepticus?

    A. Levetiracetam

    B. Oxcarbazepine

    C. Propofol

    D. Phenobarbital

E. Valproic acid

11. A 76-year-old man is admitted to the surgical ICU after undergoing open abdominal aortic aneurysm repair. Which of the following pharmacologic interventions has been reported to reduce the incidence of delirium in ICU patients?

    A. Oral haloperidol

    B. IV haloperidol

    C. Overnight IV dexmedetomidine infusion

    D. Oral quetiapine

12. A 51-year-old man with bilateral internal carotid stenosis of 90% on the left and 80% on the right is admitted to the ICU after a left carotid endarterectomy. Which of the following is the most appropriate treatment strategy for optimal cerebral blood flow?

    A. Labetalol infusion to reduce systolic blood pressure (SBP) to less than 100 mm Hg

    B. Nicardipine infusion to reduce SBP to less than 120 mm Hg

    C. Nitroglycerin infusion to reduce SBP to less than 150 mm Hg

    D. Phenylephrine infusion to maintain SBP above 140 mm Hg

    E. Norepinephrine infusion to maintain SBP above 160 mm Hg

13. A 58-year-old woman presents to the emergency department for throbbing headaches. She admits to not refilling her antihypertensive prescription. Her blood pressure is 200/120 mm Hg. She is beginning to demonstrate decreased mental status. Initial management should include

    A. IV calcium channel blocker for goal mean arterial pressure (MAP) 110 mm Hg within 1 hour for hypertensive emergency.

    B. IV beta-blocker for goal MAP 90 mm Hg within 1 hour for hypertensive emergency.

    C. oral calcium channel blocker for goal blood pressure 160/100 mm Hg within 2 to 6 hours for hypertensive urgency.

    D. oral angiotensin-converting enzyme inhibitor for goal blood pressure 120/80 mm Hg within 24 to 48 hours for hypertensive urgency.

14. A 46-year-old woman presents to the emergency department after sudden onset of confusion, seizure, and headache. Blood pressure is 146/83 mm Hg, heart rate 104 beats/min, point-of-care glucose 93

mg/dL, and oxygen saturation 94%. Head CT shows no obvious signs of bleeding or early ischemia. Her seizures continually progress. The decision is made to intubate for airway protection, and emergent brain MRI is performed, which shows bilateral white matter vasogenic edema around the posterior parieto-occipital regions. Basic metabolic panel and complete blood count show no abnormalities from known baseline. Past medical history is significant for renal failure, for which she received a kidney transplant 2 years ago; diabetes mellitus; and migraine. Her medication regimen includes tacrolimus, 1 mg twice daily; metformin, 1 g twice daily; and acetaminophen as needed for migraine. She had a hysterectomy 6 years ago. Which of the following is the most likely cause of her presentation?

A. Medication regimen

B. Elevated blood pressure

C. Abnormal renal function

D. Preeclampsia

15. A 53-year-old man is evaluated in the emergency department for acute-onset lethargy and right leg weakness. He is found to have a subarachnoid hemorrhage and is admitted to the ICU. Shortly after arrival, he develops acute respiratory failure and is intubated. He is afebrile, with blood pressure 150/75 mm Hg and heart rate 122 beats/min. Lung examination reveals bilateral crackles. Pink frothy secretions are noted with suctioning. PaO_2 is 78 mm Hg with FIO_2 1.0. Chest radiograph reveals bilateral ground-glass opacifications. Echocardiography reveals normal left ventricular ejection fraction. Which of the following is the most appropriate treatment approach for his condition?

A. Administer furosemide.

B. Administer phentolamine.

C. Administer methylprednisolone.

D. Assess volume status and maintain euvolemia.

E. Initiate empiric antibiotic therapy for aspiration pneumonia.

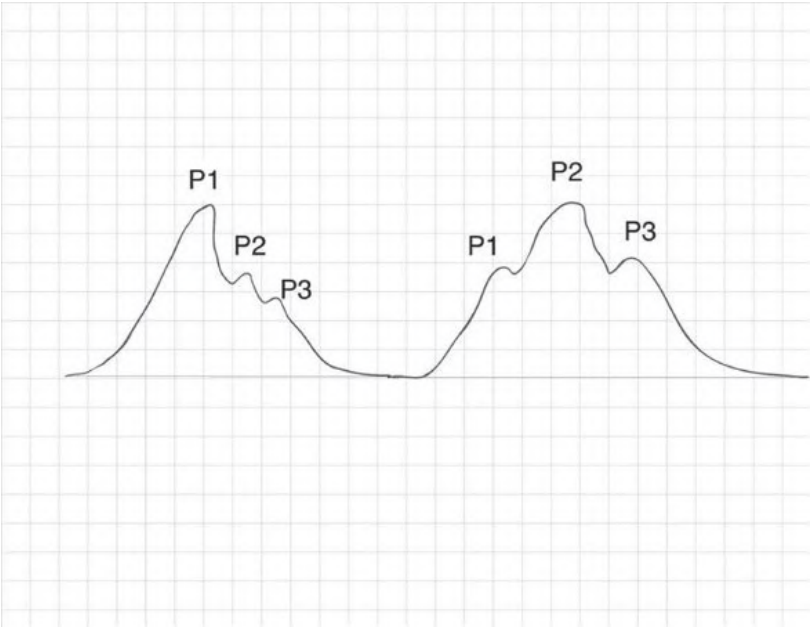
Part 10 Answers:

Neurologic Disorders

1. Rationale

Answer: A

The first intracranial pressure (ICP) waveform has 3 distinct peaks. P1 is related to the arterial pulse wave. P2 is the rebound of the arterial pulse wave and is inversely related to cerebral compliance. P3 is related to the dicrotic notch in the arterial pressure waveform and represents the closing of the aortic valve. In the ICP waveform shown, the first wave of the series shows normal ICP where P1 is the highest wave, followed by P2 and P3. In the second wave of the series, P2 is higher than P1, which demonstrates decreasing compliance in the brain. The decrease in brain compliance is likely due to elevating ICP and worsening cerebral edema. With decreased compliance, a small increase in intracranial contents such as increasing edema can cause a large change in ICP. Therefore, the most appropriate next step is to increase the sodium goal and try to further decrease edema. The current sodium goal is not at its maximum effective range so it should be maximized before moving on to different tiers of therapy such as decompressive craniectomy. Hyperventilation to decrease CO_2 is a temporizing measure and will decrease ICP but should not be used long-term because it can lead to ischemia. There is no evidence that this patient is not adequately sedated, so sedation is less likely to make a difference. Continuing to monitor him is inappropriate because there is evidence of worsening intracranial compliance with an already borderline ICP. Taking preventive steps once the waveform change is realized can prevent worsening ICP elevation.



References:

1. Di Ieva A, Schmitz EM, Cusimano MD. Analysis of intracranial pressure: past, present, and future. *Neuroscientist*. 2013 Dec;19(6):592-603.

2. Jha RM, Kochanek PM. A precision medicine approach to cerebral edema and intracranial hypertension after severe traumatic brain injury: Quo vadis? *Curr Neurol Neurosci Rep.* 2018 Nov 8;18(12):105.

2. Rationale

Answer: D

Osmotic agents are first-line treatment for elevated intracranial pressure. Hypertonic saline preserves or increases intravascular volume, while mannitol decreases intravascular volume through its diuretic effect. Thus, in a hypovolemic patient, hypertonic saline is the most appropriate osmotic agent. Dexamethasone is used to treat vasogenic edema associated with intracranial lesions such as tumors but has not been shown to be of benefit in the cytotoxic edema associated with traumatic brain injury. Pentobarbital reduces intracranial pressure but is not a first-line agent and is reserved for refractory intracranial hypertension because of its hemodynamic complications and the inability to conduct neurologic examinations on a patient in an induced coma.

Reference:

1. Koenig, MA. Cerebral edema and elevated intracranial pressure. *Continuum (Minneap Minn).* 2018 Dec;24(6):1588-1602.

3. Rationale

Answer: C

Neuroleptic malignant syndrome (NMS) is an infrequent life-threatening condition characterized by delirium, muscular rigidity, fever, autonomic nervous system dysregulation, and elevated creatinine phosphokinase (CPK) (1000-100,000 U/L) associated with the administration of typical and atypical antipsychotic drugs. Malignant hyperthermia can present with similar symptoms, but this patient was not exposed to triggering agents (succinylcholine, inhaled anesthetics). Central anticholinergic syndrome does not cause elevated CPK or muscle rigidity. Serotonin syndrome does not present with elevated CPK or muscle rigidity (although clonus and hyper-reflexivity are present), and there is no history of exposure to serotonergic agents.

Reference:

1. Tse L, Barr Am, Scarapicchia V, Vila-Rodriquez F. Neuroleptic malignant syndrome: a review forms a clinically oriented perspective. *Curr Neuropharmacol.* 2015;13(3):395-406.

4. Rationale

Answer: A

American Association of Neurological Surgeons/Congress of Neurological Surgeons guidelines recommend correcting systolic blood pressure less than 90 mm Hg and maintaining mean arterial pressure 85 to 90 mm Hg for the first 7 days after spinal cord injury. This recommendation is based on class III evidence that has consistently shown that blood pressure augmentation results in higher American Spinal Injury Association Impairment Scale scores during this time. Norepinephrine has been compared to dopamine for use in

spinal cord injury and has been found to decrease intrathecal pressure resulting in increased spinal cord perfusion pressure; therefore, norepinephrine is preferred to dopamine. This patient is borderline bradycardic at baseline so phenylephrine should be avoided.

References:

1. Altaf F, Griesdale DE, Belanger L, et al. The differential effects of norepinephrine and dopamine on cerebrospinal fluid pressure and spinal cord perfusion pressure after acute human spinal cord injury. *Spinal Cord*. 2017 Jan;55(1):33-38.
2. Ryken TC, Hurlbert RJ, Hadley MN, et al. The acute cardiopulmonary management of patients with cervical spinal cord injuries. *Neurosurgery*. 2013 Mar;72 Suppl 2:84-92.
3. Yue JK, Tsolinas R, Burke JF, et al. Vasopressor support in managing acute spinal cord injury: a knowledge update. *J Neurosurg Sci*. 2019 Jun;63(3):308-317.

5. Rationale

Answer: A

Multiple studies indicate that deep sedation has significant negative consequences, including increased hospital mortality, increased ICU length of stay, and longer course of mechanical ventilation. Shehabi demonstrated reduced 6-month survival with deep sedation. Multiple studies show that increased sedation correlates directly with more cognitive dysfunction, which is part of post-intensive care syndrome (PICS). Girard showed that increased sedation directly correlated with increased posttraumatic stress disorder and more disturbing memories of hospitalization. With deep sedation leading to a longer course of mechanical ventilation, patients are also at risk for a physical aspect of PICS known as ICU-acquired neuromuscular weakness.

References:

1. Girard TD, Kress JP, Fuchs BD, et al. Efficacy and safety of a paired sedation and ventilator weaning protocol for mechanically ventilated patients in intensive care (awakening and breathing controlled trial): a randomized controlled trial. *Lancet*. 2008 Jan 12;371(9607):126-134.

2. Kress JP, Pohlman AS, O'Connor MF, Hall JB. Daily interruption of sedative infusions in critically ill patients undergoing mechanical ventilation. *N Engl J Med*. 2000 May 18;342(20):1471-1477.

3. Shehabi Y, Bellomo R, Reade MC, et al; Sedation Practice in Intensive Care Evaluation (SPICE) Study Investigators; ANZICS Clinical Trials Group. Early intensive care sedation predicts long-term mortality in ventilated critically ill patients. *Am J Respir Crit Care Med*. 2012 Oct 15;186(8):724-731.

4. Rawal G, Yadav, Kumar R. Post-intensive care syndrome: an overview. *J Transl Int Med*. 2007 Jun 30;5(2):90-92.

5. Treggiari MM, Romand JA, Yanez ND, et al. Randomized trial of light versus deep sedation on mental health after critical illness. *Crit Care Med*. 2009 Sep;27(9):2527-2534.

6. Rationale

Answer: C

Thrombolytic therapy has been shown to improve outcome in acute ischemic stroke if administered within 4.5 hours of symptom onset, if there are no contraindications. The time requirement leaves out people who wake with stroke symptoms. Two studies have looked at extending the time frame of thrombolytic administration. The WAKE-UP trial performed MRI on patients who awoke with stroke symptoms. Thrombolytics were administered if a lesion was seen on diffusion-weighted images and not on fluid-attenuated inversion recovery MRI (signifying stroke onset within 4.5 hours). Patients receiving thrombolytics had a significantly better outcome. The EXTEND trial performed CT/MRI perfusion studies on patients presenting between 4.5 and 9 hours after stroke onset. If mismatch lesions were seen,

thrombolytic therapy was associated with significantly improved outcome. Full anticoagulation for atrial fibrillation immediately after stroke onset has not been associated with improved outcome and may cause increased intracranial bleeding. Patients administered thrombolytic therapy need blood pressure control, with a goal of less than 180/105 mm Hg. If thrombolytics are not administered, then systolic pressure as high as 220 mm Hg can be tolerated.

References:

1. Gotz T, Simonsen CZ, Foutitie F, et al; WAKE-UP Investigators. MRI-guided thrombolysis for stroke with unknown time of onset. *N Engl J Med*. 2018 Aug 16;379(7):611-622.
2. Ma H, Campbell BCV, Parsons MW, et al; EXTEND Investigators. Thrombolysis guided by perfusion imaging up to 9 hours after onset of stroke. *N Engl J Med*. 2019 May 9;380(19):1795-1803.

7. Rationale

Answer: C

Serum sodium level of 175 mEq/L precludes performance of brain death examination because electrolytes must be within or close to normal limits; otherwise, they can confound brain death examination. Brain death examination would not be precluded by core body temperature 36°C (96.8°F), discontinuation of sedating drugs for more than 5 half-lives, or hypotension requiring vasopressors to maintain systolic blood pressure above 100 mm Hg. A patient who is thought to meet the criteria for brain death determination should meet certain additional criteria before a formal determination can be made. Core temperature should be higher than 36°C (96.8°F) and all confounding medications such as sedatives and paralytics should be out of the body (more than 5 half-lives or undetectable levels in blood). Renal and hepatic functions should be considered while determining drug intoxication or the presence of drugs in the body. The patient should be hemodynamically stable (maybe even with the help of

vasopressors) and should not be hypoxic and should require minimum ventilator support (positive end-expiratory pressure 5-8 cm H_2O, $FIO_2 < 0.50$).

References:

1. Spinello IM. Brain death determination. *J Intensive Care Med*. 2015 Sep;30(6):326-337.
2. Wijdicks EF, Varelas PN, Gronseth GS, Greer DM; American Academy of Neurology. Evidence-based guideline update: determining brain death in adults: report of the Quality Standards Subcommittee of the American Academy of Neurology. *Neurology*. 2010 Jun 8;74(23):1911-1918.

8. Rationale

Answer: B

This patient has serotonin syndrome, which can be difficult to differentiate from neuroleptic malignant syndrome (NMS) and malignant hyperthermia (MH), all of which require different management strategies. None of the agents she received can precipitate NMS. Succinylcholine can trigger MH. However, end-tidal CO_2 is only minimally elevated, and MH usually presents with decreased reflexes. Of serotonin syndrome, NMS, and MH, only serotonin syndrome presents with hyperreflexia. In this patient, citalopram may have interacted with fentanyl to precipitate serotonin syndrome.

References:

1. Ailawadhi S, Sung KW, Carlson LA, Baer MR. Serotonin syndrome caused by interaction between citalopram and fentanyl. *J Clin Pharm Ther*. 2007 Apr;32(2):199-202.
2. Kirschner R, Donovan JW. Serotonin syndrome precipitated by fentanyl during procedural sedation. *J Emerg Med*. 2010 May;38(4):477-480.
3. Kitamura S, Kawano T, Kaminaga S, et al. Effects of fentanyl on serotonin syndrome-like behaviors in rats. *J Anesth*. 2016 Feb;30(1):178-182.

9. Rationale

Answer: E

This patient has a serious traumatic brain injury (TBI) with evidence of diffuse brain injury (petechiae on CT) and evidence of less-than-adequate cerebral perfusion. Mean arterial pressure of 70 mm Hg and intracranial pressure (ICP) of 32 mm Hg result in cerebral perfusion pressure (CPP) of 38 mm Hg, which is less than the target CPP of 60 to 70 mm Hg suggested by Brain Trauma Foundation guidelines. Barbiturate coma is a second-tier therapy in this context and may further reduce blood pressure. Similarly, propofol sedation to reduce ICP may compromise blood pressure and overall, negatively affect CPP. Emergent decompressive craniectomy has been shown to be ineffective in the context of TBI, as have steroids (which are probably deleterious). Hypertonic saline will reduce ICP via an osmotherapeutic effect while maintaining circulating volume and avoiding hypotension.

References:

1. Carney N, Totten AM, O'Reilly C, et al. Guidelines for the management of severe traumatic brain injury, fourth edition. *Neurosurgery*. 2017 Jan 1;80(1):6-15.

2. Cooper DJ, Rosenfeld JV, Murray L, et al; DECRA Trial Investigators; Australian and New Zealand Intensive Care Society Clinical Trials Group. Decompressive craniectomy in diffuse traumatic brain injury. *N Engl J Med*. 2011 Apr 21;364(16):1493-1502.

10. Rationale

Answer: A

According to both American Epilepsy Society and Neurocritical Care Society guidelines for treatment of status epilepticus, levetiracetam is a recommended agent for benzodiazepine-refractory seizures. Oxcarbazepine is not a first-line agent, according to the status epilepticus guidelines, and is available only for enteral administration, which is not an ideal administration route for patients with status epilepticus. Oxcarbazepine can also cause hyponatremia and is not

recommended in a patient who is already hyponatremic. According to the guidelines, propofol is a third-line therapy, so a patient should generally have failed another antiepileptic drug before propofol is considered since it requires intubation. Valproic acid has a black box warning for hepatotoxicity, and phenobarbital should be used cautiously in patients with hepatic impairment. Therefore, neither of these drugs is appropriate in this scenario as a first-line agent for a patient with liver failure.

Reference:

1. Glauser T, Shinnar S, Gloss D, et al. Evidence-based guideline: treatment of convulsive status epilepticus in children and adults: report of the Guideline Committee of the American Epilepsy Society. *Epilepsy Curr.* 2016 Jan-Feb;16(1):48-61.

11. Rationale

Answer: C

This patient is at high risk for ICU delirium and should be screened with the Confusion Assessment Method for the ICU. Nonpharmacologic interventions including reorientation, cognitive stimulation, and improving sleep should be used for all ICU patients. Emerging data indicate that nocturnal infusion of dexmedetomidine in critically ill patients can reduce the incidence of ICU delirium. Delirium guidelines do not recommend for or against this treatment; more research is needed. Neither oral nor IV haloperidol nor oral quetiapine have been found to prevent ICU delirium.

References:

1. Devlin JW, Skrobik Y, Gelinas C, et al. Clinical practice guidelines for the prevention and management of pain, agitation/sedation, delirium, immobility, and sleep disruption in adult patients in the ICU. *Crit Care Med*. 2018 Sep;46(9): e825-e873.

2. Skrobik Y, Duprey MS, Hill NS, Devlin JW. Low-dose nocturnal dexmedetomidine prevents ICU delirium. A

randomized, placebo-controlled trial. *Am J Respir Crit Care Med*. 2018 May 1;197(9):1147-1156.

12. Rationale

Answer: B

This patient is vulnerable to hyperperfusion syndrome with bilateral stenosis greater than 80%. Endarterectomy presents excessive blood flow before recovery of autoregulation with consequent risk of intracranial hypertension and hemorrhage. Nicardipine is an effective agent titrated to a moderate goal and is commonly used in this circumstance. Labetalol is not contraindicated, but a goal of 100 mm Hg is too aggressive with risk of ischemic stroke. Nitroglycerin can lead to vasodilation, augmenting any existing cerebral swelling. Phenylephrine and norepinephrine will worsen hypertension, especially to these targets, since blood pressure should ideally be less than 120 mm Hg in this age group.

References:

1. Lieb M, Shah U, Hines GL. Cerebral hyperperfusion syndrome after carotid intervention: a review. *Cardiol Rev*. 2012 Mar-Apr;20(2):84-89.

2. Nyamekye IK, Begum S, Slaney PL. Post-carotid endarterectomy cerebral hyperperfusion syndrome. *J R Soc Med*. 2005 Oct;98(10):472-474.

3. Van Mook WN, Rennenberg RJ, Schurink GW, et al. Cerebral hyperperfusion syndrome. *Lancet Neurol*. 2005 Dec;4(12):877-888.

13. Rationale

Answer: A

This patient has signs of hypertensive encephalopathy, which is considered a hypertensive emergency and for which an IV calcium channel blocker for goal mean arterial pressure (MAP) 110 mm Hg within 1 hour is most appropriate. The MAP decrease is 25% within 1 hour. An IV beta-blocker for goal MAP 90 mm Hg within 1 hour is inappropriate because the

MAP decrease is greater than 25%. Oral calcium channel blocker for goal blood pressure 160/100 mm Hg within 2 to 6 hours for hypertensive urgency is inappropriate, as is an oral angiotensin-converting enzyme inhibitor for goal blood pressure 120/80 mm Hg within 24 to 48 hours.

Reference:

1. Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH ASPC/NMA/PCNA Guideline for the prevention, detection, evaluation, and management of high blood pressure in adults: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2018 May 15;71(19): e127-e248.

14. Rationale

Answer: A

This patient has posterior reversible encephalopathy syndrome (PRES). Hypertension is normally one of the main precipitating factors for the development of PRES, and patients typically present with extreme blood pressure elevation. Abnormal renal function can exacerbate PRES and is a well-documented cause; however, she does not seem to have any exacerbations in renal function from baseline. Pregnancy and eclampsia can cause a form of PRES, but she had a hysterectomy 6 years ago. Immunosuppressants such as tacrolimus have been known to cause PRES; a supratherapeutic level would confirm tacrolimus as the likely culprit.

Reference:

1. Harirchian MH, Ghaffarpour M, Tabaeizadeh M, Siroos B. Immunosuppressive drugs, an emerging cause of posterior reversible encephalopathy syndrome: case series. *J Stroke Cerebrovasc Dis.* 2015 Aug;24(8): e191-e195.

15. Rationale

Answer: D

This patient's most likely diagnosis is neurogenic pulmonary edema, based on the clinical findings, diagnostic study results, and timing after neurologic injury. No specific therapy, including steroids or phentolamine, has proven effective for this condition. Although excessive sympathetic drive has been postulated as a pathophysiologic mechanism and clinical improvement following phentolamine has been reported in the literature, phentolamine could result in decreased cerebral perfusion. Hypovolemia should likewise be avoided for the same reason. Aspiration pneumonia is a consideration, but his clinical presentation and timing make this less likely.

References:

1. Busl KM, Bleck TP. Neurogenic pulmonary edema. *Crit Care Med*. 2015 Aug;43(8):1710-1715.

2. Davison DL, Chawla LS, Selassie L, Tevar R, Junker C, Seneff MG. Neurogenic pulmonary edema: successful treatment with IV phentolamine. *Chest*. 2012 Mar;141(3):793-795.

Part 11.

Hematologic Disorders

Instructions: For each question, select the most correct answer.

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1. A 67-year-old man presents to the emergency department with crushing back and abdominal pain. Past medical history is significant for hypertension, for which he takes lisinopril, and atrial fibrillation, for which he takes dabigatran, 150 mg twice daily, and metoprolol succinate. He has nausea, vomiting, and dizziness. Initial vital signs are blood pressure 194/110 mm Hg, heart rate 103 beats/min, and oxygen saturation 96%. Basic metabolic panel shows creatinine 2.1 mg/dL and estimated glomerular filtration rate 31 mL/min. Chest CT angiogram is performed for suspicion of possible abdominal aortic aneurysm rupture, which results confirm. He is assessed by the cardiothoracic surgical team and deemed a candidate for emergent surgery. He is administered idarucizumab, 5 g over 15 minutes, for dabigatran reversal and then undergoes surgery. Which of the following factors carries the highest risk of bleeding secondary to rebound dabigatran levels?

   A. Dabigatran dose

   B. Renal dysfunction

   C. Idarucizumab dose

   D. Lisinopril interaction

2. A 50-year-old man is admitted with acute-onset right-sided weakness and difficulty speaking. NIH Stroke Scale score is 20. Head CT reveals no intracranial hemorrhage. His last known normal time was 2 hours ago. Home medications include ibuprofen

as needed, aspirin, warfarin, amlodipine, and donepezil. Medical history includes a deep vein thrombosis 6 months ago, hypertension, early dementia, and arthritis. Laboratory findings include serum sodium 140 mmol/L, potassium 3.6 mmol/L, BUN 27 mg/dL, serum creatinine 1.8 mg/dL, glucose 265 mg/dL, hematocrit 34.6%, platelet count 245,000/mm^3, and INR 1.8. Vital signs are blood pressure 163/101 mm Hg, heart rate 99 beats/min, arterial oxygen saturation 97%, and respiratory rate 20 breaths/min. Which of the following is an exclusion for administering systemic thrombolysis with IV alteplase for his ischemic stroke?

A. Blood glucose level 265 mg/dL

B. INR 1.8

C. Systolic blood pressure greater than 160 mm Hg

D. Last known normal time 2 hours ago

E. Aspirin

3. A 60-year-old man with a history of atrial fibrillation on warfarin fell from standing and is admitted with an acute left-sided holo-hemispheric subdural hemorrhage with 12-mm midline shift. On examination, he has a sluggish left pupil and brisk right pupil, with all other brainstem reflexes intact. He is intubated and is lethargic and paretic on the right and following commands on the left. Prothrombin time is 21.6 sec and INR is 3.2. He is 182 cm tall (5 ft 11 in) and weighs 142 kg (313 lb.). The team would like recommendations for reversing warfarin for an emergent craniotomy and clot evacuation. Which of the following is the most appropriate choice for reversal of his warfarin anticoagulation?

   A. Fresh frozen plasma, 2 units, and IV vitamin K, 10 mg

B. IV 4-factor prothrombin complex concentrate, 2500 units once, and IV vitamin K, 10 mg

C. IV idarucizumab, 5 g once, and IV vitamin K, 10 mg

D. IV andexanet alfa, 400 mg bolus, followed by infusion of 4 mg/min for 120 minutes

4. A 57-year-old man was admitted overnight for hematemesis due to variceal bleeding. He has a history of severe alcohol abuse, cirrhosis, and portal hypertension. He underwent banding of 2 large esophageal varices and is intubated. He is hemodynamically stable and does not require vasopressors. He has received 2 units packed RBCs and ceftriaxone for spontaneous bacterial peritonitis prophylaxis. He is started on a proton pump inhibitor twice daily. Laboratory results today show hemoglobin 8.7 g/dL after transfusion, platelets 90×10^9/L, and WBCs 2.1×10^9/L. He also has neutropenia at 0.8×10^9/L. HIV testing is negative. Which of the following best explains his low WBC count?

   A. Proton pump inhibitor

   B. Ceftriaxone

   C. Hemorrhage

   D. Immune-mediated destruction

   E. Bone marrow suppression

5. A 67-year-old man is admitted to the ICU with acute hypoxemic respiratory failure and encephalopathy. He was previously in good health but has newly diagnosed acute myeloid leukemia. Temperature is 38.4°C (101.1°F). Oxygen saturation is 87% on room air but improves to 95% with supplemental oxygen at 6 L/min via nasal cannula. Chest radiograph demonstrates bilateral ground-glass opacification. He

is disoriented and agitated but moving all extremities spontaneously. Relevant laboratory findings include WBC count 137,000 cells/μL, hemoglobin 7.2 g/dL, platelet count 7,000 cells/μL, potassium 4.6 mEq/L, and creatinine 1.1 mg/dL. Which of the following interventions is most likely to improve his chance of survival?

A. Hydroxyurea

B. Leukapheresis

C. Induction chemotherapy

D. High-dose steroid therapy

6. A 55-year-old man was admitted to the critical care unit 7 days ago after falling from a ladder. His injuries include multiple rib fractures with flail chest and left acetabular fracture that is nonoperative. He is intubated for respiratory failure due to the flail chest. Duplex ultrasound of the left lower extremity demonstrates a proximal femoral vein thrombosis. Which of the following is the most appropriate approach to the deep vein thrombosis?

   A. Inferior vena cava (IVC) filter placement and lifelong anticoagulation with an oral agent

   B. IVC filter placement without anticoagulation

   C. Low-molecular-weight heparin (LMWH) now and lifelong anticoagulation with an oral agent

   D. LMWH now and 3 months of anticoagulation with an oral agent

   E. Aspirin, 325 mg twice daily

7. A 35-year-old woman was involved in a high-speed motor vehicle collision. In the emergency department she is found to be hypotensive. She is unresponsive to a 1-L IV crystalloid fluid bolus. Abdominal ultrasound is positive for intraperitoneal fluid. She is transported

to the operating room for a laparotomy, and a massive blood transfusion is started. She has a high-grade spleen injury and requires more than 10 units packed RBCs in addition to equal units of fresh frozen plasma and platelets. Which of the following complications is most likely as a direct result of the massive blood transfusion?

A. Coagulopathy

B. Hypocalcemia

C. Hypophosphatemia

D. Metabolic acidosis

Part 11 Answers:

Hematologic Disorders

1. Rationale

Answer: B

Idarucizumab is a monoclonal antibody that binds free dabigatran with quasi-irreversible binding and then is excreted via the kidneys. Very few patients with severe renal impairment were studied during the approval process for idarucizumab; there have been continued case reports of those receiving idarucizumab with concomitant renal failure leading to rebound dabigatran levels 6 to 24 hours after administration. Therefore, renal dysfunction carries the highest risk in this patient for rebound drug levels and hemorrhage while undergoing this emergent procedure. The 5-g dose of idarucizumab has been shown to effectively bind common FDA-approved doses. It is the excretion of the idarucizumab-dabigatran complex that leads to rebound drug concentrations. There is no considerable drug-drug interaction between dabigatran and lisinopril.

References:

1. Eikelboom JW, van Ryn J, Reilly P, et al. Dabigatran reversal with idarucizumab in patients with renal impairment. *J Am Coll Cardiol*. 2019 Oct 8;74(14):1760-1768.

2. Glund S, Coble K, Gansser D, et al. Pharmacokinetics of idarucizumab and its target dabigatran in patients requiring urgent reversal of the anticoagulant effect of dabigatran. *J Thromb Haemost*. 2019 Aug;17(8):1319-1328.

2. Rationale

Answer: B

According to American Heart Association/American Stroke Association guidelines, exclusions for the administration of

systemic thrombolysis include INR greater than 1.7. Blood glucose level of 50 to 400 mg/dL is an acceptable range for IV thrombolytic administration. Systolic blood pressure of less than 185 mm Hg and a last known normal time of 4.5 hours ago are both acceptable for IV thrombolytic administration. Aspirin is not a contraindication for alteplase.

Reference:

1. Powers WJ, Rabinstein AA, Ackerson T, et al; American Heart Association Stroke Council. 2018 guidelines for the early management of patients with acute ischemic stroke: a guideline for healthcare professionals from the American Heart Association/American Stroke Association. *Stroke*. 2018 Mar;49(3): e46-e110.

3. Rationale

Answer: B

Four-factor prothrombin complex concentrate (4F-PCC) is effective for rapidly reversing coagulopathy due to warfarin. Fresh frozen plasma takes too long because it needs to be thawed; also, a dose of 2 units is too low. Neurocritical Care Society guidelines for reversal of intracranial hemorrhage (ICH) recommend PCC over plasma. The American College of Cardiology Task Force on Expert Consensus Decision Pathways and the U.S. FDA recommend a dose of 25 units/kg 4F-PCC for INR 2 to 4 with a maximum of 2500 units plus IV vitamin K. For patients who weigh more than 100 kg (220 lb.), there is a dose cap of 2500 units. It is important to administer vitamin K in addition to 4F-PCC to prevent coagulopathy from recurring after 4F-PCC wears off. Other agents for direct oral anticoagulant reversal include idarucizumab for dabigatran and andexanet alfa for apixaban and rivaroxaban, but neither of these agents has a mechanism that is useful for the reversal of warfarin anticoagulation.

Reference:

1. Tomaselli GF, Mahaffey KW, Cuker A, et al. 2017 ACC expert consensus decision pathway on management of bleeding in patients on oral anticoagulants: a report of the

American College of Cardiology Task Force on Expert Consensus Decision Pathways. *J Am Coll Cardiol*. 2017 Dec 19;70(24):3042-3067.

4. Rationale

Answer: E

Bone marrow suppression best explains this patient's low WBC count. Leukopenia is a common finding in people with a history of chronic alcohol abuse. Heavy alcohol intake has been shown to cause hypocellular changes on bone marrow evaluation. Bone marrow changes subsequently affect several cell lines, including granulocytes. Nutritional deficiencies and splenomegaly contribute variably. There is no evidence to suggest an immune-mediated process causing leukopenia. Drug-related cytopenias are very unlikely after 1 day (in this case, ceftriaxone, and a proton pump inhibitor). Although hemorrhage may be associated with leukopenia, the more likely cause in this patient with chronic heavy alcohol abuse is alcohol-related bone marrow suppression.

References:

1. Girard DE, Kumar KL, McAfee JH. Hematologic effects of acute and chronic alcohol abuse. *Hematol Oncol Clin North Am*. 1987 Jun;1(2):321-334.

2. Latvala J, Parkkila S, Niemela O. Excess alcohol consumption is common in patients with cytopenia: studies in blood and bone marrow cells. *Alcohol Clin Exp Res*. 2004 Apr;28(4):619-624.

3. Shi X, DeLucia AL, Bao J, Zhang P. Alcohol abuse and disorder of granulopoiesis. *Pharmacol Ther*. 2019 Jun; 198:206-219.

5. Rationale

Answer: C

This patient's presentation is consistent with leukostasis, which is associated with 90% mortality at 1 week. Emergent intervention is required. Induction chemotherapy is the

preferred approach because it will result in a rapid decrease in WBC count and is most likely to improve the chance of long-term survival. Leukapheresis should be reserved for situations in which chemotherapy cannot be initiated promptly; it has not been associated with improved survival. Hydroxyurea monotherapy would be most appropriate for hyperleukocytosis (without evidence of leukostasis) when chemotherapy is not immediately available. Diffuse alveolar hemorrhage and tumor lysis syndrome are other diagnostic considerations but less likely than leukostasis in this patient.

References:

1. Korkmaz S. The management of hyperleukocytosis in 2017: Do we still need leukapheresis? *Transfus Apher Sci*. 2018 Feb;57(1):4-7.
2. Oberoi S, Lehmbecher T, Phillips B, et al. Leukapheresis and low-dose chemotherapy do not reduce early mortality in acute myeloid leukemia hyperleukocytosis: a systematic review and meta-analysis. *Leuk Res*. 2014 Apr;38(4):460-468.

6. Rationale

Answer: D

Trauma is a risk factor for venous thromboembolism (VTE), making this a provoked event. The guideline for treatment duration for provoked VTE is 3 months. It should be treated with anticoagulation when there are no contraindications to anticoagulant therapy, regardless of whether it is proximal or distal. Inferior vena cava filters do not prevent or treat venous thromboses and their sequelae (such as chronic venous insufficiency). Their only purpose is to prevent life-threatening pulmonary emboli. Thus, their utility is strictly limited to patients for whom anticoagulation is contraindicated. A common method of anticoagulation is administration of a short-acting agent such as unfractionated or low-molecular-weight heparin, followed by an oral agent such as warfarin or a novel oral anticoagulant.

Reference:

1. Kearon C, Akl EA, Ornelas J, et al. Antithrombotic therapy for VTE disease: CHEST guideline and expert panel report. *Chest*. 2016 Feb;149(2):315-352.

7. Rationale

Answer: B

While massive blood transfusion is indicated in unstable trauma patients with large-volume hemorrhage, possible complications include metabolic alkalosis, hypocalcemia, and hyperkalemia, which occur in 50% of patients when more than 5 units of blood products are transfused. Sodium citrate and citric acid are added to blood products in storage to prevent coagulation. The citrate binds ionized calcium, which leads to significant free hypocalcemia. The calcium bound to albumin is not affected. If the hypocalcemia is severe enough it can result in paresthesia and cardiac arrhythmias. Furthermore, each unit of blood can generate a total of 23 mEq/L bicarbonate as citrate is metabolized, which results in metabolic alkalosis if the kidneys cannot excrete the excess bicarbonate. Also, the alkalosis can result in hypokalemia as hydrogen ions move out of cells to compensate for the alkalosis through the hydrogen/potassium transporter. A balanced blood transfusion strategy, as in this scenario, should prevent coagulopathy. Other complications of blood transfusion include hypothermia, hypokalemia (secondary to metabolic alkalosis), hyperkalemia (secondary to potassium from rapid, high-volume infusion after prolonged blood storage), transfusion-associated lung injury, and transfusion-associated circulatory overload.

References:

1. Guerado E, Medina A, Mata MI, Galvan JM, Bertrand ML. Protocols for massive blood transfusion: when and why, and potential complications. *Eur J Trauma Emerg Surg*. 2016 Jun;42(3):283-295.

2. Sihler KC, Napolitano LM. Complications of massive transfusion. *Chest*. 2010 Jan;137(1):209-220.

Part 12.

Oncology

Instructions: For each question, select the most correct answer.

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1. A 70-year-old man with a history of hypertension and a recent diagnosis of multiple myeloma is admitted to the medical ICU with confusion and vision changes. On admission, he is found to have epistaxis and oral mucosal bleeding, hypercalcemia at 11.2 mg/dL, elevated BUN at 32 mg/dL, and creatinine level of 1.8 mg/dL. Serum viscosity is elevated at 5.3 centipoise (cP). Which of the following is the most appropriate next treatment?

   A. Methylprednisolone

   B. Lenalidomide

   C. Bortezomib

   D. Plasmapheresis

2. A 66-year-old man with a history of congestive heart failure, ejection fraction 30%, glucose-6-phosphate dehydrogenase (G6PD) deficiency, and recently diagnosed acute myeloid leukemia, subtype M4 (AML-M4), is admitted to the oncology unit for treatment. WBC count is 112×10^9/L. He is administered fludarabine and cladribine for induction chemotherapy. He is unable to get a significant amount of fluid because of preexisting heart failure. He is administered allopurinol and rasburicase for prevention and treatment of expected tumor lysis syndrome (high risk with AML-M4). The nurse enters his room at midnight to measure vital signs and notices that his oxygen saturation is 84%. There is a good pulse oximetry waveform. A rapid response is

called. The ICU team arrives and finds the patient awake and oriented, in no distress. Which of the following drugs best explains his current condition?

A. Rasburicase

B. Allopurinol

C. Fludarabine

D. Cladribine

3. A 58-year-old man is transferred to the ICU for severe hypoxemia. He received melphalan followed by autologous hematopoietic cell transplantation for multiple myeloma 14 days ago. He has been treated with vancomycin and cefepime for neutropenic fever. He has an erythematous skin rash on his torso. Chest radiograph reveals diffuse bilateral opacities. Oxygen saturation is 92%, with FIO_2 0.6 via high-flow, high-humidity nasal cannula. Significant laboratory findings are WBCs 1000 cells/mm^3 (88% neutrophils), hemoglobin 7.3 mg/dL, and platelets 28,000 cells/mm^3. Which of the following is the most likely explanation for this clinical presentation?

   A. Diffuse alveolar hemorrhage

   B. Melphalan pulmonary toxicity

   C. Transfusion-related acute lung injury

   D. Peri-engraftment respiratory distress syndrome

4. A 60-year-old man of Mediterranean ancestry with hypertension, hyperlipidemia, glucose-6-phosphate dehydrogenase (G6PD) deficiency, and recently diagnosed non-Hodgkin lymphoma presents to the emergency department with altered mental status and decreased urine output. He recently began chemotherapy for the lymphoma. Laboratory results are significant for: potassium, 6 mEq/L; total calcium, 7 mg/dL; phosphorus, 5 mg/dL; uric acid, 9 mg/dL;

and creatinine, 4.2 mg/dL. Which of the following is the most appropriate initial therapy for this patient?

A. Hemodialysis

B. Normal saline boluses

C. Rasburicase

D. Allopurinol

E. Sodium bicarbonate

5. A 32-year-old Korean woman with acute myeloid leukemia is transferred from the hematology/oncology unit to the medical ICU with tumor lysis syndrome. Treating her with allopurinol places her at risk for which of the following complications?

A. Acute typhlitis

B. Septic shock

C. Acute decompensated cardiomyopathy

D. Acute renal failure

E. Severe acute adverse cutaneous reaction

6. Checkpoint inhibitors, immunomodulatory antibodies that are used to enhance the immune system, have substantially improved the prognosis for patients with advanced malignancy. The 3 categories of these drugs are programmed cell death receptor 1 (PD-1), programmed cell death ligand 1 (PD-L1), and cytotoxic T-lymphocyte-associated antigen 4 (CTLA-4). Which of the following is an early adverse event with these agents?

A. Dermatologic toxicity

B. Nephrotoxicity

C. Pneumonitis

D. Endocrinopathy

7. A 33-year-old woman who was involved in a motor vehicle collision is found to have a 4.5-cm mass on the left adrenal gland. Trauma workup is otherwise normal. Diagnostic laboratory workup for the traumatic event is significant for serum potassium 2.2 mEq/L. Blood pressure is 220/105 mm Hg. She had been having headaches before this event. Initial workup of this incidental adrenal mass should include

   A. abdominal MRI.

   B. image-guided fine needle aspiration.

   C. plasma aldosterone concentration-to-plasma renin activity ratio.

   D. positron emission tomography with fludeoxyglucose.

   E. fluoroscopic image-guided vein sampling.

Part 12 Answers:

Oncology

1. Rationale

Answer: D

Multiple myeloma is the second most common hematologic cancer in which neoplastic plasma cells accumulate in the bone marrow. It is characterized by hypercalcemia, renal insufficiency, anemia, and painful lytic bone lesions. Treatment includes immunomodulatory drugs such as thalidomide and lenalidomide, proteasome inhibitors such as bortezomib, and steroids. Elevated levels of serum immunoglobulins can coat blood cells, causing increased blood viscosity, or hyperviscosity syndrome, which may be manifested as spontaneous bleeding, neurologic deficits, and vision changes. Hyperviscosity syndrome is an oncologic emergency and can result in congestive heart failure, acute kidney failure, pulmonary edema, seizures, and death. Treatment is urgent and requires plasmapheresis and occasionally repeat plasmapheresis for refractory episodes.

References:

1. Higdon ML, Higdon JA. Multiple myeloma. BMJ 2013 Jun 26;346: f3863. Higdon ML, Higdon JA. Treatment of oncologic emergencies. *Am Fam Physician*. 2006 Dec 1;74(11):1873-1880.
2. Mikhael J, Ismaila N, Cheung MC, et al. Treatment of multiple myeloma: ASCO and CCO joint clinical practice guideline. *J Clin Oncol*. 2019 May 10;37(14):1228-1263.

2. Rationale

Answer: A

This patient has leukemia treated with induction and consolidation chemotherapy. He cannot have standard chemotherapy because of his heart failure, which precludes

anthracyclines (daunorubicin/daunomycin) or idarubicin. Patients with low ejection fraction can be treated with fludarabine or topotecan. One of the complications of chemotherapy is tumor lysis syndrome (TLS), which is more likely to develop in tumors with high tumor burden, rapid cell turnover, and increased sensitivity to chemotherapeutic agents and is thus most often seen in acute leukemias with high WBC counts at presentation and diffuse Burkitt-type non-Hodgkin lymphoma. M4-type acute myeloid leukemia is considered a high-risk malignancy in terms of precipitating TLS. Successful management of TLS in patients presenting with malignancy involves: 1) risk assessment, 2) hydration with fluids, 3) medications (allopurinol and rasburicase), 4) urinary alkalinization, 5) leukapheresis, and 6) dialysis. Rasburicase is a recombinant urate oxidase enzyme. Urate oxidase converts uric acid to water-soluble allantoin, which can then be freely excreted in the urine. Rasburicase has been reported to cause both methemoglobinemia and hemolytic anemia as a result of increased oxidative stress inside the erythrocyte, typically in people with glucose-6-phosphate dehydrogenase (G6PD) deficiency. When rasburicase converts uric acid to allantoin, the byproduct is hydrogen peroxide, an oxidizing agent. Methemoglobin is formed when the iron moiety in the heme molecule is oxidized from the ferrous to the ferric state, which cannot bind oxygen molecules. In addition, the remaining ferrous atoms in the heme tetramer have increased oxygen affinity and thus are less likely to release oxygen molecules in the periphery. This results in a left shift in the oxyhemoglobin dissociation curve and produces a functional anemia. The increased oxidative stress also leads to Heinz body formation, RBC rigidity, and erythrocyte destruction by macrophages in the reticuloendothelial system, producing hemolytic anemia. The combination of this functional and hemolytic anemia can result in large drops in serum hemoglobin levels and significant impairment in oxygen delivery to vital organs. Methemoglobinemia is typically treated with methylene blue, which catalyzes one of the reactions in which methemoglobin is reduced back to hemoglobin. However, in people with

G6PD deficiency, this pathway is inactive and methylene blue cannot catalyze the reaction but instead becomes oxidized itself, further increasing the oxidative stress on the cell. These G6PD-deficient patients can be treated with ascorbic acid, 1 g daily. They may also need double-volume RBC exchange transfusion.

References:

1. American Cancer Society. Typical treatment of most types of acute myeloid leukemia (except APL). Last revised September 2, 3030. Accessed March 4, 2022. https://www.cancer.org/cancer/acute-myeloid-leukemia/treating/typical-treatment-of-aml.html
2. Sherwood GB, Paschal RD, Adamski J. Rasburicase-induced methemoglobinemia: case report, literature review, and proposed treatment algorithm. *Clin Case Rep*. 2016 Feb 3;4(4):315-319.
3. Will A, Tholouli E. The clinical management of tumor lysis syndrome in hematological malignancies. *Br J Haematol*. 2011 Jul;154(1):3-13.

3. Rationale

Answer: D

This patient has all 3 major criteria for peri-engraftment respiratory distress syndrome: noninfectious fever, skin rash, and noncardiogenic pulmonary edema. He was not transfused, making diffuse alveolar hemorrhage and transfusion-related acute lung injury unlikely. Acute respiratory distress syndrome is a potential manifestation of melphalan toxicity; however, the rash is not characteristic of melphalan toxicity.

References:

1. Cornell RF, Hari P, Drobyski WR. Engraftment syndrome after autologous stem cell transplantation: an update unifying the definition and management approach. *Biol Blood Marrow Transplant*. 2015 Dec;21(12):2061-2068.

2. Vande Vusse LK, Madtes DK. Early onset noninfectious pulmonary syndromes after hematopoietic cell transplantation. *Clin Chest Med*. 2017 Jun;38(2):233-248.

4. Rationale

Answer: B

This patient has tumor lysis syndrome (TLS). IV fluid resuscitation is the primary goal. Rasburicase is a recombinant urate oxidase enzyme, which converts uric acid to an inactive and soluble metabolite. It can be administered prophylactically and with established TLS but is contraindicated in patients with glucose-6-phosphate dehydrogenase (G6PD) deficiency. Allopurinol decreases uric acid formation and is helpful in the management of TLS but does not reduce preexisting serum uric acid, which needs to be lowered in this patient. The role of urinary alkalization is controversial. Previously urinary alkalization has been recommended to increase uric acid solubility, but there are no data to support this practice and it may lead to deposition of calcium phosphate in organs. It can be considered when metabolic acidosis is present but should be avoided when hyperphosphatemia is present. Hemodialysis has similar indications for other types of acute kidney injury, although at lower thresholds, given the potential rapid rise in potassium.

References:

1. Higdon ML, Atkinson CJ, Lawrence KV. Oncologic emergencies: recognition and initial management. *Am Fam Physician*. 2018 Jun 1;97(11):741-748.

2. Larson RA, Pui CH. Tumor lysis syndrome: prevention and treatment. UpToDate. Last updated July 1, 2021. Accessed March 4, 2022. https://www.uptodate.com/contents/tumor-lysis-syndrome-prevention-and-treatment

5. Rationale

Answer: E

The Clinical Pharmacogenetics Implementation Consortium guidelines for human leukocyte antigen B (HLA-B) genotype and allopurinol dosing suggest that people of Korean descent are at greatest risk of developing severe cutaneous adverse reactions because of mutations in the HLA-B*58:01 allele. The American College of Rheumatology recommends genotyping patients of Korean descent before starting allopurinol.

References:

1. Khanna D, Fitzgerald JD, Khanna PP, et al; American College of Rheumatology. 2012 American College of Rheumatology guidelines for management of gout. Part 1: systematic nonpharmacologic and pharmacologic therapeutic approaches to hyperuricemia. *Arthritis Care Res* (Hoboken). 2012 Oct;64(10):1431-1446.
2. Saito Y, Stamp LK, Caudle KE, et al; Clinical Pharmacogenetics Implementation Consortium. Clinical Pharmacogenetics Implementation Consortium (CPIC) guidelines for human leukocyte antigen B (HLA-B) genotype and allopurinol dosing: 2015 update. *Clin Pharmacol Ther.* 2016 Jan;99(1):36-37.

6. Rationale

Answer: A

Checkpoint inhibitors are immunomodulatory antibodies used to enhance the immune system. They have substantially improved prognoses for patients with advanced malignancy. Nivolumab and pembrolizumab target programmed cell death receptor 1 (PD-1). Atezolizumab, avelumab, and durvalumab all target programmed cell death ligand 1 (PD-L1). Ipilimumab targets cytotoxic T-lymphocyte-associated antigen 4 (CTLA-4). Adverse events are thought to be due to general immunologic enhancement. Temporary immunosuppression with corticosteroids, tumor necrosis factor-alpha antagonists, mycophenolate mofetil, or other agents can be effective treatments. Dermatologic toxicity is the earliest adverse event, with onset an average of 3.6 weeks after treatment initiation. The onset of hepatotoxicity most commonly occurs 8 to 12

weeks into treatment. Diarrhea/colitis most commonly occurs approximately 6 weeks into treatment. Nephrotoxicity is an uncommon side effect with these agents. Pneumonitis is an uncommon but potentially severe or fatal complication with checkpoint inhibitor immunotherapy. The overall incidence of clinically significant endocrinopathies is approximately 10% of patients treated with checkpoint inhibitors.

References:

1. Brahmer JR, Lacchetti C, Schneider BJ, et al; National Comprehensive Cancer Network. Management of immune-related adverse events in patients treated with immune checkpoint inhibitor therapy: American Society of Clinical Oncology Clinical practice guideline. *J Clin Oncol* 2018 Jun 10;36(17):1714-1768.
2. Champiat S, Lambotte O, Barreau E, et al. Management of immune checkpoint blockade dysimmune toxicities: a collaborative position paper. *Ann Oncol.* 2016 Apr;27(4):559-574.
3. Naidoo J, Page DB, Li BT, et al. Toxicities of the anti-PD-1 and anti-PD-L1 immune checkpoint antibodies. *Ann Oncol.* 2015 Dec;26(12):2375-2391.
4. Puzanov I, Diab A, Abdallah K, et al; Society for Immunotherapy of Cancer Toxicity Management Working Group. Managing toxicities associated with immune checkpoint inhibitors: consensus recommendations from the Society for Immunotherapy of Cancer (SITC) Toxicity Management Working Group. *J Immunother Cancer.* 2017 Nov 21;5(1):95.

7. Rationale

Answer: C

Biochemical diagnosis is the initial route for diagnosing adrenal tumors. This patient has hypertension at a relatively young age coupled with hypokalemia, suggestive of hyperaldosteronism. Fine needle aspiration is contraindicated because of the possibility that the tumor is a pheochromocytoma and could

initiate a hypertensive crisis. Positron emission tomography and fluoroscopic vein imaging are further down the diagnostic algorithm when a lesion is difficult to identify.

References:

1. Fassnacht M, Arlt W, Bancos I, et al. Management of adrenal incidentalomas: European Society of Endocrinology clinical practice guideline in collaboration with the European Network for the Study of Adrenal Tumors. *Eur J Endocrinol*. 2016 Aug;175(2): G1-G34.

2. Sosa JA, Udelsman R. The adrenal glands. In: Townsend CM, Beauchamp RD, Evers BM, Mattox KL, eds. *Sabiston Textbook of Surgery: The Biological Basis of Modern Surgical Practice*. 20th ed. Elsevier; 2017:963-995.

Part 13.

Surgery and Trauma

Instructions: For each question, select the most correct answer.

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1. A 24-year-old woman who is 22 weeks pregnant presents to the emergency department for a 1-week history of generalized weakness. She says that she was in good health last week and has no past medical history. She denies headache, fever, chills, nausea, or vomiting. Blood pressure is 165/115 mm Hg. She is not in distress. Laboratory results are significant only for creatinine level 2.1 mg/dL. Which of the following is the most appropriate next step?

   A. Admit her to the ward.

   B. Monitor both mother and fetus intermittently.

   C. Reduce diastolic blood pressure by 20% in the first 60 minutes.

   D. Reduce systolic blood pressure to 120 mm Hg within the first 60 minutes.

   E. Observe and treat when systolic blood pressure is above 220 mm Hg.

2. A 45-year-old woman was started on meloxicam by her primary care physician 2 weeks ago for rheumatoid arthritis. She presents to the emergency department with a 1-day history of sloughing of the skin. She has had fatigue and fever up to 38.6°C (101.4°F). She has crusting of her eyes and oral mucosa. The skin over her entire trunk and extremities is sloughing and extremely painful. Vital signs are temperature 39°C (102.2°F), heart rate 125 beats/min, and blood pressure 100/60 mm Hg. After

replenishing fluid losses, which of the following is the most appropriate next step in management?

A. Skin biopsy

B. Transfer to specialized burn center

C. Steroids

D. Broad-spectrum antibiotics

3. A 64-year-old man underwent a laparoscopic colon resection. Epidural anesthesia was used for part of the procedure. The next day, he was reintubated for pulmonary edema, using etomidate. He was eventually discharged to inpatient rehabilitation and, 6 days after the procedure, he is evaluated in the emergency department for abdominal pain, vomiting, hypotension, tachycardia, and altered mental status. Lactic acid is 8.4 mmol/L. Because of the lack of airway protection and expected clinical course, he is intubated. CT with oral contrast reveals contrast in the peritoneal space. In addition to operatively addressing the anastomotic leak, which of the following resuscitation strategies is most appropriate?

   A. Hypertonic saline

   B. Isotonic crystalloids

   C. Colloids

   D. Cortisol replacement therapy

4. A 59-year-old man with a history of tobacco use, chronic obstructive pulmonary disease, and heart failure with preserved ejection fraction presents to the emergency department for sudden onset of chest and back pain. Vital signs are heart rate 92 beats/min, blood pressure 198/116 mm Hg, respiratory rate 18 breaths/min and oxygen saturation 88% on 2 L/min oxygen via nasal cannula. ECG shows normal sinus rhythm. Chest radiograph shows a possible widened

mediastinum. Chest CT shows descending thoracic aortic dissection. Which of the following is the most appropriate initial treatment for this patient?

A. IV esmolol

B. IV nicardipine

C. Emergent surgical consultation

D. Emergent consultation for angiography

5. A 31-year-old man with no past medical history is brought to the emergency department after being involved in a high-speed motor vehicle collision. On arrival, he receives 4000 mL normal saline, 4 units packed RBCs, and 4 units fresh frozen plasma for the treatment of massive bleeding with hemorrhagic shock. He is endotracheally intubated and placed on mechanical ventilation. Laboratory studies are pending. CT reveals a large left-sided subdural hematoma and a large left-sided hemothorax with multiple rib fractures. Vital signs are blood pressure 88/58 mm Hg, heart rate 122 beats/min, temperature 35.4°C (95.7°F), oxygen saturation 94% on FIO_2 0.7, and respiratory rate 22 breaths/min. Bleeding is ongoing despite these interventions. Which of the following is the most appropriate treatment?

   A. Platelets

   B. Packed RBCs

   C. Fresh frozen plasma

   D. Normal saline

   E. Human albumin

6. A 26-year-old man with no significant past medical history develops shortness of breath with PaO_2 59 mm Hg on room air, confusion, and a petechial rash over the upper torso and conjunctiva 2 days after sustaining an isolated right femoral fracture. Which of

the following is the most likely expected finding on chest CT?

A. Diffuse, patchy ground-glass opacities with intralobular septal thickening and small, centrilobular nodules in a peripheral and upper lobe distribution

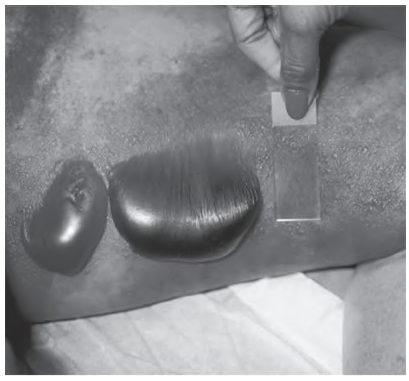
B. Bilateral symmetrical septal lines with bronchial cuffing, vascular engorgement, and pleural effusions

C. Intraluminal filling defects in the pulmonary artery with soft tissue attenuation

D. Isolated single-lung lobe consolidation

7. A 46-year-old morbidly obese woman with a history of poorly managed diabetes presents to the emergency department with progressive, burning pain and swelling in her left upper leg. She says that the skin on the affected leg has become more mottled and that large fluid-filled blisters began forming today. She reports fatigue, malaise, fevers, and chills. Temperature is 38.8°C (101.8°F), heart rate in the 130s beats/min, and blood pressure 90/64 mm Hg. She is agitated and ill-appearing. Her left thigh is exquisitely tender to palpation and to passive extension. Laboratory findings include sodium 126 mEq/L, bicarbonate 14 mg/dL, BUN 70 mg/dL, creatinine 2.1 mg/dL, glucose 280 mg/dL, and WBC count 22,000/mm^3. Radiograph and physical findings are shown below. In addition to broad-spectrum antibiotics and fluid resuscitation, which of the following is the most appropriate initial management of this patient?





A. Antibiotic ointment

B. Tissue debridement

C. Percutaneous drainage

D. Hyperbaric oxygen therapy

8. A 22-year-old man sustained a single gunshot to the right upper quadrant 2 weeks ago. He underwent emergent exploratory laparotomy and was found to have an isolated grade III injury to the right lobe of the liver, which was controlled with temporary packing and an argon beam coagulator. He underwent a second operation where bleeding became better controlled; thus, full closure of his abdomen was performed. Overnight he was transferred to the ICU with massive hematemesis and reduction in hemoglobin level from 13.5 g/dL to 9 g/dL during the past 4 hours. INR is 1.6 and clotting time is normal. Total bilirubin is double the upper limit of normal. He is hemodynamically normal. Esophagogastroduodenoscopy shows only a blood clot adhering to his ampulla. Which of the following is the most appropriate treatment for his haemobilia?

   A. Medical management with proton pump inhibitor and octreotide

B. Endoscopic retrograde cholangiopancreatography with sphincterotomy and possible stent

C. Prothrombin complex concentrate

D. Interventional radiology angioembolization

E. Fresh frozen plasma

9. A 67-year-old man is admitted to the ICU with acute chest pain. He undergoes cardiac catheterization that discloses severe multivessel coronary artery disease. Coronary artery bypass grafting is performed. About 24 hours later, he has severe left leg pain, which worsens with palpation or moving his toes. The endoscopic vein-harvesting incisions are intact. There is no groin hematoma or significant edema, and distal pulses are intact. Relevant laboratory results include creatinine 2.4 mg/dL (baseline 1.5 mg/dL) and creatine phosphokinase 13,758 mg/dL. Which of the following is the most likely diagnosis?

   A. Femoral vein thrombosis

   B. Cholesterol embolization

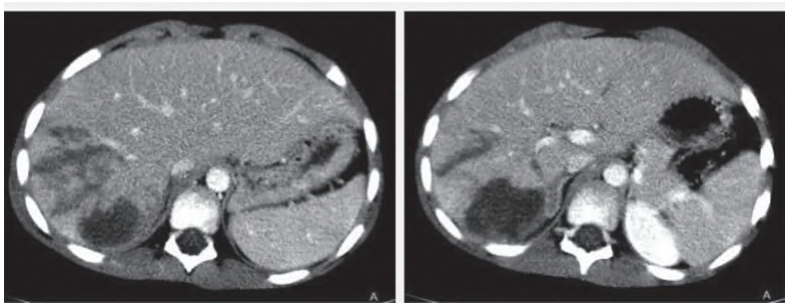
   C. Femoral artery thrombosis

   D. Acute compartment syndrome

   E. Statin-induced rhabdomyolysis

10. An 18-year-old man presents to the emergency department after being hit by a car at moderate speed while riding his bicycle. He had no loss of consciousness, and his Glasgow Coma Scale score is 15. Primary survey reveals that he can speak without difficulty, has equal breath sounds bilaterally, and his femoral pulse is 90 beats/min. Secondary survey reveals diffuse truncal abrasions and an abdomen that is diffusely tender but without rebound, rigidity, or guarding. Focused assessment with sonography in trauma (FAST) examination is negative in all 4

quadrants. Initial chemistry profile and CBC are unremarkable. Abdominal CT is shown below. Which of the following treatment strategies is indicated for initial management of this patient?



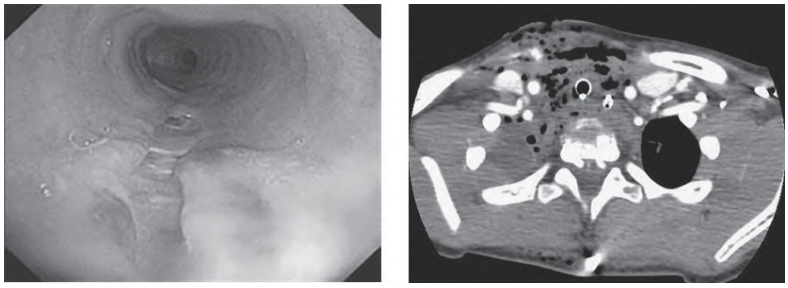
A. Exploratory laparotomy

B. Angioembolization of right hepatic artery

C. Serial abdominal examination with repeat abdominal CT if he becomes hypotensive

D. Percutaneous drainage of the hematoma

11. A 40-year-old woman with a history of severe traumatic brain injury undergoes percutaneous tracheostomy. Following the procedure, she develops subcutaneous emphysema of the neck and anterior chest wall. She also becomes progressively tachycardic, tachypneic, and febrile to 39°C (102.2°F). Chest CT and bronchoscopy are shown below. Which of the following is the most appropriate initial management of this patient?



A. Surgical exploration of the neck and superior mediastinum

B. Tracheostomy revision

C. Placement of a cuffed airway below the level of the injury

D. Tracheal stenting

12. A 36-year-old man sustains a crush injury of the right lower extremity in an industrial accident. On examination, he has a severely comminuted fracture of the right tibia and fibula. He is hemodynamically stable and has palpable distal pedal pulses in the right foot. Initial chemistry panel and CBC are unremarkable. Physical and radiographic findings are shown below. He is admitted to the hospital and splinted, pending orthopedic evaluation. Over the following 12 hours, he develops progressive oliguria with scant, very dark urine. Repeat chemistry panel shows creatinine 3.2 mg/dL. Which of the following is the most appropriate initial management of this patient?





A. Broad-spectrum antibiotics

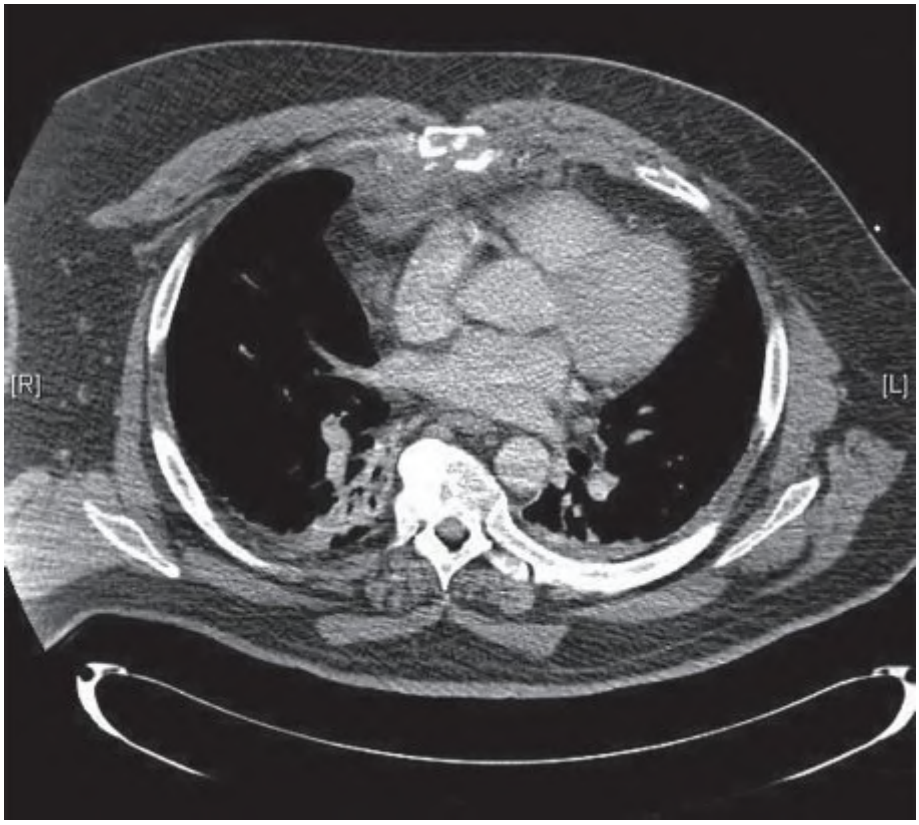
B. Renal replacement therapy

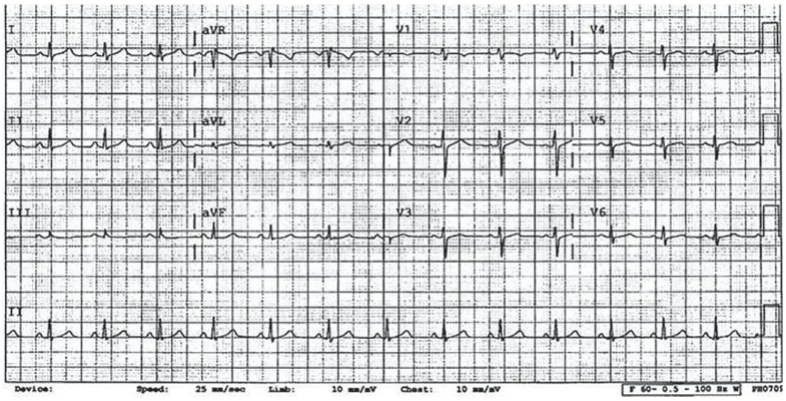
C. Anticoagulation

D. IV fluid administration

13. A 56-year-old woman with obesity and a history of alcohol abuse was the unrestrained driver in a high-speed motor vehicle collision. On extrication, the vehicle's steering wheel and column were significantly

deformed. In the emergency department, she is alert but agitated. Her primary symptom is severe chest pain. Primary survey shows a patent airway, symmetrical breath sounds bilaterally, and tachycardia (heart rate in the 130s beats/min). She remains normotensive. ECG monitor shows sinus tachycardia with occasional premature ventricular contractions. Secondary survey shows prominent contusions over the central chest with tenderness to sternal palpation. The remainder of her physical examination is unremarkable. Chemistry profile, CBC, and INR are unremarkable. Initial troponin I level is 0.15 ng/mL. CT and ECG are shown below. Her findings are most consistent with





A. blunt cardiac injury.

B. pericardial tamponade.

C. diaphragmatic rupture.

D. blunt aortic injury.

14. A 37-year-old man lost control of his bicycle. The trajectory caused him to flip over the handlebars and land on his head and face. He was wearing a helmet, which bystanders said was severely damaged by the accident. They also reported that he had a 2-minute loss of consciousness at the scene. On arrival in the emergency department, he is awake and alert. Glasgow Coma Scale score is 15. Blood pressure is 125/73 mm Hg, respiratory rate 13 breaths/min and nonlabored, heart rate 79 beats/min, and oxygen saturation 99% on room air. He reports pain to his face, especially near his mouth where 2 teeth are missing. Multiple abrasions are noted. He also reports pain to his right knee, which has an obvious deformity and edema. He has mid-back pain that is not tender to palpation. He has difficulty swallowing his saliva and needs to suction himself every 10 to 15 seconds. Which of the following is the primary concern in this patient with multisystem trauma?

A. Facial lacerations/abrasions

B. Back pain

C. Knee pain

D. Head injury/loss of consciousness

E. Difficulty swallowing

15. A 42-year-old man fell off a roof at a construction site. He is hemodynamically stable and alert and oriented. His primary symptom is left-sided severe chest pain and tenderness. He also has left lower extremity pain at the upper thigh. Imaging reveals a closed subtrochanteric femur fracture and several left-sided rib fractures with an underlying moderate pulmonary contusion and hemothorax. The remainder of his trauma evaluation is negative for any other injury. Vital signs are temperature 37.9°C (100.2°F), heart rate 114 beats/min, blood pressure 120/62 mm Hg, respiratory rate 28 breaths/min, and oxygen saturation as measured by pulse oximetry 92% on 8 L via nasal cannula. He has no significant medical history and is admitted for ICU monitoring because of hypoxia and femur fracture. Which of the following is an indication for performing open reduction and internal fixation of his rib fractures?

    A. 4 nondisplaced fractures with continued worsening of the pulmonary contusion as seen on chest radiograph and with ongoing hypoxia on hospital day 3

    B. 4 nondisplaced fractures as part of the femur repair procedure on hospital day 2

    C. 4 severely displaced fractures with resolution of chest wall pain after starting multimodal pain regimen

    D. 4-rib flail segment on hospital day 1 with ongoing hypoxia despite multimodal pain regimen

    E. 4 severely displaced rib fractures with worsening hypoxia requiring mechanical ventilation on hospital day 6

Part 13 Answers:

Surgery and Trauma

1. Rationale

Answer: C

This patient meets the American College of Obstetricians and Gynecologists' (ACOG) criteria for hypertensive emergency. Systolic blood pressure and diastolic blood pressure are above 160- and 110-mm Hg, respectively, and she has evidence of end-organ damage (creatinine 2.1 mg/dL). ACOG recommends admitting patients with hypertensive emergency to the ICU for continuous monitoring of both mother and fetus. It is recommended to reduce blood pressure slowly to normal range within 24 to 48 hours, not within the first hour. It is also recommended to reduce diastolic blood pressure by 20% or to 100 to 110 mm Hg within 30 to 60 minutes. Systolic blood pressure higher than 180 mm Hg or diastolic blood pressure higher than 120 mm Hg is the definition of a hypertensive emergency according to the American Heart Association. These blood pressure levels are less tolerated by obstetric patients.

References:

1. Marik PE, Rivera R. Hypertensive emergencies: an update. *Curr Opin Crit Care*. 2011 Dec;17(6):569-580.
2. Vadhera RB, Simon M. Hypertensive emergencies in pregnancy. *Clin Obstet Gynecol*. 2014 Dec;57(4):797-805.

2. Rationale

Answer: B

This patient has toxic epidermal necrolysis (TEN), which involves detachment of greater than 30% of body surface area. Stevens-Johnson syndrome (SJS) is the less severe condition, involving less than 10% of body surface area. Medications are a leading cause of SJS/TEN; it typically occurs during the first 8

weeks of treatment. Allopurinol, lamotrigine, oxicam nonsteroidal anti-inflammatory steroidal drugs, and antibacterial sulfonamides have all been implicated. SJS/TEN is a clinical diagnosis; skin biopsy is an adjunct. The offending agent should be discontinued. Treatment is supportive. The principles are the same as for major burns—wound care, fluid and electrolyte management, nutritional support, temperature management, pain control, and monitoring or treatment of infections. The Score of Toxic Epidermal Necrolysis (SCORTEN) is used to risk-stratify mortality. Independent prognosis factors include age 40 years or older, malignancy, body surface area detached 10% or greater, tachycardia 120 beats/min or more, serum urea greater than 10 mmol/L, serum glucose greater than 14 mmol/L, and serum bicarbonate less than 20 mmol/L. Several studies have found improved mortality when patients are transferred to specialized burn centers. The use of steroids is controversial and has not been evaluated in clinical trials. Patients with SJS/TEN are at high risk of infection, and sepsis is a prominent cause of death. Prophylactic systemic antibiotics are not indicated.

References:

1. Guégan S, Bastuji-Garin S, PoszepczynskaGuignéE, Roujeau JC, Revuz J. Performance of the SCORTEN during the first five days of hospitalization to predict the prognosis of epidermal necrolysis. *J Invest Dermatol*. 2006 Feb;126(2):272-276.

2. Palmieri TL, Greenhalgh DG, Saffle JR, et al. A multicenter review of toxic epidermal necrolysis treated in U.S. burn centers at the end of the twentieth century. *J Burn Care Rehabil*. 2002 Mar-Apr;23(2):87-96.

3. Trent JT, Kirsner RS, Romanelli P, Kerdel FA. Use of SCORTEN to accurately predict mortality in patients with toxic epidermal necrolysis in the United States. *Arch Dermatol*. 2004 Jul;140(7):890-892.

3. Rationale

Answer: B

In patients with anastomotic bowel leak, postoperative fluid and electrolyte imbalances are associated with worse outcomes. Common causes of hypotension in the postoperative period are intra-abdominal (IAB) catastrophe (necrotic bowel, anastomotic leak), bleeding, adrenal insufficiency, and IAB sepsis. Hypotension in these patients is caused by large fluid shifts to the extravascular space, leading to hypovolemia. This patient also has IAB sepsis due to the IAB catastrophe. The magnitude of hypovolemia is usually underappreciated. Volume resuscitation with isotonic crystalloids (either lactated Ringer solution or normal saline) to predefined goals is the mainstay of therapy. Central venous pressure goals are often stated but correlate poorly with volume responsiveness. There is no definitive evidence to suggest the use of colloids in this setting. The Saline Versus Albumin Fluid Evaluation Study showed no difference in 28-day outcome in critically ill patients resuscitated with saline versus albumin. Numerous randomized trials have compared crystalloid and colloid resuscitation. A recent Cochrane review demonstrated no survival benefit to colloids versus crystalloids. Epidural or spinal anesthesia is a relatively common cause of hypotension due to vasodilation caused by loss of sympathetic tone; it will respond to decreasing the dose of anesthetic or narcotic and modest volume resuscitation, which typically occurs in the immediate postoperative period. Etomidate suppresses 11-beta-hydroxylase production in the adrenal gland for 24 to 72 hours, thus reducing cortisol synthesis, but it does not seem to lead to an increased need for cortisol replacement therapy. This patient is well outside this time frame. Judicious use of crystalloids is important to avoid causing abdominal compartment syndrome.

Reference:

1. Kao CC, Kao LS, Todd SR. Critical care issues in the postoperative period. In: Society of Critical Care Medicine. *Comprehensive Critical Care: Adult*. Society of Critical Care Medicine; 2012: chap 44.

4. Rationale **Answer: A**

The 2 types of thoracic aortic aneurysm dissections require different treatments. Ascending aortic dissection requires emergent surgical treatment. Descending aortic dissection requires medical management, especially if the patient is hypertensive. This patient has a descending thoracic aortic dissection, so surgical intervention is not required at this time, especially since he is significantly hypertensive. Angiography is not a treatment option for this patient. Both nicardipine and esmolol are good treatment options, but first-line treatment is a beta-blocker. Since esmolol has a shorter half-life than other beta-blockers, it is appropriate because of his history of chronic obstructive pulmonary disease and heart failure.

Reference:

1. Hirtazka LF, Bakris GL, Bekman JA, et al; American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines; American Association for Thoracic Surgery; American College of Radiology; American Stroke Association; Society of Cardiovascular Anesthesiologists; Society for Cardiovascular Angiography and Interventions; Society of Interventional Radiology; Society of Thoracic Surgeons; Society for Vascular Medicine. 2010 ACCF/AHA/AATS/ACR/ASA/SCA/SCAI/SIR/STS/SVM guidelines for the diagnosis and management of patients with thoracic aortic disease. *J Am Coll Cardiol*. 2010 Apr 6;55(14): e27-e129.

5. Rationale

Answer: A

In adult patients with severe trauma, practice guidelines recommend transfusing equal amounts (1:1:1) of packed RBCs, fresh frozen plasma (FFP), and platelets during the early empiric phase of resuscitation. Maintaining this transfusion ratio minimizes dilution of circulating clotting factors and platelets. This patient has received equal amounts of packed RBCs and FFP but has not received any platelets. Normal saline and human albumin should not be administered because

resuscitation should be limited to maintain systolic blood pressure of approximately 90 mm Hg to prevent further dilutional coagulopathy and renewed bleeding. In patients with traumatic brain injuries, fluid resuscitation with albumin has been associated with higher mortality rates versus resuscitation with saline.

References:

1. Cannon JW, Khan MA, Raja AS, et al. Damage control resuscitation in patients with severe traumatic hemorrhage: a practice management guideline from the Eastern Association for the Surgery of Trauma. *J Trauma Acute Care Surg*. 2017 Mar;82(3):605-617.
2. Holcomb JB, Jenkins D, Rhee P, et al. Damage control resuscitation: directly addressing the early coagulopathy of trauma. *J Trauma*. 2007 Feb;62(2):307-310.
3. SAFE Study Investigators; Australian and New Zealand Intensive Care Society Clinical Trials Group; Australian Red Cross Blood Service; George Institute for International Health, et al. Saline or albumin for fluid resuscitation in patients with traumatic brain injury. *N Engl J Med*. 2007 Aug 30;357(9):874-884.
4. Spahn DR, Bouillon B, Cerny V, et al. The European guideline on management of major bleeding and coagulopathy following trauma: fifth edition. *Crit Care*. 2019 Mar 27;23(1):98. doi.

6. Rationale

Answer: A

Fat embolism syndrome is characterized by respiratory distress, confusion, and a petechial rash over the anterior upper torso, axillary regions, and conjunctiva. Typical findings on chest CT are diffuse, patchy ground-glass opacities with intralobular septal thickening (crazy-paving pattern) and small (< 10 mm) ill-defined centrilobular nodules in a predominately peripheral and upper lobe distribution often located in the subpleural regions and along the interlobular septa (thought to represent

the vasculogenic origin of mechanical obstruction by fat globules). Bilateral symmetrical septal lines with bronchial cuffing, vascular engorgement, and pleural effusions are characteristic of pulmonary edema. Pulmonary artery intraluminal filling defects are seen with pulmonary embolism, while lobar consolidation is seen with pneumonia. While all 4 are possible, the timing and symptoms suggest fat embolism syndrome as the most likely diagnosis.

Reference:

1. Newbigin K, Souza CA, Torres C, et al. Fat embolism syndrome: state-of-the-art review focused on pulmonary imaging findings. *Respir Med*. 2016 Apr; 113:93-100.

7. Rationale

Answer: B

This patient's findings of septic shock in the context of physical, laboratory, and imaging findings suggestive of a complex necrotizing soft tissue infection should prompt immediate resuscitation, initiation of broad-spectrum IV antibiotics, and aggressive surgical debridement to obtain source control. Source control with aggressive surgical debridement is essential to successful management of necrotizing soft tissue infections. The role of hyperbaric oxygen therapy in the treatment of necrotizing soft tissue infection is unclear.

References:

1. Pasternack MS, Swartz MN. Cellulitis, necrotizing fasciitis, and subcutaneous tissue infections. In: Bennett JE, Dolin R, Blaser MJ, eds. *Mandell, Douglas, and Bennett's Principles and Practice of Infectious Diseases*. 8th ed. Elsevier; 2015:1194-1215.

2. Stevens DL, Bryant AE. Necrotizing soft-tissue infections. *N Engl J Med*. 2017 Dec 7;377(23):2253-2265.

8. Rationale

Answer: D

Haemobilia is an uncommon complication of liver trauma occurring in less than 3% of patients with liver trauma. It is immediately life threatening. Management is different than for most other upper gastrointestinal bleeding that can often be treated with endoscopic interventions. Haemobilia most commonly presents with gastrointestinal bleeding and may be associated with jaundice depending on how rapid the presentation is. In the setting of recent trauma, CT angiography may be performed before endoscopic evaluation but the finding of blood at the ampulla should raise concern, although it is not always found on endoscopy. The mainstay of initial treatment is transcatheter arterial embolization, which has a success rate of 80% to 100%. Endoscopic retrograde cholangiopancreatography may still have a role if biliary clearance is needed or if the etiology dictates (eg, perihilar invasive hepatocellular carcinoma).

References:

1. Cathcart S, Birk JW, Tadros M, Schuster M. Hemobilia: an uncommon but notable cause of upper gastrointestinal bleeding. *J Clin Gastroenterol.* 2017 Oct;51(9):796-804.
2. Navuluri R. Hemobilia. *Semin Intervent Radiol.* 2016 Dec;33(4):324-331.

9. Rationale

Answer: D

Acute compartment syndrome is the diagnosis most likely to produce this patient's constellation of findings. The presence of intact distal pulses makes arterial thrombosis unlikely. Cholesterol embolization is less likely because of the absence of skin findings. Venous thrombosis is unlikely to result in severe pain in the absence of significant edema. Statin-induced rhabdomyolysis usually occurs weeks to months after the initiation of therapy and is unlikely to produce focal muscle pain.

References:

1. Te Kolste HJ, Balm R, de Mol B. Acute compartment syndrome of the lower leg after coronary artery bypass grafting: a silent but dangerous complication. *Thorac Cardiovasc Surg*. 2015 Jun;63(4):300-306.

2. Von Keudell AG, Weaver MJ, Appleton PT, et al. Diagnosis and treatment of acute extremity compartment syndrome. *Lancet*. 2015 Sep 26;386(10000):1299-1310.

10. Rationale

Answer: C

For a hemodynamically stable patient, conservative, nonoperative management of blunt solid organ injury is appropriate, even for high-grade injuries. This patient's abdominal CT shows a grade III liver laceration, and he is hemodynamically stable. Observation with serial abdominal examinations is appropriate. Repeat imaging is indicated for hemodynamic instability, persistent systemic inflammatory response, increasing abdominal pain, jaundice, or unexpected decrease in hemoglobin. Immediate laparotomy is not indicated in a hemodynamically stable patient. Angioembolization would be indicated initially for active contrast extravasation on abdominal CT or for persistent hemorrhage leading to hemodynamic instability. Percutaneous drainage of a subcapsular or intrahepatic hematoma is not indicated.

Reference:

1. Stassen NA, Bhullar I, Cheng JD, et al; Eastern Association for the Surgery of Trauma. Nonoperative management of blunt hepatic injury: an Eastern Association for the Surgery of Trauma practice management guideline. *J Trauma Acute Care Surg*. 2012 Nov;73(5 Suppl 4): S288-S293.

11. Rationale

Answer: C

This patient's findings of abrupt onset of subcutaneous emphysema, pneumomediastinum, and signs of respiratory distress following tracheostomy should alert the clinician to possible tracheal injury. Although uncommon, laceration of the

posterior membranous trachea, and potentially the underlying esophagus, can be a devastating complication. Placement of a cuffed airway below the level of the injury excludes continued air leakage under positive pressure. Broad-spectrum IV antibiotics should be administered to mitigate potential mediastinitis. Esophagoscopy is mandatory to exclude esophageal injury. Exploration of the neck may be needed if conservative management of tracheal injury fails or if there is a concurrent esophageal injury. IV antibiotics alone will not address a tracheal injury. Tracheal stenting is an emerging but unproven methodology for initial management of tracheal injury.

References:

1. Cunningham LC, Jatana KR, Grischkan JM. Conservative management of iatrogenic membranous tracheal wall injury: a discussion of 2 successful pediatric cases. *JAMA Otolaryngol Head Neck Surg*. 2013 Apr;139(4):405-410.

2. Schneider T, Storz K, Dienemann H, Hoffmann H. Management of iatrogenic tracheobronchial injuries: a retrospective analysis of 29 cases. *Ann Thorac Surg*. 2007 Jun;83(6):1960-1964.

12. Rationale

Answer: D

This patient's findings are suspicious for myoglobinuria secondary to traumatic rhabdomyolysis. To mitigate the nephrotoxic effects of myoglobinuria, brisk diuresis is promoted with isotonic fluid administration, and alkaline urine is achieved with bicarbonate administration. While compartment syndrome necessitating fasciotomy can occur with crush injuries, the findings do not suggest compartment syndrome. Renal replacement therapy is required for refractory acute kidney injury but is not typically an early intervention. Antibiotic therapy and anticoagulation are inappropriate currently.

References:

1. Bosch X, Poch E, Grau JM. Rhabdomyolysis, and acute kidney injury. *N Engl J Med*. 2009 Jul 2;361(1):62-72.

2. Torres PA, Helmstetter JA, Kaye AM, Kaye AD. Rhabdomyolysis: pathogenesis, diagnosis, and treatment. *Ochsner J.* 2015 Spring;15(1):58-69.

13. Rationale

Answer: A

The findings of blunt anterior chest wall injury combined with the cross-sectional imaging findings of sternal fracture, abnormal ECG findings (ectopy, ST depression, T wave inversion), and cardiac enzyme elevation are most consistent with the diagnosis of blunt cardiac injury. Pericardial tamponade, diaphragmatic rupture, and aortic injury can occur following blunt thoracic trauma, but the physical findings and imaging are not consistent with these injuries.

References:

1. Asensio JA, et al. Trauma to the heart. In: Feliciano DV, Mattox KL, Moore EE. *Trauma*. 6th ed. McGraw Hill; 2008.

2. Clancy K, Velopulos C, Bilaniuk JW, et al; Eastern Association for the Surgery of Trauma. Screening for blunt cardiac injury: an Eastern Association for the Surgery of Trauma practice management guideline. *J Trauma Acute Care Surg*. 2012 Nov;73(5 Suppl 4): S301-S306.

3. Schultz JM, Trunkey DD. Blunt cardiac injury. *Crit Care Clin*. 2004 Jan;20(1):57-70.

14. Rationale

Answer: E

Despite Glasgow Coma Scale score of 15, excellent oxygen saturations, and nonlabored respirations, this patient's difficulty swallowing could indicate either an esophageal injury or airway injury. The first step in multisystem trauma is to assess the ABCDEs: airway, breathing, circulation, disability, and exposure. Evaluation of his difficulty swallowing is the priority.

Reference:

1. Henry S. *ATLS Advanced Trauma Life Support Student Course Manual*. 10th ed. American College of Surgeons; 2018:7.

15. Rationale

Answer: D

Surgical stabilization of rib fractures (SSRF) continues to evolve as an effective approach to avoid respiratory failure and improve long-term quality-of-life indicators for patients with displaced rib fractures. Guidelines center on early SSRF and recommend that only flail segment fracture patterns be routinely repaired while other fracture types be considered, but there is no evidence to support SSRF in these cases. The term "early" is within 72 hours of injury. Further research is ongoing to identify the most appropriate patients for SSRF and indications may expand in the future but patients whose pain resolves and those undergoing other procedures should not routinely undergo repair. Nondisplaced fractures rarely require SSRF.

References:

1. Brasel KJ, Moore EE, Albrecht RA, et al. Western Trauma Association critical decisions in trauma: management of rib fractures. *J Trauma Acute Care Surg*. 2017 Jan;82(1):200-203.

2. Senekjian L, Nirula R. Rib fracture fixation: indications and outcomes. *Crit Care Clin*. 2017 Jan;33(1):153-165.

Part 14.

Transplantation

Instructions: For each question, select the most correct answer.

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1. A 48-year-old man develops hypotension on postoperative day 4 after uncomplicated orthotopic liver transplantation for alcoholic cirrhosis (preoperative Model for End-stage Liver Disease [MELD] score 22). Vital signs are temperature 38.0°C (100.4°F), heart rate 112 beats/min, blood pressure 80/40 mm Hg, and oxygen saturation as measured by pulse oximetry 98% on room air. His hemodynamic profile shows central venous pressure 8 mm Hg, pulmonary artery pressure 34/14 mm Hg, and cardiac index 3.4 L/min. Transthoracic echocardiography shows a hyperdynamic left ventricle with normal right ventricular function and size. There is no evidence of valvular abnormalities. Laboratory abnormalities are WBCs 22,000/μL, hematocrit unchanged from immediate postoperative level of 32%, and lactate 14.0 mg/dL. Liver function testing shows: AST 1100 U/L, ALT 1200 U/L, alkaline phosphatase 600 U/L, and bilirubin 3.5 mg/dL. Which of the following is the most likely diagnosis?

   A. Hepatic artery thrombosis

   B. Portal venous thrombosis

   C. Hemorrhagic shock

   D. Acute pulmonary embolism

2. A 55-year-old-woman with a history of stage D heart failure secondary to viral cardiomyopathy underwent orthotopic heart transplantation. In the operating room, she initially had severe hypotension and shock

requiring norepinephrine, 0.15 μg/kg/min, and dobutamine, 10 μg/kg/min. On arrival in the cardiovascular ICU, hemodynamics include heart rate 110 beats/min, mean arterial pressure 65 mm Hg, pulmonary artery pressure 55/40 mm Hg, cardiac index 1.8 L/min/m^2, systemic vascular resistance 800 dynes-sec-cm^{-5}, and left atrial pressure 15 mm Hg. Which of the following medications is most indicated at this point?

A. Bosentan

B. Inhaled nitric oxide

C. IV epoprostenol

D. Tadalafil

E. Nitroprusside

3. A 30-year-old woman with type 1 diabetes mellitus underwent a pancreatic transplant without complications. She needed only a brief period of vasopressor support intraoperatively. Urine output throughout the case was adequate. Afterward, she is admitted to the surgical ICU, where insulin infusion at 1 unit/hr is continued. After 8 hours, serum glucose continues to rise, and her insulin requirement quickly escalates to 10 units/hr. Urine output, hemoglobin, and hematocrit levels are unchanged. Which of the following is the most appropriate next step?

   A. Administer 2 liters of crystalloids.

   B. Administer IV furosemide, 80 mg.

   C. Notify the transplant team for emergent operative intervention.

   D. Obtain CT.

4. A 56-year-old man was admitted for status epilepticus after phenytoin failure. The seizures were controlled, and he was switched from phenytoin to levetiracetam. His remaining home medications were restarted, including oral lisinopril, 20 mg daily; oral lansoprazole, 15 mg daily; oral prednisone, 20 mg daily; and oral tacrolimus, 15 mg twice daily. Two days later, he is found to have severe acute kidney injury (AKI) requiring renal replacement therapy. Which of the following is the most likely cause of his AKI?

   A. Lisinopril

   B. Tacrolimus

   C. Levetiracetam

   D. Acute illness

5. A 25-year-old woman is in the lCU after receiving a living related donor renal transplant from her sister. She has never required dialysis and does not have diabetes. Immunosuppression consists of azathioprine, 3 mg/kg/day, and methylprednisolone, 1 g preoperatively, followed by oral prednisone, 60 mg daily. She has done well in the lCU postoperatively and is awake and alert. Urine output is 500 mL/h, which was managed by milliliter-for-milliliter urine output replacement with 5% dextrose, 0.45% sodium chloride solution with potassium chloride, 20 mEq/L. The next morning, she is slightly disoriented and has weakness and muscle spasms in her extremities. Cardiac rhythm strip shows a slightly prolonged QT interval with broadened T waves and U waves. Vital signs are within normal limits and urine output is 130 mL/hr. Morning laboratory values are: hemoglobin 10.2 g/dL, WBCs 5000/μL, sodium 137 mEq/L, potassium 4.7 mEq/L, calcium 9.6 mg/dL (ionized calcium 4.3 mg/dL), magnesium 1.7 mg/dL, phosphorus 2.5 mg/dL, and blood glucose 180

mg/dL. The most appropriate initial management includes

A. forced diuresis with furosemide.

B. magnesium sulfate, 1 g IV.

C. supplemental calcium, 100 mg IV.

D. potassium chloride, 10 mEq supplements every hour for 3 hours.

E. IV phosphate infusion, 0.02 mg/kg/hr.

Part 14 Answers:

Transplantation

1. Rationale

Answer: A

Major complications of orthotopic liver transplantation (OLT) can be intraoperative, acute, or delayed. Intraoperative complications include hyperacute graft dysfunction, pulmonary embolism, hemorrhagic shock, reperfusion injury, and life-threatening reactions to blood or blood products. Acute complications include primary graft dysfunction, hepatic artery thrombosis (HAT), portal venous thrombosis, septic shock, and organ system dysfunction or failure from perioperative complications. Delayed complications include delayed graft dysfunction and biliary anastomotic leak. HAT is a very severe early complication of OLT; it can lead to catastrophic necrosis of the newly implanted liver. HAT usually presents with fever, leukocytosis, hypotension, and acutely increased hepatic enzymes (AST, ALT, and alkaline phosphatase). Lactate is also frequently elevated. Bilirubin is usually only minimally increased in the first 12 to 24 hours of HAT but rapidly increases afterward. Because HAT frequently leads to rapid, fulminant liver failure symptoms, patients become critically ill, have a high mortality rate, and are frequently listed for emergency re-transplantation. Pulmonary embolism is typically associated with hypoxemia and evidence of right ventricular dysfunction, eg, elevated central venous pressure, or evidence of right ventricular distention on echocardiography. Hemorrhagic shock is typically associated with tachycardia and evidence of low cardiac output with decreased central venous pressure and pulmonary artery pressure. Hematocrit is frequently decreased. Biliary anastomotic leak is associated with increased bilirubin and hypotension, like septic shock; however, generally hepatic enzymes increase only mildly. Portal venous thrombosis may lead to bowel ischemia and hyperbilirubinemia but hemodynamic abnormalities and significantly increased

hepatic enzymes are not a normal finding after liver transplantation.

References:

1. Moreno R, Berenguer M. Post-liver transplantation medical complications. *Ann Hepatol*. 2006 Apr-Jun;5(2):77-85.

2. Mourad MM, Liossis C, Gunson BK, et al. Etiology and management of hepatic artery thrombosis after adult liver transplantation. *Liver Transpl*. 2014 Jun;20(6):713-723.

3. Pareja E, Cortes M, Navarro R, Sanjuan F, Lopez R, Mir J. Vascular complications after orthotopic liver transplantation: hepatic artery thrombosis. *Transplant Proc*. 2010 Oct;42(8):2970-2972.

2. Rationale

Answer: B

This patient has pulmonary hypertension, with mean pulmonary arterial pressure 45 mm Hg. Bosentan has not been studied for perioperative pulmonary hypertension and is not appropriate for acute use. Inhaled epoprostenol has been shown to be like inhaled nitric oxide in reducing pulmonary hypertension and right ventricular dysfunction; however, IV formulations are not specific and can carry a risk of systemic hypotension. Tadalafil has been studied for use in group I (idiopathic) and group II (left ventricular associated) pulmonary hypertension; however, it is not endorsed by the International Society of Heart and Lung Transplantation for use in the perioperative setting. Nitroprusside can help lower pulmonary pressures, particularly by unloading the left ventricle; however, this patient is on norepinephrine and has low systemic vascular resistance, which would have opposing effects.

References:

1. Ardehali A, Hughes K, Sadeghi A, et al. Inhaled nitric oxide for pulmonary hypertension after heart transplantation. *Transplantation*. 2001 Aug;72(4):638-641.

2. Costanzo MR, Dipchand A, Starling R, et al; International Society of Heart and Lung Transplantation Guidelines. The International Society of Heart and Lung Transplantation guidelines for the care of heart transplant recipients. *J Heart Lung Transplant*. 2010 Aug;29(8):914-956.

3. Khan TA, Schnickel G, Ross D, et al. A prospective, randomized, crossover pilot study of inhaled nitric oxide versus inhaled prostacyclin in heart transplant and lung transplant recipients. *J Thorac Cardiovasc Surg*. 2009 Dec;138(6):1417-1424.

3. Rationale

Answer: C

This patient has thrombosis of the pancreatic venous graft, which is the most common vascular complication of pancreatic transplant. Thrombosis is usually venous, with an incidence of 5% to 8%; arterial thrombosis is less common. Acute venous thrombosis usually starts with a sudden rise in serum glucose, which occurred when her insulin requirement rose from 1 to 10 units/hr. Treatment for pancreatic graft thrombosis is early operative intervention. She had already received 2 liters of crystalloid, so more fluids are not needed. She does not have oliguria since her urine output is unchanged, so furosemide is not indicated. Hemoglobin and hematocrit are unchanged, which suggests that she is not bleeding, so transfusion of blood products is not indicated. For patients with suspected pancreatic venous graft thrombosis, CT is not the diagnostic test of choice, but perfusion imaging of the graft should be performed.

Reference:

1. Kaufman, DB. Complications of pancreatic transplantation. In: Mulholland MW, Doherty GM. *Complications in Surgery*. 2nd ed. Lippincott Williams and Wilkins; 2011; chap 49.

4. Rationale

Answer: B

While angiotensin-converting enzyme inhibitors can cause an increase in serum creatinine, this usually presents within the first week of initiating therapy when angiotensin II levels are rapidly reduced. There is a cytochrome P450 interaction between phenytoin and tacrolimus. As a cytochrome inducer, phenytoin induces the metabolism of tacrolimus, thereby decreasing its level. Patients on phenytoin and tacrolimus require higher doses of tacrolimus to maintain therapeutic levels. On discontinuing phenytoin, the tacrolimus dose must be decreased, and levels closely monitored. Levetiracetam is not associated with nephrotoxicity. While acute illness can be contributory, drug-induced acute kidney injury is more likely in this patient.

References:

1. Christians U, Jacobsen W, Benet LZ, Lampen A. Mechanisms of clinically relevant drug interactions associated with tacrolimus. *Clin Pharmacokinet.* 2002;41(11):813-851.

2. Hollenberg NK, Swartz SL, Passan DR, Williams GH. Increased glomerular filtration rate after converting-enzyme inhibition in essential hypertension. *N Engl J Med.* 1979 Jul 5;301(1):9-12.

5. Rationale

Answer: B

This patient has a change in mental status 24 hours after a living related renal transplant. Significant posttransplant diuresis (500 mL/hr) is frequently associated with tubular dysfunction caused by a brief period of warm ischemia and short renal preservation. This diuresis may lead to electrolyte loss including hypokalemia, hypomagnesemia, and hypophosphatemia. Mental status changes are nonspecific for many of the electrolyte disturbances that can occur in the ICU. Neuromuscular weakness and irritability may be associated with hypocalcemia, hypomagnesemia, and hypokalemia. Her urine output is appropriate, and furosemide could accentuate the symptomatic electrolyte abnormalities. A serum magnesium

level at the lower limits of normal may not reflect a normal ionized magnesium level; there is no clinically available ionized magnesium determination. Even a normal serum total magnesium level may mask a magnesium deficiency manifested clinically by low levels of ionized magnesium. The ECG changes showing prolonged QT interval with broadened T waves and the presence of U waves, while classically associated with hypokalemia, can also be seen with hypomagnesemia. The normal ionized calcium level makes symptomatic hypocalcemia unlikely. The potassium level was normal, and she was receiving a supplemental potassium infusion. Hypophosphatemia can occur similarly under these circumstances, but symptoms are unusual unless the serum phosphorus concentration is less than 2 mg/dL.

References:

1. Barke RA. Fluid and electrolytes. In: Abrams JH, Cerra FB, eds. *Essentials of Surgical Critical Care: Clinical Cases and Practical Solutions*. Quality Medical Publishing; 1993:488.

2. Carroll HJ, Oh MS. *Water, Electrolyte, and Acid-Base Metabolism: Diagnosis and Management*. 2nd ed. Lippincott Williams & WIlkins; 1989.

3. Grusner RWG. Critical care problems in kidney transplant recipients. In: Rippe JM, Erwin RS, Fink MP, Cerra FB, eds. *Intensive Care Medicine*. 3rd ed. Little Brown; 1996:1372.

Part 15.

Environmental Injury and Disaster Management

Instructions: For each question, select the most correct answer.

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1. A 42-year-old man is airlifted to a hospital after a failed winter summit attempt of a 4000-m (13,000-ft) peak. His feet are hard and cold with hemorrhagic bullae. He is awake and shivering and answers questions appropriately. Core temperature is 36.8°C (98.2°F), and the rest of his vital signs are normal. Which of the following therapies is most likely to lead to the best possible outcome for this patient?

   A. Acetazolamide

   B. Dexamethasone

   C. Enoxaparin

   D. Methylene blue

   E. Phentolamine

2. A 22-year-old man with no past medical history presents after hiking in the mountains. He and his friends are hiking in the area for a few days. His friends note that he began to "walk funny" soon after they started hiking and has become progressively more confused and lethargic. On examination, he is afebrile, normotensive, and not tachycardic. Oxygen saturation is 95% on room air. Examination is remarkable for truncal ataxia; he cannot ambulate without assistance. Noncontrast head CT findings are normal. Laboratory results show WBCs 9400/L, hemoglobin 13.5 g/dL, sodium 137 mEq/L, and creatinine 0.9 mg/dL. Alcohol level and urine drug

screen are both negative. Which of the following is the most appropriate next step?

A. Brain MRI

B. CT angiogram of head and neck

C. Dexamethasone

D. Acetazolamide

E. Serum B12 level

3. A 75-kg (165-lb), 45-year-old woman is admitted to the burn unit 4 hours after sustaining a 25% total body surface area burn. She is receiving 400 mL/hr lactated Ringer solution. Urine output has been 20 mL/hr for the past 2 hours, although her mean arterial blood pressure has remained within the normal range of 70 to 90 mm Hg. Which of the following is the most appropriate next step in management?

   A. Add 5% albumin, 30 mL/hr, to her fluids.

   B. Add hypertonic saline, 3% at 40 mL/hr.

   C. Increase her IV fluids to 500 mL/hr.

   D. Administer a 1-L bolus of lactated Ringer solution.

   E. Transfuse 2 units packed RBCs.

Part 15 Answers:

Environmental Injury and Disaster Management

1. Rationale

Answer: C

Microvascular thrombosis is a major complication of frostbite. Treatment includes ibuprofen, low-molecular-weight heparin, and consideration of intra-arterial tissue plasminogen activator. These have been shown to reduce the need for amputation. Other beneficial therapies may include vasodilators such as nitroglycerine, hydration, debridement, pain medication, elevation, and tetanus prophylaxis.

References:

1. McIntosh SE, Freer L, Grissom CK, et al. Wilderness Medical Society clinical practice guidelines for the prevention and treatment of frostbite: 2019 update. *Wilderness Environ Med*. 2019 Dec;30(4S): S19-S32.
2. Russell K, Cochran A. Cold Injury. In: Rodway GW, Weber DC, McIntosh SE. *Mountain Medicine, and Technical Rescue*. Carreg; 2016:192-195.

2. Rationale

Answer: C

This patient has signs of high-altitude cerebral edema (HACE), which is characterized by headache, ataxia, confusion, lethargy, or altered mental status that can progress to coma and death. While stroke is a consideration, his history and age make this less likely. The treatment for HACE is dexamethasone. Acetazolamide can be used in mild cases and as prophylaxis; however, he is lethargic and altered.

References:

1. Luks AM, Swenson ER, Bärtsch P. Acute high-altitude sickness. *Eur Respir Rev.* 2017 Jan 31;26(143):160096.

2. West JB. High-altitude medicine. *Am J Respir Crit Care Med*. 2012 Dec 15;186(12):1229-1237.

3. Rationale

Answer: C

Adequate IV fluid resuscitation, generally measured by urine output in burn patients, is of critical importance. Several formulas are used to calculate fluid requirements in the first 24 hours, with half administered in the first 8 hours. Most burn unit protocols recommend increasing the rate of fluid administration by 20% or 100 mL/hr as the first step in patients with diminished urine output. Although albumin has been used in burn patients, it has not been shown to improve mortality and is generally not administered within the first 8 hours of burn resuscitation. Hypertonic saline is generally avoided in the acute resuscitation of burn patients. There is no indication for transfusion in the absence of ongoing blood loss or hypotension. Because of the concern for over-resuscitation, bolus IV fluids are avoided unless there is hypotension.

References:

1. Mosier M, Cancio LC, Amani H, et al. Burn resuscitation. In: American Burn Association Consensus Statements.2013. Accessed March 6, 2022. https://apps.dtic.mil/sti/pdfs/ADA619608.pdf

2. Department of Surgical Education, Orlando Regional Medical Center. Acute adult burn resuscitation. Revised November 26, 2019. Accessed June 10, 2020. http://www.surgicalcriticalcare.net/Guidelines/Burn%20Resuscitation%202019.pdf

3. Milner S, Caffrey JC. Medical management of the burn patient. In: Cameron JL, Cameron AM. *Current Surgical Therapy*. 12th ed. Elsevier; 2016.

Part 16.

Pharmacology and Toxicology

Instructions: For each question, select the most correct answer.

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1. An 83-year-old woman with a past medical history of hypertension, Child-Pugh B cirrhosis, and stage 3 chronic kidney disease is admitted to the medical ICU for the treatment of hypertensive emergency. She is drowsy and cannot reliably swallow oral medications. Given her underlying cirrhosis and kidney disease, which of the following is the most appropriate medication for her hypertensive emergency?

   A. Clevidipine

   B. Labetalol

   C. Nicardipine

   D. Sodium nitroprusside

   E. Enalaprilat

2. A 34-year-old man was admitted to the trauma ICU 16 days ago after a go-cart accident. He has a past medical history of a seizure disorder for which he takes levetiracetam and valproic acid, which have both been continued during his admission. His initial injuries include a portal vein injury, pancreatic transection, colonic mesenteric injury, and a traumatic flank hernia. On hospital day 1, he underwent an emergent exploratory laparotomy with portal vein repair, right colectomy, and pancreaticoduodenectomy. His hospital course has been complicated by development of severe infected necrotizing pancreatitis and intolerance of enteral nutrition, necessitating parenteral nutrition support. On hospital day 16, he develops septic shock and is

empirically started on meropenem, vancomycin, and micafungin. Which of the following significant drug-drug interactions is most concerning and should be mitigated by which strategy to reduce the risk of harm to this patient?

A. Because both vancomycin and micafungin are hepatotoxic, avoid this combination. Administer linezolid instead of vancomycin.

B. Because meropenem decreases valproic acid concentrations, avoid this combination if possible. If meropenem must be used, administer an alternative antiepileptic medication.

C. Because meropenem increases valproic acid concentrations, empirically reduce the valproic acid dose and closely monitor valproic acid concentration.

D. Because micafungin decreases meropenem concentrations, use high-dose, extended-infusion meropenem to maintain adequate concentrations.

3. Which of the following patients is most at risk for developing augmented renal clearance during a hospital stay?

A. 52-year-old man in the medical ICU for diabetic ketoacidosis

B. 27-year-old man in the trauma ICU with a traumatic brain injury after a motor vehicle accident

C. 62-year-old woman in the medical ICU for severe chronic obstructive pulmonary disease exacerbation

D. 75-year-old woman with non-ST segment elevation myocardial infarction

4. A 48-year-old man with a past medical history of a seizure disorder presents to the medical ICU intubated

and mechanically ventilated for treatment of severe acute respiratory distress syndrome from community-acquired staphylococcal pneumonia. Despite receiving appropriate early antimicrobial coverage with clindamycin, his hypoxemia steadily progresses. He is deeply sedated with a midazolam infusion, neuromuscular blockade is induced with rocuronium, and he is administered inhaled nitric oxide. A nicardipine infusion is started for hypertension, and he is restarted on his outpatient carbamazepine for his seizure disorder. Effective neuromuscular blockade cannot be obtained despite frequent rocuronium doses. Which of his medications most likely explains his response to rocuronium?

A. Clindamycin

B. Midazolam

C. Nitric oxide

D. Nicardipine

E. Carbamazepine

5. A 23-year-old woman with a past medical history of depression, anxiety, and gangrenous cholecystitis underwent an uncomplicated open cholecystectomy under general anesthesia. Afterward, on arrival in the surgical ICU, she is nauseated and receives metoclopramide and droperidol. She is restarted on her outpatient medications of gabapentin, acetaminophen, and fluoxetine. She reports severe abdominal pain and receives scheduled tramadol for analgesia, but her pain does not improve. Which of the following medications most likely explains her response to tramadol?

   A. Metoclopramide

   B. Droperidol

   C. Gabapentin

D. Acetaminophen

E. Fluoxetine

6. A 36-year-old woman is admitted to the ICU after being intubated out of hospital for airway protection. She is found to have a profound anion gap metabolic acidosis with lactate level greater than 4 mmol/dL that minimally responds to fluid resuscitation. Her blood pressure has been normal. She takes the following medications: dolutegravir/abacavir/lamivudine, escitalopram, insulin glargine, levothyroxine, pantoprazole, and sitagliptin. Which of these may have contributed to her lactatemia?

   A. Dolutegravir/abacavir/lamivudine

   B. Escitalopram

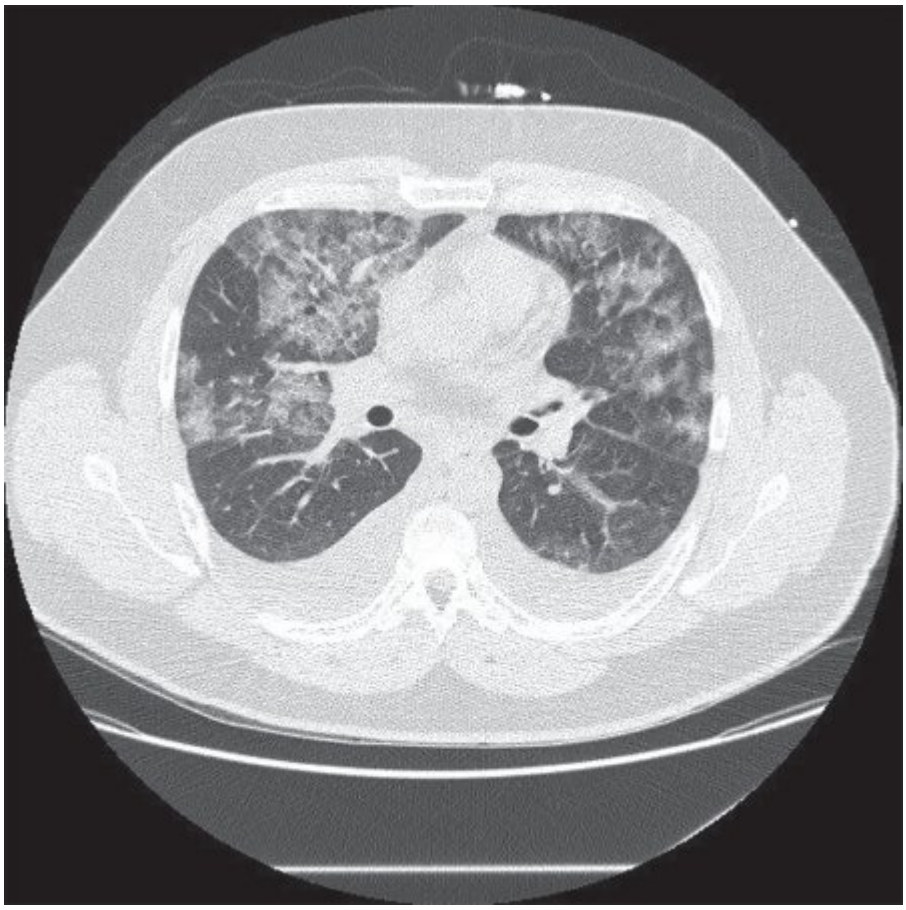
   C. Insulin glargine

   D. Levothyroxine

   E. Pantoprazole

7. A 65-year-old woman with a history of uncontrolled diabetes, end-stage renal failure for which she undergoes chronic hemodialysis, and IV drug abuse presents with a lower extremity ulcer and is found to have osteomyelitis. Blood cultures grow methicillin-resistant *Staphylococcus aureus*. Echocardiography shows evidence of vegetation on the tricuspid valve and left ventricular ejection fraction is 25%. Because of antibiotic susceptibility, she is started on daptomycin. Her course is complicated by atrial fibrillation with rapid ventricular response. She does not tolerate diltiazem because of bradycardia; she is transitioned to amiodarone and started on a therapeutic dose of enoxaparin. She does well on the medical ward until 12 days later when she develops shortness of breath, cough, fever, and chest pains with inspiration. Laboratory findings are WBCs 26,000/μL (65%

neutrophils, 20% lymphocytes, 10% eosinophils, 5% basophils), hemoglobin 10 g/dL, hematocrit 30%, platelets 95,000/μL, creatine kinase 50 IU/L, serial troponin I level 0 for all 3 values, and brain natriuretic peptide 700 pg/mL. ECG shows sinus bradycardia. Antinuclear antibody, antineutrophil cytoplasmic antibodies, complement levels, and glomerular basement antibody levels are normal. Chest CT angiogram is shown below. She receives furosemide and diuresis 1 L urine with only mild improvement of symptoms. Cefepime is started. She becomes progressively hypoxemic and is transferred to the ICU and intubated. Bronchoscopy with bronchoalveolar lavage is performed with serial aliquots of saline with return of clear fluid that is sent for culture. In addition to starting steroids, which of the following measures is most likely to result in rapid improvement of her condition?



A. Stop enoxaparin.

B. Stop amiodarone.

C. Stop daptomycin.

D. Begin continuous infusion of furosemide.

E. Begin rituximab.

8. Which of the following early interventions has been associated with a mortality benefit in the management of anaphylactic shock?

   A. Corticosteroids

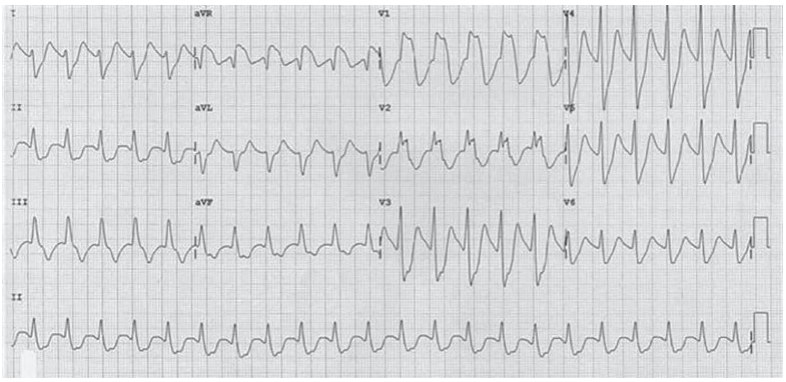
   B. Epinephrine

   C. Histamine-1 receptor antagonists

   D. Histamine-2 receptor antagonists

9. A 32-year-old woman with a history of depression is brought to the emergency department by her family after being found unresponsive at home with an empty pill bottle next to her. Temperature is 37.8°C (100.1°F), heart rate 110 beats/min, blood pressure 100/55 mm Hg, and oxygen saturation as measured by pulse oximetry 98% on room air. She is unresponsive to painful stimuli. Pupils are dilated and sluggishly reactive. Muscle tone and deep tendon reflexes are normal. There is no clonus. Her abdomen is slightly distended, and bowel sounds are diminished. Her skin is flushed and dry. Laboratory findings are WBC count 10,000/μL, hemoglobin 12 g/dL, platelet count 375,000/μL, sodium 141 mEq/L, potassium 4.7 mEq/L, chloride 111 mEq/L, bicarbonate 23 mEq/L, BUN 19 mg/dL, creatinine 0.9 mg/dL, calcium 9 mg/dL, magnesium 2.4 mEq/L, glucose 104 mg/dL, and creatinine kinase 130 U/L. Serum toxicology screen is negative for salicylates, ethanol, and acetaminophen. Liver function tests are within normal

limits. ECG is shown below. Which of the following is the most appropriate IV medication?



A. Cyproheptadine

B. Sodium bicarbonate

C. Magnesium

D. Naloxone

10. A 64-year-old man with a past medical history of hypertension and polysubstance abuse was found unconscious at home. He is admitted to the ICU for hypercarbic respiratory failure and altered mental status. A vasopressor and continuous venovenous hemodiafiltration therapy are initiated. Which of the following best describes a drug that is most likely to be highly eliminated by continuous venovenous hemodiafiltration therapy?

    A. Large molecule, high protein binding, large volume of distribution

    B. Small molecule, high protein binding, small volume of distribution

    C. Small molecule, low protein binding, small volume of distribution

    D. Large molecule, low protein binding, small volume of distribution

11. A 24-year-old man with a past medical history of depression is found unresponsive in his garage with a

vehicle running. He was last seen well by a friend approximately 3 hours before presentation. On arrival in the emergency department, temperature is 36.9°C (98.5°F), heart rate 95 beats/min, blood pressure 100/70 mm Hg, respiratory rate 22 breaths/min, and oxygen saturation 97% on room air. He localizes pain with his upper extremities but does not follow commands. His mucous membranes are dry, and his skin is cherry-red in appearance. Naloxone is administered with no improvement, and he is intubated. ECG shows normal sinus rhythm with QRS interval of 90 msec. Which of the following is the most appropriate treatment for his condition?

A. Additional naloxone

B. Hyperbaric oxygen therapy

C. Pralidoxime

D. IV sodium bicarbonate

12. A 35-year-old woman with a history of depression presents to the emergency department 5 hours after ingesting a large quantity of immediate-release verapamil tablets. On examination, she has decreased level of consciousness and is ill appearing. Heart rate is 30 beats/min, blood pressure 60/30 mm Hg, and respiratory rate 28 breaths/min. A 2-L bolus of IV crystalloid is administered and a high-dose norepinephrine infusion is started, but she remains hypotensive, with systolic blood pressure 75 mm Hg and heart rate 35 beats/min. IV atropine, 1 mg, is administered, with no change in heart rate or blood pressure. Central venous access is obtained, and a calcium chloride infusion is started, but she remains hypotensive. Which of the following is the most appropriate next intervention?

A. Extracorporeal removal of verapamil with hemodialysis

B. Administration of high-dose IV insulin and IV dextrose

C. Administration of activated charcoal by mouth

D. Administration of additional IV atropine

13. A 45-year-old man underwent partial small bowel resection for mesenteric ischemia due to superior mesenteric vein thrombosis and was treated for 7 days with enoxaparin while hospitalized. Three weeks later, he again presents with abdominal pain. CT scan shows bowel wall thickening with intestinal pneumatosis consistent with ischemic bowel and again shows superior mesenteric vein thrombosis. IV piperacillin/tazobactam and vancomycin are started, and an exploratory laparotomy with bowel resection and loop ileostomy is performed. Postoperatively, IV heparin infusion is started. On the day after surgery, he has shortness of breath and develops hypoxemia requiring high-flow nasal cannula therapy. Venous duplex scan shows a new left femoral vein thrombosis. Helical chest CT scan shows segmental pulmonary embolism. Laboratory findings are shown below. Which of the following is the most appropriate treatment approach?

Laboratory test	At Admission	24 Hours postadmission	36 Hours Postadmission	Reference range
WBC count, $x10^3/\mu L$	20	16	13	4-11
Hemoglobin, g/dL	12	9	8.8	13.5-17.5
Platelet Count, $10^3/\mu L$	420	565	65	150-400
International normalized ratio (INR)	1.1		1.0	<1.1
Partial thromboplastin time, sec	34		33	25-35

A. Continue heparin for new thrombosis.

B. Replace heparin with argatroban for heparin-induced thrombocytopenia.

C. Replace piperacillin/tazobactam with meropenem for antibiotic-induced thrombocytopenia.

D. Transfuse platelets for bleeding risk.

14. A 67-year-old woman is in the ICU with acute respiratory distress syndrome (ARDS) secondary to pneumonia. She is mechanically ventilated and now requires initiation of a neuromuscular blocking agent for ARDS. Her target Richmond Agitation-Sedation Scale goal is changed from –1 to –5. Which of the following analgesic/sedative combinations is most appropriate for this patient?

    A. Fentanyl infusion plus propofol infusion

    B. Dexmedetomidine infusion plus fentanyl infusion

    C. Fentanyl intermittent boluses plus propofol infusion

    D. Propofol infusion plus midazolam infusion

15. A 114-kg (251-lb), 29-year-old man is admitted for polytrauma. On 4, he is diagnosed with ventilator-associated pneumonia. He has no medication allergies. Laboratory results are AST 33 U/L, ALT 15 U/L, and serum creatinine 1.2 mg/dL (estimated creatinine clearance > 146 mL/min). Hemodynamics are stable, and urine output is greater than 0.5 mL/kg/h. Appropriate antibiotics are initiated to provide a sufficient spectrum of activity. Which of the following is the optimal pharmacokinetic/pharmacodynamic approach?

    A. Meropenem, 2 g over 30 minutes delivered every 8 hours

    B. Gentamicin, 2.5 mg/kg adjusted body weight over 30 minutes delivered every 8 hours

    C. Vancomycin, 2900 mg over 180 minutes, followed by 2200 mg every 12 hours over 120 minutes

    D. Piperacillin/tazobactam, 4.5 g over 30 minutes, followed by 4.5 g every 6 hours over 180 minutes

Part 16 Answers:

Pharmacology and Toxicology

1. Rationale

Answer: A

Hypertensive emergencies are characterized by severe blood pressure elevation accompanied by end-organ dysfunction. While prompt reduction in blood pressure is necessary to avoid organ injury, excessive hypotension from inappropriate dosage or prolonged duration of action is potentially dangerous. The 4 primary pharmacokinetic processes are absorption, distribution, metabolism, and elimination. Liver dysfunction from cirrhosis reduces drug metabolism via hepatic enzymes and reduces drugs' protein binding from hypoalbuminemia. The kidneys eliminate most drugs and/or their metabolites from the body. A decrease in renal function is associated with reduced renal excretion of certain drugs. Normal aging is associated with an increased volume of distribution from the proportional increase in body fat relative to skeletal muscle. Dose reductions should be considered in elderly patients or in those with hepatic/renal impairment. Clevidipine is a short-acting dihydropyridine calcium channel blocker that undergoes rapid hydrolysis by blood and extravascular tissues to inactive metabolites. Its pharmacokinetics are least likely to be affected by aging, renal dysfunction, or hepatic dysfunction. Labetalol is a beta-blocker that undergoes hepatic metabolism. It has increased bioavailability and reduced elimination in elderly patients. Nicardipine is a calcium channel blocker that undergoes extensive first-pass hepatic metabolism. Its clearance may be decreased in patients with hepatic and renal impairment. Sodium nitroprusside is metabolized by combining with hemoglobin. The use of sodium nitroprusside in patients with renal impairment may lead to the accumulation of thiocyanate (one of sodium nitroprusside's metabolites) and subsequent toxicity. Enalaprilat is an angiotensin-converting enzyme

inhibitor that is excreted by the kidneys. Its dose should be reduced in patients with renal dysfunction.

References:

1. Amarapurkar D, Jalan R, Guan R, Kwo P. Improving survival in patients with decompensated cirrhosis. *Int J Hepatol*. 2011; 2011:565108.
2. Espinosa A, Ripollés-Melchor J, Casans-Francés R, et al; Evidence Anesthesia Review Group. Perioperative use of clevidipine: a systematic review and meta-analysis. *PLoS One*. 2016 Mar 28;11(3): e0150625.
3. Padilla Ramos A, Varon J. Current and newer agents for hypertensive emergencies. *Curr Hypertens Rep*. 2014 Jul;16(7):450.
4. Reeve E, Trenaman SC, Rockwood K, Hilmer SN. Pharmacokinetic and pharmacodynamic alterations in older people with dementia. *Expert Opin Drug Metab Toxicol*. 2017 Jun;13(6):651-668.

2. Rationale

Answer: B

Vancomycin is nephrotoxic, not hepatotoxic; therefore, there is no additive toxic effect to the liver with concomitant administration of vancomycin and micafungin. Co-administration of meropenem and valproic acid produces a clinically significant rapid decline in valproic acid concentrations, which can lead to breakthrough seizures. If the combination cannot be avoided, an alternative antiepileptic medication should be administered. There are no documented reports of drug-drug interactions between micafungin and meropenem.

References:

1. Coves-Orts FJ, Barras-Blasco J, Navarro-Ruiz A, Murcia-Lopez A, Palacios-Ortega F. Acute seizures due to a probable interaction between valproic acid and meropenem. *Ann Pharmacother*. 2005 Mar;39(3):533-537.

2. Haroutiunian S, Ratz Y, Rabinovich B, Adam M, Hoffman A. Valproic acid plasma concentration decreases in a dose-independent manner following administration of meropenem: a retrospective study. *J Clin Pharmacol*. 2009 Nov; 49(11):1363-1369.

3. Nacarkucuk E, Saglam H, Okan M. Meropenem decreases serum level of valproic acid. *Pediatr Neurol*. 2004 Sep;31(3):232-234.

4. Spriet I, Goyens J, Meersseman W, Wilmer A, Willems L, van Paesschen W. Interaction between valproate and meropenem: a retrospective study. *Ann Pharmacother*. 2007 Jul;41(7):1130-1136.

3. Rationale

Answer: B

Studies have shown that patients exhibiting augmented renal clearance (ARC) tend to be younger than age 50 years, male, and have a recent history of trauma and lower critical illness severity scores such as Sequential Organ Failure Assessment (SOFA) score, Simplified Acute Physiology Score (SAPS) II, or Acute Physiology and Chronic Health Evaluation (APACHE) II. The median age of patients with ARC is between 34 and 50 years. ARC is more closely aligned with trauma than with medical causes of illness. The 27-year-old man with a traumatic brain injury is more at risk for ARC than the other patients.

Reference:

1. Bilbao-Meseguer I, Rodriguez-Gascon A, Barrasa H, Isla A, Solinis MA. Augmented renal clearance in critically ill patients: a systematic review. *Clin Pharmacokinet*. 2018 Sep;57(9):1107-1121.

4. Rationale

Answer: E

Resistance to nondepolarizing neuromuscular blockers (eg, rocuronium, vecuronium) occurs with upregulation of skeletal muscle acetylcholine receptors, which occurs in the setting of

burns, chronic disuse, denervation, direct muscle trauma, or certain medications. Patients who chronically take carbamazepine or phenytoin have upregulation of their skeletal muscle acetylcholine receptors. When rocuronium is administered, shorter durations of neuromuscular blockade can occur, and higher infusion rates may be necessary to achieve the desired clinical effect. Clindamycin causes end-plate ion channel blockade and potentiates the action of rocuronium. Calcium channel blockers such as nicardipine have been shown to increase the neuromuscular effect of depolarizing and nondepolarizing neuromuscular blockers. Midazolam and nitric oxide do not impact the paralytic response to rocuronium.

References:

1. Capuano A, Sullo MG, Rafaniello C, et al. Complete resistance after maximal dose of rocuronium. *J Pharmacol Pharmacother.* 2015 Jul-Sep;6(3):175-178.
2. Lee JH, Lee SI, Chung CJ, et al. The synergistic effect of gentamicin and clindamycin on rocuronium-induced neuromuscular blockade. *Korean J Anesthesiol.* 2013 Feb;64(2):143-151.
3. Lee SY, Kim YH, Ko YK, et al. Effects of nicardipine on the onset time and intubation conditions of rocuronium-induced neuromuscular blockade. *J Clin Anesth*. 2016 Aug; 32:112-118.

5. Rationale

Answer: E

Tramadol is a prodrug that is metabolized by cytochrome P450 enzymes CYP2D6 and CYP3A4 to its more potent opioid analgesic metabolites. Because of this, the analgesic potency of a given dose of tramadol is influenced by a patient's cytochrome P450 activity, with poor metabolizers experiencing little conversion to the active metabolite and patients with a high metabolic profile experiencing the greatest analgesic effects. Fluoxetine is a strong inhibitor of CYP2D6 and thus significantly reduces the metabolism of tramadol to its active metabolite. Metoclopramide is a dopamine receptor antagonist;

it has been shown to provide a synergistic analgesic effect when co-administered with tramadol. Gabapentin is a structural analog of γ-aminobutyric acid (GABA) that offers a synergistic analgesic effect when co-administered with tramadol. Acetaminophen activates descending central nervous system serotonergic inhibitory pathways and does not reduce tramadol's effectiveness. Droperidol is an antiemetic that blocks dopamine stimulation in the chemoreceptor trigger zone. It has not been shown to decrease tramadol's effectiveness.

References:

1. Frost DA, Soric MM, Kaiser R, Neugebauer RE. Efficacy of tramadol for pain management in patients receiving strong cytochrome P450 2D6 inhibitors. *Pharmacotherapy*. 2019 Jun;39(6):724-729.

2. Gong L, Stamer UM, Tzvetkov MV, Altman RB, Klein TE. PharmGKB summary: tramadol pathway. *Pharmacogenet Genomics*. 2014 Jul;24(7):374-80.

3. McKeon GP, Pacharinsak C, Long CT, et al. Analgesic effects of tramadol, tramadol-gabapentin, and buprenorphine in an incisional model of pain in rats (Rattus norvegicus). *J Am Assoc Lab Anim Sci*. 2011 Mar;50(2):192-197.

4. Miotto K, Cho AK, Khalil MA, Blanco K, Sasaki JD, Rawson R. Trends in tramadol: pharmacology, metabolism, and misuse. *Anesth Analg*. 2017 Jan;124(1):44-51.

5. Pang W, Liu YC, Maboudou E, et al. Metoclopramide improves the quality of tramadol PCA indistinguishable to morphine PCA: a prospective, randomized, double blind clinical comparison. *Pain Med*. 2013 Sep;14(9):1426-134.

6. Rationale

Answer: A

Many medications can cause lactic acidosis, most notably the biguanide class of antihyperglycemics. Another common cause is nucleoside reverse transcriptase inhibitors (NRTIs) used to treat HIV infection. Abacavir and lamivudine are both NRTIs

that are commonly included in combination anti-HIV formulations.

References:

1. Luft, FC. Lactic acidosis update for critical care clinicians. *J Am Soc Nephrol*. 2001 Feb;12 Suppl 17: S15-S19.
2. Pham AQ, Xu LH, Moe OW. Drug-induced metabolic acidosis. *F1000Res*. 2015 Dec 16;4: F1000.
3. Smith ZR, Horng M, Rech MA. Medication-induced hyperlactatemia and lactic acidosis: a systematic review of the literature. *Pharmacotherapy*. 2019 Sep;39(9):946-963.

7. Rationale

Answer: C

This patient has acute eosinophilic pneumonia due to daptomycin. Her renal failure was likely a risk factor. CT angiogram shows peripheral, patchy infiltrates with central sparing, which is a classic finding in eosinophilic pneumonia, but radiographic findings can vary. Peripheral blood count shows significant absolute count of eosinophils. While amiodarone-induced pneumonitis may cause similar findings, it usually presents months after initiation and does not resolve rapidly after discontinuation of the drug because of its long half-life. Based on the CT findings and the absence of bloody fluid on serial aliquots of saline during bronchoscopy, alveolar hemorrhage is less likely. The CT findings are also inconsistent with pulmonary edema, and additional furosemide is unlikely to have any benefit. While eosinophilic vasculitides are in the differential diagnosis, the most likely diagnosis is drug toxicity. Starting rituximab is inadvisable in the setting of an unremarkable autoimmune panel and absence of alveolar hemorrhage. In addition, the therapeutic effect of rituximab occurs after 4 to 6 weeks whereas most patients with daptomycin pneumonitis improve within 24 hours to a week after its discontinuation.

References:

1. Higashi Y, Nakamura S, Tsuji Y, et al. Daptomycin-induced eosinophilic pneumonia, and a review of the published literature. *Intern Med*. 2018 Jan 15;57(2):253-258.

2. Schwaiblmair M, Berghaus T, Haeckel T, Wagner T, von Scheidt W. Amiodarone-induced pulmonary toxicity: an under-recognized and severe adverse effect? *Clin Res Cardiol*. 2010 Nov;99(11):693-700.

3. Sweidan AJ, Singh NK, Dang N, Lam V, Datta J. Amiodarone-induced pulmonary toxicity: a frequently missed complication. *Clin Med Insights Case Rep*. 2016 Oct 9; 9:91-94.

4. Uppal P, LaPlante KL, Gaitanis MM, Jankowich MD, Ward KE. Daptomycin-induced eosinophilic pneumonia: a systematic review. *Antimicrob Resist Infect Control*. 2016 Dec 12; 5:55.

8. Rationale

Answer: B

Epinephrine has been retrospectively associated with cardiac arrest prevention in the management of anaphylactic shock. In addition to restoring vascular tone, epinephrine is lifesaving because it also increases cardiac output and stabilizes mast cells from further degranulation. Antihistamines (H1 or H2 antagonists) may reduce symptom burden but have not been definitively associated with a mortality benefit. Glucocorticoids may confer a reduction in late anaphylactic symptoms in the setting of a biphasic anaphylactic episode, but evidence is lacking to substantiate benefit. In addition to epinephrine, large volumes of IV fluids should be administered.

Reference:

1. Simons FER, Ardusso LRF, Bilo MB, et al; World Allergy Organization. World Allergy Organization anaphylaxis guidelines: summary. *J Allergy Clin Immunol*. 2011 Mar;127(3):587-593.

9. Rationale

Answer: B

This patient likely has overdosed on a tricyclic antidepressant (TCA). Signs of TCA poisoning include sedation, anticholinergic toxicity (hyperthermia, flushing, dry skin, ileus, urinary retention, dilated pupils), hypotension, and seizures. Cardiac toxicity may be evident on ECG as sinus tachycardia (due to anticholinergic effects), QRS interval prolongation longer than 100 msec (due to sodium channel blockade), prominent R wave in lead aVR of greater than 3 mm, and QT interval prolongation. QRS prolongation longer than 100 msec or R wave greater than 3 mm in lead aVR have been associated with increased risk for ventricular arrhythmias and seizures. ECG shows typical signs of TCA toxicity, including sinus tachycardia, prolonged QRS of approximately 124 msec, prominent R wave in lead aVR, and prolonged QT interval. QRS interval longer than 100 msec is generally considered an indication for sodium bicarbonate treatment, the benefit of which may be related to increased extracellular sodium or increased serum pH, which tends to reduce binding of TCAs to sodium channels. The sodium bicarbonate is often administered as an initial IV bolus of 1 to 2 mEq/kg followed by a continuous IV infusion of 150 mEq of sodium bicarbonate mixed in 5% dextrose. Narrowing of the QRS interval and decrease in the R wave amplitude in lead aVR may be seen after the initial sodium bicarbonate bolus. The pH should be followed closely during sodium bicarbonate treatment; a goal pH of 7.50 to 7.55 is typically targeted. IV cyproheptadine is used for potential serotonin syndrome. Although she does have some features of serotonin syndrome (flushing, dilated pupils), she does not have other classic findings, such as clonus and hyperreflexia, so cyproheptadine is not indicated. Although the QT interval is prolonged, her serum magnesium level is within normal limits, so magnesium would be of little benefit. Sodium bicarbonate is a better first option considering the ECG changes consistent with TCA toxicity. She does not have typical signs of opioid toxicity, so naloxone is not indicated.

References:

1. Kerr GW, McGuffie AC, Wilkie S. Tricyclic antidepressant overdose: a review. *Emerg Med J*. 2001 Jul;18(4):236-241.

2. Mokhlesi B, Leikin JB, Murray P, Corbridge TC. Adult toxicology in critical care: part II: specific poisonings. *Chest*. 2003 Mar;123(3):897-922.

10. Rationale

Answer: C

The drugs most likely to be eliminated by continuous venovenous hemodiafiltration therapy are small molecules with low protein binding and small volume of distribution. Small molecules are more easily cleared through the filters than are large molecules. Low protein binding increases the unbound fraction of drug in the blood for removal during renal replacement therapy. Large molecules, high protein binding, and large volume of distribution all prevent a drug from being eliminated via renal replacement therapy. Drugs with a large molecular size are less likely than drugs with a small molecular size to pass through the filter and be eliminated by renal replacement therapy.

Reference:

1. Choi G, Gomersall CD, Tian Q, Joynt GM, Freebairn R, Lipman J. Principles of antibacterial dosing in continuous renal replacement therapy. *Crit Care Med*. 2009 Jul;37(7):2268-2282.

11. Rationale

Answer: B

Acute carbon monoxide poisoning causes restricted diffusion of the globus pallidus, manifesting as hypodensities of basal ganglia on brain CT. Brain MRI shows hyperintense signals in the basal ganglia on diffusion-weighted imaging with corresponding hypo-intensity on apparent diffusion coefficient, fluid attenuated inversion recovery (FLAIR), and T2-weighted imaging. The most appropriate treatment is hyperbaric oxygen therapy. It seems to be most effective when used early (within the first 6 hours) and may be beneficial in patients with a

markedly elevated carbon monoxide level (eg, > 25%) or in those with loss of consciousness, altered mental status, severe metabolic acidosis, or cardiac ischemia. Pralidoxime is the treatment of choice for organophosphate poisoning; however, this patient does not have any of the expected signs of cholinergic toxicity (bradycardia, miosis, salivation, bronchorrhea, emesis, diarrhea). IV sodium bicarbonate may be considered for salicylate poisoning or tricyclic overdose. However, his ECG does not suggest tricyclic toxicity. Although high-dose naloxone may be required for some opiate overdoses, his presentation does not suggest an opiate toxidrome, which would be characterized by miosis and a low respiratory rate.

References:

1. Rose JJ, Wang L, Xu Q, et al. Carbon monoxide poisoning: pathogenesis, management, and future direction of therapy. *Am J Respir Crit Care Med*. 2017 Mar 1;195(5):596-606.

2. Zibrak EA. Carbon monoxide poisoning. *N Engl J Med*. 1998 Nov 26;339(22):1603-1608.

12. Rationale

Answer: B

High-dose IV insulin and IV dextrose should be administered. Calcium channel blockers inhibit voltage-gated L-type calcium channel opening in myocardial and vascular smooth muscle cells, resulting in negative myocardial inotropy and chronotropy and peripheral vasodilation. Calcium channel blockers also inhibit L-type calcium channels in pancreatic islet cells, reducing insulin secretion, resulting in hyperglycemia and impaired cardiac glucose use. Initial treatment of symptomatic calcium channel blocker overdose involves supportive care with IV fluid administration. IV atropine can be administered to patients with bradycardia and hypotension but is rarely effective. Additional atropine is unlikely to improve this patient's condition. Vasopressor support with agents that have both alpha and beta-1 adrenergic effects, such as norepinephrine or epinephrine, should be initiated in patients

with shock. IV calcium infusions are often administered to patients with calcium channel blocker overdose but are often ineffective at restoring hemodynamic stability. Calcium channel blockers are highly protein bound and typically have a large volume of distribution, making extracorporeal removal by dialysis ineffective. Activated charcoal should be administered to patients with calcium channel blocker overdose but it is maximally effective when administered within 1 hour after ingestion and should not be administered to patients with decreased level of consciousness who may not be able to effectively protect the airway from aspiration. There is a growing body of evidence from case reports and case series demonstrating successful use of rescue hyper-insulin euglycemia therapy in patients with calcium channel blocker overdose who do not respond to initial supportive care. The negative inotropic effects of verapamil may result not only from calcium channel blockade but also from impaired carbohydrate use by the myocardium. High-dose insulin (typically 0.5-2 units/kg/h of regular IV insulin) may improve hemodynamics.

References:

1. Patel NP, Pugh ME, Goldberg S, Eiger G. Hyperinsulinemic euglycemia therapy for verapamil poisoning: a review. *Am J Crit Care*. 2007 Sep;16(5):498-503.

2. St-Onge M, Dubé PA, Gosselin S, et al. Treatment of calcium channel blocker poisoning: a systematic review. *Clin Toxicol (Phila)*. 2014 Nov;52(9):926-944.

13. Rationale

Answer: B

This patient previously took enoxaparin, with a time course consistent with rapid-onset heparin-induced thrombocytopenia (HIT). Treatment of HIT requires discontinuation of heparin products and initiation of a non-heparin anticoagulant to treat the immediate hypercoagulable state. Direct thrombin inhibitors, fondaparinux, and direct oral anticoagulants are effective. Patients should continue taking non-heparin

anticoagulants until platelet count is normal. HIT type I (less than or equal to 20% of patients with HIT) is caused by heparin-induced platelet clumping with no increased risk of thrombosis. HIT type II results from platelet activation by anti-heparin/platelet factor 4 antibody, resulting in a 30-fold increased risk of thrombosis and platelet consumption. Treatment with heparin is more likely than low-molecular-weight heparin to cause HIT (3% vs. 1%). Among patients with HIT, platelet counts typically begin to decrease by at least 50% from baseline 5 to 14 days after heparin exposure. Rapid onset HIT occurs in patients who previously took heparin. The HIT antibody that formed with prior heparin exposure remains active and platelets decrease within 24 hours of heparin re-exposure. When HIT is suspected, the 4Ts score is used to determine the pretest probability of HIT. Scoring is based on timing and potential magnitude of platelet count decrease, presence of thrombosis, and alternative causes of thrombocytopenia. Patients with a 4Ts score below 3 have less than a 5% risk of HIT and require no further testing. Patients with a 4Ts score of 4 or greater should undergo HIT antibody testing.

References:

1. Arepally GM. Heparin-induced thrombocytopenia. *Blood*. 2017 May 25;129(21):2864-2872.
2. Arepally GM, Ortel TL. Clinical practice. Heparin-induced thrombocytopenia. *N Engl J Med*. 2006 Aug 24;355(8):809-817.
3. Lo GK, Juhl D, Warkentin TE, Sigouin CS, Eichler P, Greinacher A. Evaluation of pretest clinical score (4 T's) for the diagnosis of heparin-induced thrombocytopenia in two clinical settings. *J Thromb Haemost*. 2006 Apr;4(4):759-765.

14. Rationale

Answer: A

The pain, agitation/sedation, delirium, immobility, and sleep disruption guidelines recommend an analgesia-first approach to sedation, so an analgesic agent should be incorporated into this

patient's sedation regimen. Dexmedetomidine does not reliably provide deep sedation. The combination of propofol and midazolam is inappropriate because these are both sedatives. Fentanyl and propofol infusions are the most appropriate analgesic/sedative combination for this patient. Fentanyl intermittent boluses would not provide deep sedation/analgesia.

References:

1. Devlin JW, Skrobik Y, Gélinas C, et al. Clinical practice guidelines for the prevention and management of pain, agitation/sedation, delirium, immobility, and sleep disruption in adult patients in the ICU. *Crit Care Med*. 2018 Sep;46(9): e825-e873.
2. Ruokonen E, Parviainen I, Jakob SM, et al; Dexmedetomidine for Continuous Sedation Investigators. Dexmedetomidine versus propofol/midazolam for long-term sedation during mechanical ventilation. *Intensive Care Med*. 2009 Feb;35(2):282-290.

15. Rationale

Answer: D

Optimization of antimicrobials requires an understanding of the pharmacokinetics (what the body does to the drug: absorption, distribution, metabolism, and excretion) and pharmacodynamics (what the drug does to the body: determinants of response) of each medication. This patient is at particularly high risk for rapid elimination of renally eliminated drugs because of factors associated with augmented renal clearance (age, estimated creatinine clearance, and recent traumatic brain injury). Therefore, more frequent dosing is needed for drugs with time-dependent bacterial killing. Beta-lactam antimicrobials exhibit time-dependent bacterial killing, which means the pharmacokinetic/pharmacodynamic (PK/PD) target of interest is the fraction of time during the dosing interval that the drug concentration exceeds the organism's minimum inhibitory concentration (MIC). For this reason, shorter dosing intervals or prolonged infusions for this

medication class are preferred relative to higher doses delivered less frequently. Vancomycin is a time-dependent antimicrobial with the trough concentration or area under the inhibitory curve (AUIC) as the preferred PK/PD targets. High peaks are associated with greater toxicity, so relatively lower doses delivered more frequently are preferred in patients with robust kidney function. Gentamicin is a concentration-dependent antibiotic with a prolonged post-antibiotic effect. The degree to which the maximum concentration exceeds the organism's MIC better predicts the outcome than the trough levels. Persistently elevated trough levels are associated with increased risk of nephrotoxicity. Thus, traditional aminoglycoside dosing several times daily is generally less preferred than pulse dosing, which delivers a higher dose at a less frequent interval to maximize PK/PD.

References:

1. Roberts JA, Lipman J. Pharmacokinetic issues for antibiotics in the critically ill patient. *Crit Care Med*. 2009 Mar;37(3):840-851.

2. Rybak MJ, Le J, Lodise TP, et al. Therapeutic monitoring of vancomycin for serious methicillin-resistant Staphylococcus aureus infections: a revised consensus guideline and review by the American Society of Health-System Pharmacists, the Infectious Diseases Society of America, the Pediatric Infectious Diseases Society, and the Society of Infectious Diseases Pharmacists. *Am J Health Syst Pharm*. 2020 May 19;77(11):835-864.

3. Udy AA, Varghese JM, Altukroni M, et al. Subtherapeutic initial β-lactam concentrations in select critically ill patients: association between augmented renal clearance and low trough drug concentrations. *Chest*. 2012 Jul;142(1):30-39.

Part 17.

Obstetrics

Instructions: For each question, select the most correct answer.

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1. Which of the following physiologic parameters decreases during pregnancy?

   A. Heart rate

   B. Central venous pressure

   C. Erythrocyte volume

   D. Systemic vascular resistance

2. Which of the following tests best distinguishes between acute fatty liver and hemolysis, elevated liver enzymes, and low platelets (HELLP) syndrome in pregnancy?

   A. Serum ammonia level

   B. Serum bilirubin level

   C. Ferritin level

   D. Disseminated intravascular coagulation profile

   E. Liver biopsy

3. A 30-year-old pregnant woman who has had no prenatal care presents to the hospital at 34 weeks' gestation with pedal edema and hypertension and is admitted to the ICU. Initial laboratory findings are BUN 22 mg/dL, creatinine 1.6 mg/dL, ALT 350 U/L, bilirubin 5 mg/dL, ammonia 70 mg/dL, glucose 50 mg/dL, leukocytes 19,000/μL, hemoglobin 9.6 g/dL, platelets 120,000/μL, prothrombin time 20.2 sec, and partial thromboplastin time 45 sec. On physical examination, she is slightly lethargic, with

scattered bilateral basilar rales. A normal-sized liver is appreciated on percussion and an appropriately sized uterus is palpable and nontender. Fetal ultrasonography is unremarkable except for an unexpected placenta previa. Which of the following is the most likely diagnosis?

A. Acute viral hepatitis

B. Life-threatening preeclampsia

C. Acute hemolysis, elevated liver enzymes, and low platelets (HELLP) syndrome

D. Intrahepatic cholestasis of pregnancy

E. Acute fatty liver of pregnancy

Part 17 Answers:

Obstetrics

1. Rationale

Answer: D

Cardiac output and plasma volume increase during normal pregnancy, accompanied by decreases in systolic, diastolic, and mean blood pressure due to decreased systemic vascular resistance.

Reference:

1. Miller RD, Eriksson LI, Fleisher LA, Wiener-Kronish JP, Cohen NH, Young WL. *Miller's Anesthesia*. 8th ed. Saunders. 2014.

2. Rationale

Answer: A

During pregnancy, women may develop acute fatty liver, eclampsia, or hemolysis, elevated liver enzymes, and low platelets (HELLP) syndrome; they may also have overlapping conditions. Distinguishing among acute fatty liver and other conditions is important because subcapsular hematoma and rupture are common and can quickly progress to fulminant liver failure. Moreover, acute fatty liver can affect the fetus with significant muscle dysfunction and is thought to be associated with sudden infant death syndrome. Extreme vigilance is required. Since usual onset is around 36 weeks' gestation, treatment is delivery of the fetus. Increased ammonia level is highly suspicious for acute fatty liver when accompanied by clinical progression. Disseminated intravascular coagulation, anemia, and hyperbilirubinemia are universally common in acute fatty liver, eclampsia, and HELLP syndrome.

References:

1. Guntupalli SR, Steingrub J. Hepatic disease and pregnancy: an overview of diagnosis and management. *Crit Care Med*.

2005 Oct;33(10 Suppl): S332-S339.

2. Vigil-De Gracia P. Acute fatty liver and HELLP syndrome: two distinct pregnancy disorders. *Int J Gynaecol Obstet*. 2001 Jun;73(3):215-220.

3. Rationale

Answer: E

It can be difficult to distinguish among the possible etiologies of liver disease in pregnancy. This patient's laboratory findings are most consistent with acute fatty liver of pregnancy, which often presents in a fulminant manner. Some of the laboratory findings in acute fatty liver, eclampsia/preeclampsia, and hemolysis, elevated liver enzymes, and low platelets (HELLP) syndrome are shown below. In acute fatty liver of pregnancy, the liver is normal-sized or small, and AST levels rise to higher levels than in eclampsia/preeclampsia or HELLP syndrome. Signs of hepatic failure, such as higher bilirubin level, hypoglycemia, elevated ammonia level, and disseminated intravascular coagulation (DIC), are often present. Hepatitis in pregnancy presents with laboratory findings like those of a nonpregnant patient, with higher levels of transaminases, absence of DIC, and often an enlarged liver. Intrahepatic cholestasis of pregnancy is a benign condition associated with pruritus, jaundice, elevated transaminase levels, elevated bilirubin level, and normal prothrombin time. The most difficult distinction in a pregnant patient is liver disease due to eclampsia/severe preeclampsia and HELLP syndrome. Ammonia level may be particularly helpful in these situations. Consultation with a maternal fetal specialist is warranted for consideration of immediate delivery.

Table 1. Differentation of liver disease

Test	Acute fatty liver	Eclampsia/pre-eclampsia	HELLP
Liver size	Small	Normal or ↑	Normal or ↑
ALT (usual range)	300 U/L	60 U/L	150 U/L
Bilirubin	↑	Normal, mild ↑	Normal, mild ↑
Glucose	↓	Normal	Normal
Ammonia	↑	Normal	Normal
DIC	75%	Rare	20-40%

References:

1. Guntupalli SR, Steingrub J. Hepatic disease and pregnancy: an overview of diagnosis and management. *Crit Care Med*. 2005 Oct;33(10 Suppl): S332-S339.

2. Rahman TM, Wendon J. Severe hepatic dysfunction in pregnancy. *QJM*. 2002 Jun;95(6):343-357.

3. Wakim-Fleming J, Zein NN. The liver in pregnancy: disease vs benign changes. *Cleve Clin J Med*. 2005 Aug;72(8):713-721.

Part 18.

Research, Administration, and Ethics

Instructions: For each question, select the most correct answer.

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1. Post-intensive care syndrome (PICS) encompasses impairments in cognition, mental health, and physical health in survivors of critical illness. Families can also experience PICS, in which case it is called PICS-F and includes adverse psychological outcomes such as anxiety, acute stress disorder, posttraumatic stress disorder, depression, and complicated grief. Which of the following is the most appropriate strategy for reducing PICS-F?

   A. The family should not be allowed to participate in the patient's care because increased responsibility results in stress and anxiety.

   B. The family should be given frequent updates in a language they understand and from consistent team members, including inviting them to be part of interdisciplinary rounds.

   C. Visiting hours should be restricted to limit the family's exposure to the stressful critical care environment.

   D. The clinical team's decisions should include the patient's input, if possible, but not family input, to avoid overwhelming the family.

2. An ICU multi-professional team recognizes the importance of high-quality sleep to patients' recovery process. The team decides to review current evidence and develop a multicomponent sleep hygiene protocol or bundle to implement in the ICU. Which of the

following strategies should be considered when developing this protocol or bundle?

A. Infusing propofol to improve sleep at night in mechanically ventilated patients

B. Routinely using physiologic monitoring to determine whether a patient is awake or asleep

C. Limiting interruptions at night

D. Relying on clinician observation of patients' sleep and wake times to determine sleep efficacy

3. A researcher would like to compare the 30-day risk of death between patients undergoing continuous hemodialysis and those undergoing intermittent hemodialysis. Which of the following statements regarding this survival analysis is correct?

A. Patients whose survival status is unknown at the end of the study period should be included in the analysis.

B. Kaplan-Meier curves can be used to examine risk-adjusted survival time.

C. The appropriate statistical test of difference for survival curves is one-way analysis of variance.

D. Logistic regression allows additional covariates to be added to survival models.

E. The log-rank test and the Cox proportional hazards model assume that the hazard ratio is constant over time.

4. A 62-year-old man with amyotrophic lateral sclerosis (ALS) was admitted to the ICU for acute respiratory failure secondary to aspiration pneumonitis. He was intubated and on mechanical ventilator support for 36 hours. The day after extubation his physician approaches him to discuss the progression of his ALS and options for medical care at the end of life. Which

of the following end-of-life care topics is most appropriate for the physician to prioritize in this discussion?

A. Advance care planning

B. Withholding and/or withdrawing life-sustaining treatment

C. Physician-assisted suicide/euthanasia

D. Do-not-resuscitate order

5. In the emergency department, 2 patients with COVID-19 require ICU resources, including mechanical ventilation. Only 1 ICU bed is available. One patient is an 84-year-old woman who has lived in a nursing facility since having a stroke 4 years ago. Her husband died 10 years ago, and she has 2 children, aged 54 and 58, who live in the area and visit her monthly. She has significant cardiac and pulmonary disease. She is normally cognitively intact but cannot walk because of a dense hemiparesis and paralysis from the stroke. The other patient is a 32-year-old woman with no previous medical issues. She has a husband and 3 children under age 14. She is an elementary school teacher who is very active in her church and community. Which of the following is the most appropriate course of action?

   A. Admit the 84-year-old woman because she has a poorer quality of life, is less likely to survive, and needs the ICU resources more.

   B. Admit the 32-year-old woman because she has young children and more productive years to live and is more likely to survive with or without the ICU resources.

   C. Contact the hospital's triage team to make the decision based on the hospital's established allocation criteria.

D. Assign a priority score to each patient based on likelihood of survival and preexisting conditions and admit the patient with the higher score.

Part 18 Answers:

Research, Administration, and Ethics

1. Rationale

Answer: B

Post-intensive care syndrome (PICS) is increasingly recognized as a condition occurring in patients who survive an ICU stay. PICS can affect both patients and their families. Every step of ICU care plays a role in the long-term effects of the ICU stay on both patient and families. A comprehensive approach to the physical, social, and psychological outcomes is critical, including an evaluation of routine practices related to families in the ICU.

References:

1. Davidson JE, Jones C, Bienvenu OJ. Family response to critical illness: post-intensive care syndrome - family. *Crit Care Med*. 2012 Feb;40(2):618-624.
2. Iwashyna T, Netzer G. The burdens of survivorship: an approach to thinking about long-term outcomes after critical illness. *Semin Respir Crit Care Med*. 2012 Aug;33(4):327-338.

2. Rationale

Answer: C

Routine use of physiologic sleep monitoring of critically ill adults is not recommended because of the very low quality of evidence to support it and high costs of resources to implement it. However, clinicians should routinely inquire about patients' sleep using a validated assessment tool (eg, Richards-Campbell Sleep Questionnaire) or an informal bedside assessment (asking patients about their sleep). Propofol is not recommended for promoting sleep because of the low quality of evidence to support it. A randomized controlled trial comparing propofol to placebo demonstrated no improvement in sleep. Although

studies of sleep protocols and bundles varied in their interventions, a conditional recommendation was made based on the potential benefits (eg, improved quality of sleep, reduced delirium) and the minimal anticipated harm. Interventions that may be beneficial include those that target patient-reported factors of sleep disruption such as limiting nighttime interruptions (eg, nursing care, procedures, medication administration, vital sign measurement, noise, light).

References:

1. Devlin JW, Skrobik Y, Gélinas C, et al. Clinical practice guidelines for the prevention and management of pain, agitation/sedation, delirium, immobility, and sleep disruption in adult patients in the ICU. *Crit Care Med*. 2018 Sept;46(9): e825-e873.
2. Watson PL, Pandharipande P, Gehlback BK, et al. Atypical sleep in ventilated patients: empirical electroencephalography findings and the path toward revised ICU sleep scoring criteria. *Crit Care Med*. 2013 Aug;41(8):1958-1967.
3. Weinhouse G, Skrobik Y. The importance of sleep for recovery from critical illness. In: Posa P, Singh J, Stollings J, eds. *ICU Liberation*. 2nd ed. Society of Critical Care Medicine; 2020.

3. Rationale

Answer: E

Survival analyses are used to compare the risks of death or other events between 2 or more groups where the risk changes with time. In these analyses, the study period must be clearly defined, and cases with missing end points are censored. Kaplan-Meier curves can be used to estimate survival but are not risk adjusted. Survival curves are compared between groups using the log-rank test rather than one-way analysis of variance. The Cox proportional hazards model allows additional covariates to be added to the model, while logistic regression uses a basic logistic function to model a binary dependent variable. Both the log-rank test and the Cox proportional

hazards model assume that the hazard ratio comparing 2 groups remains constant over the study period.

Reference:

1. Bewick V, Cheek L, Ball J. Statistics review 12: survival analysis. *Crit Care.* 2004 Oct;8(5):389-394.

4. Rationale

Answer: A

The most appropriate topic to prioritize is advance care planning, which supports patient autonomy, facilitates decision-making, and promotes better end-of-life care regardless of age or health status. Advance care planning allows this patient to identify who he would want to make decisions for him should he lose decision-making capacity, consider which factors contribute to his quality of life, identify critical abilities he is not willing to live without, and express his wishes based on these values with respect to specific medical interventions. A proactive discussion about specific medical interventions allows clinicians to address patient concerns and expectations. Any misunderstandings about specific medical conditions, prognoses, or interventions can be clarified. Advance care planning often results in the execution of an advance directive, in which the patient expresses wishes for end-of-life care, appoints surrogate decision-makers, and completes documentation that allows or disallows specific medical interventions. Patients who have appropriate decision-making capacity have the right to decline medical treatment or request that it be stopped. If a patient loses decision-making capacity, his/her surrogate may request termination of medical treatment. When an intervention will not improve the patient's condition or no longer assists in achieving the patient's goals for care or quality of life, it is ethically appropriate for a physician to recommend that the intervention be withheld or withdrawn. While amyotrophic lateral sclerosis is a progressive life-limiting disease, this patient has been liberated from mechanical ventilator support, and a conversation to elicit his wishes should be attempted before suggesting withholding treatment.

Permitting a physician to engage in assisted suicide or euthanasia would ultimately cause more harm than good. Physician-assisted suicide and euthanasia are fundamentally incompatible with the physician's role as healer. It would be impossible to control and would pose serious societal risks. Instead of engaging in assisted suicide or euthanasia, physicians must aggressively respond to patients' end-of-life needs. Physicians should not abandon a patient once a cure is determined to be impossible, must respect patient autonomy, must provide good communication and emotional support, and must provide appropriate comfort care and adequate pain control. Do-not-resuscitate (DNR) orders can be appropriate for any patient medically at risk of cardiopulmonary arrest, regardless of the patient's age or health status. Do-not-attempt-resuscitation (DNAR) orders apply in any care setting, in or out of hospital, within the constraints of applicable law. If a patient has a cardiopulmonary arrest when there is no DNAR order in the medical record, resuscitation should be attempted if it is medically appropriate. If it is found afterward that the patient would not have wanted resuscitation, the attending physician should order that resuscitation efforts be stopped. Physicians should address the potential need for resuscitation early in the disease course, while the patient has decision-making capacity, and should encourage the patient to choose a surrogate. A DNAR order could be discussed during this conversation, but this recommendation should be informed by the patient's wishes for end-of-life care.

Reference:

1. American Medical Association. Code of Medical Ethics. Chapter 5. Opinions on caring for patients at the end of life. Opinion 5.1. Advance care planning. Opinion 5.3. Withholding or withdrawing life-sustaining treatment. Opinion 5.4 Orders not to attempt resuscitation (DNAR). Opinion 5.7. Physician-assisted suicide. Opinion 5.8. Euthanasia. 2016. Accessed March 8, 2022. https://www.ama-assn.org/sites/default/files/media-browser/code-of-medical-ethics-chapter-5.pdf

5. Rationale Answer: C

The COVID-19 pandemic has stressed hospitals and ICUs throughout the world. The number of patients requiring ICU resources has challenged clinicians needing to make decisions about resource use in a fair and ethical manner. Recommendations from clinical ethics experts outlining best practices for decision-making include 1) decision-making by a team not involved in clinical care of either patient, 2) evaluation of likelihood of survival, preexisting conditions, and saving the most life-years, and 3) preexisting policies for triage in extenuating circumstances. The most appropriate course of action incorporates the triage team. While assigning a priority score based on likelihood of survival and preexisting conditions to each patient and admitting the patient with the higher score is consistent with evaluating for likelihood of survival and most life-years, this scoring should not be done by the treating clinician because it can result in biased decision-making and moral distress. Admitting the 84-year-old woman based on poorer quality of life and less likelihood of survival or admitting the 32-year-old woman based on her age, number and age of her children, and productive years left to live is based on preexisting value judgements about the worth of life, prediction of need, and survival. While these facets should be considered, the clinician who will ultimately cares for the admitted patient should not make these decisions because it is not a systematic, objective approach and it may cause the clinician moral distress. Separation of the triage role from the clinician role enhances objectivity, avoids conflict of commitment, and minimizes clinician moral distress.

References:

1. Biddison ELD, Gwon HS, Schoch-Spana M, et al. Scarce resource allocation during disasters: a mixed-method community engagement study. *Chest*. 2018 Jan;153(1):187-195.

2. National Institute for Health and Care Excellence. COVID-19 rapid guideline: critical care in adults. Last

updated February 12, 2021. Accessed March 8, 2022. https://www.nice.org.uk/guidance/ng159

3. Sprung CL, Joynt GM, Christian MD, Truog RD, Rello J, Nates JL. Adult ICU triage during the coronavirus disease 2019 pandemic: Who will live and who will die? Recommendations to improve survival. *Crit Care Med*. 2020 Aug;48(8):1196-1202.

4. White DB, Lo B. A framework for rationing ventilators and critical care beds during the COVID-19 pandemic. *JAMA*. 2020 May 12;323(18):1773-1774.

5. White DB, Katz MH, Luce JM, Lo B. Who should receive life support during a public health emergency? Using ethical principles to improve allocation decisions. *Ann Intern Med*. 2009 Jan 20;150(2):132-138.

Part 19.

Critical Care Ultrasound

Instructions: For each question, select the most correct answer.

1. A 67-year-old man is undergoing radiofrequency ablation for persistent atrial fibrillation. During the ablation he develops tachycardia and hypotension. Heart chamber pressures are right atrial pressure 19 mm Hg, right ventricular diastolic pressure 17 mm Hg, and pulmonary artery wedge pressure 15 mm Hg. Echocardiography is most likely to show

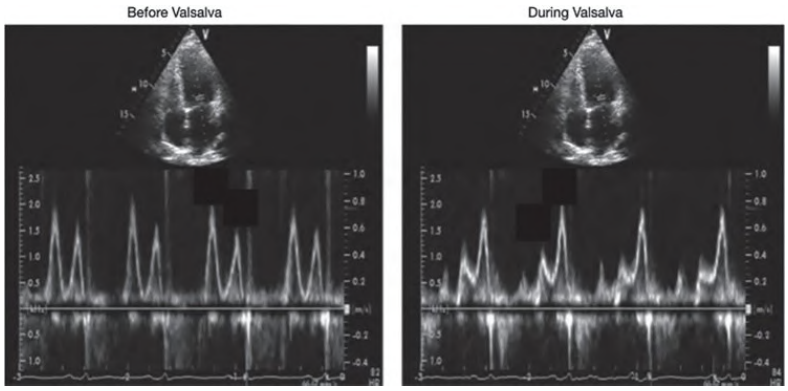
   A. right atrial diastolic collapse.

   B. right atrial systolic collapse.

   C. left atrial diastolic collapse.

   D. left atrial systolic collapse.

2. A 67-year-old woman presents with dyspnea on exertion. On transthoracic echocardiography, left ventricular ejection fraction is 60%. Additional images are shown below. She most likely has which of the following diastolic dysfunction grades and left atrial pressures?



   A. Grade 1 diastolic dysfunction with elevated left atrial pressures

B. Grade 1 diastolic dysfunction with normal left atrial pressures

C. Grade 2 diastolic dysfunction with elevated left atrial pressures

D. Grade 2 diastolic dysfunction with normal left atrial pressures

E. Grade 3 diastolic dysfunction with normal left atrial pressures

Part 19 Answers:

Critical Care Ultrasound

1. Rationale

Answer: A

Equalization of central pressures occurs during cardiac tamponade due to increased pericardial pressure. On echocardiogram, the right atrium will show collapse or inversion during diastole due to the low chamber pressure at this time. Because of its relatively higher pressures and the variability by which it is enclosed by the pericardium, the left atrium inverts much less.

Reference:

1. Appleton C, Gilliam L, Koulogiannis K. Cardiac tamponade. *Cardiol Clin*. 2017 Nov;35(4):525-537.

2. Rationale

Answer: C

Diastolic dysfunction (DD) occurs when the left ventricular myocardium is noncompliant and cannot accept blood return normally from the left atrium. DD is evaluated by multiple parameters on echocardiography, specifically mitral inflow velocity, mitral valve tissue velocity (by tissue Doppler), left atrial size, pulmonary vein flow, and pulmonary artery systolic pressure. Mitral inflow velocity is assessed by pulsed wave Doppler on apical 4-chamber view, positioning the sample volume at the tip of the mitral leaflets. The upward deflections are identified as the E and A waves during diastole when blood fills the ventricle. The first peak is called the E wave for early diastolic filling and the second peak is called the A wave for atrial contraction. There are 3 grades of DD, based on mitral inflow velocity: Grade I (impaired relaxation) is a normal finding, occurring in nearly everybody by age 60. The E wave velocity is reduced, causing E/A reversal (ratio < 1.0). Left atrial pressures are typically normal. Grade II (pseudo-normal)

is pathologic and results in elevated left atrial pressures. The E/A ratio is normal (0.8 ± 1.5). An E/A ratio less than 1 with Valsalva is a major clue to the presence of grade II DD with elevated left atrial pressures. Grade III (restrictive) results in significantly elevated left atrial pressures. The E/A ratio is greater than 2 and can change to less than 1.0 with Valsalva; in fixed restrictive, there are no mitral inflow velocity changes with Valsalva. This patient's initial mitral inflow pattern shows an E/A ratio greater than 1, which is inconsistent with grade I DD. Patients with grade I DD typically have normal left atrial pressures. The mitral inflow pattern goes from normal to grade 1 with Valsalva, consistent with pseudo-normal DD with elevated left atrial pressures. Because of the change in mitral inflow velocity with Valsalva, left atrial pressure is elevated. The E/A ratio is not greater than 2.0; restrictive DD has elevated left atrial pressure.

Reference:

1. Nagueh SF, Smiseth OA, Appleton CP, et al. Recommendations for the evaluation of left ventricular diastolic function by echocardiography: an update from the American Society of Echocardiography and the European Association of Cardiovascular Imaging. *Eur Heart J Cardiovasc Imaging*. 2016 Dec;17(12):1321-1360.

Part 20.

Epidemiology/Outcomes

Instructions: For each question, select the most correct answer.

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1. Which of the following statistical tests should be performed before considering the statistical significance of any conclusion of a study using a retrospective observational cohort design of a common disease in a diverse population with many subjects?

   A. Wilcoxon rank sum test for stratification

   B. Multiple logistic regression to control for confounding variables

   C. Chi-square analysis to assess for similarities between groups

   D. t test to assess for background changes over time

   E. Cronbach alpha to assess intersubject agreement for validity

2. As part of a research project, 100 critically ill patients with sepsis are prospectively grouped into those who require renal replacement therapy and those who do not. Approximately 90 days after hospital admission, the patients are contacted by phone to evaluate their functional status. The results of these telephone conversations are compared between the two groups. Which of the following best describes the type of clinical trial that is being used to test the study hypothesis?

   A. Randomized controlled trial

   B. Crossover study

C. Case-control study

D. Cohort study

E. Cross-sectional study

Part 20 Answers:

Epidemiology/Outcomes

1. Rationale

Answer: B

Regression is often used to control for confounders when study design prevents prospective randomization. Logistic regression is appropriate for dichotomous or categorical variables, and linear regression is used for continuous variables. Multiple logistic regression is ideal to control for the presence of multiple confounders in retrospective studies. Wilcoxon rank sum test (and most tests with someone's name) are used for nonparametric statistical analyses in studies with a small number of participants, typically 30 or less. Chi-square analyses compare the counts of categorical variables. The t test compares the means of 2 continuous variables. Cronbach alpha is used to measure the internal validity of a scale score.

Reference:

1. Motulsky H. *Intuitive Biostatistics: A Nonmathematical Guide to Statistical Thinking*. 4th ed. Oxford University Press; 2017:379-394.

2. Rationale

Answer: D

The planning and conduct of research studies should include a consideration of appropriate study design. In general, research studies can be classified as either observational or interventional. Observational studies involve merely the observation of defined outcomes in specific study groups. Interventional studies involve the application of an intervention and then an evaluation of an outcome derived from that intervention. Observational studies include cohort studies, case-control studies, cross-sectional studies, case series, and case reports. This study is a cohort study. In the Roman legions, a cohort was a group of soldiers who all lived and fought

together in war and peacetime. Cohort studies are prospective research studies in which subjects in a group all have a defined condition or receive a specific treatment. Cohorts are followed over time and the study outcome is compared between different cohorts. In this study, patients with sepsis are grouped into 2 cohorts, one that requires renal replacement therapy and one that does not. The study outcome, functional status, is then compared between the 2 cohorts. A randomized controlled trial is an interventional study that has a control group (no intervention) and the subjects are randomized to either the intervention or control group to reduce the potential for confounding bias. A crossover study is another type of interventional study, typically of 2 or more different interventions, where subjects receive 1 intervention followed by the second intervention. In case-control studies, subjects with different clinical outcomes are first identified, then factors that may have contributed to those outcomes are retrospectively evaluated. Cross-sectional studies evaluate a large population at a single point in time to evaluate which exposures are present.

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